Behavioral Asset Pricing

PhD Lecture Notes

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Chapter 1

Introduction

Every model of asset prices makes claims that data can check. The frictionless rational model—agents with correct beliefs and sensible preferences, trading in markets without impediments—delivers a set of sharp predictions: aggregate market prices should track the rational forecast of future cash flows, expected returns in the cross-section should line up with risk, and investors should hold diversified portfolios and trade only on information. This chapter assembles the facts against which those predictions are judged. None of the facts, taken alone, is a verdict against the rational framework; rational interpretations exist for essentially all of them, and we will state the leading one in each case. What the facts establish, jointly, is the agenda of the rest of this book: a large body of empirical regularities that are awkward for the frictionless rational benchmark, a set of investor behaviors that plausibly generate them, and a set of constraints that prevent rational traders from eliminating them.

The chapter proceeds in four steps. Section 1.1 defines behavioral finance and situates it among the paradigms of modern finance. Section 1.2 states the two classical objections to the field and the responses to each. Section 1.3 lays out the road ahead, and Sections 1.4–1.7 then present the facts: the aggregate stock market (Section 1.4), the cross-section of stock returns (Section 1.5), bubbles (Section 1.6), and the trading and portfolio behavior of individual and institutional investors (Section 1.7).

1.1 What Is Behavioral Finance?

Definition 1.1 (Deviation from full rationality)**.** *Behavioral finance investigates whether some financial phenomena are the result of less than fully rational thinking.*

Definition 1.2 (Psychological realism)**.** *Behavioral finance seeks to understand financial phenomena using models that are* psychologically more realistic *than their predecessors.*

These two definitions, in the spirit of the surveys by Barberis and Thaler (2003) and Ritter (2003), are complementary rather than competing. Definition 1.1 characterizes the field by its object of study—departures from full rationality—while Definition 1.2 characterizes it by its method: whatever the phenomenon, the modeling commitment is to assumptions about human judgment and preference that experimental and survey evidence can support.

To see what is at stake, it helps to place behavioral finance among the three paradigms that organize modern finance. *Traditional finance* studies agents who are fully rational and operate in markets without frictions; its workhorses are the capital asset pricing model (CAPM), the efficient market hypothesis, and the consumption-based pricing framework. *Frictional finance* keeps agents

fully rational but introduces market imperfections—transaction costs, borrowing and short-sale constraints, information asymmetries—and is institutionally more realistic than the traditional paradigm. *Behavioral finance* relaxes the rationality of some agents and is psychologically more realistic. Table 1.1 summarizes the comparison. The categories are not mutually exclusive: realistic models routinely combine frictions with non-rational agents, and much of the field's progress consists of showing that the two ingredients interact—frictions are often the reason irrational demand survives in prices.

Table 1.1: Three paradigms in finance

Paradigm	Agents	Market frictions	Source of realism
Traditional finance	Fully rational	None	—
Frictional finance	Fully rational	Present	Institutional
Behavioral finance	Some less than fully rational	Present or absent	Psychological

Notes: Behavioral models typically also contain frictions; limits to arbitrage (Section 1.2) are the reason non-rational demand can move prices.

What exactly does "fully rational" mean? In the standard usage, rationality has two components. First, agents *update beliefs correctly* on receipt of new information—that is, according to Bayes' rule. Second, agents *make decisions according to sensible preferences*—that is, maximizing expected utility (EU), where the utility function is defined over wealth or consumption. Behavioral finance, correspondingly, studies two families of departures: less than fully rational *beliefs*, meaning systematic deviations from Bayes' rule such as extrapolation and overconfidence; and less than fully rational *preferences*, meaning either non-expected-utility preferences (prospect theory is the leading example) or expected utility defined over arguments other than final wealth or consumption. This two-by-two map—beliefs versus preferences, at the level of individual judgment and at the level of market prices—organizes the applications in the second half of this book.

1.2 Two Challenges for Behavioral Finance

1.2.1 The Arbitrage Critique

The oldest objection, associated with Friedman (1953), is that rational agents will quickly undo any effect that irrational agents have on prices. If noise traders push a stock above fundamental value, rational traders have an incentive to sell it short, and in doing so they push the price back. In this view, irrational traders may lose money, but prices stay correct.

The modern response is that this argument assumes away the costs and risks of trading against mispricing. Arbitrage—in the sense used here, trading on price discrepancies that involve risk and capital—is limited: the mispricing can widen before it closes, the arbitrageur's capital is finite and often borrowed, and the ultimate owners of that capital may withdraw it precisely when the trade looks worst. Shleifer and Vishny (1997) formalize these *limits to arbitrage*, and we treat them in depth in the next chapter. The practical implication for this chapter is methodological: when we observe a puzzling pattern in prices, the question is not only "is there a behavioral mechanism that generates it?" but also "why do rational arbitrageurs not eliminate it?" The answer usually involves the frictions of Table 1.1.

1.2.2 The Lack-of-Discipline Critique

The second objection, pressed most forcefully by Fama (1998), is that behavioral finance is undisciplined: the psychology literature is vast, and after the fact it is too easy to find some bias that fits any observed pattern. A field that can explain everything explains nothing.

The response is methodological. First, behavioral models earn credibility by generating *new* predictions—relationships in data that were not used to motivate the model—and these predictions must survive out-of-sample tests, just as in any other part of empirical economics. Second, the discipline is enforced collectively and quantitatively: the sheer number of published cross-sectional predictors has forced the field to confront multiple testing and publication bias directly (Harvey, Liu, and Zhu, 2016; McLean and Pontiff, 2016; Hou, Xue, and Zhang, 2020), a development we return to in Section 1.5.9. The facts surveyed in this chapter should be read with this standard in mind: a fact becomes interesting not when some bias can be cited after the fact, but when a model that predicted it ex ante also predicts something else that turns out to be true.

1.3 The Road Map

The empirical core of this chapter consists of two families of facts. First, *asset pricing*: the aggregate stock market, the cross-section of stock returns, and bubbles. Second, *investor trading and portfolio choice*: how individual investors and institutional investors actually behave. For each fact we ask the same pair of questions: what would the frictionless rational model predict, and what are the leading candidates—rational and behavioral—for explaining the gap?

The chapter closes with the plan for the book. Part II develops limits to arbitrage, the supply side of the story: why mispricing, once created, is not instantly eliminated. Part III develops the psychological foundations and their finance applications: non-standard preferences, biased beliefs, heterogeneous beliefs, and bounded rationality, including psychology-free models that reproduce behavioral pricing patterns with fully rational agents.

1.4 The Aggregate Stock Market

1.4.1 Time-Series Predictability

The first fact is that aggregate stock market returns are predictable in the time series. Valuation ratios such as the price–dividend ratio (P/D) and the cyclically adjusted price–earnings ratio (CAPE) forecast subsequent market returns at long horizons (Campbell and Shiller, 1988; Fama and French, 1988). Figure 1.1 shows the raw pattern for the U.S. market: each point is a month between 1881 and 2016, with the Shiller CAPE ratio on the horizontal axis and the annualized real return on the S&P 500 over the following ten years on the vertical axis. When CAPE is low, subsequent ten-year real returns tend to be high; when CAPE is high, they tend to be low, ranging over this sample from 20% to below -6% per year. The fitted line is $y = 13.0 - 0.38x$ with an R^2 of 0.23: a ten-point increase in CAPE is associated with almost four percentage points per year of lower subsequent real return. This appears to contradict the simplest reading of the efficient market hypothesis, under which expected returns should not vary in a way that public information predicts.

Predictability per se does not refute rationality: in a rational model, expected returns can vary over time if the price of risk or the quantity of risk varies. The leading rational explanations are exactly of this form. In *rare disaster* models, shifts in the perceived probability of a macroeconomic catastrophe move discount rates even if realized consumption is smooth (Rietz, 1988; Barro, 2006); in *long-run risk* models, small persistent movements in the growth rate and its volatility carry large

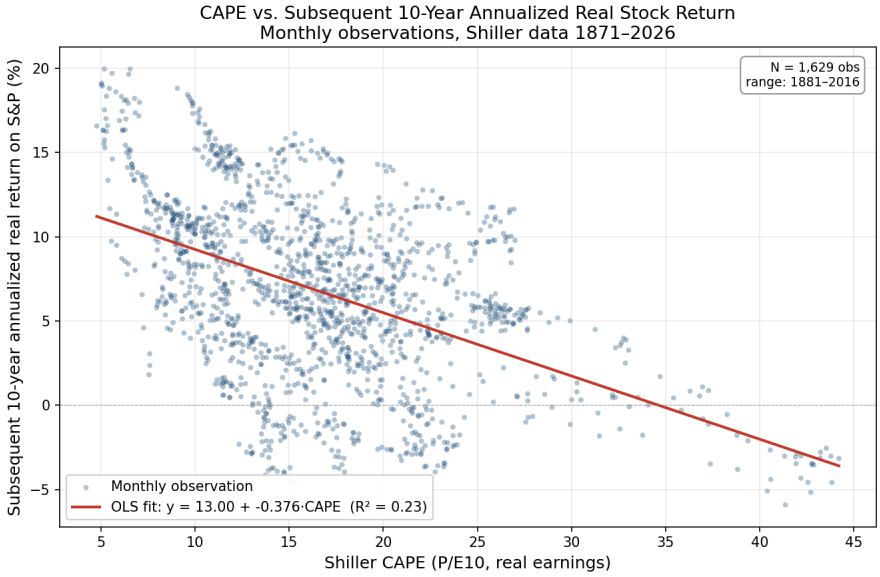


Figure 1.1: Subsequent ten-year returns decline with the starting valuation ratio. Scatter of the Shiller CAPE ratio against the subsequent ten-year annualized real return on the S&P 500, 1,629 monthly observations, 1881–2016 (the ten-year forward window ends the sample in 2016). The fitted line is $y = 13.0 - 0.38x$, $R^2 = 0.23$. *Source:* Author's calculations based on Robert Shiller's *Irrational Exuberance* data.

pricing implications (Bansal and Yaron, 2004); in *habit formation* models, risk aversion moves with consumption relative to habit, generating time-varying discount rates (Campbell and Cochrane, 1999). The behavioral alternative is that expected-return variation reflects time-varying sentiment—investors extrapolate past returns, so prices are high precisely when optimism is high, and subsequent returns are low (Barberis and Thaler, 2003). Distinguishing these accounts empirically is difficult, because both predict the same regression sign; the discriminating evidence, discussed in Part III, comes from surveys of investor expectations: extrapolative beliefs imply that expected returns as reported by investors should be *high* when valuation ratios are high, whereas rational time-varying discount rates imply they should be *low*.

1.4.2 Excess Volatility

The second fact is that the level of stock prices moves too much to be justified by rationally-varying forecasts of the cash flows paid to investors. In an influential test, Shiller (1981) compared the real S&P Composite price index with the *ex post rational price*—the present value of the dividends actually paid subsequently, discounted at a constant rate. If prices equal the rational forecast of discounted future dividends, the realized price should be an unbiased forecast of the ex post rational price and should therefore be *less* volatile, since a forecast averages out the shocks that the realization contains. The data show the opposite. Figure 1.2 reproduces the comparison: the actual price (solid line) swings over a range several times wider than the smoothly drifting ex post rational price (dashed line), with peaks at 1901, 1929, and the mid-1960s.

The volatility and predictability facts are two readings of a single phenomenon, a point made transparent by the present-value identity: a high P/D ratio must forecast higher subsequent dividend growth, a lower expected return, or some combination. In the data, a high P/D ratio does *not* predict higher dividend growth; it predicts a lower price in the future—that is, a lower expected return. Virtually all of the action in valuation ratios comes through the expected-return channel,

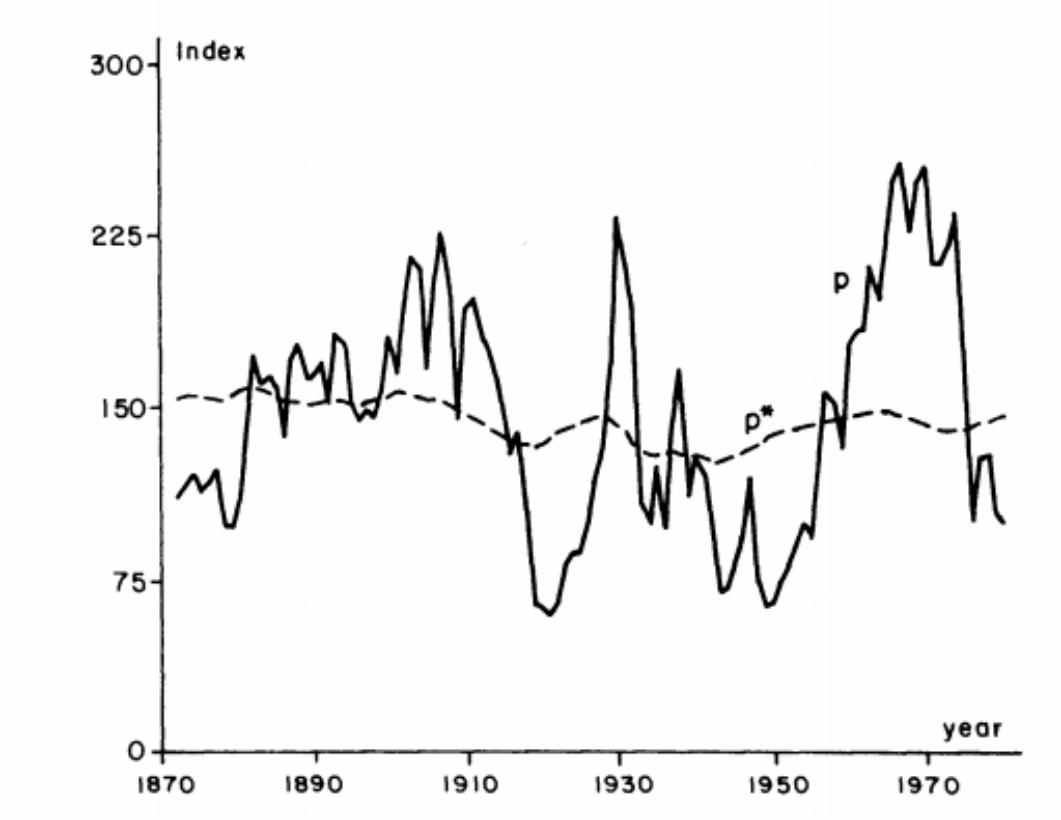


FIGURE 1

Note: Real Standard and Poor's Composite Stock Price Index (solid line p) and *ex post* rational price (dotted line p^*), 1871–1979, both detrended by dividing a long-run exponential growth factor. The variable p^* is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 1, Appendix.

Figure 1.2: Excess volatility. The real S&P Composite price index (solid line) against the ex post rational price (dashed line), defined as the present value of subsequently realized real dividends, both detrended, 1871–1979. The actual index is far more volatile than the present value it is supposed to forecast. *Source:* Shiller (1981).

which is why the time-series predictability of returns and the excess volatility of prices are the same fact stated in levels and in changes. The interpretive battle is therefore also the same: habit, disaster, and long-run-risk models generate discount-rate variation with rational beliefs, while belief-based models generate it with sentiment swings.

1.4.3 The Equity Premium Puzzle

The third fact concerns the *level* of expected returns rather than their variation. Mehra and Prescott (1985) document that the historical U.S. equity premium—the average return on stocks in excess of the risk-free rate, about 6% per year over their long sample—is far larger than a simple rational frictionless model with power utility can reproduce. The consumption-based benchmark makes the tension precise. In the lognormal approximation, the risk premium on any asset i satisfies

$$E_t[r_{i,t+1}] - r_{f,t} + \frac{\sigma_{i,t}^2}{2} = \gamma \operatorname{Cov}_t(r_{i,t+1},\, \Delta c_{t+1})\,,$$

where γ is the coefficient of relative risk aversion. The crux is the covariance term: the representative investor's consumption growth is almost uncorrelated with stock returns, so in the consumption-based model the stock market does not look very risky to the representative investor, and the model predicts a very small equity premium. Matching the observed premium requires a coefficient of relative risk aversion far outside the range that independent evidence on risk taking supports; matching the risk-free rate at the same time makes the tension worse.

As with predictability, the rational responses change the risk or the price of risk—rare disasters (Rietz, 1988; Barro, 2006), long-run risks (Bansal and Yaron, 2004), and habit (Campbell and Cochrane, 1999) all raise the model-implied premium by making the relevant risk bigger or the relevant risk aversion state-contingent. The behavioral responses change the preference object: with prospect-theoretic loss aversion and frequent (myopic) evaluation of gains and losses, investors behave as if the stock market is much riskier than consumption risk implies (Benartzi and Thaler, 1995; Barberis, Huang, and Santos, 2001). We develop both routes in Part III.

1.5 The Cross-Section of Stock Returns

We now move from the market as a whole to differences in expected returns *across* stocks—the cross-section. Nearly all of the facts below are constructed by the same procedure: sort stocks into portfolios on a firm characteristic measured at date t, hold the portfolios, and compare subsequent returns. Under the frictionless rational benchmark, only exposure to systematic risk should earn a premium; a characteristic that predicts returns after controlling for risk, with no risk interpretation, is an *anomaly*. For each anomaly below we state the fact, and then the state of play between the risk-based and mispricing-based readings.

1.5.1 Long-Term Reversal, Momentum, and Short-Term Reversal

Long-term reversal. A stock's cumulative return over the past three to five years *negatively* predicts its future return in the cross-section (De Bondt and Thaler, 1985). Figure 1.3 shows the classic evidence: starting from the formation date, the portfolio of past losers (top line) earns steadily higher cumulative returns than the market, while the portfolio of past winners (bottom line) earns steadily lower returns, with the gap approaching 40 percentage points by the end of the five-year horizon. De Bondt and Thaler (1985) read this as *overreaction*: extreme past performance prompts extreme expectations, which subsequently correct. A risk-based reading is also possible—perhaps

past losers have become genuinely riskier, so their higher returns compensate risk. The difficulty for the risk reading is the gradual, continuing drift in returns long after formation, which is hard to square with a one-time change in risk.

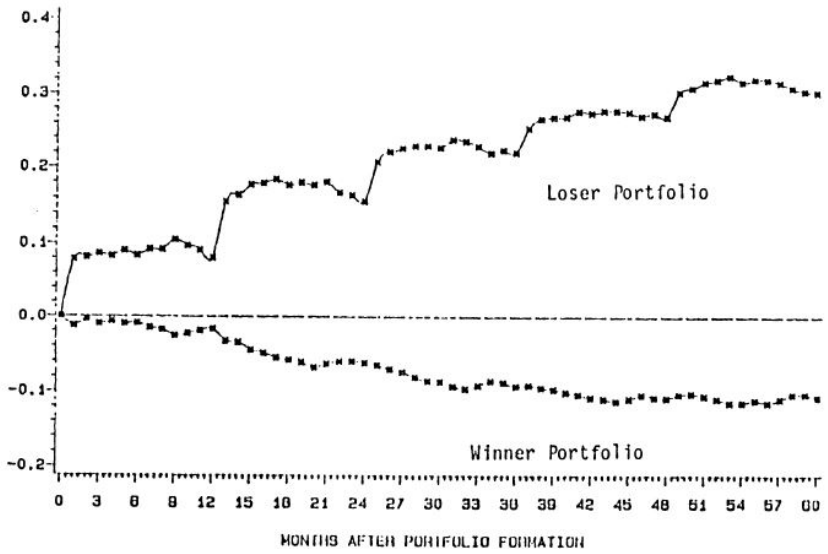


Figure 1.3: Long-term reversal. Cumulative returns of portfolios of past "losers" and past "winners" over 60 months following portfolio formation. Prior losers subsequently outperform prior winners by a wide and persistent margin. *Source:* De Bondt and Thaler (1985).

Momentum. At intermediate horizons the sign flips. A stock's return over the past k months, for k between 3 and 12, *positively* predicts its future return in the cross-section (Jegadeesh and Titman, 1993). Figure 1.4 shows average monthly returns of value-weighted decile portfolios sorted on past 12-month returns over 1927–2024: average returns rise almost monotonically across deciles, with the past-winner decile earning 1.46% per month against 0.09% for the past-loser decile—a spread of 1.37 percentage points per month ($t = 5.6$), one of the largest and most persistent regularities in the cross-section.

Momentum and long-term reversal together are the central interpretive challenge of this literature: a parsimonious account must produce initial continuation and eventual reversal from the same mechanism. The behavioral models of the late 1990s were built for exactly this purpose. In Barberis, Shleifer, and Vishny (1998), news is incorporated slowly because of conservatism and then over-extrapolated because of representativeness, generating underreaction at short horizons and overreaction at long horizons. In Daniel, Hirshleifer, and Subrahmanyam (1998), overconfident investors overreact to private information, and the eventual public revelation of truth produces momentum, reversal, and the predictable pattern of event-driven returns. In Hong and Stein (1999), news diffuses slowly across investors, and heterogeneous agents trading at different horizons again generate momentum followed by reversal. Rational accounts exist but must work harder: a single risk-based mechanism that prices the same characteristic with opposite signs at two horizons is difficult to construct.

Short-term reversal. At the shortest horizon the sign flips back. A stock's return over the past week or month *negatively* predicts its future return (Jegadeesh, 1990; Lehmann, 1990). The leading

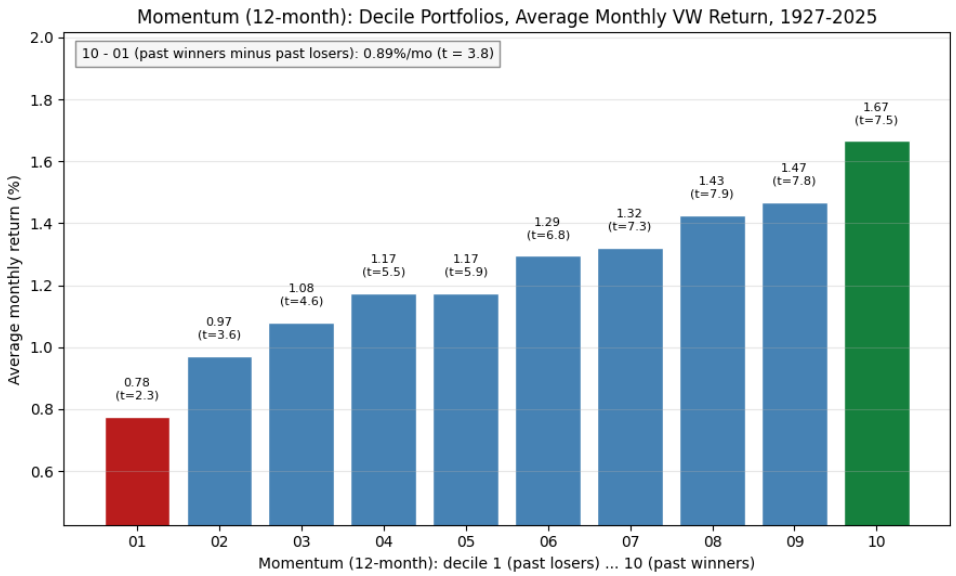


Figure 1.4: Momentum. Average monthly value-weighted returns (%) of decile portfolios sorted on past 12-month returns (Mom12m), 1927–2024. Decile 1 holds past losers, decile 10 past winners; the winner-minus-loser spread averages 1.37% per month ($t = 5.6$). *Source:* Author's calculations based on Open Source Asset Pricing decile portfolio data (Chen and Zimmermann, 2022).

interpretation of this anomaly is largely rational: compensation for liquidity provision. Short-term reversal strategies buy stocks whose prices were pushed down by liquidity demanders and sell stocks pushed up, in effect selling immediacy; the returns should therefore be high exactly when providing liquidity is expensive. Nagel (2012) confirms a sharp prediction of this view: the returns of short-term reversal strategies comove strongly with the VIX index of implied volatility, spiking in the turbulent episodes of 1998, 2002, and 2008–2009, when the marginal value of immediacy was highest (Figure 1.5). Short-term reversal is thus the anomaly most comfortably interpreted not as mispricing but as a price for a service—an instructive boundary case for the field.

1.5.2 The Value Premium

Ratios of fundamentals to price—book-to-market (B/M), earnings-to-price (E/P), and cash flow-to-price (CF/P)—positively predict future returns: cheap ("value") stocks outperform expensive ("growth") stocks. Figure 1.6 shows average monthly returns of value-weighted decile portfolios sorted on B/M over 1951–2024: average returns rise almost monotonically across deciles, from 0.88% per month for the lowest-B/M (growth) decile to 1.36% for the highest-B/M (value) decile—a premium of 0.48 percentage points per month ($t = 3.2$), or nearly six percentage points per year. The value premium motivated the Fama–French three-factor model, in which the return on a high-minus-low B/M portfolio is treated as a priced risk factor (Fama and French, 1993): on this reading, value stocks earn more because they are riskier in a dimension the market prices (Fama and French, 1996). The behavioral alternative is extrapolation: investors extrapolate past growth, leaving the stocks of disappointing firms too cheap and those of successful firms too dear, so that value earns a premium as the expectations gap closes (Lakonishok, Shleifer, and Vishny, 1994). The two readings remain unresolved and structure much of modern cross-sectional research.

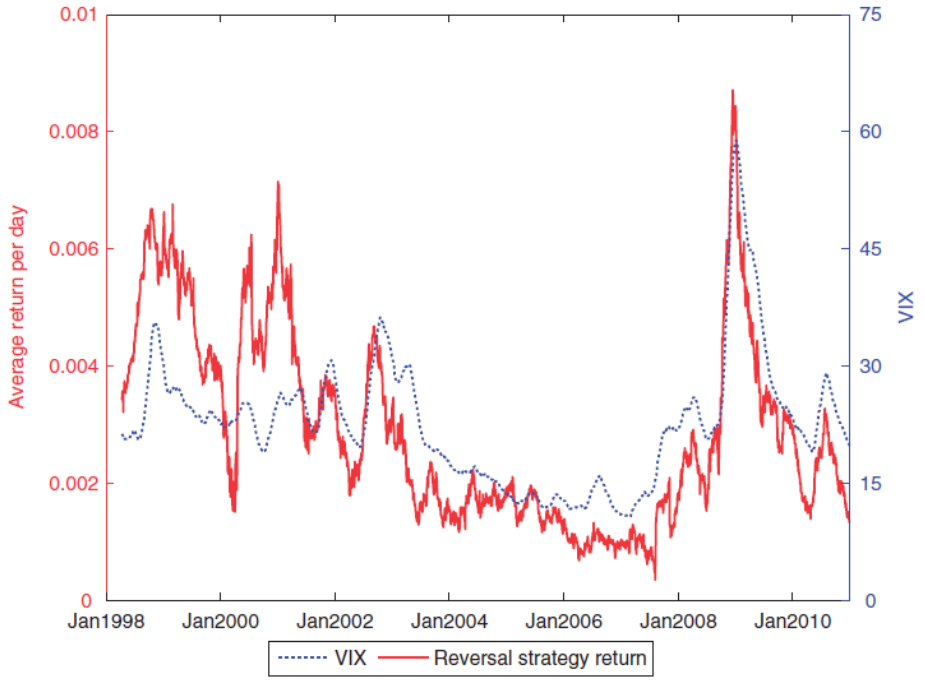


Figure 1
Three-month moving averages of daily return-reversal strategy returns and the CBOE S&P 500 implied volatility index (VIX)
Each day t, the reversal strategy returns are calculated as the average of returns from five reversal strategies that weight stocks proportional to the negative of market-adjusted returns on days $t-1$, $t-2$, ..., $t-5$, with weights scaled to add up to \$1 short and \$1 long. Returns are calculated from daily CRSP closing transaction prices, and returns are hedged against conditional market factor exposure.

Figure 1.5: Short-term reversal and liquidity provision. Three-month moving averages of daily returns on short-term reversal strategies (left axis) track the VIX index of implied volatility (right axis), 1998–2011. Reversal returns are large precisely when liquidity provision is most valuable. *Source:* Nagel (2012).

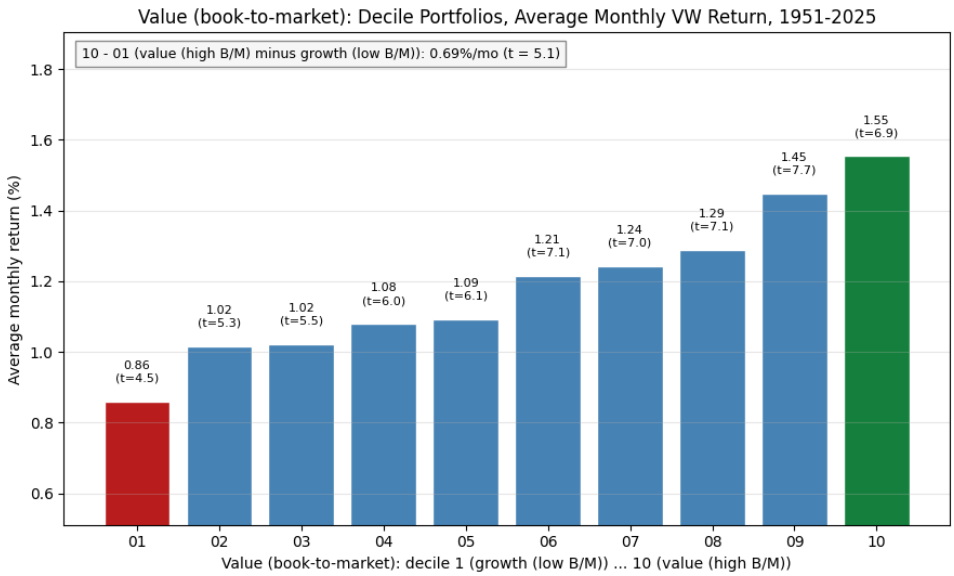


Figure 1.6: The value premium. Average monthly value-weighted returns (%) of decile portfolios sorted on book-to-market, 1951–2024. Decile 1 holds growth (low B/M) stocks, decile 10 value (high B/M) stocks; the value-minus-growth spread averages 0.48% per month ($t = 3.2$). *Source:* Author's calculations based on Open Source Asset Pricing decile portfolio data (Chen and Zimmermann, 2022).

1.5.3 Value and Momentum Everywhere

The value and momentum effects are not confined to U.S. individual stocks. Asness, Moskowitz, and Pedersen (2013) document both effects, with the same structure, within most developed equity markets, across country equity indices, across government bond indices, across currencies, and across commodities. Figure 1.7 shows cumulative returns of value, momentum, and combined ("combo") strategies in four of these asset classes. Two features stand out. First, value and momentum premia are positive in every asset class examined. Second, the value and momentum strategies are strongly *negatively* correlated everywhere—correlations from about -0.46 to -0.35 in the classes shown—so the combined strategy has a far higher Sharpe ratio than either leg alone. This universality is itself evidence: whatever generates the two effects, it is not an artifact of one market's institutions or data, and the robust negative correlation between the two premia is a restriction that successful theories must reproduce. It also provides a practical motivation for the multi-asset perspective that recurs throughout this book.

1.5.4 Issuance, Repurchase, and Asset Growth

The issuance/repurchase anomaly. Firms' own trading decisions predict their stock returns. Firms that issue equity—through initial public offerings (IPOs) or seasoned equity offerings (SEOs)—subsequently underperform a control group of non-issuing firms (Loughran and Ritter, 1995); firms that repurchase equity subsequently outperform (Ikenberry, Lakonishok, and Vermaelen, 1995). Figures 1.8 and 1.9 show the two sides of the pattern: in every one of the five years after issuance, firms that issued equity earn lower returns than non-issuers, with the SEO gap especially large in the first two years. The natural behavioral reading is that managers exploit mispricing—issuing when their stock is overvalued and repurchasing when it is undervalued, with the subsequent returns reflecting correction. A rational reading is that issuance and repurchase respond to investment

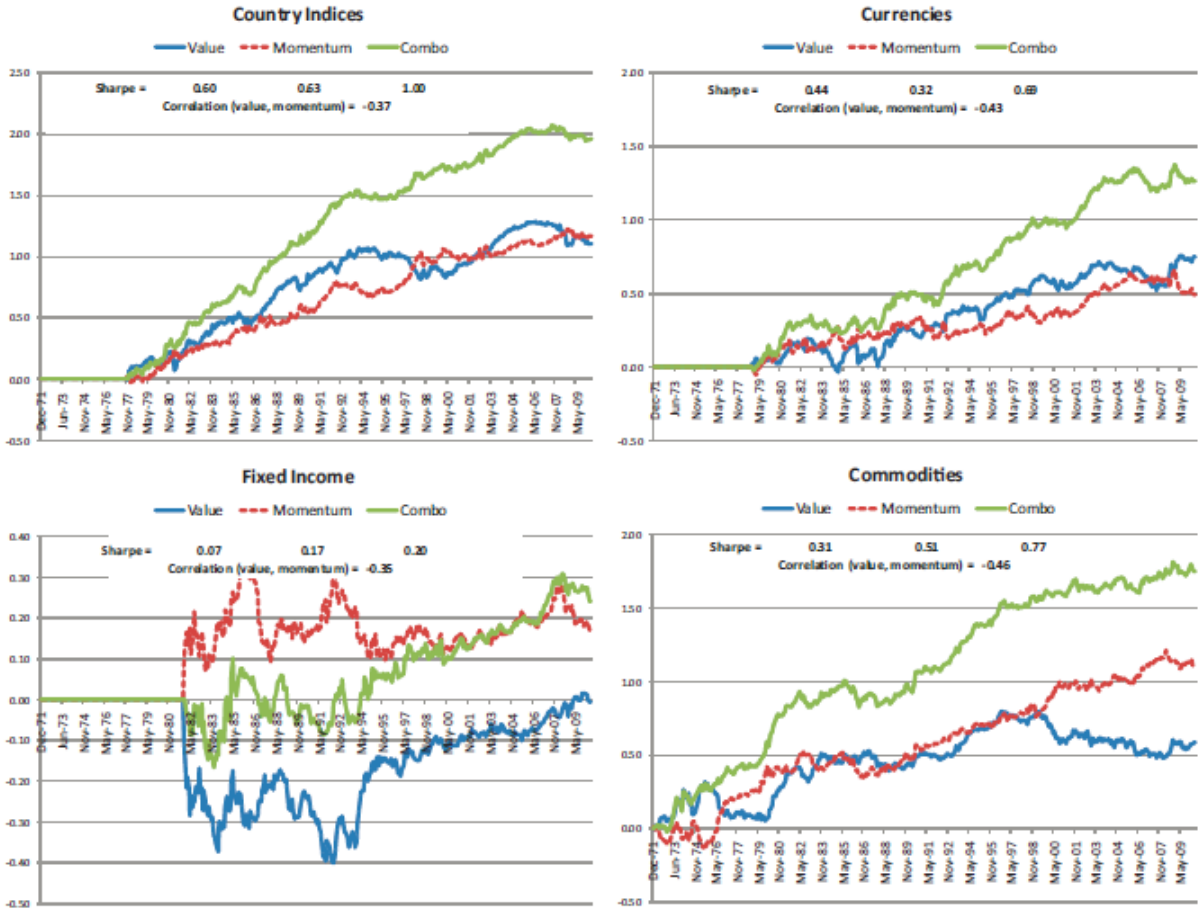


Figure 1.7: Value and momentum everywhere. Cumulative returns of value, momentum, and combined strategies in country equity indices, currencies, fixed income, and commodities, with Sharpe ratios annotated. Value–momentum correlations range from about −0.46 to −0.35 across asset classes. *Source:* Asness, Moskowitz, and Pedersen (2013).

opportunities and the cost of capital, so the same pattern could arise without any mispricing; distinguishing the two requires evidence on whether issuance is timed to valuation per se.

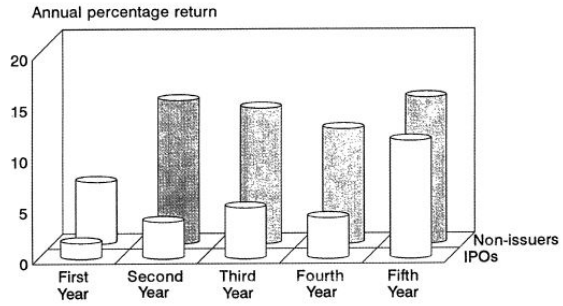


Figure 1.8: The new issues puzzle, IPOs. Annual percentage returns of IPO firms versus non-issuers in each of the five years after issuance. IPO firms underperform in every year. *Source:* Loughran and Ritter (1995).

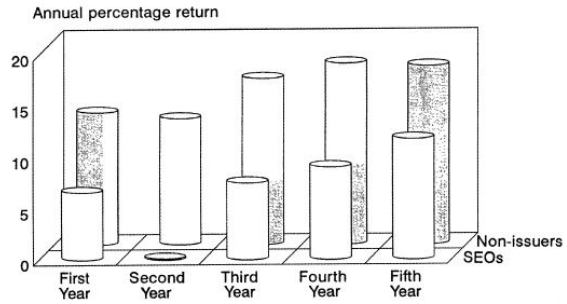


Figure 1.9: The new issues puzzle, SEOs. Annual percentage returns of SEO firms versus non-issuers in each of the five years after issuance. *Source:* Loughran and Ritter (1995).

Net share issuance. The firm-level events generalize to a clean accounting signal: net share issuance itself. Daniel and Titman (2006) and Pontiff and Woodgate (2008) show that the change in a firm's shares outstanding negatively predicts its subsequent returns, both in the U.S. and internationally. Figure 1.10 shows the pattern in quintile portfolios: for one-year net issuance (1927–2024), the low-issuance quintile earns 1.41% per month against 0.87% for the high-issuance quintile, a spread of 0.54 percentage points per month ($t = 6.1$); for five-year net issuance (1931–2024) the comparable spread is 0.44 percentage points per month ($t = 5.0$). Issuance is thus among the most robust cross-sectional predictors, and it is the same fact as the IPO and SEO evidence stated in a form observable for every firm in every year.

The asset growth anomaly. The issuance pattern generalizes to investment itself: firms that grow assets more quickly, or invest more, earn lower subsequent returns (Titman, Wei, and Xie, 2004; Cooper, Gulen, and Schill, 2008). Figure 1.11 shows mean monthly returns of decile portfolios sorted on asset growth, aligned in event time around the sorting year. Before the sort, high-asset-growth firms are the strong performers—their returns rise toward 4% per month at year -2—which is why they invest. After the sort, the ranking inverts: the high-growth decile drops toward the bottom of the cross-section and stays there for years, while low-growth deciles persistently outperform. The behavioral interpretation continues the extrapolation theme—firms invest most when investors have over-extrapolated their prospects. The rational interpretation runs through the neoclassical q

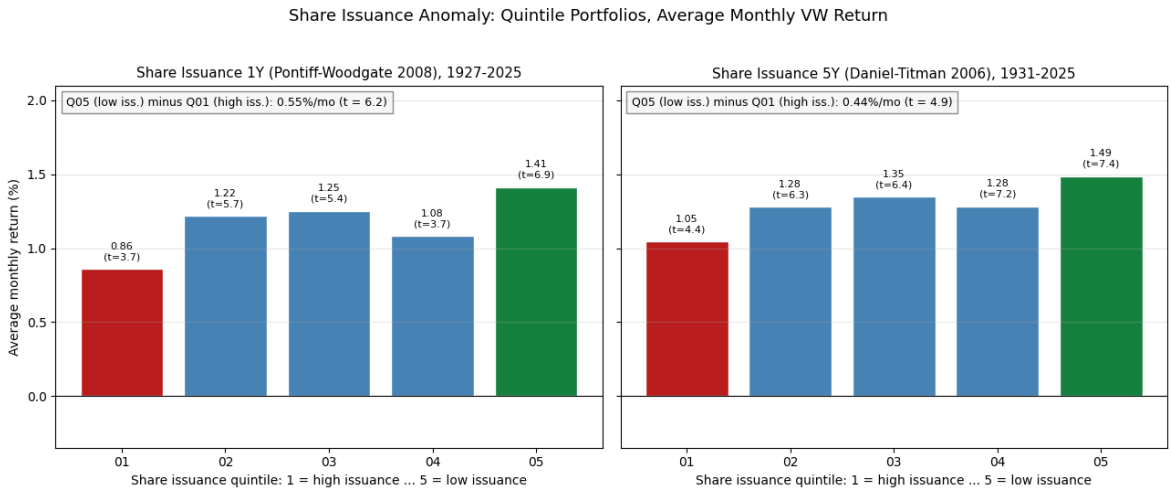


Figure 1.10: Net share issuance. Average monthly value-weighted returns (%) of quintile portfolios sorted on one-year (left; Pontiff and Woodgate, 2008) and five-year (right; Daniel and Titman, 2006) net share issuance. Quintile 1 holds high-issuance firms, quintile 5 low-issuance firms. The low-minus-high-issuance spread averages 0.54% per month ($t = 6.1$) for one-year issuance, 1927–2024, and 0.44% per month ($t = 5.0$) for five-year issuance, 1931–2024. *Source:* Author's calculations based on Open Source Asset Pricing quintile portfolio data (Chen and Zimmermann, 2022).

theory of investment: high investment signals low costs of capital, i.e. low expected returns, and investment-based factor models built on this logic absorb a substantial fraction of the anomaly zoo (Hou, Xue, and Zhang, 2015; Fama and French, 2015).

1.5.5 Profitability, Quality, and Distress

Profitability and quality. More profitable firms earn *higher* future returns, a fact initially surprising under the intuition that "good firms are good stocks"—in equilibrium, quality should be priced. Figure 1.12 shows average monthly returns of value-weighted quintile portfolios sorted on gross profitability over 1951–2025: returns rise from 0.77% per month for the least profitable quintile to 1.20% for the most profitable, a spread of 0.44 percentage points per month ($t = 4.1$). Profitability is related to a broader set of *quality* characteristics: Asness, Frazzini, and Pedersen (2019) construct a quality measure combining profitability, recent growth in profits, low return and earnings volatility, and low leverage, and show that high-quality stocks outperform junk stocks with otherwise similar characteristics. In the rational factor-model tradition, profitability and investment factors are now central: they appear in both the five-factor model of Fama and French (2015) and the q-factor model of Hou, Xue, and Zhang (2015), where a profitability premium is a priced risk or a reflection of discount-rate differences. The behavioral reading of quality connects it again to extrapolation: investors underprice the persistence of profitability.

Financial distress. If the CAPM held, stocks with higher systematic risk would earn higher expected returns, and financial distress—when it raises risk—would be rewarded. The data contradict this logic: financially distressed stocks earn *lower* alphas (Campbell, Hilscher, and Szilagyi, 2008). Figure 1.13 shows alphas and mean excess returns of portfolios sorted on model-predicted failure probability: all three measures decline roughly monotonically with distress, and for the most distressed decile the three-factor alpha is around −24% per year. Distress risk, in other words, is

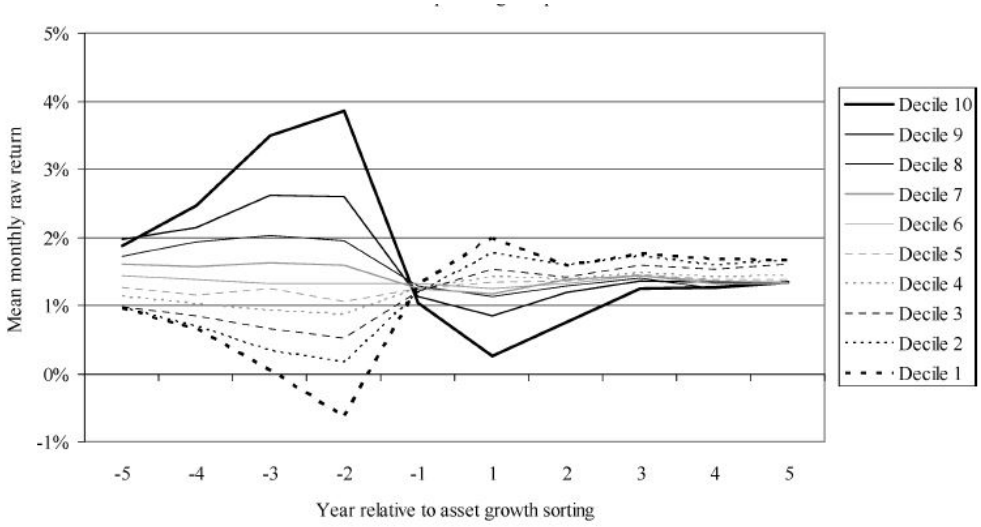


Figure 1.11: The asset growth anomaly. Mean monthly returns of decile portfolios sorted on asset growth, by year relative to the portfolio formation year. High-asset-growth firms earn high returns before formation but low returns after; low-growth firms the reverse. *Source:* Titman, Wei, and Xie (2004).

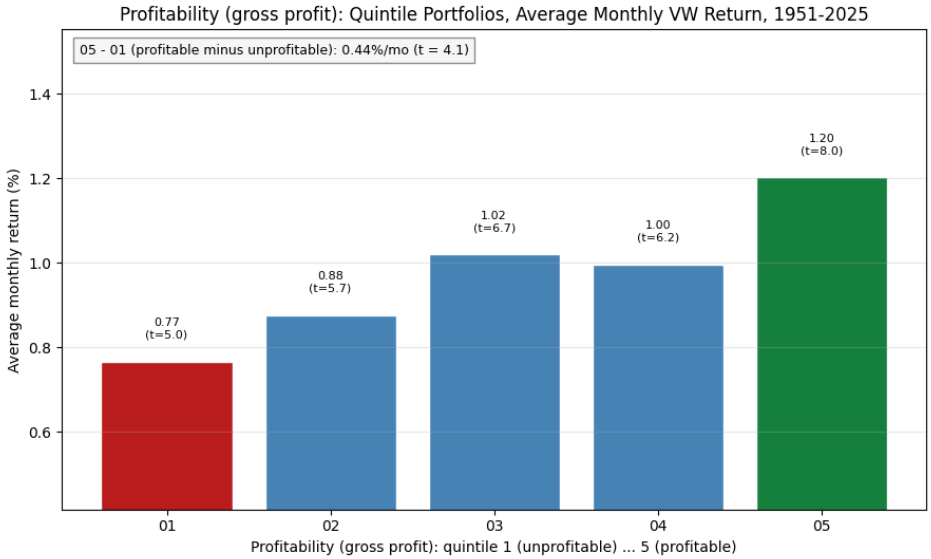


Figure 1.12: The profitability anomaly. Average monthly value-weighted returns (%) of quintile portfolios sorted on gross profitability, 1951–2025. Returns rise from 0.77% per month in the least profitable quintile to 1.20% in the most profitable (spread 0.44% per month, $t = 4.1$). *Source:* Author's calculations based on Open Source Asset Pricing portfolio data (Chen and Zimmermann, 2022).

priced with the wrong sign relative to any consumption- or market-beta-based model. Candidate explanations include the lottery-like payoff profile of distressed stocks (Section 1.5.8 develops the demand side) and the difficulty of shorting exactly those stocks, which leaves them overpriced—another point where frictions and sentiment interact.

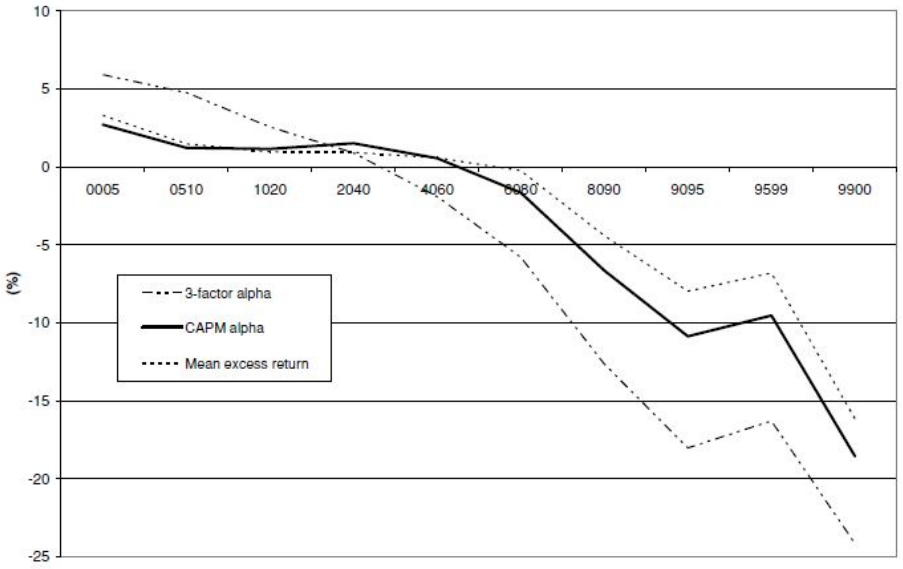


Figure 2. Alphas of distressed stock portfolios. The figure plots the annualized mean excess return over the market, CAPM alpha, and Fama–French three-factor alpha for the 10 distress risk-sorted portfolios from 1981 to 2003. Portfolios are formed at the beginning of January every year using the model-predicted probability of failure.

Figure 1.13: Distress risk is priced with the wrong sign. CAPM alpha, Fama–French three-factor alpha, and mean excess return of portfolios sorted on model-predicted failure probability, 1981–2003. All three decline sharply with distress; the most distressed portfolio earns a three-factor alpha of roughly -24%. *Source:* Campbell, Hilscher, and Szilagyi (2008).

The same conclusion emerges from an older, accounting-based distress measure. Dichev (1998) sorts stocks on the O-score, an accounting-based measure of bankruptcy probability, and finds that high-O-score (distressed) stocks subsequently *underperform* safe stocks, again with the wrong sign for a risk story. Figure 1.14 shows the Open Source Asset Pricing version of this sort, which is binary rather than decile-based: over 1972–2024, the high-O-score portfolio earns 0.24% per month—statistically indistinguishable from zero ($t = 0.7$)—against 1.16% per month ($t = 5.2$) for the low-O-score portfolio, a long-short spread of 0.92 percentage points per month ($t = 4.2$). Whether distress is measured by a model-predicted failure probability or by a raw accounting score, distressed stocks are not rewarded; they are penalized.

1.5.6 Accruals and Net Operating Assets

The anomalies above act through prices; the accrual anomaly acts through the accounting statement that feeds prices. Sloan (1996) starts from the decomposition of earnings into cash flow and accruals,

$$E_t = \underbrace{CF_t}_{\substack{\text{cash flow,}\\ \text{persists}}} + \underbrace{Acc_t}_{\substack{\text{accruals,}\\ \text{mean-revert}}} ,$$

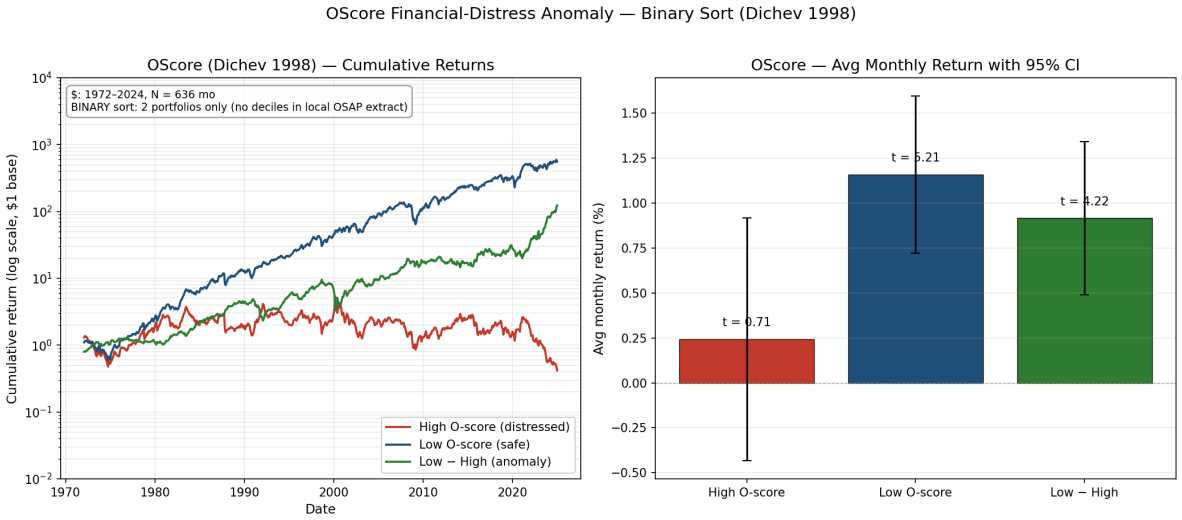


Figure 1.14: The O-score financial-distress anomaly. Left: cumulative returns (log scale, \$1 base) of high-O-score (distressed), low-O-score (safe), and low-minus-high portfolios, 1972–2024 (636 months). Right: average monthly returns with 95% confidence intervals. The distressed portfolio earns 0.24% per month ($t = 0.7$) against 1.16% ($t = 5.2$) for the safe portfolio; the long-short spread averages 0.92% per month ($t = 4.2$). *Source:* Author's calculations based on Open Source Asset Pricing portfolio data (Chen and Zimmermann, 2022), following Dichev (1998).

where the cash portion of earnings tends to persist while the non-cash portion—receivables, inventory, and similar items—tends to reverse. Investors who extrapolate total earnings without making this distinction overprice high-accrual firms and underprice low-accrual firms; as the accruals reverse, prices correct. In Sloan's sample, a strategy long low-accrual and short high-accrual stocks earns roughly 10% per year of abnormal return.

Hirshleifer, Hou, et al. (2004) extend the logic from the flow to the stock, ranking firms by net operating assets,

$$\text{NOA} = (\text{TA} - \text{Cash}) - (\text{TL} - \text{Debt}),$$

the accumulated balance-sheet position that past accruals have built up. A bloated NOA is a durable footprint of past earnings management, and it is less noisy than a single year of accruals, which ordinary real expansions also generate; accordingly, high-NOA firms underperform low-NOA firms by roughly 15–17% per year, and the stock measure subsumes much of the one-year flow effect. Figure 1.15 shows both patterns in the Open Source Asset Pricing data: the high-accrual decile earns 0.85% per month against 1.24% for the low-accrual decile (spread 0.38% per month, $t = 6.1$, 1952–2024), and the high-NOA decile earns 0.51% against 1.25% for the low-NOA decile (spread 0.74% per month, $t = 6.1$, 1963–2024). The raw-spread magnitudes are smaller than the abnormal returns reported in the original studies, but the sign and the statistical significance agree.

1.5.7 Post-Earnings-Announcement Drift

The surprise in a firm's most recent earnings announcement positively predicts future returns: after good earnings news, returns drift upward for weeks and months; after bad news, they drift downward. The pattern was present at the very birth of empirical capital markets research (Ball and Brown, 1968) and has been quantified in the event-study tradition ever since; Figure 1.16 shows the canonical picture. Cumulative excess returns across ten earnings-surprise deciles fan out monotonically over the 90 trading days after the announcement: the most positive surprise

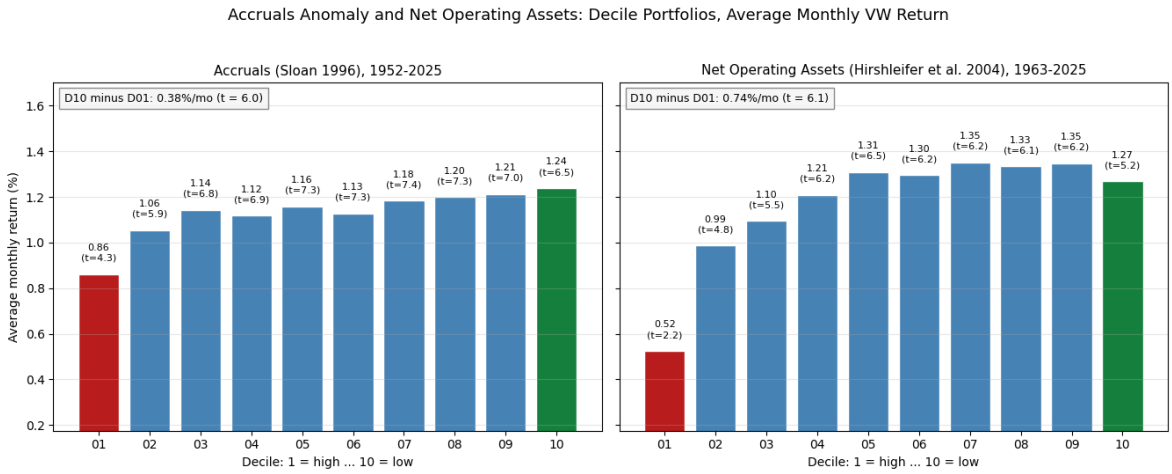


Figure 1.15: The accrual and net operating asset anomalies. Average monthly value-weighted returns (%) of decile portfolios sorted on accruals (left; Sloan, 1996, 1952–2024) and net operating assets (right; Hirshleifer, Hou, et al., 2004, 1963–2024). Decile 1 holds high-accrual (high-NOA) firms, decile 10 the opposite. High-accrual firms earn 0.85% per month against 1.24% for low-accrual firms (spread 0.38%, $t = 6.1$); high-NOA firms earn 0.51% against 1.25% for low-NOA firms (spread 0.74%, $t = 6.1$). *Source:* Author's calculations based on Open Source Asset Pricing decile portfolio data (Chen and Zimmermann, 2022).

decile has drifted about 8% higher and the most negative about 7.5% lower by day 90. Related findings hold for revisions in analysts' consensus forecasts and recommendation changes, which also positively predict subsequent returns.

Drift is the cleanest expression of *underreaction*: prices respond to public news too slowly, and the slow response is itself predictable from the news. The joint-hypothesis caveat applies—an apparent underreaction can be compensation for risk that happens to correlate with news surprises—but drift following an announcement that is public, dated, and of known sign strains that reading. Underreaction to news also fits naturally into the slow-information-diffusion models cited above (Hong and Stein, 1999) and into the disposition effect, to which we return in Section 1.7.5: if investors sell winners prematurely after good news, the price impact of the news is delayed (Frazzini, 2006).

1.5.8 Low-Risk Anomalies: Idiosyncratic Volatility and Beta

The idiosyncratic volatility puzzle. Stocks with higher idiosyncratic volatility have *lower* subsequent alphas (Ang et al., 2006)—the opposite of any compensation logic, since idiosyncratic risk should not be priced at all under the CAPM. Figure 1.17 shows the long-horizon implication in the low-volatility version of the fact: one dollar invested in 1968 in the lowest-volatility quintile of U.S. stocks compounds to several tens of dollars by 2008, while a dollar in the highest-volatility quintile ends the forty-year period roughly where it started. Baker, Bradley, and Wurgler (2011) stress why arbitrage does not correct this: benchmarked institutional investors face constraints on leverage and relative performance, so underpriced boring stocks are unattractive to them while overpriced volatile stocks are not painful enough to short—benchmarks themselves are a limit to arbitrage.

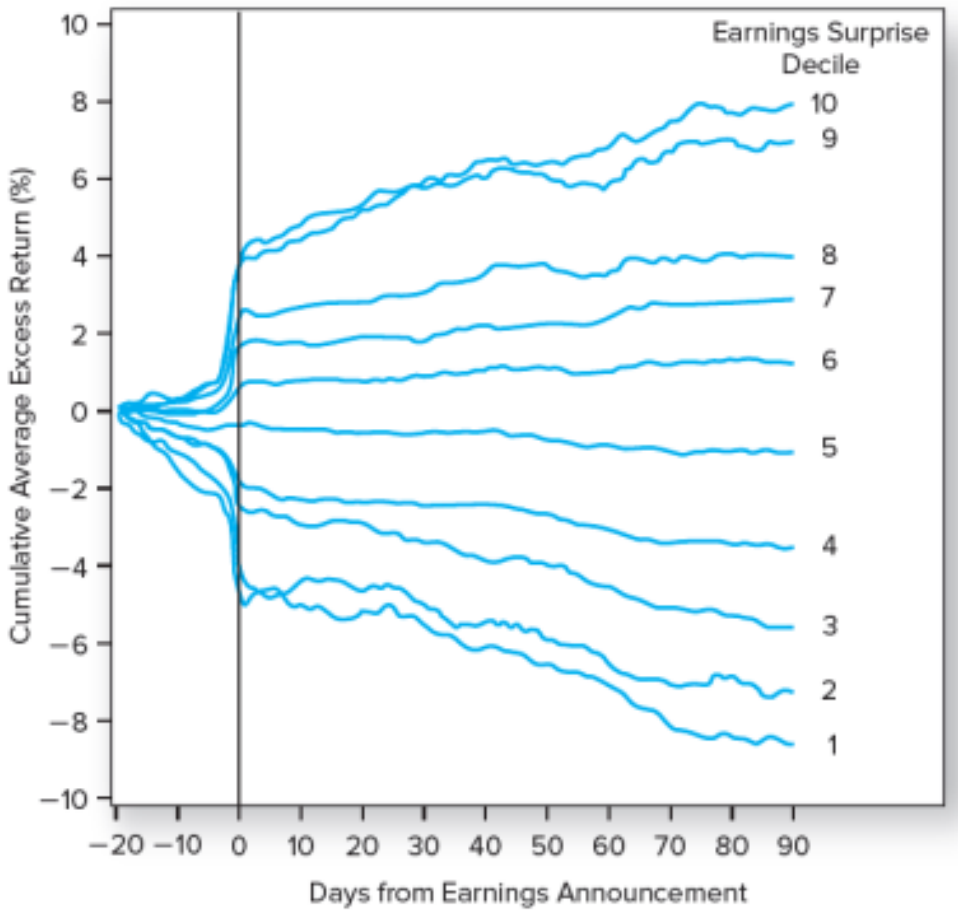


Figure 11.5 Cumulative abnormal returns in response to earnings announcements

Source: Reprinted from R. J. Rendleman Jr., C. P. Jones, and H. A. Latané, "Empirical Anomalies Based on Unexpected Earnings and the Importance of Risk Adjustments," *Journal of Financial Economics* 10 (1982), pp. 269–287.

Figure 1.16: Post-earnings-announcement drift. Cumulative average excess returns of earnings-surprise deciles around the announcement (day 0), from 20 days before to 90 days after. The initial reaction at the announcement is followed by continued drift in the direction of the surprise, ordered monotonically across deciles. *Source:* Rendleman, Jones, and Latané (1982).

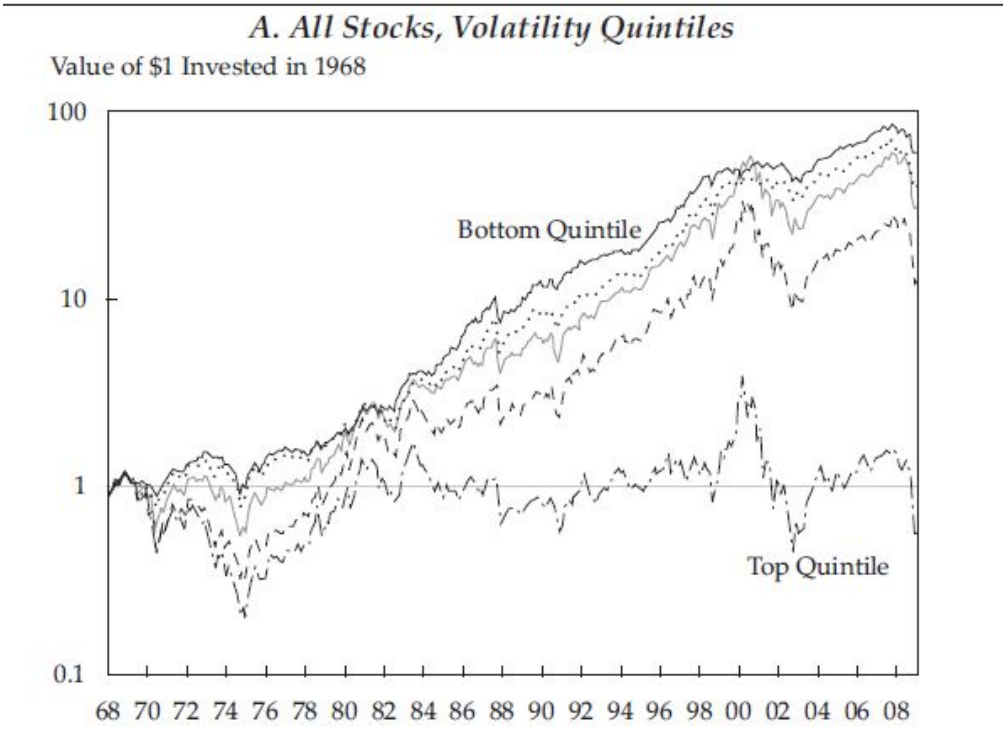


Figure 1.17: The low-volatility anomaly. Growth of one dollar invested in volatility-quintile portfolios, 1968–2008, log scale. The lowest-volatility quintile compounds to several tens of dollars; the highest-volatility quintile ends the period near one dollar. *Source:* Baker, Bradley, and Wurgler (2011).

The beta anomaly. The same flattening appears along the beta dimension. Stocks with higher betas have similar raw returns and lower CAPM alphas than low-beta stocks: the empirical security market line is far too flat relative to the CAPM's prediction (Fama and French, 2004). Figure 1.18 plots average returns against beta for ten beta-sorted portfolios over 1928–2003 together with the CAPM-predicted line; the portfolios' average returns cluster between roughly 11% and 14% across a beta range over which the CAPM predicts a doubling of expected return. Frazzini and Pedersen (2014) show that the pattern extends far beyond U.S. stocks—their "betting against beta" (BAB) factor, long leveraged low-beta securities and short high-beta securities, earns positive alphas in U.S. and international equities, Treasury bonds of different maturities, credit indices sorted on maturity and rating, commodities, country bonds, and foreign exchange (Figure 1.19).

Figure 2
Average Annualized Monthly Return versus Beta for Value Weight Portfolios Formed on Prior Beta, 1928–2003

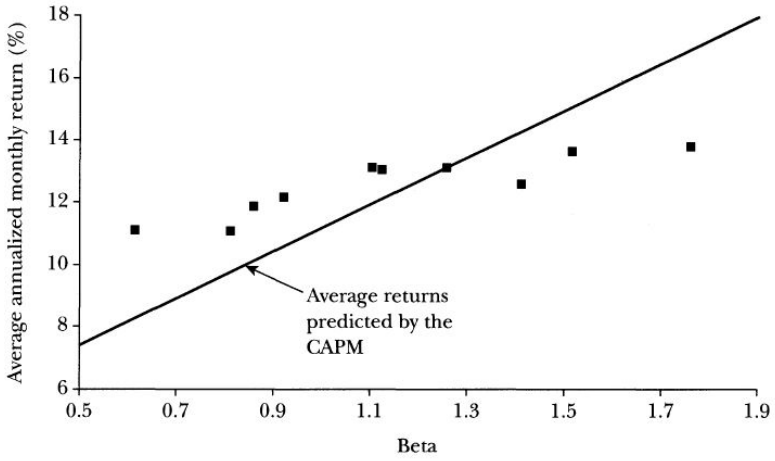


Figure 1.18: The empirical security market line is too flat. Average annualized monthly returns of ten value-weight portfolios sorted on prior beta, 1928–2003, against the average returns predicted by the CAPM. Low-beta portfolios earn more, and high-beta portfolios less, than the CAPM implies. *Source:* Fama and French (2004).

The low-risk anomalies illustrate how frictions and preferences blur. Frazzini and Pedersen (2014) generate the flat security market line with fully rational agents who face leverage and benchmark constraints: unable to lever, investors reach for beta, bidding up high-beta assets. Baker, Bradley, and Wurgler (2011) add the institutional benchmark structure. A complementary behavioral force is demand for lottery-like payoffs—preferences over the shape of the return distribution, not just its moments—which directly prices high-volatility, high-skewness stocks (Bali, Cakici, and Whitelaw, 2011). Note the structural lesson: several of the field's most robust cross-sectional facts now have candidate explanations that are *psychology-free*, a theme of Part III.

1.5.9 How Many Anomalies Are There?

The phenomena above are not exhaustive; they are the subset with the longest history and the most influence. Table 1.2 lists, in summary form, firm-level characteristics with documented predictive power for the cross-section of stock returns and the sign of the relation—reversal, momentum, earnings surprise, size, value ratios, issuance, accruals, both volatility measures, and profitability. The full modern catalog runs to hundreds of published predictors, which raises the discipline problem

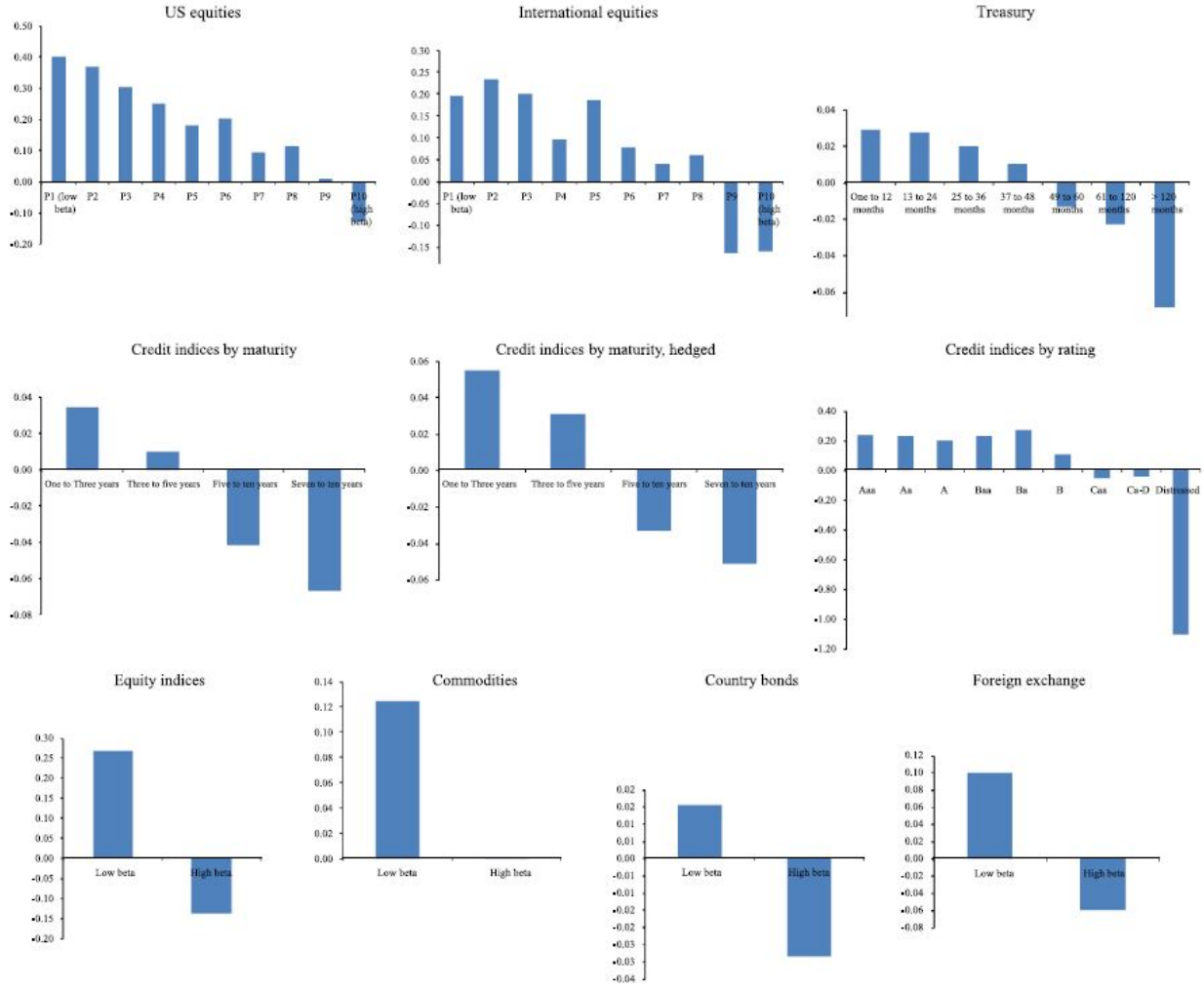


Figure 1.19: Betting against beta. Alphas of low-beta versus high-beta portfolios across asset classes: U.S. and international equities (beta-sorted deciles), Treasury bonds by maturity, credit indices by maturity and rating, equity indices, commodities, country bonds, and foreign exchange. Low-beta portfolios earn positive alphas and high-beta portfolios negative alphas almost everywhere. *Source:* Frazzini and Pedersen (2014).

of Section 1.2 in quantitative form. Harvey, Liu, and Zhu (2016) argue that with so many factors tried, conventional t-statistics of 2 are no longer adequate: the hurdle for a new factor should be closer to 3. McLean and Pontiff (2016) document that anomaly returns decline materially after publication—consistent with both data mining and arbitrage—and Hou, Xue, and Zhang (2020) replicate hundreds of published anomalies in a common sample and find that a majority are explained by a four-factor model built on investment and profitability.

The upshot is not that the anomaly literature collapses, but that its claims must be, and now are, held to replication standards—and on that score the news is good. Chen and Zimmermann (2022) construct the Open Source Cross-Sectional Asset Pricing dataset, an open replication of the published catalog, and find that the catalog largely survives. Figure 1.20 plots the t-statistic of each predictor's replicated long-short return against the t-statistic reported in the original study: the points cluster along the 45-degree line, with a fitted relation $t_{\text{reproduction}} = 0.2 + 0.89\, t_{\text{original}}$ and an R^2 of 84%—replicated signals come back about one-tenth weaker, on average, but overwhelmingly present. Jensen, Kelly, and Pedersen (2023) extend the exercise globally and reach the same verdict: most anomalies replicate out of sample in international markets, and the alpha magnitudes closely align with those in the U.S. Figure 1.21 shows their central comparison, U.S. factor alphas against world ex-U.S. alphas; factors that earn high alphas in the U.S. tend to earn high alphas elsewhere, a strong restriction on data-mining stories, since a purely mined pattern has no reason to travel across markets. Both datasets are publicly available—Open Source Asset Pricing for the U.S. market and the Global Factor Returns data of Jensen, Kelly, and Pedersen (2023) for international markets—and every fact in this section can be re-examined with them.

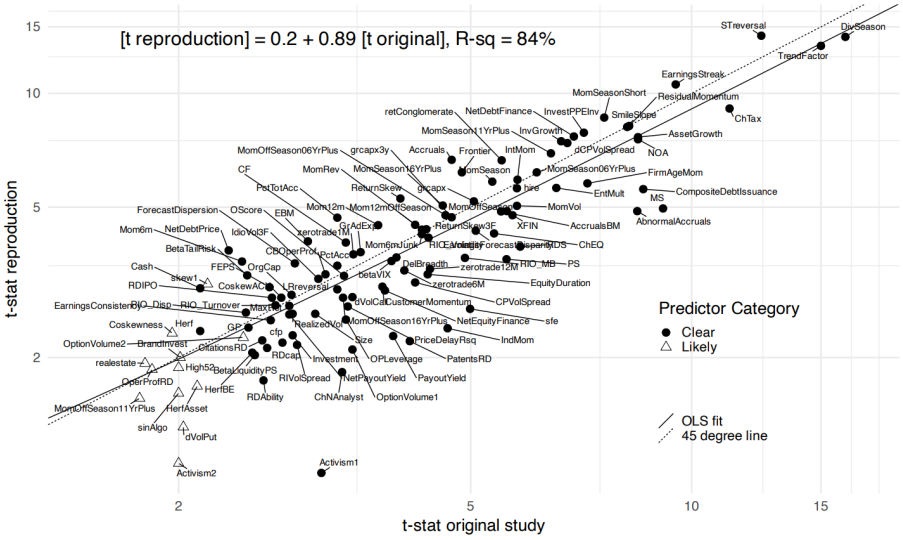


Figure 1.20: Open source replication. t-statistic of each predictor's replicated long-short return against the t-statistic reported in the original study, Open Source Asset Pricing dataset (October 2025 release). Each point is one published predictor, colored by predictor category; the dotted line is the 45-degree line. The fitted relation is $t_{\text{reproduction}} = 0.2 + 0.89\, t_{\text{original}}$ with $R^2 = 84\%$. *Source:* Chen and Zimmermann (2022), Open Source Asset Pricing project.

1.6 Bubbles

A *bubble* is a situation in which, for a significant period of time, an asset trades at a price that appears to be substantially higher than its rational or frictionless price. Bubbles appear to exhibit

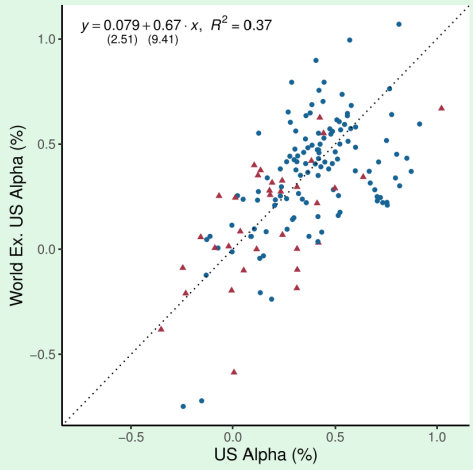


Figure 7. U.S. factor alphas versus world ex.-U.S. The figure compares OLS alphas for U.S. factors versus their international counterpart. Each world ex.-U.S. factor is a capitalization-weighted average of the factor in all other countries of our sample. Blue circles correspond to factors that were significant in the original study, while red triangles are those for which the original paper did not find a significant effect (or did not study the factor in terms of average return significance). The dotted line is the 45° line. The figure also shows a regression of world ex.-U.S. alpha on U.S. alpha. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 1.21: Is there a replication crisis in finance? OLS alphas of U.S. factors against their world ex-U.S. counterparts, each world factor a capitalization-weighted average of the factor across the other sample countries. Blue circles mark factors significant in the original study; red triangles those for which the original paper did not find a significant effect on average returns. The dotted line is the 45-degree line; the fitted regression is $y = 0.79 + 0.67x$ with $R^2 = 0.37$. *Source:* Jensen, Kelly, and Pedersen (2023), Figure 7.

Table 1.2: Firm characteristics with predictive power for stock returns in the cross-section

Characteristic	Predictive sign
Past three-year return	−
Past six-month return	+
Past one-month return	−
Earnings surprise	+
Market capitalization	−
Price-to-fundamentals ratio (B/M, E/P, CF/P)	−
Equity issuance	−
Net share issuance	−
Accruals / net operating assets	−
Systematic volatility (beta)	−
Idiosyncratic volatility	−
Profitability	+

Notes: The sign is the direction of the relation between the characteristic and subsequent returns. The table is illustrative, not exhaustive; for the replication status of the modern anomaly catalog see Harvey, Liu, and Zhu (2016), McLean and Pontiff (2016), Hou, Xue, and Zhang (2020), and Chen and Zimmermann (2022). *Source:* EF8083 course materials.

a set of recurring characteristics:

(i) The price of the asset rises sharply over some period and then declines, reversing much of the rise.

(ii) During the price rise, there are many reports in the media and from sophisticated investors that the asset is overvalued.

(iii) As the price rises and reaches its peak, trading volume in the asset is abnormally high.

(iv) During the episode, many investors hold highly extrapolative expectations: their expected future return on the asset rises with the asset's past return.

(v) During the price rise, some sophisticated investors increase their exposure to the asset.

(vi) In the early stages of the price rise, there is positive news about the asset's future fundamentals.

History offers a long catalog of candidate episodes. Figure 1.22 assembles the most cited—the tulip mania of 1634–1637, the Mississippi (1718–1720) and South Sea (1719–1721) schemes, the Great Depression stock market of 1929, Japanese equities of 1982–1992, the technology bubble of 1994–2002, and the 2006–2009 financial crisis—aligning each price path at its peak. Bitcoin is shown on the same scale as of September 2026: after peaking at $126,198 in October 2025, it stood at $84,453 on September 23, 2026, roughly 30% below its peak. The figure also documents why the definition above is deliberately cautious: an episode qualifies as a bubble only after the collapse, and a price path that has risen extraordinarily but fallen only part of the way back—Bitcoin at the chart's right edge—does not yet fit the "rise and collapse" template of the historical episodes.

Two regularities from the list above are especially informative for theory. First, bubbles are accompanied by very high trading volume, which is difficult to generate in a model with common priors and rational expectations: if all investors agree the asset is overvalued, who buys? Hong and Stein (2007) argue that bubbles are naturally *disagreement* episodes—short-sale constraints prevent pessimists' views from being expressed in prices, and volume rises because optimists trade against each other as their beliefs, all extrapolative in different degrees, diverge. Figure 1.23 shows the dot-com episode in this light: monthly turnover in internet stocks ran near 100% at the peak, an order of magnitude above turnover in other stocks, while the internet sector's equal-weighted index rose roughly ten-fold from its December 1996 base and collapsed back within two years. Second, sophisticated investors often *ride* the bubble rather than pop it: Brunnermeier and Nagel (2004) show that hedge funds were heavily invested in technology stocks through the late 1990s, reduced their positions before the 2000 collapse, and earned high returns doing so—consistent with arbitrageurs weighing the risk that mispricing widens before it closes. Understanding bubbles is an important challenge because their collapse can trigger severe economic downturns; we return to the mechanics in Part II.

1.7 Investor Trading and Portfolio Choice

The price facts of Sections 1.4–1.6 describe outcomes. This section turns to behavior: how individual investors and institutional investors actually trade and choose portfolios. The relevance is direct—behavioral finance needs demand-side mechanisms, and the rational benchmark makes predictions about behavior too.

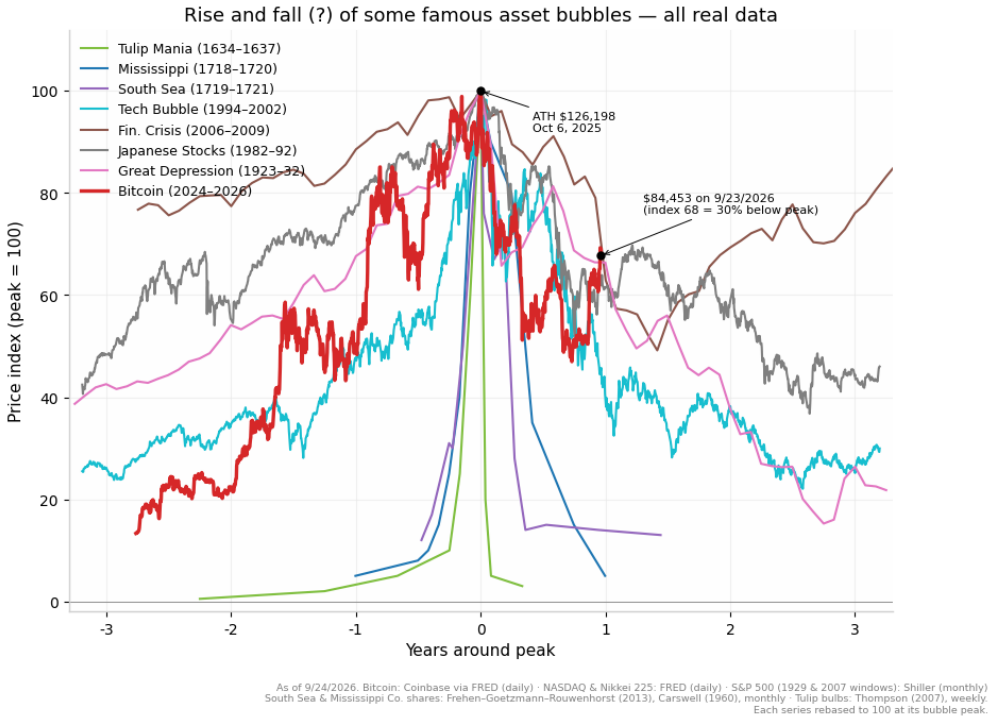


Figure 1.22: Famous bubbles in history. Price paths of eight bubble episodes—tulip mania (1634–1637), the Mississippi bubble (1718–1720), the South Sea bubble (1719–1721), the Great Depression stock market (1929), Japanese stocks (1982–1992), the technology bubble (1994–2002), the 2006–2009 financial crisis, and Bitcoin—expressed as multiples of the starting price and aligned at each episode's peak (year 0), as of September 23, 2026. *Source:* Convoy Investments, using daily data from Coinbase and FRED (Federal Reserve Economic Data).

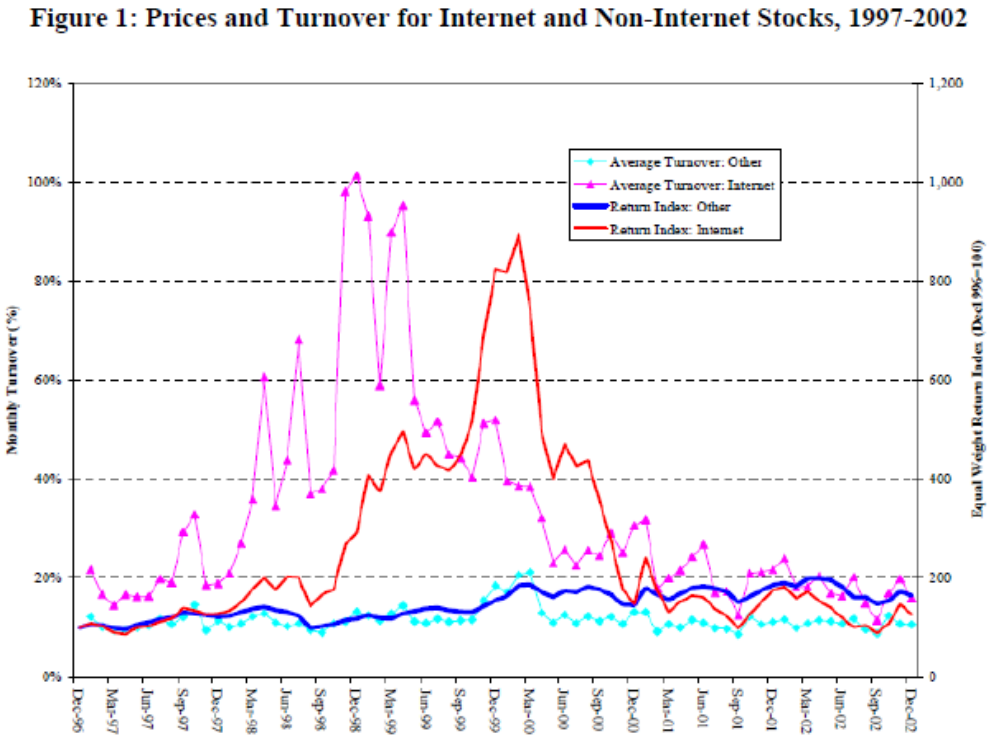


Figure 1.23: Prices and volume in the dot-com bubble. Equal-weighted return indexes (right axis, December 1996 = 100) and monthly turnover (left axis) for internet and non-internet stocks, December 1996–December 2002. Internet turnover approached 100% per month at the peak, and the internet index rose roughly ten-fold before collapsing. *Source:* Hong and Stein (2007).

1.7.1 Non-Participation

Historically, most U.S. households held no stock at all: in 1984, only 28% of households held any equity (Mankiw and Zeldes, 1991). This is a puzzle for the frictionless rational model of portfolio choice, which predicts that every household should hold at least some stock: equity carries a large premium (Section 1.4.3), and any household with savings should tilt some of it toward equities. Rational explanations are available and partly successful: fixed participation costs can deter small savers, and background risk—labor income and undiversifiable real estate exposure—can make equities unattractive for households whose human capital already correlates with market conditions. How much non-participation these forces can quantitatively explain remains debated; we return to preference-based explanations (loss aversion, narrow framing) in Part III.

1.7.2 Buying High and Selling Low

A household's actual investment experience is not the return of the assets it holds, but the return of its money: what matters is when funds flow in and out. Dichev (2007) makes this point operational by comparing *buy-and-hold* returns (the return an investor would earn by holding a fixed position) with *dollar-weighted* returns (the internal rate of return on the cash flows actually invested) across 19 major international stock exchanges. The result, reproduced in Figure 1.24, is remarkably uniform: in 18 of the 19 countries, the dollar-weighted return is below the buy-and-hold return. In the United States, the buy-and-hold return exceeds the dollar-weighted return by about one percentage point per year (11.5% versus 10.5%); the average gap across countries is about 1.5 percentage points per year. Investors as a group earn less than the assets they invest in, because capital flows in after prices have risen and out after prices have fallen—buying high and selling low in the aggregate.

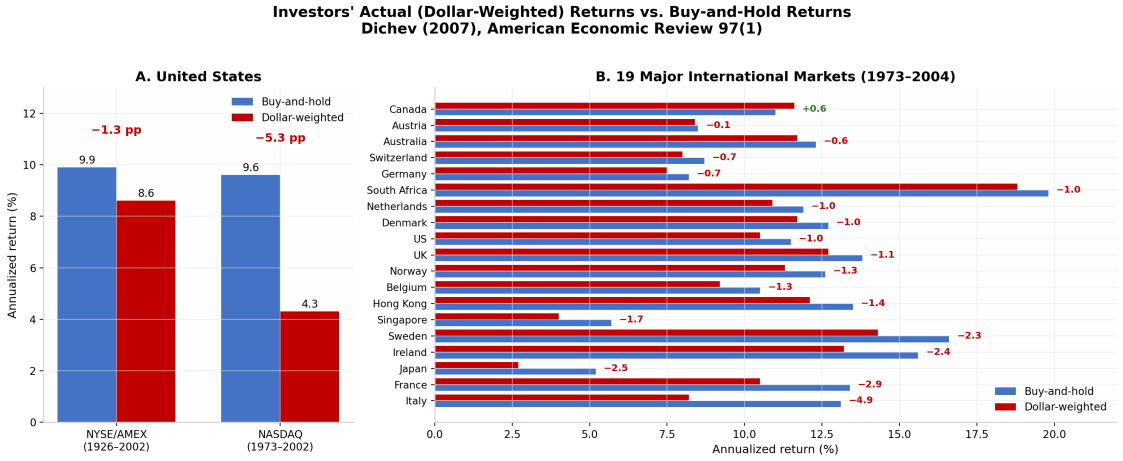


Figure 1.24: Buying high and selling low. Panel A: actual (dollar-weighted) versus buy-and-hold returns for the United States, NYSE/AMEX over 1926–2002 and NASDAQ over 1973–2002. Panel B: annualized buy-and-hold and dollar-weighted stock market returns for the major international markets of the sample, 1973–2004. Dollar-weighted returns sit below buy-and-hold returns in almost every market. *Source:* Dichev (2007), Figure 1.

1.7.3 Under-Diversification

Traditional portfolio theory predicts that households should hold well-diversified portfolios, differing at most in the split between the risk-free asset and the market portfolio. The data disagree along

four distinct margins.

Home bias. Investors allocate a surprisingly large fraction of their equity holdings to domestic equity. In 1989, households in the U.S., the U.K., and Japan invested 94%, 82%, and 98% of their equity portfolios, respectively, in domestic equities (French and Poterba, 1991)—far more than market-capitalization weights imply, and this despite the well-documented cross-country correlation of business cycles being far from perfect.

Local bias. Diversification fails at shorter distances too: the typical U.S. household holds about 30% of its portfolio in stocks headquartered within a 250-mile radius of its home. The natural rational hypothesis is an information advantage in local firms. The evidence is mixed: Ivković and Weisbenner (2005) find that households earn abnormal returns in local stocks, consistent with an information advantage; Seasholes and Zhu (2010) find that purchases of local stocks by individual investors do not systematically outperform, suggesting familiarity rather than information.

Ownership of few individual stocks. Many households hold a handful of stocks rather than a diversified fund. Curcuru et al. (2004) report that in 1989, 30% of U.S. equity-owning households allocated more than half of their equity exposure to fewer than 10 stocks. With so few names, the portfolios carry large idiosyncratic risk that could be diversified away at essentially no cost—the aggregate loss from such under-diversification is large relative to the equity premium itself.

Own-company stock. The concentration extends inside the retirement plan, where it is hardest to rationalize by information: Benartzi (2001) reports that across a sample of 154 retirement plans, participants allocated 23% of their discretionary contributions to the stock of their own employer—loading their financial wealth on the same firm that pays their labor income. Figure 1.25 shows how extreme concentration can become in large plans: among retirement plans with at least $1 billion in assets at end-2015, fifteen companies' plans held 38.6% to 65.2% of plan assets in company stock, with Sherwin-Williams at 65.2%, Scana at 64.9%, and Colgate-Palmolive at 60.7%. The leading behavioral account, developed by Benartzi (2001), is extrapolation of past company stock returns; we return to it in Part III.

1.7.4 Excessive Trading

If trading has costs and confers no informational benefit, rational investors should trade rarely. Using account-level data on 66,465 households at a large discount brokerage firm, Barber and Odean (2000) show that households that trade more earn lower net returns: the gross performance of the portfolios is broadly similar across households, but the costs of trading—commissions and the bid–ask spread—subtract from it at a rate increasing in turnover. Figure 1.26 shows the pattern in summary form: gross returns are similar across turnover groups (roughly 18% per year across the sample period), while net returns fall from about 18.5% for the lowest-turnover group to about 11.5% for the highest, with the average individual investor and an S&P 500 index fund shown for comparison. Barber and Odean (2000) attribute the pattern to overconfidence: overconfident investors overestimate the precision of their information and therefore trade too much. The interpretation is corroborated by the gender evidence of Barber and Odean (2001), who find that men—more overconfident than women in domains such as finance—trade more and thereby reduce their net returns by more than women do.

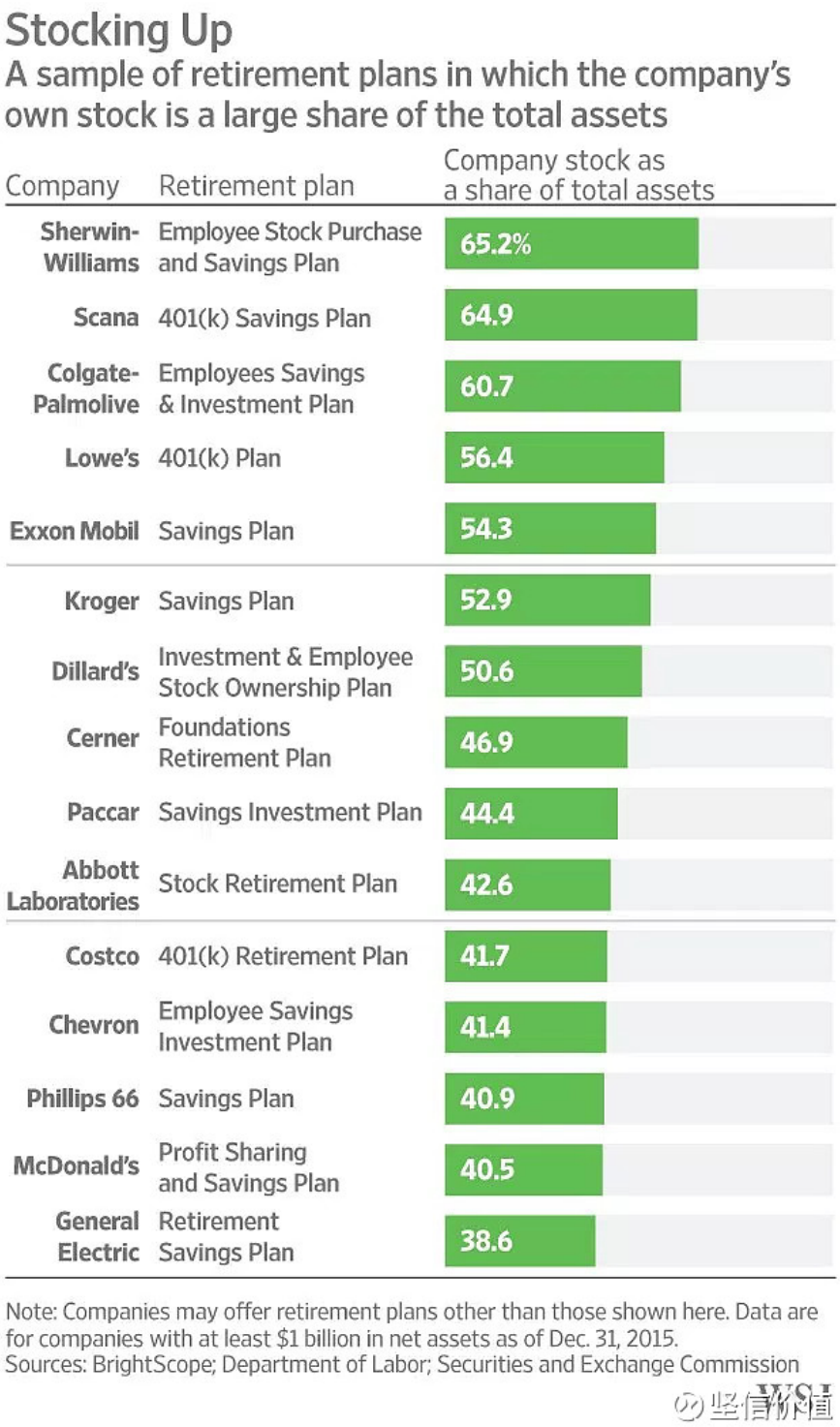


Figure 1.25: Own-company stock in retirement plans. Company stock as a percentage of total plan assets for a sample of large U.S. retirement plans (plan assets of at least $1 billion as of December 31, 2015). Concentration ranges from 38.6% (General Electric) to 65.2% (Sherwin-Williams). *Source:* "Stocking Up," *The Wall Street Journal*, based on BrightScope, Department of Labor, and SEC data.

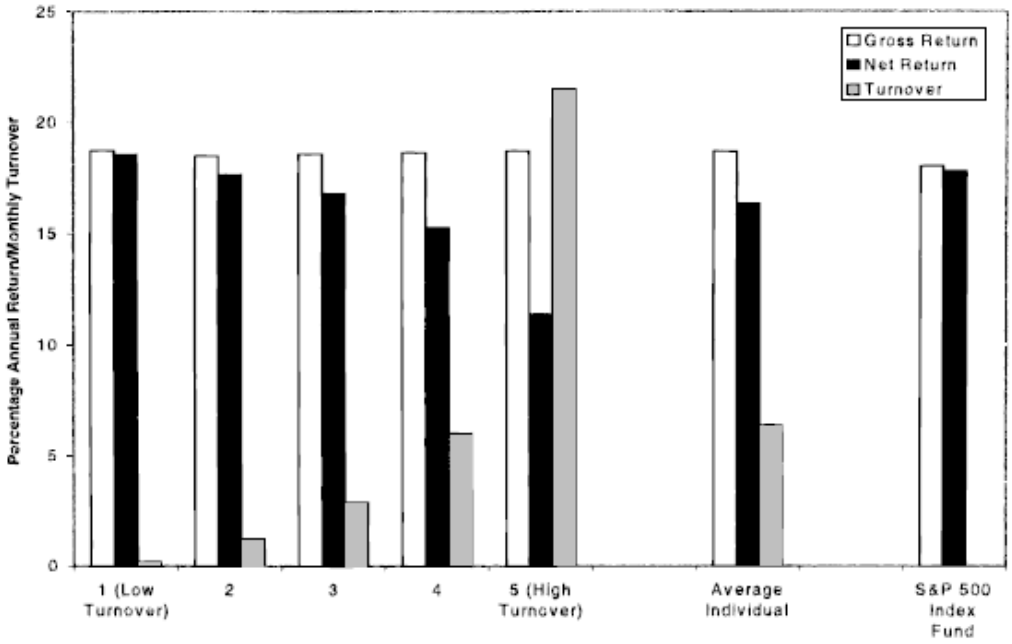


Figure 1.26: Trading is hazardous to your wealth. Gross returns, net returns, and turnover for households sorted into quintiles by turnover, together with the average individual investor and an S&P 500 index fund, following Barber and Odean (2000). Gross returns are similar across groups; net returns fall from about 18.5% per year for the lowest-turnover quintile to about 11.5% for the highest. *Source:* EF8083 course materials, following Barber and Odean (2000).

1.7.5 The Disposition Effect

The single most robust fact about individual investor trading is the *disposition effect*: investors have a greater tendency to sell stocks trading at a gain relative to their purchase price than stocks trading at a loss (Odean, 1998). The clean test uses the proportions of gains and losses that are realized. Define, for a set of accounts and a time period,

$$\mathrm{PGR} = \frac{\text{Realized Gains}}{\text{Realized Gains} + \text{Paper Gains}}, \qquad \mathrm{PLR} = \frac{\text{Realized Losses}}{\text{Realized Losses} + \text{Paper Losses}},$$

where paper gains (losses) are positions that could have been sold at a gain (loss) and were not. The disposition hypothesis is $\mathrm{PGR} > \mathrm{PLR}$.

Figure 1.27 shows the result for the full data set of Odean (1998): PGR is 0.148 against PLR of 0.098, a difference of 0.050 with a t-statistic of 35—investors realize gains at roughly 1.5 times the rate at which they realize losses. The counts behind the proportions make the asymmetry vivid: 13,883 realized gains against 79,658 paper gains, versus 11,930 realized losses against 110,348 paper losses. A notable internal check is December: PGR falls to 0.108 and PLR rises to 0.128, so that PLR exceeds PGR (difference 0.020, $t = 4.3$)—consistent with year-end tax-loss selling temporarily overriding the reluctance to realize losses.

The disposition effect is not an artifact of retail accounts or of stocks. It is present among mutual fund managers (Frazzini, 2006) and, in modified form, in the housing market, where sellers whose expected price is below their purchase price set higher reservation prices and wait longer to sell (Genesove and Mayer, 2001). Nor is it easily rationalized: the most obvious potential explanation—that sales of winners reflect informed trading—fails on the data, because the winners investors sell

PGR and PLR for the Entire Data Set

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. Realized gains, paper gains, losses, and paper losses are aggregated over time (1987–1993) and across all accounts in the data set. PGR and PLR are reported for the entire year, for December only, and for January through November. For the entire year there are 13,883 realized gains, 79,658 paper gains, 11,930 realized losses, and 110,348 paper losses. For December there are 866 realized gains, 7,131 paper gains, 1,555 realized losses, and 10,604 paper losses. The *t*-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

	Entire Year	December	Jan.–Nov.
PLR	0.098	0.128	0.094
PGR	0.148	0.108	0.152
Difference in proportions	−0.050	0.020	−0.058
t-statistic	−35	4.3	−38

Figure 1.27: The disposition effect. Proportions of gains realized (PGR) and losses realized (PLR) for the entire data set of Odean (1998): over the whole year, PGR = 0.148 versus PLR = 0.098; in December only, the pattern reverses (PLR = 0.128 versus PGR = 0.108), consistent with tax-motivated loss realization. *Source:* Odean (1998).

subsequently *outperform* the losers they hold on to (Odean, 1998). If anything, the optimal tax-motivated strategy pushes the other way (realize losses, defer gains), making the observed behavior costly on two dimensions. The effect is also more pronounced among less sophisticated investors (Dhar and Zhu, 2006), and it has pricing consequences: Frazzini (2006) links the disposition-driven selling pressure after good news to underreaction and momentum, providing one of the clearest cases where a micro-level behavioral fact predicts a market-level anomaly—exactly the kind of out-of-sample discipline Section 1.2 demands. The preference-based foundation, prospect theory and loss aversion with reference dependence at the purchase price, is the subject of Part III.

1.7.6 Who Trades Against the Anomalies?

If anomalies reflect mispricing, someone must eventually trade against it. McLean, Pontiff, and Reilly (2025) ask who trades against forecasted returns, measuring the trades of nine classes of market participants against the forecasted returns produced by 130 published anomalies. Figure 1.28 shows their central exhibit: for each participant group, average trading across stocks sorted into quintiles by forecasted return. The patterns are sharp. Retail investors trade *against* the forecasts: their trading falls from 1.24% in the lowest forecasted-return quintile to 0.01% in the highest (spread −1.23%, $t = -2.3$)—they are the buyers of the stocks predicted to do poorly. Short sellers are on the right side, tilting toward low-forecasted-return stocks (spread +0.62%, $t = 6.0$), as are hedge funds in the top forecasted-return quintile. Firms themselves are the most aggressive "smart money" of all: net firm trading (issuance and repurchase) moves +11.47 percentage points ($t = 6.2$) from the low-forecasted-return quintile to the high—firms issue when forecasted returns are low and repurchase when they are high, consistent with the market-timing interpretation of Section 1.5.4. The correction of mispricing, in other words, comes primarily from short sellers and from firms—not from the conventional asset managers to whom we now turn.

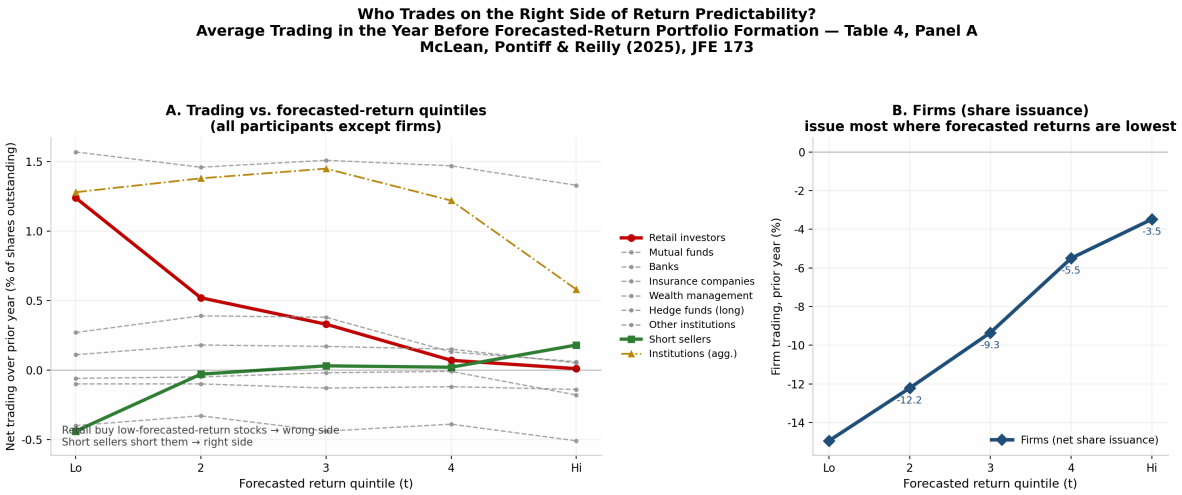


Figure 1.28: Taking sides on return predictability. Average trading in the year before forecasted-return portfolio formation. Panel A: mean trading by participant group (retail investors, mutual funds, banks, insurance companies, wealth management, hedge funds, other institutions, short sellers, and institutions in aggregate) across quintiles of forecasted stock returns built from 130 published anomalies. Panel B: firms' net share issuance across the same quintiles—firms issue most where forecasted returns are lowest. *Source:* McLean, Pontiff, and Reilly (2025), Table 4, Panel A.

1.7.7 Institutional Investors

Average performance. What should we expect from professional money managers? Berk and Green (2004) build the benchmark answer in a fully rational industry equilibrium: skilled managers exist, investors chase skill, and fund flows compete away the rents—so managers' *gross* alpha should be positive on average (they do have skill), while their *net* alpha should be zero on average (fees absorb the skill). The evidence roughly matches the second prediction and strains the first: the gross alpha of mutual funds appears to be positive, but not substantially so. One might be tempted to conclude that small average alphas imply prices are correct—no one earns abnormal profit, so no profit is left on the table. The inference is wrong, and the next two findings show why: exploiting anomalies requires taking positions against them, and that is precisely what institutions, in aggregate, fail to do.

Institutions and the anomalies. Lewellen (2011) documents that institutional investors in aggregate essentially hold the market portfolio: despite managing the majority of U.S. equity, institutions show no discernible tilt toward well-known anomalies. Edelen, Ince, and Kadlec (2016) sharpen the conclusion with a more striking finding: during the formation periods of anomaly portfolios, institutions have a strong tendency to buy the *overvalued* stocks—the short legs. Figure 1.29 shows their evidence: the change in institutional ownership over formation periods is systematically larger for short-leg stocks than long-leg stocks, with average long-minus-short differences of −3.0% (change in the number of institutions) and −1.2% (change in fraction of shares held), both highly significant, and with the notable exception of momentum, where institutions trade *with* the anomaly. Hedge funds and institutions with high turnover exploit anomalies more than other institutions do—consistent with the short-seller evidence of Section 1.7.6—but they are too small relative to the market to remove the patterns.

Institutional behavior thus closes the loop opened by the arbitrage critique in Section 1.2. The agents best positioned to correct mispricing hold, in aggregate, something close to the market

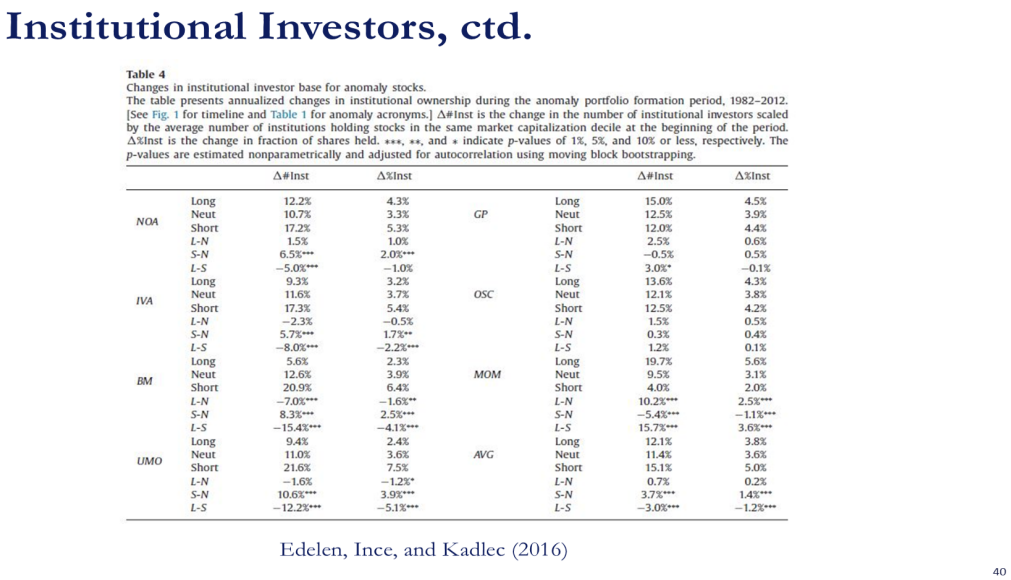


Figure 1.29: Institutions trade with the anomalies, not against them. Change in institutional ownership during anomaly portfolio formation periods, 1982–2012, for the long, neutral, and short legs of seven anomalies (NOA, IVA, BM, UMO, GP, and OSC, plus momentum (MOM)) and their average. The long-minus-short difference is negative and significant for every anomaly except momentum: institutions accumulate the overvalued (short-leg) stocks. *Source:* Edelen, Ince, and Kadlec (2016), Table 4.

portfolio—and tilt, if anything, toward the mispriced side of several anomalies. The reasons are the limits to arbitrage and agency structures of Part II: benchmarking, career risk, leverage constraints, and flows punish concentrated contrarian positions even when they are eventually right.

1.8 The Road Ahead

This chapter has assembled facts; the rest of the book is about mechanisms. Three threads run through everything above.

First, the price facts—predictability and excess volatility in the aggregate market, the anomaly families in the cross-section, and bubbles—share a common structure: each admits a rational interpretation built on time-varying risk or risk prices, and a behavioral interpretation built on belief or preference deviations. The state of knowledge differs case by case. Short-term reversal now looks mostly like compensation for liquidity provision; profitability and investment fit partly into rational factor structures; the equity premium, the low-risk anomalies, and post-earnings-announcement drift remain actively contested. The book's working method is to keep both candidate mechanisms explicit and ask what evidence would separate them.

Second, the investor facts identify the demand side. Non-participation, under-diversification, excessive trading, the disposition effect, and buying-high-and-selling-low are each hard for the frictionless model of portfolio choice to absorb, and each maps onto a specific psychological candidate—loss aversion, narrow framing, overconfidence, extrapolation—developed in Part III. The strongest results connect the two levels: the disposition effect predicts pricing patterns (Frazzini, 2006), and survey-based extrapolative expectations discriminate between sentiment and discount-rate readings

of predictability.

Third, the "who trades against it" facts identify the supply side. Firms and short sellers take the correct side of forecasted returns; conventional institutions mostly do not (Lewellen, 2011; Edelen, Ince, and Kadlec, 2016; McLean, Pontiff, and Reilly, 2025). Why professional arbitrage is limited—and what those limits imply for how far mispricing can go—is the subject of Part II.

The plan of the book follows this logic:

- **Part II. Limits to Arbitrage.** Noise trader risk, short-sale constraints, funding constraints, and agency frictions: why rational capital cannot be relied upon to keep prices at fundamental value.
- **Part III. Psychological Foundations and Applications.** Non-standard preferences (prospect theory, loss aversion, the disposition effect); biased beliefs (overconfidence, over-extrapolation); heterogeneous beliefs and disagreement; bounded rationality (limited attention, investor sentiment); and psychology-free models that reproduce behavioral pricing patterns with rational agents.

By the end, every fact in this chapter should be attached to at least one explicit mechanism, and every mechanism to at least one prediction that could have failed.

Chapter 2

Limits to Arbitrage

Section 1.2 stated the oldest objection to behavioral finance: if irrational investors push prices away from fundamental values, rational investors have every incentive to trade against them, and in doing so to push the prices back. Irrational traders may lose money, but prices stay correct. This *arbitrage critique*, associated with Friedman (1953), is the doorway through which every behavioral claim must pass. A major achievement of behavioral finance in the 1990s was to push back against it—to show that there are *limits to arbitrage*: risks and costs that prevent rational investors from correcting mispricings quickly, easily, or at arbitrary scale. Where those limits bind, irrational demand can move prices significantly and for a long time.

This chapter is the supply side of the behavioral story. Section 2.1 fixes terminology: arbitrage, the law of one price, fundamental value, and mispricing. Section 2.2 develops the theory of limited arbitrage—the risks (fundamental risk, noise trader risk, horizon risk, model risk) and the costs (trading costs, shorting costs, learning and information-processing costs) that constrain arbitrageurs, drawing on De Long et al. (1990) and Shleifer and Vishny (1997); a case study of Long-Term Capital Management shows the risks operating together on one balance sheet. Section 2.3 examines settings in which the fundamental value is nearly observable—twin shares, equity carve-outs, and index inclusions—and in which substantial, long-lived mispricings have been documented. Section 2.4 connects limits to arbitrage to the anomaly families of Chapter 1, from correlational amplification tests to quasi-experimental causal evidence, and to the costs of learning that an anomaly exists. Section 2.5 asks how limited arbitrage is, and what that implies for what follows.

2.1 Arbitrage, Fundamental Value, and Mispricing

Four pieces of terminology organize everything in this chapter. The *fundamental value* of an asset is its price in an economy with rational investors and no frictions—the price that properly reflects all available public information, the efficient market price. In an economy with frictions, or in which some investors are less than fully rational, an asset's price can depart from fundamental value; such a departure is a *mispricing*, or an *inefficiency*. Rational investors who trade against mispricings are conventionally called *arbitrageurs*; their less than fully rational counterparts are *noise traders*.

Definition 2.1 (Arbitrage)**.** *An arbitrage is an investment strategy that guarantees a positive payoff in some state of the world, with no possibility of a negative payoff, no net investment, and at arbitrary scale.*

In symbols, an arbitrage is a portfolio with payoffs V_s across future states s satisfying $V_0 \leq 0$ (zero or negative net investment today), $V_s \geq 0$ for all s (no possibility of a loss in any future

state), and $\Pr[V_s > 0] > 0$ (strictly positive probability of a gain). So defined, an arbitrage is a money machine, and its existence is inconsistent with any common notion of equilibrium: if such an opportunity were present, someone would act on it, prices would change, and the opportunity would vanish. Closely related is the *law of one price*: two assets with the same payoff in every state of the world must have the same price. The key to the law of one price is the existence of *perfect substitutes*—assets with exactly the same payoffs—and it underlies much of the design of derivatives, where replication arguments pin down prices exactly.

The textbook definition is the right benchmark, but almost nothing that professionals call "arbitrage" satisfies it. Buying an underpriced stock, shorting an overpriced one, and trading a merger's acquirer against its target all require capital, involve risk of loss, and can be scaled only so far. The professional usage refers to trading on price discrepancies that *involve* risk and capital; this chapter follows that usage. The arbitrage critique implicitly assumes arbitrage in the textbook sense—costless, riskless, unlimited. The behavioral response is that real-world arbitrage is none of these things, and that the gap between the two notions of arbitrage is exactly where noise traders live. The magnitude of that gap has been formalized in a large theoretical literature surveyed by Gromb and Vayanos (2010).

2.2 The Theory of Limited Arbitrage

2.2.1 A Taxonomy of Limits

The arbitrage critique says that correcting a mispricing is easy. In reality it is not: there are risks and costs that limit arbitrageurs' ability and willingness to take the other side of irrational demand. Table 2.1 organizes the standard taxonomy.

Table 2.1: Limits to arbitrage: risks and costs

Limit	Type	Description
Fundamental risk	Risk	Adverse news about the fundamental value
Noise trader risk	Risk	The mispricing worsens before it closes
Horizon risk	Risk	Capital is withdrawn before the correction
Model risk	Risk	The "fundamental value" itself is uncertain
Trading costs	Cost	Commissions, bid–ask spreads, price impact
Shorting costs	Cost	Lending fees, short-sale bans, market prohibitions
Learning costs	Cost	Discovering that the anomaly exists at all
Information processing costs	Cost	Acquiring and processing fundamental data

Notes: Following Shleifer and Vishny (1997) and the taxonomy in the EF8083 course materials. The risks are conceptually prior: even with zero transaction costs, each of the four risks alone can deter an arbitrageur.

2.2.2 Fundamental Risk

Fundamental risk is the risk that there will be adverse news about the fundamental value of the mispriced asset. It is usually a problem because perfect substitutes are not available for what the arbitrageur really wants to trade. An investor who believes Ford is underpriced can hedge market-wide movements by shorting GM, but GM is not Ford: news about Ford's own fundamentals—its models, its labor contracts, its litigation—arrives unhedged. The hedge strips out the priced factors

and leaves exactly the idiosyncratic risk that the trade is concentrated in. Fundamental risk is thus not a footnote to arbitrage but its central occupational hazard, and Section 2.3 shows that mispricings are indeed largest where substitutes are worst.

A two-equation decomposition makes the point exact. Let firm i and its same-industry peer j have returns

$$r_{it} = \alpha_i + \beta_i r_{I,t} + \varepsilon_{it}, \qquad r_{jt} = \alpha_j + \beta_j r_{I,t} + \varepsilon_{jt},$$

where $r_{I,t}$ is the industry factor and $\mathrm{Corr}(\varepsilon_{it}, \varepsilon_{jt}) = \rho_{ij}$ measures how close a substitute j is for i. If i is overvalued by θ_{it}, the arbitrageur goes long j and shorts β_j/β_i units of i, so that

$$r_t^{\text{arb}} = r_{jt} - \frac{\beta_j}{\beta_i} r_{it} = \Big(\alpha_j - \frac{\beta_j}{\beta_i}\alpha_i\Big) + \underbrace{\Big(\varepsilon_{jt} - \frac{\beta_j}{\beta_i}\varepsilon_{it}\Big)}_{\text{unhedgeable firm-specific risk}}.$$

The industry exposure cancels; the firm-specific risk does not. With $\beta_i = \beta_j$ and equal residual variances σ_ε^2, the arbitrage portfolio's residual variance is

$$\sigma^2_{\text{arb}} = 2\,\sigma_\varepsilon^2\,(1 - \rho_{ij}).$$

A perfect substitute drives ρ_{ij} toward one and the variance toward zero; with no close substitute, ρ_{ij} is far below one and the variance approaches $2\sigma_\varepsilon^2$. A mean–variance arbitrageur with risk aversion γ therefore takes the position

$$x^* = \frac{\theta_{it}}{\gamma\,\sigma^2_{\text{arb}}} = \frac{\theta_{it}}{2\gamma\,\sigma_\varepsilon^2\,(1-\rho_{ij})},$$

which grows with the quality of the substitute and shrinks with firm-specific risk. The chain of implication is the whole theory in miniature: no close substitute ⇒ fundamental risk remains ⇒ positions are bounded ⇒ the mispricing θ_{it} persists.

2.2.3 Noise Trader Risk

The deeper limit, emphasized by De Long et al. (1990) and Shleifer and Vishny (1997), is *noise trader risk*: the risk that the mispricing worsens in the short run, forcing the arbitrageur to close out prematurely at a loss. The force behind it is not the security but the capital structure of the arbitrage itself. Real-world arbitrageurs typically manage other people's money, and they use leverage. Both facts make their positions fragile exactly when the trade looks worst.

Consider the sequence. Arbitrageurs act against a mispricing. The mispricing persists—or widens. The clients, observing losses in an account that was sold to them as near-riskless, conclude the manager is incompetent and demand their money back. To deliver the cash, the manager must unwind the position at a loss, realizing exactly the outcome the clients feared. A manager who foresees this possibility will trade less aggressively against the mispricing from the start. The equilibrium consequence, made precise in De Long et al. (1990), is uncomfortable for the classical view: noise trader risk is priced, the mispricing survives, and noise traders are not automatically driven out of the market.

Leverage and short selling compound the problem through the lenders. A fund that borrows to buy an undervalued asset posts the position as collateral; if the mispricing worsens, the lender sees the collateral decline and may call the loan, forcing the sale. A fund that shorts an overvalued asset faces the mirror image: when the asset rises, the lender demands additional margin, and if the fund cannot meet the demand, the short position is closed out at a loss—just before, perhaps, the correction arrives. Every link in this chain—clients, lenders, margin clerks—withdraws support

at the moment of maximum drawdown, which is precisely the moment the expected profit of the trade is largest. Shleifer and Vishny (1997) call arbitrage conducted under these conditions *performance-based arbitrage*: the arbitrageur's capital base is a function of the very mispricing he is trying to correct.

2.2.4 Case Study: Long-Term Capital Management

The leverage mechanics above are not hypothetical. Long-Term Capital Management (LTCM) was founded in 1994 by John Meriwether, formerly head of fixed-income arbitrage at Salomon Brothers, and its principals included Myron Scholes and Robert Merton, who would share the 1997 Nobel Memorial Prize. The strategy was *convergence trades*: go long cheap securities, short rich ones, and bet that the spreads between them revert—swap spreads, Treasury on-the-run versus off-the-run spreads, merger arbitrage, and Russian and other emerging-market debt. At its peak in early 1998 the fund had equity of roughly $4.8 billion, a balance sheet above $100 billion, and off-balance-sheet derivatives with notionals in the trillions, for on-balance-sheet leverage above $25\times$ (far higher counting derivatives). Each individual position was small; the book as a whole was one massively correlated bet, short liquidity and short volatility across every market it touched.

The collapse played out over weeks. On August 17, 1998, Russia defaulted, triggering a global flight to liquidity and quality. Spreads diverged instead of converging; LTCM lost about $4.5 billion within weeks. Its lenders raised margin requirements, forcing the fund to delever into falling markets—a liquidity spiral. On September 23, 1998, the Federal Reserve Bank of New York brokered a $3.6 billion private recapitalization by a consortium of fourteen banks, the fund having been judged too interconnected to fail.

LTCM's trades were convergence trades, so each position had a substitute—but no perfect one, and the imperfections compounded one another. On-the-run and off-the-run Treasuries differ in liquidity, not just price, so $\sigma^2_{\text{arb}} = 2\sigma^2_\varepsilon(1-\rho_{ij}) > 0$; in a crisis ρ_{ij} collapses, and the substitute stops hedging exactly when it matters most (fundamental risk). The flight to quality moved prices *further* from fair value before any convergence, so the optimal position $x^* = \theta/(\gamma\sigma^2_{\text{arb}})$ shrank just as LTCM's own capital shrank with its losses, forcing unwinding at the worst possible moment (noise trader and wealth risk). And as margin requirements m_j rose, funding conditions entered the pricing of positions—with margin-sensitive intermediaries, an asset's expected return must compensate for the margin it ties up, $E_t[r_j] = R_f + \gamma\sigma_j^2 x_j + \mu\, m_j$ with $\mu > 0$—so forced fire sales widened spreads further and deepened the losses: a feedback loop. The lesson generalizes beyond 1998. Even a "riskless" arbitrage is exposed to substitute imperfection, funding and wealth risk, and margin spirals, and leverage amplifies all three. The market can stay irrational longer than an arbitrageur can stay liquid.

2.2.5 Arbitrage in Reality

Noise trader risk is not a theoretical curiosity, and the same technology-boom years that broke LTCM supplied a second canonical demonstration involving no exotic instruments at all. Julian Robertson's Tiger Fund and George Soros's Quantum Fund were among the most celebrated investors of their generation—global macro and value-oriented, exactly the professionals the arbitrage critique has in mind when it says rational traders will correct irrational prices. During the technology boom, both judged internet stocks to be massively overvalued and positioned against them. Figures 2.1 and 2.2 show the consequence: as the overpricing widened rather than closed, both funds sustained deep, prolonged underperformance relative to the market. The Tiger Fund, down and facing redemptions, liquidated in March 2000—within weeks of the peak its positions had anticipated. Soros's Quantum

Fund suffered its large losses in the April 2000 technology reversal and subsequently restructured away from the strategy. The bets, in the end, were right; the capital did not survive to collect. This is noise trader risk in the data, and it explains why Brunnermeier and Nagel (2004) find that many hedge funds rode the bubble rather than popped it (Section 1.6).

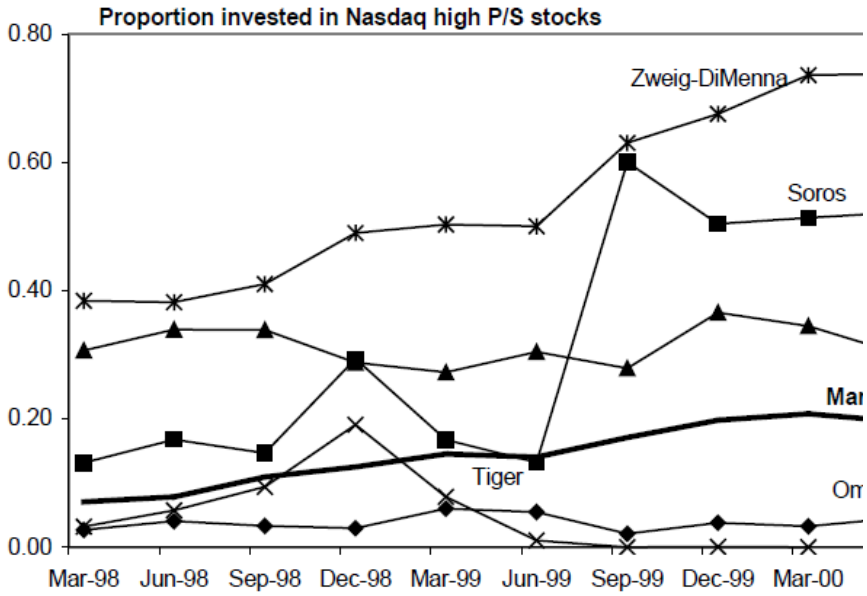


Figure 2.1: Arbitrage in reality. Cumulative performance of the Tiger Fund and George Soros's Quantum Fund relative to the aggregate stock market during the late-1990s technology boom. Positioning against the widening overpricing produced sustained losses for both funds relative to the market. *Source:* EF8083 course materials.

2.2.6 Costs: Trading, Shorting, and Information

The risks in Table 2.1 bind even in an environment with no costs; the costs bind on top of them. *Trading costs* include brokerage commissions, bid–ask spreads, and the price impact of the arbitrageur's own orders—at scale, the act of trading moves the very price discrepancy being exploited. *Shorting costs* are more binding still: short selling requires borrowing shares from a lender and paying lending fees, and those fees can be very high for particular stocks in particular periods, precisely the stocks an arbitrageur most wants to short. Short selling is also occasionally banned outright by regulators, as in a number of markets during the 2008 crisis, and some countries do not permit it at all. Because correcting overpricing requires shorting while correcting underpricing requires only cash, shorting costs make limits to arbitrage asymmetric: overpricing is harder to eliminate than underpricing, a point to which Section 2.4 returns. *Learning costs* come before all of these: an arbitrageur must first know that an anomaly exists, and that knowledge is not free. McLean and Pontiff (2016) show that academic publication itself transmits the knowledge—once an anomaly is published, its profitability declines, and Section 2.4 quantifies the decline. Finally, *information processing costs* are distinct from learning costs: even an arbitrageur who is aware of the anomaly must acquire and process the underlying data at low cost in order to trade on it, and for many anomalies that processing is substantial.

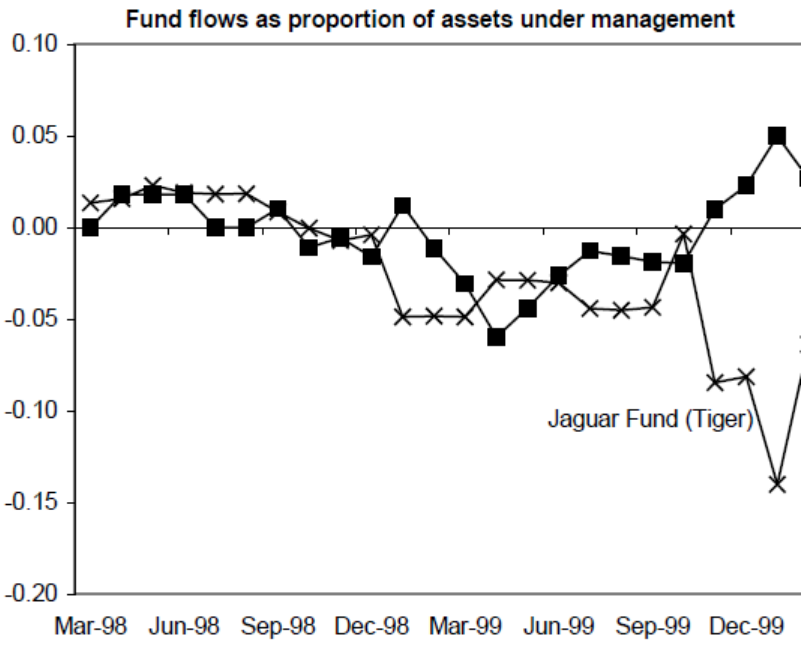


Figure 2.2: Arbitrage in reality, continued. Further performance of Soros's Quantum Fund following the period shown in Figure 2.1, as the mispricing continued to widen before the eventual reversal. *Source:* EF8083 course materials.

2.2.7 The Dynamic Arbitrage Critique

The critique has a second, dynamic version. Even if rational traders cannot quickly correct a mispricing, the argument runs, assets will *eventually* become correctly priced, and by trading sub-optimally in the meantime, irrational traders become steadily poorer. Wealth transfers from noise traders to arbitrageurs, the noise traders exit the market, and irrationality becomes irrelevant for prices. This argument generated a sizeable literature on the survival of noise traders, and it is weaker than it first appears. De Long et al. (1991) show that noise traders can survive indefinitely: if noise traders hold portfolios that are exposed to the risks other investors dislike, they can earn above-average expected returns on their noise, and the process of losing wealth to arbitrageurs can be slower than the arrival of new noise traders. The exit may take a very long time, and there is no mechanism that stops a new generation of irrational investors from entering the market. The survey evidence on extrapolative expectations assembled later in this book suggests that they reliably do.

One institutional limit deserves separate mention because it does not involve risk or cost at all. Benchmarked institutional investors are evaluated relative to an index, so holding an underpriced stock that is out of the benchmark and declining is career-dangerous even if it is eventually profitable; as Chapter 1 discussed for the low-volatility anomaly (Baker, Bradley, and Wurgler, 2011; Frazzini and Pedersen, 2014), benchmarks themselves function as limits to arbitrage. This connects the theory here to the evidence of Section 1.7.7 that institutional investors in aggregate hold something close to the market portfolio and, if anything, accumulate the overvalued sides of anomalies (Lewellen, 2011; Edelen, Ince, and Kadlec, 2016).

2.3 Evidence from Near-Arbitrage Situations

There are good theoretical reasons why arbitrage may be limited. Is there evidence that it is? The cleanest evidence would be a substantial, long-lived mispricing—a price that demonstrably departs from an observable fundamental value and stays departed. Such examples are not easy to document, precisely because fundamental value is usually unobservable and model risk (Table 2.1) lets a defender of efficiency reprice the fundamental instead of the market. The strategy of the literature is therefore to find settings where the fundamental value is nearly observable, so that the mispricing is a matter of arithmetic rather than modeling: twin shares, equity carve-outs, and index inclusions. Each setting also isolates particular frictions, which is what makes the evidence diagnostic rather than merely anecdotal.

2.3.1 Twin Shares

Twin shares are distinct shares that are claims to the same cash flows. In 1907, Royal Dutch and Shell merged their interests while remaining separate entities: Royal Dutch retained a claim to 60% of the combined group's cash flows, Shell to 40%, and the two continued to trade as separate listed companies. If prices equal fundamental values, the market value of Royal Dutch must be 1.5 times the market value of Shell, dollar for dollar, at every date:

$$\text{MV}_{RD} = 1.5 \times \text{MV}_{Shell}.$$

No hedging is required, no model of cash flows is required, and the law of one price applies with essentially perfect substitutes. Figure 2.3 shows the actual deviation from parity over nearly two decades: Royal Dutch traded at discounts to parity on the order of 30% at the extremes, with the deviation persisting for years at a time. This is about as close as field evidence gets to a controlled demonstration that prices depart from fundamentals and stay departed. Froot and Dabora (1999) add the detail that makes the case behavioral: the two shares traded in different locations (Royal Dutch largely in the United States and the Netherlands, Shell largely in the United Kingdom), and the deviation from parity correlates with relative market conditions in the two trading locations—the same cash flows are priced differently depending on whose demand is dominant where.

The same architecture appears in a modern and even starker form in the Chinese market. Dozens of Chinese companies are listed both on the mainland exchanges (A shares) and in Hong Kong (H shares). The two share classes carry identical claims, but mainland investors cannot freely buy H shares and international investors cannot freely buy A shares, so each class is priced by its own investor base. Figure 2.4 shows the result: A shares trade at a persistent premium over H shares of the same companies—the premium is typically large and has persisted for decades. Capital controls are here a legally enforced limit to arbitrage, which is what makes the case so clean: no one can arbitrage the gap because no one is allowed to.

2.3.2 Equity Carve-Outs

An equity carve-out occurs when a parent company conducts an initial public offering for shares in a subsidiary while retaining a large ownership stake. Between 1985 and 2000, Mitchell, Pulvino, and Stafford (2002) identify 70 cases in which the market value of the parent was less than the value implied by its stake in the subsidiary—cases where the market value of the parent fell below 80% of the market value of the subsidiary alone, implying that the market ascribed *negative* value to the parent's remaining assets, its "stub." A negative stub means the market's prices cannot all be right, since the sum of the parts is below one of the parts.

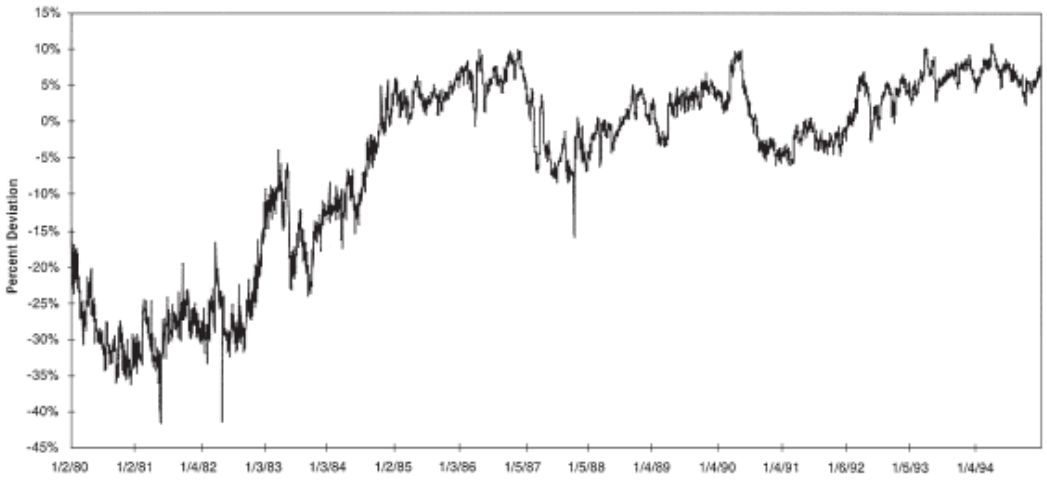


Fig. 1. Log deviations from Royal Dutch/Shell parity. Source: Froot and Dabora (1999).

Figure 2.3: Twin shares. Deviation of the Royal Dutch–Shell price ratio from the theoretical 60:40 parity, over the life of the twin-share structure. The two securities are claims to identical cash flows, yet the deviation from parity reaches tens of percent and persists for years. *Source:* EF8083 course materials, following Froot and Dabora (1999).



Figure 2.4: The A–H premium. Price ratio of mainland-listed A shares to Hong Kong-listed H shares of the same Chinese companies. Despite identical claims, A shares trade at a persistent premium over H shares; capital controls prevent the arbitrage that would close the gap. *Source:* EF8083 course materials, based on Hang Seng Stock Connect China AH Premium Index data.

The most famous case is 3Com and Palm. In March 2000, 3Com sold 5% of its subsidiary Palm in an IPO, announcing that it would spin off the remainder to its own shareholders within nine months; each share of 3Com would thereby carry a claim on 1.5 shares of Palm. The arithmetic bounds are tight: since 3Com's other businesses generated positive cash flows and 3Com had substantial net cash, the price of 3Com should have been *at least* 1.5 times the price of Palm. At Palm's post-IPO price of $95, 3Com should have traded for no less than $142. It traded at $81. The market was valuing 3Com *excluding* Palm at about negative $60 per share—an impossibility, unless one believes a profitable company with cash in the bank was worth less than zero (Lamont and Thaler, 2003).

Why was this mispricing not corrected? Every limit in Table 2.1 was present. Shorting Palm was nearly impossible: very few shares were available to borrow, so lendable Palm shares carried extraordinary fees, and the buy side of the trade (going long 3Com) carried fundamental risk—news about 3Com's remaining businesses could and did arrive. Mitchell, Pulvino, and Stafford (2002) document that in about 30% of their negative-stub cases, adverse fundamental news arrived before the gap closed, validating the fundamental-risk concern. Noise trader risk completed the triad: Palm's price moved with the technology mania, and the mispricing did not close until the spin-off approached and the mechanical link between the securities forced convergence.

2.3.3 Index Inclusions

Stocks are regularly added to the S&P 500 to replace stocks removed through merger or bankruptcy. Crucially, S&P's stated goal is to create a representative index, not to convey information about the investment value of the included stocks; inclusion is, from the standpoint of fundamentals, a nonevent. Yet both Harris and Gurel (1986) and Shleifer (1986) find that included stocks jump in price by about 3.5% on average around the announcement and execution dates. The two studies part company on what happens next: Shleifer (1986) finds that the jump is not reversed, while Harris and Gurel (1986) finds that it reverses within weeks. A nonevent that permanently moves prices implies downward-sloping demand curves for stocks—and, since the information content is nil, the persistent jump is widely viewed as a mispricing. It is evidence of limits to arbitrage because an arbitrageur could, in principle, short the included stock and buy a substitute portfolio.

What limits the correction? The question is empirical, and it has a sharp answer. Wurgler and Zhuravskaya (2002) observe that the mispricing can only be arbitraged by shorting the included stock against a hedge of substitute stocks, and the quality of the hedge varies: for some additions, close industry matches exist; for others, any hedge retains large idiosyncratic risk—fundamental risk in the language of Table 2.1. If arbitrage disciplines the price jump, the jump should be larger for stocks with worse substitutes. This is exactly what they find (Figure 2.5): included stocks with more idiosyncratic risk in the best available hedge jump significantly more upon inclusion. The magnitude of a textbook-style mispricing varies cross-sectionally with the difficulty of the arbitrage, exactly as the theory predicts.

2.3.4 The Disappearing Index Effect

The index-inclusion evidence has a modern epilogue with an important moral. Greenwood and Sammon (2025) document that the inclusion effect has essentially disappeared: the average abnormal return on addition to the S&P 500 fell from 7.4% in the 1990s to less than 1% over the past decade, even as the share of stock market assets linked to the index grew enormously (Figure 2.6). The reason is that the supply of arbitrage capital adapted: professional active investors now stand ready to provide liquidity that accommodates the demand of passive indexers. Although index trackers still buy on the order of 7–8% of a stock's float upon addition, total institutional ownership barely

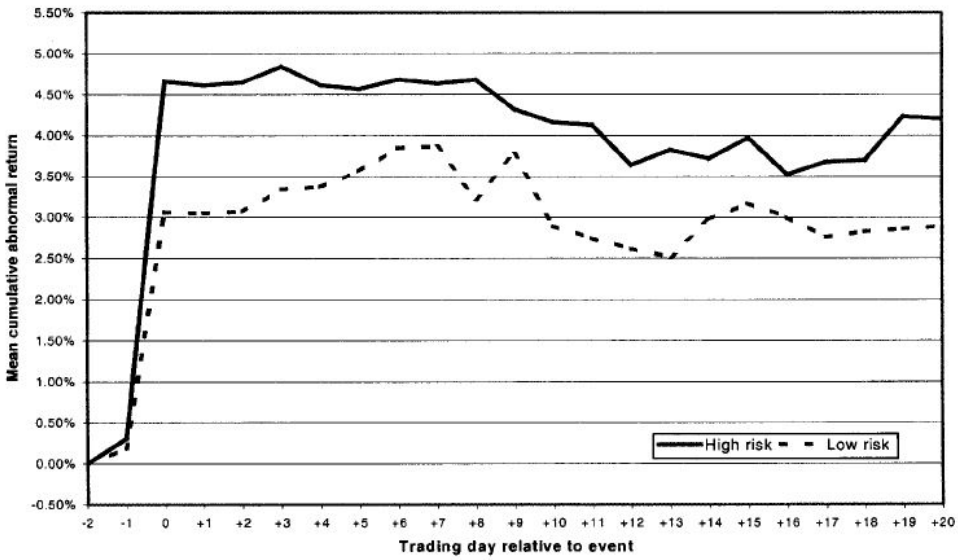


Figure 2.5: Does arbitrage flatten demand curves for stocks? The price jump upon S&P 500 inclusion is larger for added stocks whose potential arbitrage hedges carry more idiosyncratic risk, i.e. for stocks with worse substitutes. *Source:* Wurgler and Zhuravskaya (2002).

moves around index changes—the passive demand is met by active supply, at a much smaller price concession than before. Limits to arbitrage, in other words, are endogenous: they can bind tightly in one decade and relax in the next as the arbitrage industry develops. The finding is doubly useful: it confirms the demand-based reading of the original effect (demand pressure moves prices), while cautioning against treating any measured limit as a constant of nature.

2.4 Limits to Arbitrage in the Cross-Section of Anomalies

The near-arbitrage situations of Section 2.3 establish that limits to arbitrage exist. The anomaly families of Chapter 1—Section 1.5.9 in particular—raise the complementary question: if anomalies reflect mispricing, they should be *more pronounced* among stocks where the limits are more severe. This prediction generates cross-sectional tests, and, more recently, quasi-experimental evidence.

2.4.1 Amplification Tests

Two proxies have carried most of the weight. Nagel (2005) uses institutional ownership as a proxy for short-sale constraints—institutions are the natural lenders of shares, so stocks with low institutional ownership are hard and expensive to short—and finds that the short legs of anomalies are stronger among stocks with low institutional ownership. Stambaugh, Yu, and Yuan (2015) use idiosyncratic volatility as a proxy for arbitrage risk—an arbitrageur who shorts an overpriced stock must hedge, and idiosyncratic volatility is exactly the risk the hedge cannot remove—and find that mispricing is amplified among stocks with the highest idiosyncratic volatility (Figure 2.7). Their central finding sharpens the asymmetry intuition of Section 2.2: the amplification is concentrated on the short side. Anomalies' overpriced legs are strong precisely where shorting is difficult, while the underpriced legs are comparatively insensitive—consistent with arbitrage being harder on the short side, where trading requires borrowing shares, than on the long side, where it requires only cash.

Figure 2. Average index effect by year.

For each event, we compute the cumulative abnormal return over the period of interest. Lines represent 10-year moving averages of the yearly averages.

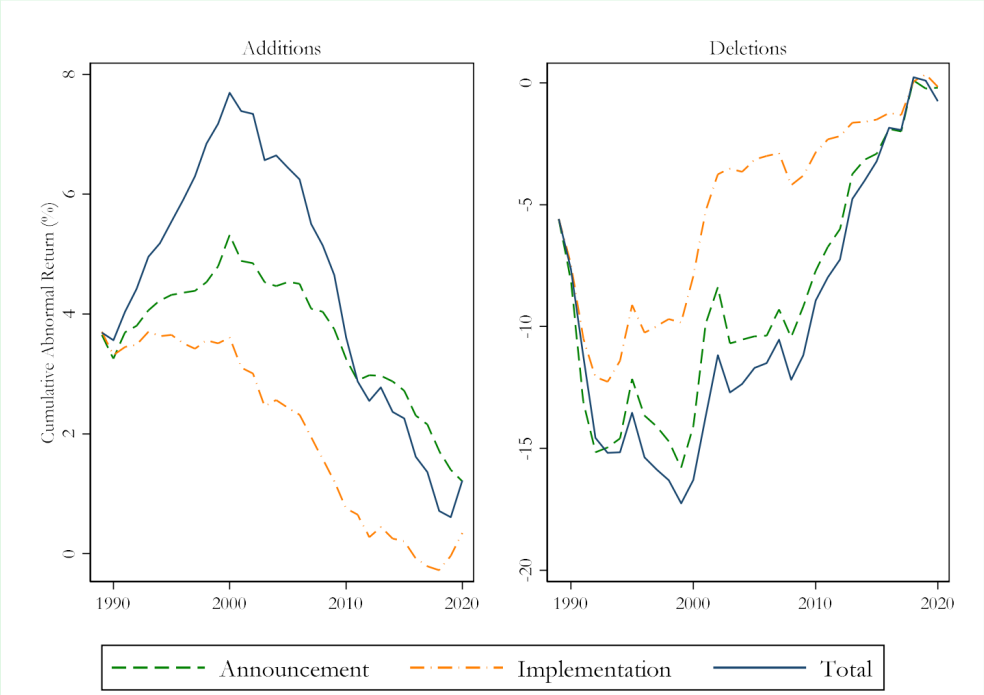


Figure 2.6: The disappearing index effect. Cumulative abnormal returns of stocks added to the S&P 500 around the effective date of the index change, by decade, from the 1970s through the 2020s. The price impact shrinks from peaks of roughly 4–7% in the 1970s–1990s to approximately zero in the two most recent decades, even as the share of market assets linked to the index grew. *Source:* Greenwood and Sammon (2025).

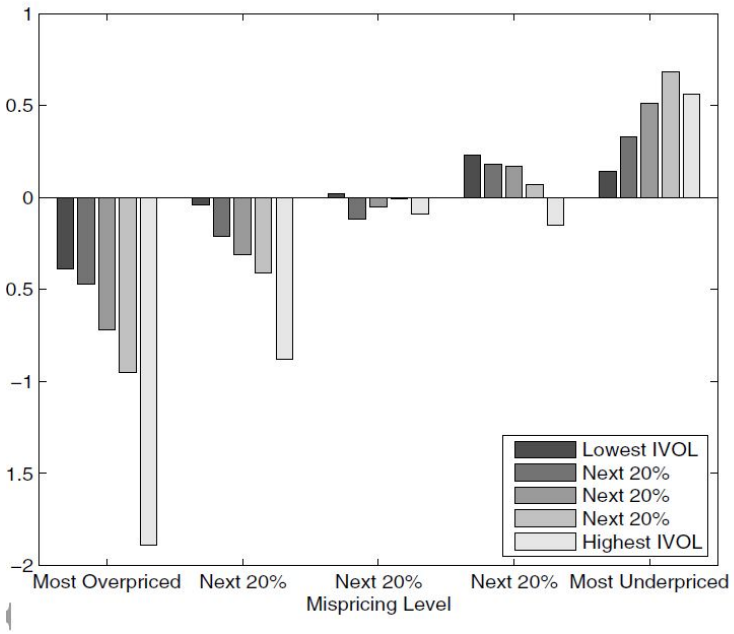


Figure 2.7: Arbitrage asymmetry and the idiosyncratic volatility puzzle. Anomaly-based mispricing scores (net of size and book-to-market effects) across quintiles of idiosyncratic volatility. Overpricing increases sharply with idiosyncratic volatility; underpricing is comparatively insensitive, consistent with the short side of anomalies being harder to arbitrage. *Source:* Stambaugh, Yu, and Yuan (2015).

2.4.2 Causal Evidence: Regulation SHO

The amplification tests are correlational—severity of limits and severity of mispricing move together, but both could proxy for a third factor. Chu, Hirshleifer, and Ma (2020) provide causal evidence using Regulation SHO as a quasi-experiment. Between May 2, 2005, and August 6, 2007, the SEC removed short-sale price tests (the uptick rule and related restrictions) for a randomly selected group of pilot stocks within the Russell 3000 index. Random selection delivers clean identification under a differences-in-differences design: construct anomaly long–short portfolios separately for pilot and non-pilot stocks, and compare the return spread for pilot firms relative to control firms, before and during the pilot period. The estimating equation is

$$r_{it} = \alpha_t + \beta_1 \, Pilot_i + \beta \, Pilot_i \times During_t + \epsilon_{it}, \tag{2.1}$$

where r_{it} is the monthly return of portfolio i—the long leg, the short leg, or the long–short portfolio of an anomaly—in month t; α_t are month fixed effects, which absorb the main effect of $During_t$; $Pilot_i$ indicates portfolios constructed from pilot stocks; and $During_t$ equals one for the full months spanned by the pilot program (June 2005 through July 2007). The coefficient of interest is β on the interaction.

The results confirm that limits to arbitrage have causal force over the anomalies (Figure 2.8). Anomalies become significantly weaker for pilot stocks relative to non-pilot stocks during the pilot period, and the effect comes predominantly from the short leg—exactly where relaxed short-sale constraints should bind. Moreover, the anomalies that weaken the most tend to be those that were most profitable to begin with: the estimated treatment effect increases with each anomaly's in-sample return, so the relaxation of a single limit moves measured mispricing in the predicted direction, on the predicted side of the trade, roughly in proportion to the mispricing there was to move.

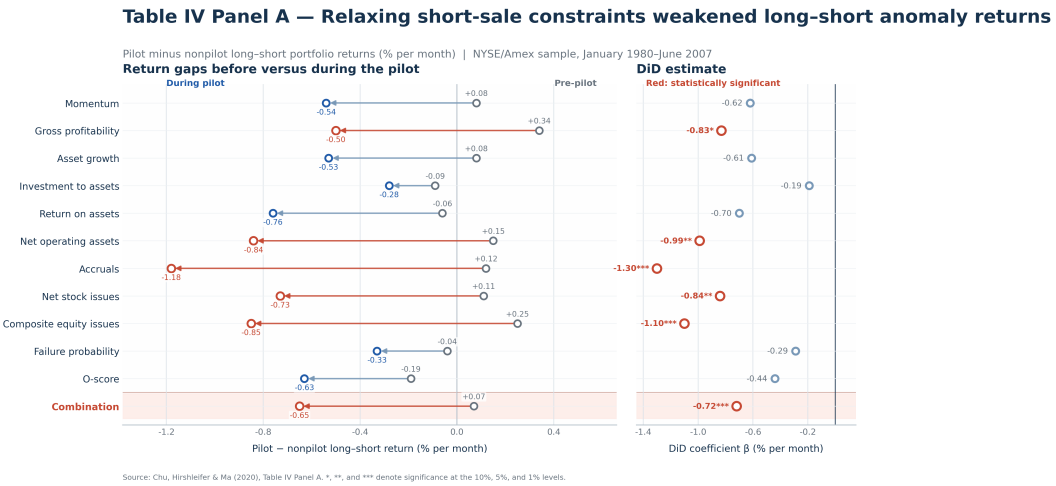


Figure 2.8: The causal effect of limits to arbitrage. The estimated change in monthly anomaly long–short returns for Regulation SHO pilot stocks relative to non-pilot stocks (the interaction coefficient in Equation 2.1), plotted against each anomaly's in-sample monthly return. Anomalies that were more profitable before the experiment weaken more when short-sale constraints are relaxed for pilot stocks. *Source:* Chu, Hirshleifer, and Ma (2020).

2.4.3 Learning Costs

The costs in Table 2.1 that bind earliest are the learning costs: before any anomaly can be arbitraged away, arbitrageurs must first become aware that it exists. McLean and Pontiff (2016) measure this by tracking the average monthly long–short returns of 97 predictor portfolios across three phases of their lives: 0.58% during the in-sample period used to discover the predictor, 0.40% out of sample but before academic publication, and 0.26% after publication (Figure 2.9). Two declines are visible, and they have different interpretations. The in-sample to out-of-sample decline reflects, in part, the statistical bias that comes from searching across many candidate predictors until something works. The post-publication decline is the learning cost operating in reverse: publication disseminates knowledge of the anomaly to arbitrageurs, whose trading then eats into the profitability. Relative to the in-sample period, roughly half of the return is gone once the knowledge is public—evidence that arbitrageurs respond to information about anomalies, but only after they have received it.

2.4.4 Information Processing Costs

The final limit in Table 2.1—the cost of acquiring and processing information—has direct evidence as well. Bowles et al. (2024) show that anomaly returns are concentrated in the first month after the anomaly-relevant information is released: a stock's anomaly characteristics are largely fixed by dated public disclosures (earnings announcements above all), and it is in the window immediately following those disclosures that the predictable returns accrue, with returns decaying quickly thereafter (Figure 2.10). If exploiting an anomaly required only capital and nerve, the returns should not be so concentrated in the days when the underlying information is new. Their concentration instead suggests that both investors and arbitrageurs process information slowly, and that the mispricing is harvested by whoever gets to it first—time, as much as capital, is the binding constraint. This is the pricing-side counterpart of the limited-attention evidence developed in Chapter 5.

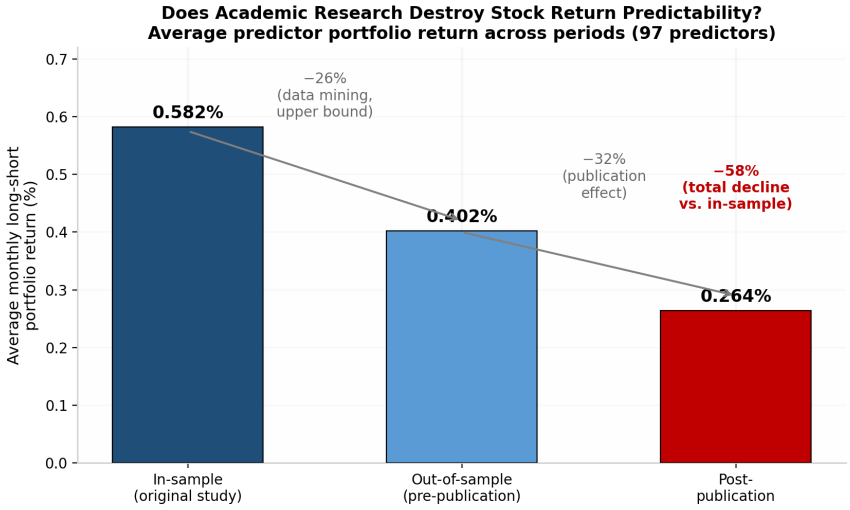


Figure 2.9: Learning costs and anomaly profitability. Average monthly returns to long–short predictor portfolios (in percent) in windows around academic publication: the in-sample discovery period, the out-of-sample period before publication, and the post-publication period, with two- and five-year post-publication follow-ups shown separately. Returns decline out of sample and fall by roughly half after publication. *Source:* McLean and Pontiff (2016).

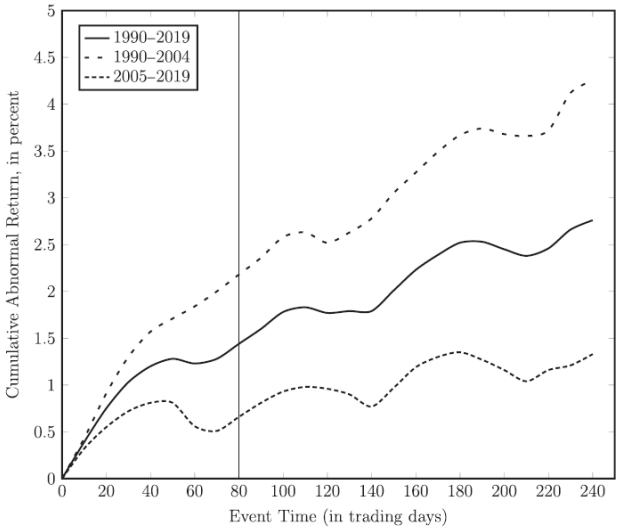


Figure 1. Anomaly returns in event time. The figure shows the profile of abnormal returns for the average anomaly (equally weighted) in event time. The figure shows the returns across the entire sample (1990 to 2019), across the early period of our sample (1990 to 2004), and across the recent years of our sample (2005 to 2019). The solid vertical bar at $t = 80$ trading days indicates when the FF92 rebalancing date occurs, on average.

Figure 2.10: Anomaly time. Returns to anomaly strategies are concentrated in the first month after anomaly-relevant information is released and decay quickly thereafter, consistent with information processing costs binding on both noise traders and arbitrageurs. *Source:* Bowles et al. (2024).

2.4.5 My Research: Analyst Bias as a Limit to Arbitrage

The limits in this chapter are usually framed as standing outside the arbitrageur: costs she pays and risks she bears. Guo, Li, and Wei (2020) ask whether the sell side itself can be one of the frictions. The starting point is the distribution of consensus recommendations across the mispricing spectrum. Sorting stocks into quintiles by a composite mispricing score of the type constructed by Stambaugh, Yu, and Yuan (2015), the paper finds that consensus recommendations in the U.S. are systematically more favorable for stocks classified as overvalued—the average recommendation rises monotonically across the quintiles (Figure 2.11)—and these stocks subsequently underperform significantly. Investors who trade on recommendations are steered toward precisely the overpriced side of anomalies.

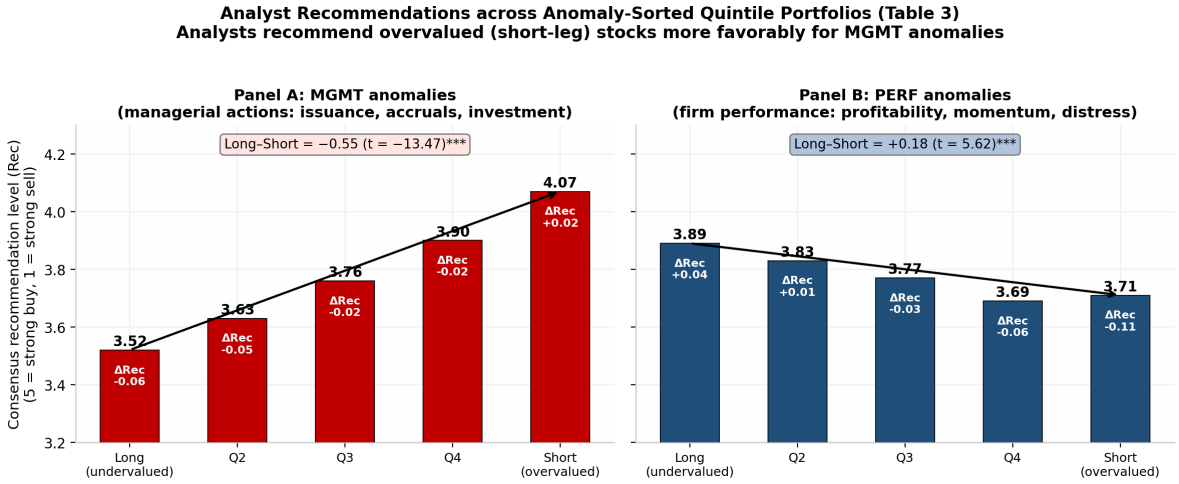


Figure 2.11: Analyst recommendations across mispricing quintiles. Mean consensus recommendation (1 = strong sell, 5 = strong buy) across quintiles of a composite mispricing score, for China (2000–2016) and the U.S. (1994–2016). In the U.S., the average recommendation rises monotonically across the quintiles, from 3.72 in quintile 1 to 3.95 in quintile 5, so the most favorable recommendations attach to the most mispriced stocks; in China the gradient runs the other way, declining from 3.55 to 3.32. *Source:* Guo, Li, and Wei (2020).

The second result links the bias directly to anomaly returns. For each anomaly, stocks are split according to whether the consensus recommendation is *consistent* with the anomaly signal (favorable on the cheap long side, unfavorable on the expensive short side) or *inconsistent* with it. If recommendations were merely noise, the split would not matter. It does: anomaly returns are amplified in the inconsistent group (Figure 2.12). When the professionals who aggregate and broadcast opinions lean against the anomaly signal, the mispricing that the anomaly measures is larger, not smaller. The implication inverts the usual reading of this chapter: biased recommendations are themselves a limit to arbitrage. Correcting an anomaly requires someone to take the other side of the demand, and one of the loudest voices in the market is systematically on the wrong side of it.

Taken together with Chapter 1, the picture is coherent from both ends. On the demand side, retail investors trade against forecasted returns (McLean, Pontiff, and Reilly, 2025); on the supply side, the correction of mispricing comes primarily from short sellers and firms, not from conventional asset managers (McLean, Pontiff, and Reilly, 2025; Edelen, Ince, and Kadlec, 2016); and the mispricing that survives is concentrated exactly where the frictions of this chapter are most severe (Nagel, 2005; Stambaugh, Yu, and Yuan, 2015; Chu, Hirshleifer, and Ma, 2020; Bowles et al., 2024).

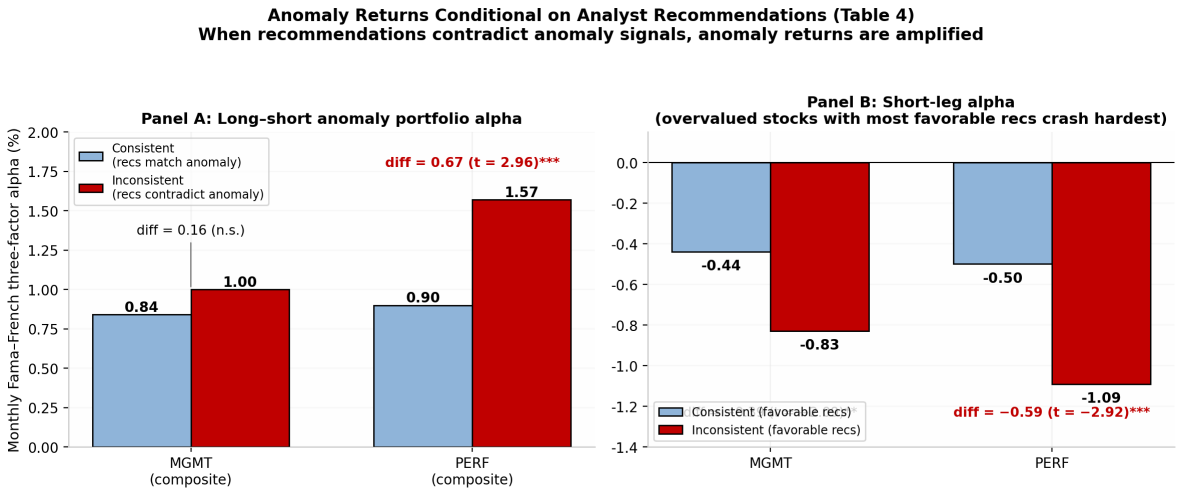


Figure 2.12: Anomaly returns when recommendations agree and disagree with the anomaly signal. Monthly Fama–French three-factor alphas (in percent) of 16 anomaly long–short portfolios, computed separately for stocks whose consensus recommendation is consistent with the anomaly signal (light bars) and stocks whose recommendation contradicts it (dark bars). Anomaly returns are substantially larger in the inconsistent group for most anomalies. *Source:* Guo, Li, and Wei (2020).

2.5 How Limited Is Arbitrage?

The literature on limits to arbitrage has been influential to the point of consensus: there is now wide agreement among academics and practitioners that arbitrage is limited, though disagreement remains about *how* limited. That success was one reason behavioral finance took off in the 1990s—it removed the standing objection that irrational demand could not matter.

Two qualifications keep the consensus honest. First, the limits are real but not boundless, and they move. The disappearing index effect (Greenwood and Sammon, 2025) shows arbitrage capital adapting to a repeatable, well-understood demand shock until the price impact nearly vanishes; the anomaly literature's post-publication declines (McLean and Pontiff, 2016) plausibly reflect the same adaptation. A mispricing that is large, salient, and cheap to trade against will attract its own correction. What the limits reliably protect are mispricings that are small relative to transaction and shorting costs, slow to close relative to capital horizons, unhedgeable, or hard to identify—conditions met by the short legs of anomalies among small, volatile stocks with low institutional ownership, not by index additions in 2025.

Second, limits to arbitrage are a necessary condition for behavioral asset pricing, not a sufficient one. They explain why mispricing *survives*; they have almost nothing to say about which direction prices are pushed in the first place. For that, the demand side is required: the preferences, beliefs, and limits to attention that generate systematic trading pressure. That is the subject of the remainder of this book. The discipline this chapter imposes on everything that follows is the reverse of the discipline the arbitrage critique imposed on behavioral finance: whenever an argument is made that irrational investors' demand affects prices, ask not only what generates the demand, but whether arbitrage is truly limited enough to let it.

Chapter 3

Preferences Under Uncertainty

The benchmark model of investor preferences is *expected utility* (EU): agents rank risky prospects by the expectation of a concave utility function defined over final wealth, and the axioms of Neumann and Morgenstern (1944) justify this ranking. EU is the workhorse of the consumption-based asset pricing literature and the implicit preference assumption behind nearly every quantitative model of asset returns. It also turns out to fit badly. Laboratory experiments run since the 1950s document systematic violations—preferences that are sensitive to framing, asymmetric around a reference point, sensitive to probability in ways that EU does not capture, and resistant—more precisely, averse—to ambiguity. Kahneman and Tversky (1979) proposed *prospect theory* as a descriptive alternative; Tversky and Kahneman (1992) sharpened it into a version, *cumulative prospect theory*, that is now the standard behavioral model of choice under risk. This chapter develops that model, surveys the modifications needed to apply it to financial markets, and reviews the asset pricing applications for which it does, and does not, deliver quantitative predictions.

The chapter proceeds in three parts. Sections 3.1–3.2 lay out the rational benchmark and the experimental evidence that breaks it. Sections 3.3–3.4 develop prospect theory: the four pillars (reference dependence, loss aversion, probability weighting, diminishing sensitivity), the fourfold pattern of risk attitudes they imply, and the additional ingredient of *narrow framing* that prospect theory needs to make contact with finance. Sections 3.5–3.8 trace the applications: trading behavior (Section 3.5), the cross-section of stock returns (Section 3.6), the aggregate stock market (Section 3.7), and ambiguity aversion as a separate psychological departure from EU (Section 3.8). Section 3.9 catalogues the limits of the theory, and Section 3.10 closes the chapter.

3.1 Expected Utility: The Rational Benchmark

Definition 3.1 (Expected utility)**.** *An agent has expected-utility preferences if, for any two prospects X and Y with the same probability space,*

$$X \succ Y \iff \sum_i p_i U(x_i) > \sum_i q_i U(y_i),$$

where U is a real-valued function, increasing, concave, and defined on outcomes.

Definition 3.1 assumes that the agent can assign probabilities to outcomes—the *risky* case in the Knight (1921) sense. Two properties follow from concavity that drive most of the asset pricing implications. First, U concave means the agent is *risk averse*: the agent prefers the expected value of a gamble to the gamble itself. Second, U is defined over *final wealth*, not over changes in wealth:

agents evaluate the new risk together with their pre-existing portfolio and consumption plan, and ask whether the combination is an improvement.

The axiomatic foundation is Neumann and Morgenstern (1944). Their four axioms (completeness, transitivity, continuity, and independence) imply that preferences over lotteries can be represented as the expectation of a utility function; the axioms are minimal in the sense that dropping any one allows preferences that EU cannot represent. Of the four, *independence* is the one that the empirical record attacks most consistently, and the one that prospect theory modifies most decisively. Independence says that if X is preferred to Y, then the prospect $pX + (1-p)Z$ is preferred to $pY + (1-p)Z$ for any common Z and any $p \in [0,1]$: preferences between two lotteries are unchanged by mixing either of them with the same third lottery.

Concavity of U, combined with its definition over final wealth, also commits EU to a specific prediction about how risk attitudes depend on the sign of the gamble: a risk-averse agent with concave U dislikes *all* mean-preserving spreads, both upside and downside. EU with diminishing marginal utility of wealth therefore predicts risk aversion over both gains and losses, and risk neutrality in the limit of small stakes. Both predictions are violated in experimental data.

3.2 Where Expected Utility Breaks

3.2.1 The Allais Paradox

The cleanest violation of independence is Allais (1953). Subjects are offered two paired choices:

Choice A.

(A) A sure \$1,000.

(B) A 10% chance of \$5,000, an 89% chance of \$1,000, and a 1% chance of \$0.

Choice B.

(C) A 10% chance of \$5,000 and a 90% chance of \$0.

(D) A 11% chance of \$1,000 and an 89% chance of \$0.

The modal choice in A is the sure \$1,000; the modal choice in B is the 11% chance of \$1,000. But the modal pattern violates independence: the prospects in B are obtained from those in A by replacing the common 89% chance of \$1,000 with an 89% chance of \$0, so an EU agent who chose A's first option should choose B's second. The contradiction is robust, replicable, and quantitative: in the Tversky and Kahneman (1992) calibration, the violation appears for nearly every parameter choice consistent with the descriptive fit of prospect theory.

3.2.2 Common Ratio and Reflection Effects

Two related violations are the *common ratio effect* and the *reflection effect*. Consider the two prospects

$$(\$4{,}000,\ 0.80) \quad \text{vs.} \quad (\$3{,}000,\ \text{certain}),$$

with a common ratio $p = 0.8$. The gamble's expected value, \$3,200, exceeds the sure amount, but EU with U concave implies that the second—the certain \$3,000—is preferred: that is what risk aversion means. In Kahneman and Tversky (1979)'s Problem 3, 80% of subjects do choose the sure \$3,000. Now multiply both probabilities by 0.25:

$$(\$4{,}000,\ 0.20) \quad \text{vs.} \quad (\$3{,}000,\ 0.25).$$

Many subjects now reverse and prefer the first, the \$4,000 gamble—even though the same risk-aversion logic should still apply. The reversal is the common ratio effect: probability weights are not linear in p.

The reflection effect is the sign analogue. Take the loss-domain counterparts of the first pair:

$$(-\$4{,}000,\ 0.80) \quad \text{vs.} \quad (-\$3{,}000,\ \text{certain}).$$

Many subjects prefer the gamble, not the certain loss—the opposite of risk aversion. EU with concave U over final wealth cannot produce risk-seeking preferences over losses.

3.2.3 Sign, Reference, and Probability: Four Robust Patterns

Kahneman and Tversky (1979) summarize the violations as four robust patterns:

(i) Reference dependence: outcomes are evaluated as gains and losses, not final states.

(ii) Loss aversion: losses loom larger than equal-sized gains.

(iii) Probability weighting: small probabilities are overweighted, large probabilities underweighted.

(iv) Diminishing sensitivity: the marginal response to changes in outcome diminishes with distance from the reference point, on both sides.

These are the four pillars of prospect theory. Each is well-documented in laboratory data; each is at odds with EU defined over final wealth; and each, as Sections 3.5–3.7 show, has asset pricing implications.

3.3 Prospect Theory

3.3.1 The Framework

Kahneman and Tversky (1979) propose the following modification of EU. A prospect is a sequence (x_i, p_i) of outcomes x_i and their objective probabilities p_i. To evaluate a prospect, the agent first rewrites the outcome space around a *reference point* (typically the status quo or recent expectation), separating it into gains and losses. The sensory analogue is instructive: hold one hand in hot water and the other in cold water for half a minute, then place both in the same lukewarm water—the hot hand feels it as cold and the cold hand as hot. Perception is keyed to the change from the prior state, not to the absolute level, and the value function inherits that property. The agent then applies two transformations:

(i) A *value function* $v(\cdot)$, defined on gains and losses, with $v(0) = 0$, v concave over gains, convex over losses, and steeper for losses than for gains at every absolute level.

(ii) A *decision-weight function* $w(\cdot)$ that transforms the cumulative probabilities of gains and losses separately into *decision weights*, which then play the role of probabilities in the value computation.

The prospect $(x, p; y, q)$, with $x \geq 0 \geq y$ and $p + q \leq 1$, has prospect-theory value

$$V \;=\; w(p)\, v(x) \;+\; w(q)\, v(y).$$

The agent accepts the prospect if $V > v(0) = 0$, i.e. if the decision-weighted gain and loss values favor taking it.

The framework is *descriptive*: it is meant to capture how agents *actually* decide, not how they *should*. The shift from normative to descriptive is the move that opens the door to finance: a descriptive theory can be wrong about welfare and still be right about behavior, and behavior is what moves prices.

3.3.2 The Value Function

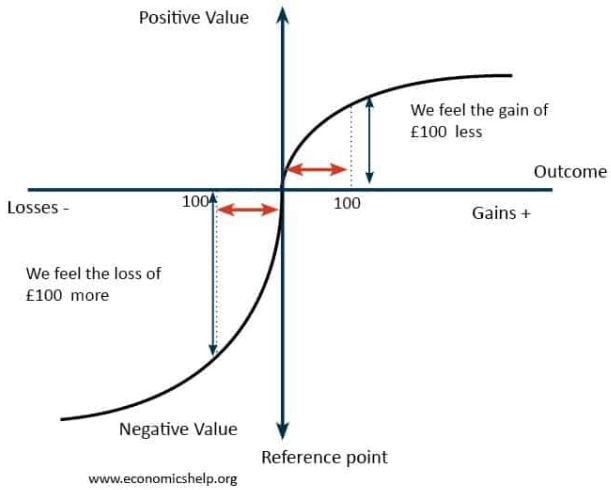


Figure 3.1: The prospect-theory value function $v(\cdot)$, with the reference point at zero and a kink at the origin. The function is concave over gains, convex over losses, and steeper for losses than for equal-sized gains: a one-unit loss reduces value more than a one-unit gain of the same magnitude increases it. *Source:* EF8083 course materials, after Kahneman and Tversky (1979).

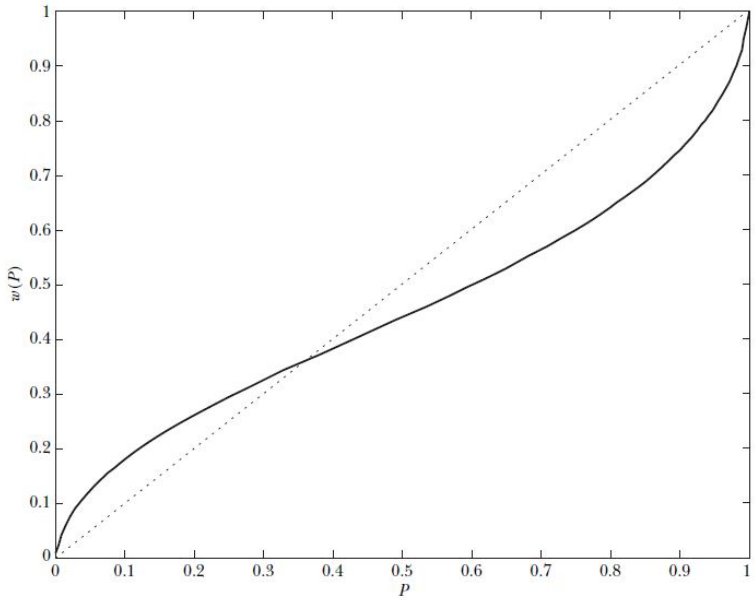
Figure 3.1 shows the canonical shape. Three features matter. First, $v(0) = 0$ and the function is S-shaped: concave over gains, convex over losses. Second, v is steeper on the loss side than on the gain side: the ratio of slopes at $+x$ and $-x$ is approximately 2:1 in the Tversky and Kahneman (1992) calibration. Third, the function is defined on gains and losses, not on final wealth: agents code outcomes relative to a reference point rather than to absolute levels. The first feature (concavity/convexity) delivers risk aversion over moderate-probability gains and risk seeking over moderate-probability losses; the second delivers loss aversion in the sense of Section 3.2; the third makes preferences sensitive to framing in ways that EU defined over final wealth does not allow.

3.3.3 Probability Weighting

The decision-weight function $w(\cdot)$ is the second key departure from EU. Figure 3.2 shows the canonical form from Tversky and Kahneman (1992): small probabilities are overweighted, large probabilities are underweighted, and the function is inverse-S-shaped. A 1-in-1,000 chance of winning \$5,000 carries a decision weight near 0.01, not 0.001; the 0.95 chance of winning carries a decision weight near 0.82, not 0.95.

Two clarifications are important for the finance applications. First, decision weights are not beliefs. Subjects who overweight a 1-in-1,000 chance in a one-shot gamble are not claiming that the event has 1% probability; they are claiming that the event matters *to them* as if it had 1% probability. The distinction matters because asset pricing applications of prospect theory should not be confused with subjective-belief models. Second, the overweighting of small probabilities explains

Figure 2
The Probability Weighting Function



Notes: The graph plots the probability weighting function proposed by Tversky and Kahneman (1992) as part of cumulative prospect theory, namely $w(P) = P^{\delta}/(P^{\delta} + (1-P)^{\delta})^{1/\delta}$, where P is an objective probability, for two values of δ. The solid line corresponds to $\delta = 0.65$, the value estimated by the authors from experimental data. The dotted line corresponds to $\delta = 1$, in other words, to linear probability weighting.

Figure 3.2: The probability weighting function $w(\cdot)$ of cumulative prospect theory. The solid line is the Tversky and Kahneman (1992) estimate, $w(P) = P^{\delta}/(P^{\delta} + (1-P)^{\delta})^{1/\delta}$ with $\delta = 0.65$; the dotted line is $w(P) = P$. Small probabilities are overweighted (note $w(0.05) \approx 0.12$); moderate-to-high probabilities are underweighted (note $w(0.95) \approx 0.82$). *Source:* Tversky and Kahneman (1992), Figure 2.

the simultaneous purchase of lotteries and insurance: the lottery is appealing because $w(0.001)$ is much larger than 0.001; the insurance is appealing because $w(0.001)$ on the loss side is also large, and the loss-side value function makes the loss salient. Both purchases are puzzling under EU.

3.3.4 Loss Aversion in Pictures

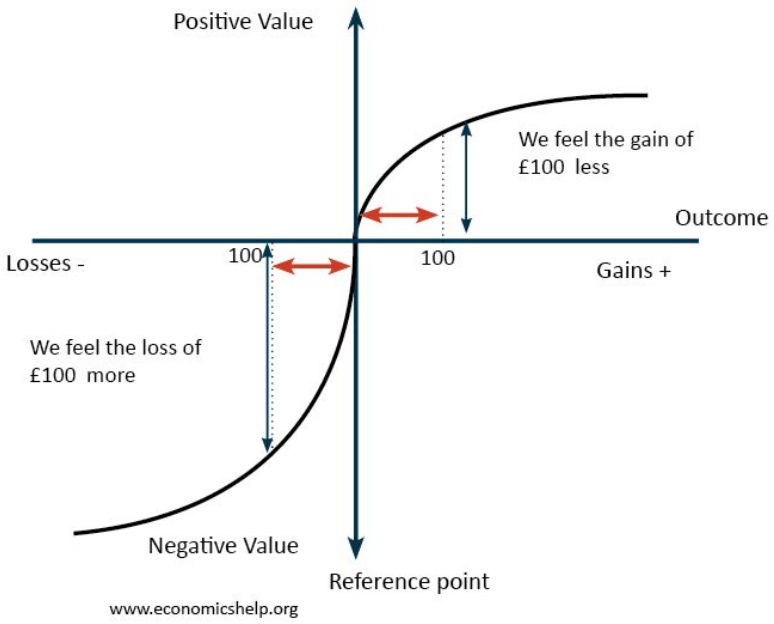


Figure 3.3: Loss aversion illustrated. A \$100 loss reduces value by more than a \$100 gain increases it; the ratio at equal magnitude is approximately 2:1 in the standard calibration. The asymmetry is the source of the reluctance to realize losses—a central feature of investor behavior. *Source:* EF8083 course materials, after Kahneman and Tversky (1979).

Figure 3.3 makes the loss-aversion feature concrete. A \$100 gain raises the value function by some amount; a \$100 loss lowers it by approximately twice as much. The asymmetry is not a quirk of small stakes: it is present at the magnitudes that matter for trading behavior, and it is the same asymmetry that generates the *disposition effect* of Section 3.5. Genesove and Mayer (2001) document the same asymmetry in the housing market (Section 3.5 returns to it).

3.3.5 The Fourfold Pattern

Combining the four pillars produces what Tversky and Kahneman (1992) call the *fourfold pattern* of risk attitudes. For prospects involving gains, agents are risk averse over moderate probabilities (the concavity of v) but risk seeking over small probabilities (the overweighting of small p by w). For prospects involving losses, agents are risk seeking over moderate probabilities (the convexity of v) but risk averse over small probabilities. Each combination is well-documented in the experimental record, and none of them is consistent with EU over final wealth.

The pattern also drives a set of asset-pricing predictions that we develop in Sections 3.5–3.7. Risk aversion over gains, combined with overweighting of small probabilities, makes investors dislike safe assets more than EU implies and like lottery-like assets more than EU implies. Risk seeking

over moderate-probability losses makes investors reluctant to cut losers, and that reluctance has a price impact when it is widespread.

3.4 Narrow Framing

The conceptual question prospect theory raises for finance is: *what is the argument of* $v(\cdot)$*?* EU has a clean answer: final wealth, or aggregate consumption. Prospect theory's value function is defined over gains and losses; the answer is whatever the agent treats as the relevant gain or loss. In a fully integrated portfolio, the agent would evaluate a new risky asset by combining it with pre-existing positions and computing the gain or loss in total wealth. In practice, most of us do not.

Definition 3.2 (Narrow framing)**.** *An agent is said to evaluate a risk in* narrow framing *if she evaluates the risk separately from her other exposures, rather than integrating it into a single evaluation of total wealth.*

Barberis, Huang, and Thaler (2006) survey the principal evidence. The standard rejection of the gamble $(\$110, 1/2; -\$100, 1/2)$, ubiquitous in laboratory studies, is evidence not only of loss aversion but also of narrow framing: an EU agent with substantial existing wealth would accept the gamble (the gain and loss are nearly symmetric and the stake is small relative to total wealth), and only a narrow-framed evaluation of the gamble on its own makes the rejection rational. Barberis, Huang, and Santos (2001) show that narrow framing is the ingredient that turns a descriptive model of risk attitudes into a quantitative asset pricing model. With loss aversion defined over total wealth and integrated evaluation, the model cannot reproduce the equity premium; with narrow framing, the model can.

Narrow framing is closely related to *mental accounting*: investors treat different assets, accounts, and horizons as separate books. The framing assumption is descriptive of how people actually trade; it is the empirical fact that turns prospect theory into an asset pricing model. The price we pay is that prospect theory's asset pricing implications depend on how investors define their reference point and frame their risks, and these choices are partly endogenous and partly outside the model.

3.5 Prospect Theory and Investor Trading Behavior

The first application is the closest to the laboratory. Loss aversion and the asymmetric value function predict that investors hold losers longer than they hold winners, because realizing a loss crystallizes the negative part of the value function while deferring the loss keeps it as a paper loss that may reverse.

3.5.1 The Disposition Effect

Definition 3.3 (Proportion of gains and losses realized)**.** *For an investor's account over a period, define*

$$\text{PGR} = \frac{\textit{Realized Gains}}{\textit{Realized Gains} + \textit{Paper Gains}}, \qquad \text{PLR} = \frac{\textit{Realized Losses}}{\textit{Realized Losses} + \textit{Paper Losses}}.$$

The disposition hypothesis is $\text{PGR} > \text{PLR}$*.*

Figures 3.4 and 3.5 show the value-function geometry behind the asymmetry. With v steeper on the loss side, the cost of crystallizing a loss is larger than the cost of holding an equivalent paper

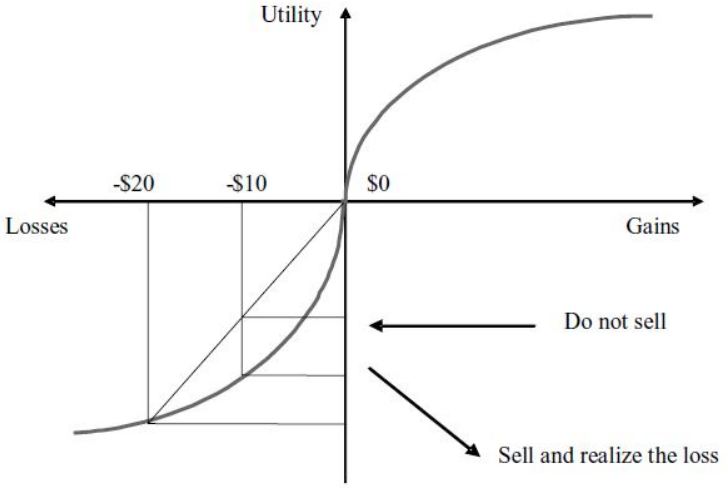


Figure 3.4: Disposition effect, losses. With a value function steeper for losses than for gains, the disutility of realizing a \$10 or \$20 loss exceeds the disutility of holding an equivalent paper loss. The diagram shows the value comparisons at stake: an investor holding a stock at $-\$10$ or $-\$20$ relative to the purchase price is reluctant to realize because doing so crystallizes the loss-side curvature. *Source:* EF8083 course materials, after Odean (1998).

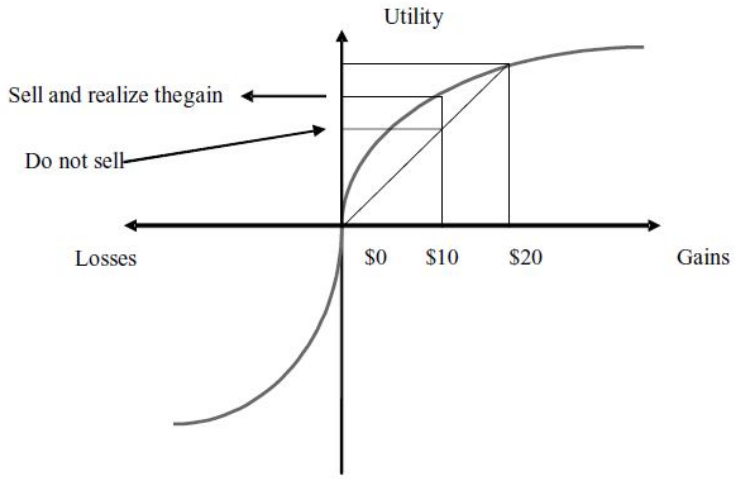


Figure 3.5: Disposition effect, gains. With a value function concave over gains, the marginal utility of additional gains falls as the gain grows; the diagram shows the value comparison an investor faces when deciding whether to realize a \$10 or \$20 gain. *Source:* EF8083 course materials, after Odean (1998).

loss; with v concave over gains, the cost of crystallizing a moderate gain is also smaller than the cost of holding the equivalent paper gain. Both effects push in the same direction: sell winners, hold losers.

Odean (1998) tests the prediction on a large discount brokerage dataset. Over the full sample, PGR = 0.148 and PLR = 0.098, a difference of 0.050 with a t-statistic of 35. The pattern is not driven by tax trading or information; Odean (1998) shows that the winners investors sell subsequently *outperform* the losers they hold, so the asymmetry is not simply informed selling. Dhar and Zhu (2006) shows that the asymmetry is larger among less sophisticated investors, consistent with the prediction that the underlying preference is a property of the agent, not of the trading environment. Frazzini (2006) extends the result to mutual fund managers, and Genesove and Mayer (2001) to the housing market, where sellers with expected prices below their purchase price set higher reservation prices and wait longer to sell—the same disposition effect in a different setting.

The disposition effect is the cleanest case where a micro-level behavioral fact generates a macro-level prediction. Frazzini (2006) links the disposition-driven eagerness to sell winners to underreaction to news: when a stock rises on good news, disposition-motivated investors want to sell and lock in the gain, but they can only sell at the depressed price, which moves further after the news; the reverse happens on bad news. The aggregate selling pressure depresses prices at the news and pushes subsequent returns in the direction of the news, producing the post-earnings-announcement drift of Chapter 1.

3.5.2 Disposition Effect and Momentum

The same logic, applied to past returns, generates momentum. Grinblatt and Han (2005) formalize the argument in a discrete-time model in which investors form mental accounts around the purchase price and update them as new information arrives. The model produces a momentum effect on the order of magnitude observed in the data, and the channel is the disposition-driven underreaction of prices to news. Frazzini (2006) provides complementary evidence: stocks with high past returns, where disposition-motivated selling pressure should be most active, exhibit the strongest post-earnings-announcement drift.

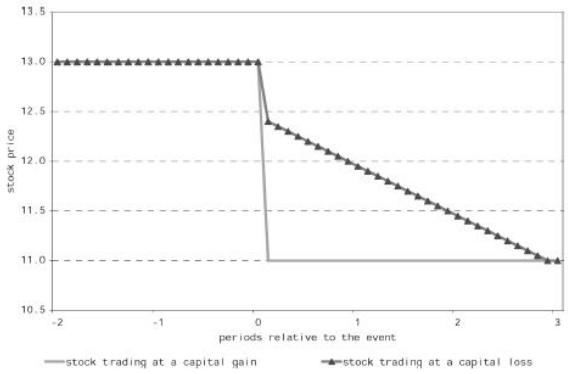


Figure 3. An example of stock price response to negative news. This figure shows an example of a stock price response to negative news. The initial stock price is $13. At date 0, public news reveals a fundamental value of $11.

Figure 3.6: Stock price response to negative news, capital-loss case. An example stock falls from $13 to a fundamental value of $11. Investors holding at a capital loss are reluctant to realize; selling pressure is muted and the price drifts slowly toward fundamental. *Source:* Frazzini (2006), Figure 3.

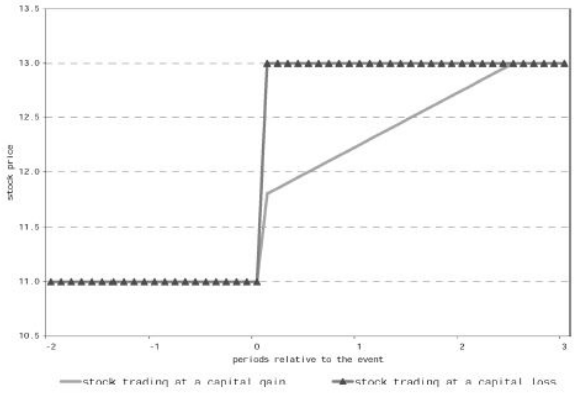


Figure 4. An example of stock price response to positive news. This figure shows an example of a stock price response to positive news. The initial stock price is $11. At date 0, public news reveals a fundamental value of $13.

Figure 3.7: Stock price response to positive news, capital-gain case. The same stock rises from $11 to a fundamental value of $13. Investors holding at a capital gain are eager sellers: the disposition effect pushes them to realize, and the price rises more slowly than a fully rational benchmark would imply. *Source:* Frazzini (2006), Figure 4.

Figures 3.6 and 3.7 show the geometry. In the capital-loss case (Figure 3.6), investors who bought at $13 and face a value of $11 are reluctant to sell; aggregate selling pressure is muted and the price drifts only gradually toward the new fundamental. In the capital-gain case (Figure 3.7), investors who bought at $11 and face a value of $13 are eager sellers; aggregate selling pressure is heavy and the price drifts only gradually up. The asymmetry is the momentum effect, and it is a direct consequence of the asymmetric value function combined with narrow framing.

3.6 Prospect Theory and the Cross-Section of Stock Returns

3.6.1 Lottery Stocks and the Skewness Premium

The second application takes the four pillars to asset prices. Barberis and Huang (2008) build a single-period model with a risk-free asset and J risky assets with multivariate Normal payoffs; all agents have cumulative prospect theory (CPT) preferences and identical beliefs. They then introduce a small, independent, positively skewed security into the economy and ask what the equilibrium price is. EU predicts that the new security earns a positive excess return only insofar as it adds to market risk; with Normal payoffs elsewhere, the new security is a diversifiable addition and EU predicts a price close to the present value of its expected payoff.

CPT predicts something different. Because agents overweight small probabilities, a small chance of a very large payoff is disproportionately valuable to them; some agents will tilt their portfolios toward the new security, accept the loss of diversification, and in equilibrium hold an undiversified position. The pricing consequence is that the new security must earn a *negative* excess return: the equilibrium price is high, the expected return is low, and investors accept the lower expected return because of the positive skewness the security adds.

Figure 3.8 shows the utility from a tilted position for two different mean returns on the new security. The investor is willing to take an undiversified position precisely because the skewness is large; for a Normal security the optimal position would be a near-zero allocation. The model generates both a low premium on positively skewed stocks and a high premium on the aggregate

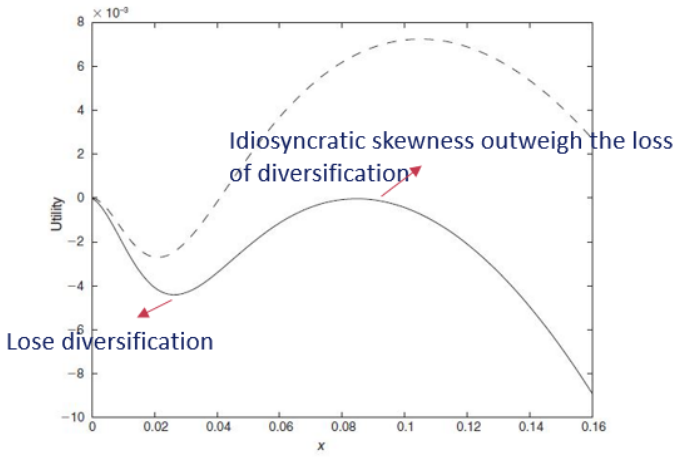


FIGURE 3. A HETEROGENEOUS HOLDINGS EQUILIBRIUM

Notes: The figure shows the utility that an investor with cumulative prospect theory preferences derives from adding a position in a positively skewed security to his current holdings of a Normally distributed market portfolio. The skewed security is highly skewed. The variable x is the fraction of wealth allocated to the skewed security relative to the fraction of wealth allocated to the market portfolio. The two lines correspond to different mean returns on the skewed security.

Figure 3.8: A heterogeneous holdings equilibrium. The figure plots the utility an investor with CPT preferences derives from adding a highly positively skewed security to a portfolio of Normally distributed assets, as a function of the share x allocated to the new security. The dashed curve corresponds to a higher mean return on the new security. The investor can choose to lose diversification by taking a large position; the optimal position balances the diversification cost (the dip on the left) against the lottery value of the skewness (the rise on the right). *Source:* Barberis and Huang (2008), Figure 3.

market, which is negatively skewed—a feature to which Section 3.7 returns.

3.6.2 Tests of the Skewness Channel

The model's central prediction is that skewness is priced in the cross-section. The empirical challenge is that *past* skewness is not a good measure of *expected future* skewness: realized skewness is noisy and time-varying. Three responses have been developed.

Regression-based expected skewness. Boyer, Mitton, and Vorkink (2010) estimate a regression model of expected idiosyncratic skewness using stock characteristics and form portfolios on the prediction. The high-minus-low spread is large and statistically significant, and the prediction sorts returns in the cross-section.

Maximum daily returns. Bali, Cakici, and Whitelaw (2011) construct a very simple measure: the maximum daily return over the past month (MAX). The economic intuition is that large daily jumps are the events CPT investors overweight, and MAX picks them up.

Figure 3.9 shows the basic result in the Open Source Asset Pricing data, which extend the Bali, Cakici, and Whitelaw (2011) sample by four decades. Stocks with the highest maximum daily returns earn the lowest subsequent returns, consistent with the lottery-demand story: CPT investors overweight the extreme upside day, bid the stock up, and the overpricing is worked off subsequently. Bali, Cakici, and Whitelaw (2011) further show that MAX exhibits strong negative return predictability and helps resolve the idiosyncratic volatility puzzle of Chapter 1—idiosyncratic volatility and MAX are correlated, but MAX captures the part of volatility that is driven by extreme upside moves.

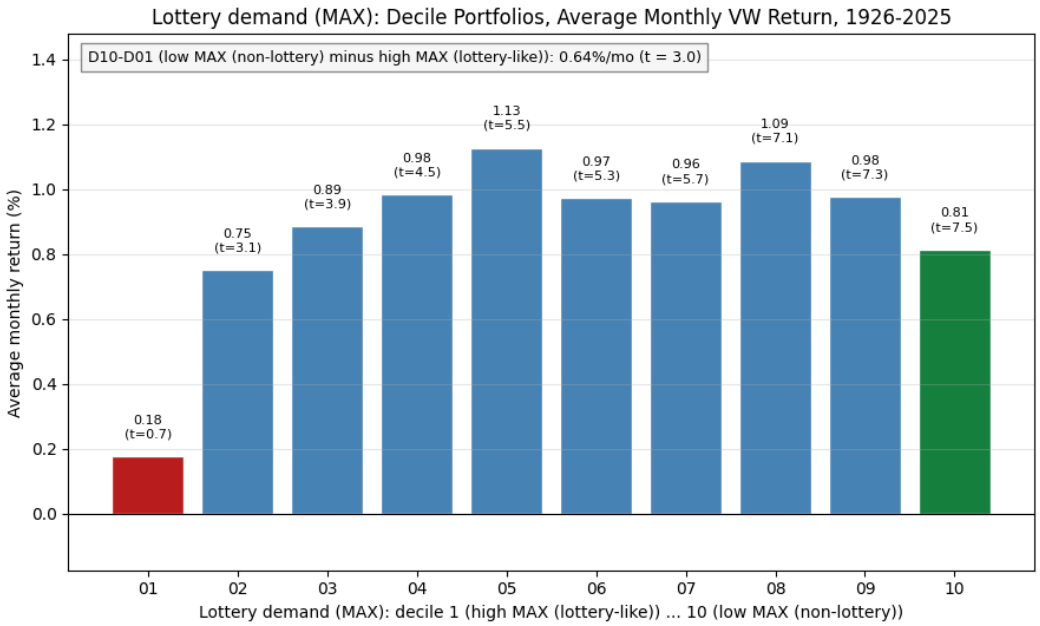


Figure 3.9: MAX-sorted decile portfolios: average monthly value-weighted returns, 1926–2024. Portfolios are sorted monthly on MAX, the maximum daily return over the prior month; decile 1 (red) is high MAX (lottery-like) and decile 10 (green) is low MAX. The high-MAX decile earns the lowest average return, 0.20% per month ($t = 0.8$); the low-MAX decile earns 0.83% per month ($t = 7.5$). The low-MAX-minus-high-MAX spread is 0.63% per month ($t = 2.9$). *Source:* Open Source Asset Pricing replication (Chen and Zimmermann, 2022) of the MAX sort in Bali, Cakici, and Whitelaw (2011).

Option-implied skewness. Conrad, Dittmar, and Ghysels (2013) use option prices to back out the implied skewness of the underlying stock and form portfolios on the prediction. The high-minus-low return spread is negative and significant, in line with CPT. The advantage over realized-skewness proxies is that option-implied skewness is forward-looking and not subject to the noise of realized skewness estimates.

3.6.3 Lottery Stocks in the Cross-Section

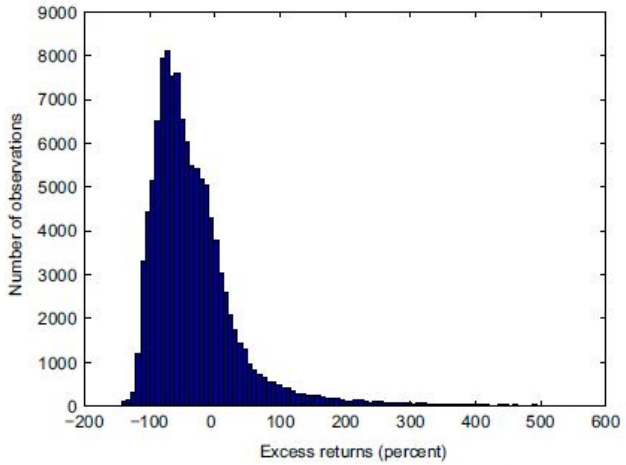


Fig. 1. Histogram of annual market-adjusted returns. The figure shows the frequency of annual market-adjusted returns for the sample of over-the-counter stocks with $0.01 minimum price and $50,000 minimum volume. Market-adjusted returns greater than 600% (0.4% of the observations) are omitted for readability.

Figure 3.10: Histogram of annual excess returns of over-the-counter stocks. The distribution has a pronounced right tail: many large positive returns, with returns above 100% truncated for display. OTC stocks are heavily held by retail investors with a taste for lottery-like payoffs, and earn low subsequent returns. *Source:* Eraker and Ready (2015), Figure 1.

The skewness story has supporting evidence across a number of asset classes. Eraker and Ready (2015) document that financially distressed stocks and over-the-counter (OTC) stocks earn low subsequent returns; both classes have positively skewed return distributions (Figure 3.10) and both are heavily held by retail investors with lottery preferences. Green and Hwang (2012) show that IPOs predicted to have higher expected skewness have higher first-day returns and lower long-run returns—the lottery-feature reading of the new issues puzzle of Chapter 1.

Figure 3.11 shows the IPO evidence. The high-skewness subsample earns higher first-day returns in every subperiod, with the spread largest in 1995–2000. Boyer and Vorkink (2014) examine equity options and find that stocks predicted to have more positively skewed payoffs have lower returns—a parallel result in a different asset class. Bali, Brown, et al. (2017) document that high-beta stocks, which have positively skewed returns, earn lower CAPM alphas than low-beta stocks; the beta anomaly of Chapter 1 and the lottery-demand story are linked through the same skewed-return channel.

Figure 1. Expected skewness and average initial returns across subperiods.

This figure reports average initial returns on IPOs across subperiods. The initial return is calculated as the return from the offer price to closing market price on the first day of trading for the IPO. We calculate estimates of expected skewness and assign IPOs to the high-/medium/low-expected skewness portfolio as described in Table 1.

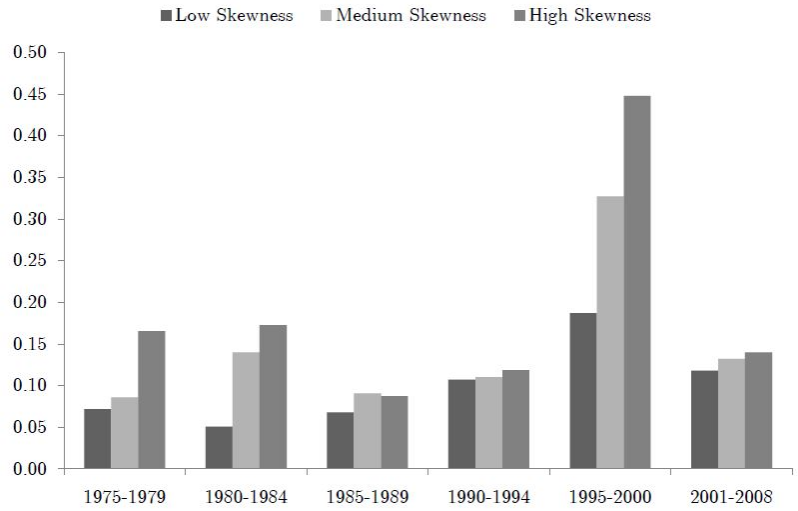


Figure 3.11: Expected skewness and average initial returns of IPOs across subperiods. In the high-skewness subsample, first-day returns are systematically higher in every subperiod and the spread is largest in the 1995–2000 technology boom. *Source:* Green and Hwang (2012), Figure 1.

3.6.4 Lottery Preference: Evidence from Holdings

The model also has predictions about *who* holds lottery-like stocks. Kumar (2009) shows that the demand for lottery-type stocks correlates with the demand for state lotteries: in U.S. counties with higher per-capita lottery sales, retail investors overweight positively skewed stocks (Figure 3.12). Socioeconomic factors associated with lottery expenditure (low income, low education, high unemployment) are also associated with higher investment in lottery-type stocks. Chen, Hwang, and Peng (2025) use textual analysis to study why analysts and writers recommend short-leg securities and find that the dominant justification is lottery-like features—around 68% of sell-side analyst reports and 69% of retail-facing articles frame their recommendation in terms of the upside potential, not safety or recent strong performance.

3.6.5 Mutual Fund Flows and Revealed Preference

A second stream of revealed-preference evidence asks whether prospect theory can predict flows into and out of mutual funds. Han, Sui, and Yang (2026) compute the prospect-theory value of each fund's return distribution over the prior period and relate it to subsequent flows. Funds whose past returns generate higher prospect-theory value attract significantly larger future flows—the demand-side analogue of the asset-pricing results. The advantage of the revealed-preference design is that flows are observed at high frequency and the analysis does not depend on a particular equilibrium model.

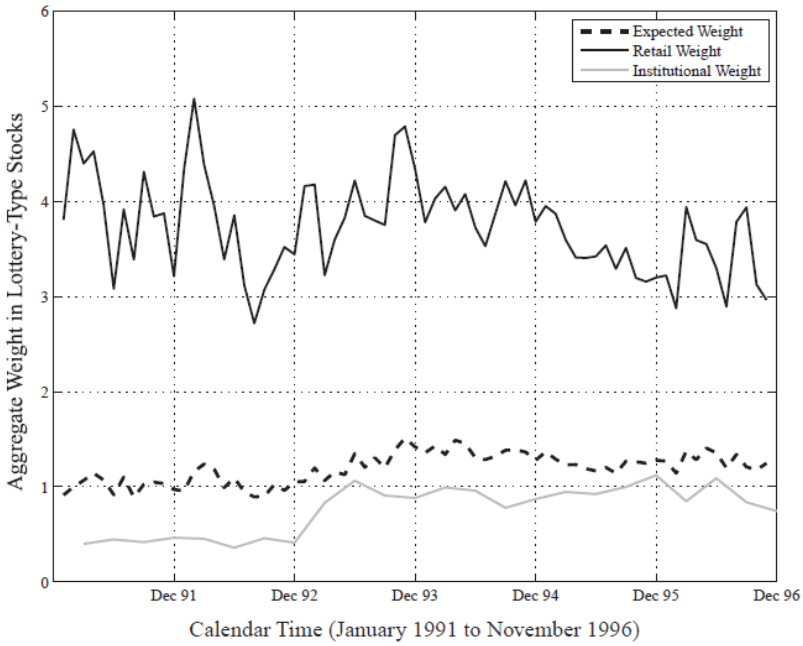


Figure 1. Aggregate weight in lottery-type stocks over time. This figure shows the time series of the actual weights allocated to lottery-type stocks in the aggregate individual and institutional investor portfolios. The expected lottery weight time series, which reflects the weight

Figure 3.12: Aggregate weight in lottery-type stocks over time, 1991–1996. The retail weight (solid black line) is consistently above the institutional weight (solid grey line) and the expected-lottery weight (dashed line). The retail weight spikes in late 1991 and again in 1993–1994, consistent with increased demand for lottery-like payoffs when economic conditions deteriorate. *Source:* Kumar (2009), Figure 1.

3.6.6 Under-Diversification

The lottery preference also predicts that retail investors should hold under-diversified portfolios concentrated in positively skewed stocks. Mitton and Vorkink (2007) document that under-diversified individuals hold stocks that are more positively skewed than the average stock, exactly the lottery-feature story. Uppal and Wang (2003) note that the same pattern arises under EU if investors are uncertain about which model to use (model misspecification), so the finding is not by itself evidence for prospect theory over its alternatives.

3.6.7 Quantitative Direct Tests

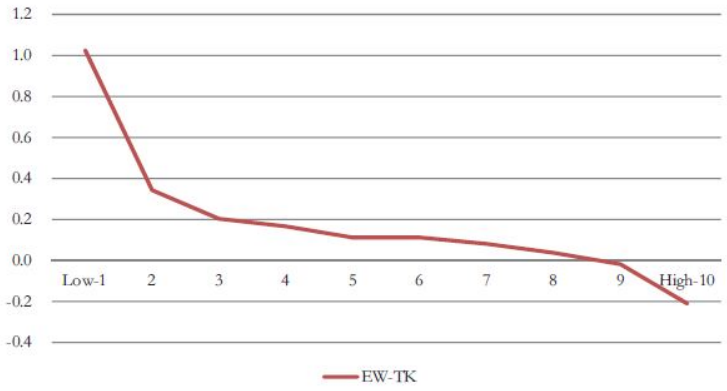


Figure 3.13: Barberis–Mukherjee–Wang (2016). The figure plots a measure of equal-weighted Tversky–Kahneman (TK) prospect-theory value across deciles of that value. Stocks with the highest PT value are the most appealing to investors, are bid up, and earn low subsequent returns; stocks with the lowest PT value earn high subsequent returns. The high-minus-low spread is roughly -1.2% per month in equal-weighted returns. *Source:* Barberis, Mukherjee, and Wang (2016).

The most direct test of prospect theory in the cross-section is to compute the prospect-theory value of each stock using its historical return distribution and ask whether high-PT-value stocks underperform subsequently. Barberis, Mukherjee, and Wang (2016) do exactly this, computing the Tversky–Kahneman PT value of each stock's prior return distribution and sorting on it. Figure 3.13 shows the basic pattern: high-PT-value stocks earn lower subsequent returns, low-PT-value stocks earn higher subsequent returns, and the high-minus-low spread is roughly -1.2% per month. The interpretation is that CPT investors find high-PT-value stocks appealing (high skewness, high probability of a large gain), tilt toward them, and thereby bid them up; the stocks become overvalued and earn low subsequent returns.

The test rests on a *joint hypothesis*: some investors mentally *represent* a stock by the distribution of its past returns, and they *evaluate* that distribution of gains and losses according to cumulative prospect theory. Valuation is the easy step—Tversky and Kahneman (1992) supply the formulas and the parameters ($\alpha = 0.88$, $\lambda = 2.25$, $\gamma = 0.61$, $\delta = 0.69$). Representation is the hard step: how does an investor summarize a stock in their mind? Barberis, Mukherjee, and Wang (2016) answer: by the distribution of *monthly returns in excess of the market over the past five years*—60 excess returns, each assigned probability $1/60$—a choice guided by how price charts appear in investment handbooks, where monthly fluctuations over roughly a five-year window are discernible and daily or weekly moves are not, and by the judgment-and-decision finding that people passively accept the representation put in front of them. Applying cumulative prospect theory to this distribution, with

gains and losses weighted separately by the weighting functions w^+ and w^-, produces the stock's *TK* (Tversky–Kahneman) value. Two properties of the construction are worth noting. First, TK does not depend on the *time order* of the 60 returns: it is a quick, passive reaction to the chart as an integral whole rather than a computed statistic. Second, with the estimated parameters the value function is close to piecewise linear, so the concavity and convexity of v play only a minor role; the measure is powered principally by loss aversion and the overweighting of tail outcomes.

The pricing model behind the test is deliberately spare. A mean-variance economy contains two types of investors: traditional mean-variance investors who hold the tangency portfolio, and prospect-theory investors who start from the tangency portfolio and tilt toward stocks with above-average TK values and away from stocks with below-average TK values. In equilibrium, the tilt depresses the expected returns of appealing stocks. If η measures the weight of prospect-theory traders, $\overline{\mathrm{TK}}$ the cross-sectional average of TK, s_j^2 stock j's idiosyncratic variance, and β_{TK} the market beta of the TK-tilted portfolio, then, under a diagonal covariance structure,

$$\frac{\mu_j - r_f}{\mu_m - r_f} \;=\; \beta_j \;-\; \frac{\eta\left(\mathrm{TK}_j - \overline{\mathrm{TK}}\right) s_j^2}{\sigma_m^2\left(1 - \eta\,\beta_{\mathrm{TK}}\right)}.$$

The first term is the CAPM; the second is the prospect-theory correction, proportional to how far the stock's TK exceeds the average. Stocks whose past return distributions are attractive to prospect-theory investors earn lower expected returns than beta alone implies—the prediction that the sorting evidence confirms. A multiperiod extension also implies that the expected *change* in a stock's TK (a very good return 60 months ago rolling out of the window) should predict its return, but simulations show this second prediction is fragile to small misspecifications; the one-period prediction—high TK, low subsequent returns—is robust, and is the one the paper tests.

3.6.8 Quantitative Performance: Barberis, Jin, and Wang (2021)

Barberis, Jin, and Wang (2021) push the test further. Using the same parameters as Tversky and Kahneman (1992) for v and w, the authors compute the prospect-theory value of each stock's return distribution and ask whether the model can predict the cross-sectional alpha of 23 anomaly-sorted decile portfolios. The result is mixed.

For four prominent anomalies—momentum, failure probability, idiosyncratic volatility, and gross profitability—the model's predicted alphas line up with the realized cross-sectional pattern (Figure 3.14). For seven others—size, value, long-term reversal, short-term reversal, accruals, asset growth, and investment—the model performs poorly (Figure 3.15). The authors conclude that PT captures the spirit of the anomalies that are most directly linked to skewness and probability weighting, but the broader anomaly zoo needs more than one preference departure. The result is also a useful methodological lesson: the discipline of Section 1.2 applies to behavioral models just as it does to rational ones. A model that explains everything in qualitative terms but matches the quantitative cross-section poorly is underdisciplined by the same standard.

3.6.9 The Risk–Return Trade-Off

Figures 3.16 and 3.17 show the geometry behind a preference that standard theory cannot produce: given the same mean payoff, investors prefer *less* volatile stocks in the domain of capital gains and *more* volatile stocks in the domain of capital losses. Expected utility with concave utility over final wealth delivers the first preference unconditionally and the second never.

The asymmetric value function also has implications for the cross-sectional risk–return relation. Wang, Yan, and Yu (2017) measure the capital gain overhang (CGO) of each stock—the unrealized

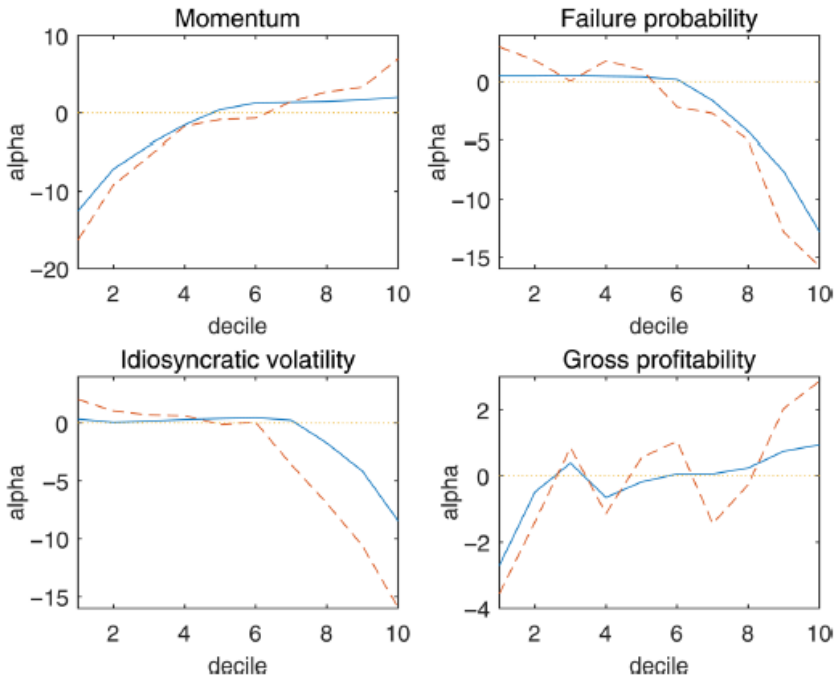


Figure 5. Model-predicted alphas for four prominent anomalies. The graphs present results for four of the anomalies for which the prospect theory model performs well: the momentum, failure probability, idiosyncratic volatility, and gross profitability anomalies. The dashed lines in the graphs plot the historical annual value-weighted alpha of each anomaly decile. The sample period is 1963 to 2015 except for the failure probability anomaly for which it is 1974 to 2015 due to data availability. The solid lines plot the alphas predicted by the model. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 3.14: Anomalies for which the prospect-theory model performs well. The dashed lines are realized annual value-weighted alphas across deciles of four anomalies (momentum, failure probability, idiosyncratic volatility, gross profitability); the solid lines are the model's predicted alphas. The model captures the qualitative shape and, for some anomalies, the magnitude of the cross-sectional alpha pattern. *Source:* Barberis, Jin, and Wang (2021), Figure 5.

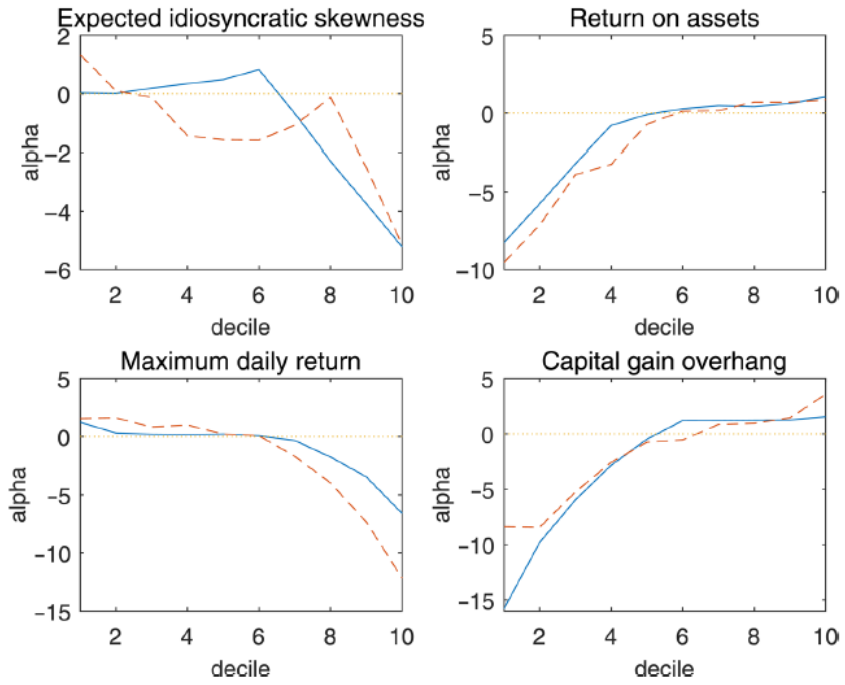


Figure 6. Model-predicted alphas for four prominent anomalies. The graphs present results for four of the anomalies for which the prospect theory model performs well: the expected idiosyncratic skewness, return on assets, maximum daily return, and capital gain overhang anomalies. The dashed lines in the graphs plot the historical annual value-weighted alpha of each anomaly decile. The sample period is 1963 to 2015. The solid lines plot the alphas predicted by the model. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 3.15: Anomalies for which the prospect-theory model performs poorly. Same construction as Figure 3.14 for expected idiosyncratic skewness, return on assets, maximum daily return, and capital gain overhang. The model's predictions have the right sign for some anomalies but the wrong shape or magnitude. *Source:* Barberis, Jin, and Wang (2021), Figure 6.

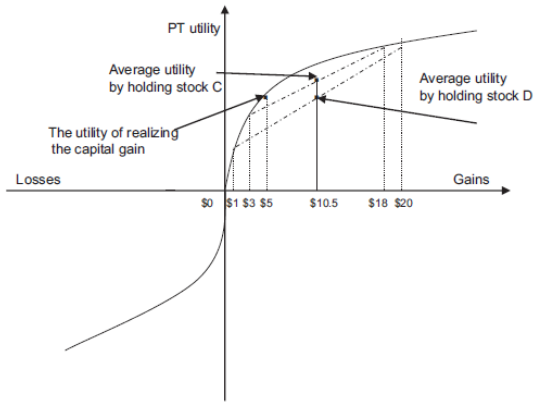


Fig. 3. Prospect theory (PT) and the risk–return trade-off utility: capital gains. Assume that investors purchased one share of stocks C and D, each at a price of $20, and the price is now $25 for each. Thus, investors are facing capital gains and are risk averse. One period later, stock C has a price of $38 or $23 with equal probability, and stock D has a price of $40 or $21 with equal probability. The figure shows the utility gain and loss of holding stocks C and D.

Figure 3.16: PT utility and the risk–return trade-off: capital gains. Investors bought one share each of stocks C and D at $20; the price of each is now $25, so they face capital gains and are risk averse. One period later, stock C is worth $38 or $23 and stock D $40 or $21, each with equal probability: the two stocks have the same mean next-period price but different volatility. In the concave gain region of the value function, the investor holding at a gain prefers the less volatile stock. *Source:* Wang, Yan, and Yu (2017), Figure 3.

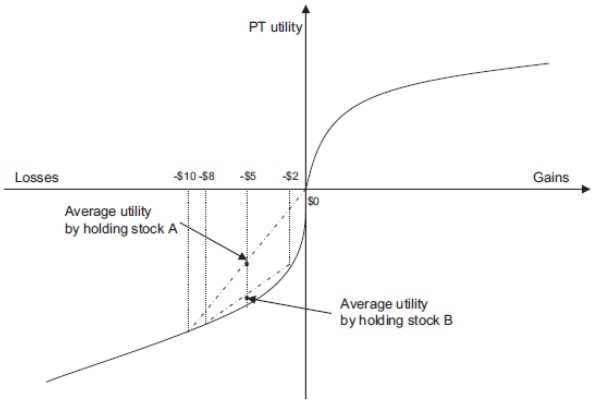


Fig. 2. Prospect theory (PT) and the risk–return trade-off utility: capital losses. Assume that investors purchased one share of stocks A and B, each at a price of $20, and the price is now $15 for each. One period later, the price of stock A can be either $20 or $10 with equal probability, and the price of stock B can be either $18 or $12 with equal probability. The figure shows the utility gain and loss of holding stocks A and B.

Figure 3.17: PT utility and the risk–return trade-off: capital losses. Investors bought one share each of stocks A and B at $20; the price of each is now $15, so they face capital losses and are risk seeking. One period later, stock A is worth $20 or $10 and stock B $18 or $12, each with equal probability—again the same mean, different volatility. In the convex loss region, the investor holding at a loss prefers the more volatile stock: only the wide distribution offers a chance of getting back to the reference point. *Source:* Wang, Yan, and Yu (2017), Figure 2.

Table 3: Return vs. risk proxy, by CGO group — slope negative for losers, positive for winners (monthly value-weighted excess returns, %; 1964–2014)

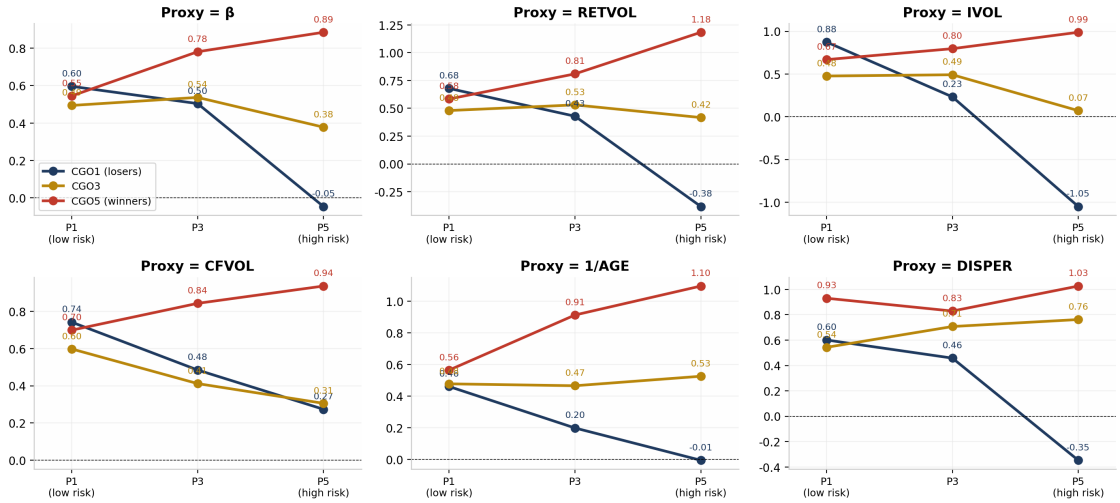


Figure 3.18: Reference-dependent preferences and the risk–return trade-off. The table reports estimates of risk–return trade-off coefficients across capital gain overhang (CGO) portfolios. The trade-off is positive for high-CGO stocks, in which investors face prior gains and are risk averse, and negative for low-CGO stocks, in which investors face prior losses and are risk seeking; the difference is statistically significant. *Source:* Wang, Yan, and Yu (2017), Table III.

gain or loss of the average investor—and ask how the cross-sectional risk–return relation depends on the sign of the overhang. The preference pair in Figures 3.16 and 3.17 predicts exactly a sign flip. Figure 3.18 shows the result: the trade-off is positive among stocks where investors face prior gains (and are therefore risk averse) and negative among stocks where investors face prior losses (and are therefore risk seeking). The two-pronged effect breaks the linear positive risk–return relation that the CAPM predicts, and offers a PT-based reading of the high-risk-low-alpha puzzle of Chapter 1.

3.6.10 My Research: Stock Splits as Behavioral Signals

The loss-aversion machinery of this chapter also speaks to corporate events, not only to portfolio choice. Cui, Li, Pang, et al. (2026) ask why stock splits can serve as a credible positive signal to the market even though the direct costs of implementing a split are trivial. The question bites because standard signaling logic requires a signal to be costly: a nearly free signal should be sent by everyone and convey nothing. Their answer is loss aversion on the investor side. Retail investors treat a stock split as good news, and the split raises their expectations about future firm fundamentals and price appreciation. Because investors evaluate subsequent outcomes relative to that elevated reference point, missing the raised expectations triggers a disproportionately large price decline: the split is mechanically cheap, but its expected cost in the loss domain is large. In equilibrium, only managers with favorable information find it worthwhile to split—loss aversion is what makes the cheap signal costly in expectation, and therefore credible.

The evidence comes from China. Investors turn more optimistic after splits, higher split ratios bring stronger market reactions, and the prices of splitting firms fall harder when the firms subsequently miss the elevated expectations. The connection to this chapter is direct: splits work as signals *because* investors evaluate outcomes as gains and losses relative to an elevated reference point, which is precisely the reference-dependent value function of Section 3.3.

3.7 Prospect Theory and the Aggregate Stock Market

3.7.1 Myopic Loss Aversion and the Equity Premium

Benartzi and Thaler (1995) propose a parsimonious model of the equity premium puzzle. The three ingredients are (i) prospect-theory preferences, (ii) evaluation of gains and losses *frequently* rather than at the end of an investor's lifetime (the *myopia*), and (iii) narrow framing, so that gains and losses are evaluated separately for each asset rather than aggregated across the portfolio. With these ingredients, the historical distribution of annual U.S. stock market returns has approximately the same prospect-theory value as the historical distribution of annual bond returns. The greater volatility of stocks lowers the stock market's prospect value; the much higher mean raises it. The two forces approximately cancel.

The mechanism is intuitive. EU agents with concave utility over consumption evaluate a stream of returns by the discounted expected utility of consumption, and the relatively small variation in consumption makes stocks look only modestly riskier than bonds. PT agents with frequent evaluation, narrow framing, and steep loss aversion see a stream of annual gains and losses; the loss-aversion parameter $\lambda \approx 2$ makes each loss hurt twice as much as an equivalent gain pleases, and the result is a much larger effective equity premium. The stock market earns a high average return to compensate investors for bearing what prospect theory says is a much more painful stream of year-to-year losses.

3.7.2 The Role of Narrow Framing

Barberis, Huang, and Santos (2001) extend the analysis to a fully specified consumption-based model with a representative agent who has prospect-theory preferences (without probability weighting). The authors find that narrow framing is a critical ingredient: with loss aversion defined over total-wealth fluctuations and integrated evaluation, the model cannot reproduce the equity premium; with narrow framing, the model can. The intuition is that the stock market has low correlation with the other risks investors face, and an integrated-evaluation EU agent would essentially treat the equity premium as a small premium for a small amount of additional risk. With narrow framing, the stock market risk is evaluated in its own mental account, and the loss aversion that applies to that account can be large. In related work, Barberis, Huang, and Thaler (2006) show that narrow framing also helps explain why many households do not participate in the stock market at all.

3.7.3 The Role of Probability Weighting

De Giorgi and Legg (2012) add probability weighting to the analysis. The aggregate stock market is negatively skewed: it is subject to occasional large crashes, and the probability of a large crash is small. The sign is the opposite of the individual-stock case, and the contrast is worth a moment. Individual stocks are mostly positively skewed—occasional large upward jumps—yet the market they aggregate into is left-tailed. The reason is asymmetric correlation. In normal times, correlations across stocks are modest, so the positive jumps of individual stocks arrive at different times and diversify away; they do not aggregate into market-level jumps. In a crisis, correlations rise toward one and every stock falls together, so the crashes *do* aggregate. Upside extremes cancel across stocks; downside extremes compound. Under probability weighting, the small probability of the compounding event is overweighted and the prospect value of the stock market falls. Adding probability weighting on top of loss aversion substantially raises the model's predicted equity premium, and it does so without sacrificing the consumption-side fit of the consumption-based model.

3.7.4 Reading the Application

Two features of the equity-premium application deserve emphasis. First, the prospect-theory reading of the equity premium is one of several—rare disasters (Barro, 2006), long-run risks (Bansal and Yaron, 2004), and habit formation (Campbell and Cochrane, 1999) all have candidate mechanisms. The behavioral reading is competitive with the rational ones on this question, not superior to them. Second, prospect theory's strength is that it is also consistent with the cross-sectional evidence (Section 3.6); a model that explains the equity premium but not the cross-section is doing less work than one that explains both.

3.8 Ambiguity Aversion

3.8.1 Risk versus Uncertainty

The psychological departures from EU surveyed so far assume that the agent can assign probabilities to outcomes. Knight (1921) drew the famous distinction between *risk*, where probabilities are known, and *uncertainty*, where they are not. EU was built for risk; the experimental evidence suggests that agents are averse to uncertainty in ways that EU cannot represent.

3.8.2 The Ellsberg Paradox

The cleanest demonstration is Ellsberg (1961). Subjects are shown two urns. Urn C contains 100 balls, 50 red and 50 black. Urn U contains 100 balls of unknown composition, with the number of balls of each color unknown to the subject. Subjects are asked to choose between bets on each urn:

> **Bet R1.** Draw from urn C; win \$100 if red. **Bet R2.** Draw from urn U; win \$100 if red.
>
> **Bet B1.** Draw from urn C; win \$100 if black. **Bet B2.** Draw from urn U; win \$100 if black.

The modal pattern is R1 $\succ$ R2 and B1 $\succ$ B2. In words: subjects prefer to bet on red from the known urn rather than the unknown urn, and the same for black. But the conjunction R1 $\succ$ R2 and B1 $\succ$ B2 implies, under the axioms of subjective expected utility, that the subject behaves as if urn U contains fewer than 50 red balls *and* fewer than 50 black balls—subjective probabilities that sum to less than one, which additive probability rules out. The modal pattern is inconsistent with subjective expected utility and is robust across subject pools and stake sizes.

3.8.3 Models of Ambiguity Aversion

Models of ambiguity aversion modify EU by relaxing the assumption of a single, additive probability measure. The leading approaches are:

Choquet expected utility. Schmeidler (1989) drops the additivity axiom of subjective expected utility and replaces the probability measure with a non-additive *capacity*. Agents weight events by capacities that can be sub-additive on ambiguous events, and the resulting preferences are averse to ambiguity.

Maxmin expected utility. Gilboa and Schmeidler (1989) replace the single prior with a set of priors and evaluate each prospect by its worst-case expected utility. The agent is ambiguity averse in the strongest possible sense—she acts as if the worst possible prior is the truth.

Smooth ambiguity aversion. Klibanoff, Marinacci, and Mukerji (2005) combine a set of priors with a second-order utility function that captures the agent's distaste for ambiguity. The model nests EU as a special case and maxmin EU as a limiting case, and it is among the most tractable for asset pricing applications.

Each model can reproduce the Ellsberg modal pattern; they differ in their out-of-sample predictions, which Section 3.8.4 develops.

3.8.4 Applications to Asset Prices

Epstein and Schneider (2010) survey the asset-pricing applications of ambiguity aversion; we summarize three.

Non-participation. Dow and Werlang (1992) observe that the distribution of stock-market returns is more ambiguous than the distribution of returns on bank deposits or Treasury bills, in the sense that there is a wider range of plausible probability models for the stock market than for the default-free short rate. An ambiguity-averse agent with the option of holding deposits may

rationally choose not to participate in the stock market, even when the stock market's expected return is higher. The application is a preference-based candidate for the non-participation puzzle of Chapter 1.

Equity premium. Maenhout (2004) builds a recursive utility model in which the agent fears that the reference model of consumption growth is misspecified and chooses portfolios that are robust to such misspecification—a form of ambiguity aversion. Concern about misspecification raises the agent's effective risk aversion, so the model can produce a large equity premium without requiring an implausibly large baseline coefficient of risk aversion.

Under-diversification. Uppal and Wang (2003) show that under model misspecification or ambiguity, agents view the distribution of returns on stocks whose payoffs they can better estimate as less ambiguous than the distribution of returns on stocks whose payoffs they cannot. Investors accordingly overweight familiar stocks: their own company's stock, local stocks, domestic stocks relative to foreign stocks. The result is a model of *home bias* and *local bias* as ambiguity-averse portfolio choice. The behavioral reading is complementary to the information-based reading discussed in Chapter 1: ambiguity aversion provides a candidate for the part of home bias that information alone cannot explain.

3.9 What Remains Contested

Two strands of evidence keep the behavioral reading honest.

First, the quantitative success of prospect theory in the cross-section is uneven. Barberis, Jin, and Wang (2021) show that the standard Tversky–Kahneman parameterization, when fed the empirical cross-section of anomaly-sorted decile portfolios, predicts the right sign for many anomalies but the right shape for only a handful (Section 3.6). The prospect-theory reading of skewness, probability weighting, and lottery preference fits a narrow but coherent part of the anomaly zoo; the broader zoo requires additional departures or a different architecture.

Second, the aggregate-market applications are competitive with rational explanations. The equity premium puzzle admits prospect-theory, rare-disaster, long-run-risk, and habit-formation resolutions; distinguishing them empirically is difficult because all four predict a large premium and a low risk-free rate. The prospect-theory reading has the comparative advantage that it ties the aggregate market to the cross-section—the same probability weighting that prices lottery-like stocks also pushes the equity premium up because the aggregate market is negatively skewed—but the comparison is a matter of relative fit, not of decisive evidence.

The methodological lesson is the same as in Chapter 1: a behavioral model earns credibility when it predicts something not used to fit it, and when that prediction survives out-of-sample tests. Prospect theory's strongest out-of-sample predictions are the disposition effect and the lottery-related cross-sectional facts (low IPO returns, low returns on MAX stocks, low returns on distressed stocks, the capital-gain-overhang pattern in the risk–return trade-off). The aggregate-market predictions are competitive with rational alternatives and have less discriminative power.

3.10 The Road Ahead

This chapter has developed a non-standard preference model and traced its asset pricing applications. The rest of the book's psychological core develops two complementary departures from the rational benchmark.

Chapter 4 develops *biased beliefs*: systematic departures from Bayes' rule—overconfidence, over-extrapolation, conservatism, representativeness—and their asset pricing implications. The combination of prospect theory with biased beliefs is what most of the survey literature in behavioral finance actually deploys. The chapter also covers heterogeneous-beliefs models and the relationship between preference and belief departures.

Chapter 5 develops *bounded rationality* and *memory-based* approaches: limited attention, salience, the role of narrative, and the way in which investors form beliefs from the stories they remember rather than from a Bayesian aggregation of evidence. These models retain rational beliefs in a thin sense but relax the assumption that all available information is processed.

Together with Chapter 1 on the empirical facts, Chapter 2 on limits to arbitrage, and the present chapter on preferences, the book develops the building blocks of behavioral asset pricing. The remaining chapters assemble them into specific empirical applications.

Chapter 4

Biased Beliefs

The benchmark model of investor beliefs is Bayes' rule. Agents hold prior probabilities over states of the world, observe signals whose likelihoods they know, and update to posterior probabilities by Bayes' formula. Combined with the expected-utility benchmark of Chapter 3, rational Bayes–expected-utility agents trading in frictionless markets deliver the predictions reviewed in Chapter 1—prices track rational forecasts of cash flows, expected returns in the cross-section line up with risk, and investors hold diversified portfolios and trade only on information. Like the expected-utility benchmark, the Bayes benchmark is the wrong descriptor of how investors actually form beliefs, and the descriptive gaps are large enough to matter for prices. Laboratory experiments, survey data, and field evidence document systematic departures from Bayes' rule—representativeness, base-rate neglect, sample-size neglect, overconfidence, over-extrapolation, conservatism, anchoring, belief perseverance—and these departures show up in trading behavior, in expectations surveys, and in the cross-sectional and time-series properties of asset returns.

This chapter develops that evidence. The first half (Sections 4.1–4.4) is the *psychology of beliefs*: a survey of the cognitive biases that distort investors' probability assessments, together with the asset pricing applications each bias has been claimed to explain. The second half (Sections 4.5–4.8) is the *heterogeneous beliefs* literature: rather than assuming a single representative investor with biased beliefs, the second half dispenses with the representative assumption and asks what cross-sectional variation in beliefs does to prices—especially when combined with short-sale constraints. Section 4.9 asks whether people ever learn, Section 4.10 catalogues the limits of the biased-beliefs program, and Section 4.11 closes the chapter.

4.1 Representativeness

Definition 4.1 (Representativeness heuristic)**.** *A person judging the probability that object A belongs to class B, or that event A was generated by process B, evaluates the probability by the degree to which A resembles B—the* degree to which A reflects the essential characteristics of B *(Kahneman and Tversky, 1974).*

The representativeness heuristic is useful in many settings: the prediction that a small, dense, red fruit is more likely an apple than a cherry is exactly the kind of inference the heuristic supports. The bias enters when the resemblance judgment is made *without* taking into account base rates and sample sizes, the two pieces of statistical information a Bayesian calculation would weight heavily. Kahneman and Tversky (1974) document two robust violations.

4.1.1 Base-Rate Neglect

The standard demonstration is the "Steve" problem. Subjects read a personality sketch of a fictional person—very shy and withdrawn, invariably helpful, but with little interest in people or in the world of reality; a meek and tidy soul with a need for order and structure and a passion for detail—and are asked whether Steve is more likely to be a librarian or a lawyer. The modal answer is librarian, because the description resembles the librarian stereotype. But there are many more lawyers than librarians in the U.S., and a Bayesian would let that base rate dominate. The question asks about the posterior odds on the left of

$$\frac{p(\text{librarian} \mid \text{data})}{p(\text{lawyer} \mid \text{data})} = \frac{p(\text{data} \mid \text{librarian})}{p(\text{data} \mid \text{lawyer})} \cdot \frac{p(\text{librarian})}{p(\text{lawyer})},$$

and representativeness judges answer using only the likelihood ratio, the first term on the right. The prior odds are dropped, and with them the information that should dominate the answer. Base-rate neglect is the neglect of the second factor in a Bayes calculation the judge never performs.

4.1.2 Sample-Size Neglect

The second violation is the failure to weight evidence by the sample size from which it is drawn. A sequence of three heads from a fair coin is taken as strong evidence that the coin is biased toward heads, even though the posterior probability of bias after three flips is still close to the prior. The same judgment applied to a sequence of thirty heads would be more defensible. Gilovich, Vallone, and Tversky (1985) document the consequence in basketball: fans and players judge a player who has made a few shots in a row to be "hot," and adjust their expectations of his next shot upward. The empirical evidence in large samples is that the hot hand is largely a myth—shot-level success rates are essentially uncorrelated with the recent history of makes and misses—although Miller and Sanjurjo (2018) later identified a selection bias in the standard tests that works against finding streaks, softening the original verdict. Gilovich, Vallone, and Tversky (1985) read the phenomenon as representativeness: fans expect a short streak of makes to be representative of a player's true ability, and they expect the next shot to reflect that true ability. In finance, the same bias is the source of over-extrapolation, to which Section 4.3 returns.

4.2 Overconfidence

Definition 4.2 (Overconfidence)**.** *Overconfidence is the tendency to believe one's judgments are more accurate than they actually are. The literature distinguishes* overprecision *(confidence in the accuracy of one's forecasts),* overplacement *(belief that one is better than others), and* overestimation *(unrealistically positive self-assessments of ability).*

The cleanest laboratory evidence is on overprecision. Alpert and Raiffa (1982) asked subjects to give 98% confidence intervals for unknown quantities (the area of Manhattan, the number of piano tuners in Chicago). Across many studies, the realized coverage rate of these intervals is closer to 60% than 98%. Subjects are wildly miscalibrated: the ranges they assign are too narrow for the information they have.

4.2.1 CFOs and the Stock Market

The overprecision finding extends to professional investors. Ben-David, Graham, and Harvey (2013) surveyed chief financial officers of U.S. corporations and asked them to report probability

distributions for the one-year-ahead return on the S&P 500. The CFOs' distributions are very tight: their 80% confidence intervals span about 28 percentage points of expected return, centered on a relatively narrow range of point estimates. The realized market return, over the same horizon, falls inside the CFOs' 80% interval only about 36% of the time. The professional respondents are not only miscalibrated, they are miscalibrated in the same direction as the student subjects of Alpert and Raiffa (1982): their intervals are too narrow by a factor of two or more.

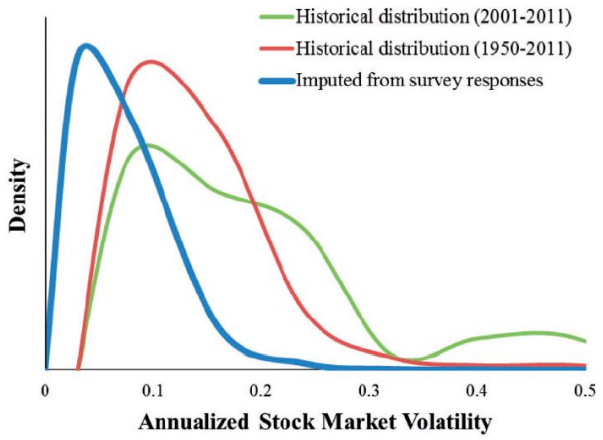


FIGURE IIA

One-Year Imputed Individual S&P 500 Volatility Versus Historical Distribution

The kernel density of the one-year S&P 500 return volatility for the period of 1950–2000 (or 2001–2011), and the kernel density of the imputed one-year S&P 500 individual volatility derived from CFOs' predictions.

Figure 4.1: CFO overprecision. Kernel densities of the one-year S&P 500 return volatility implied by CFOs' survey responses, plotted against the historical distribution of realized one-year volatility over 1950–2000 and 2001–2011. The CFO-implied densities sit well inside the historical ones: executives' subjective return distributions are far too tight, which is why the realized return falls inside their 80% confidence intervals only about 36% of the time. *Source:* Ben-David, Graham, and Harvey (2013).

4.2.2 Overplacement

A separate strand of evidence concerns *overplacement*—the belief that one is above average. Svenson (1981) surveyed drivers in the U.S. and Sweden and found that 93% of American respondents (and a large majority of Swedish respondents) rate themselves as above-average drivers. The figure is mechanically impossible: the average driver is, by definition, average. The same pattern appears in finance: in surveys of investor confidence, retail investors systematically report that their own portfolio will outperform the average portfolio of their peers.

4.2.3 Excessive Trading

Overconfidence has direct trading implications. Odean (1999) showed that the average household in a large discount-brokerage sample turns over roughly 80% of its portfolio each year and that net returns fall sharply with turnover: the gross return on the portfolio is roughly uncorrelated with trading intensity, but trading costs—commissions and bid–ask spreads—rise with it. The households that trade most earn less than the households that trade least, and the average household earns less than a passive index fund. Hong and Stein (2007) document the matching aggregate fact that trading volume is far in excess of what any rational model with symmetric information can generate.

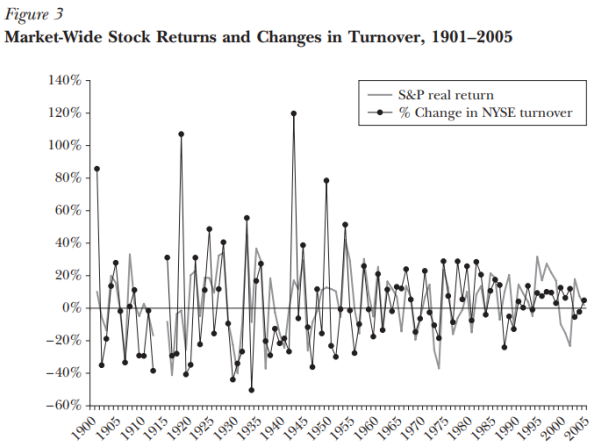


Figure 4.2: Excessive trading. The figure shows that turnover in equity markets is far higher than would be implied by liquidity-motivated or rebalancing trades alone, and that the volume is concentrated in periods when investor beliefs diverge. The dispersion in beliefs is the natural reading of the volume pattern. *Source:* EF8083 course materials, after Hong and Stein (2007).

4.2.4 Gender Differences

Barber and Odean (2001) show that the overconfidence bias is not gender-neutral. In their brokerage dataset, men trade roughly 1.5 times as often as women and earn correspondingly lower net returns; the gender gap in net returns is largest among singles and shrinks (but does not disappear) among married households, when household portfolios are pooled. Lundeberg, Fox, and Punćcohaŕ (1994) document the matching psychological result: in laboratory studies of confidence judgments, men report higher confidence than women on tasks of comparable difficulty, with the gap widening on tasks at which both perform poorly. The two strands of evidence are mutually reinforcing: the gender gap in trading costs is consistent with the gender gap in confidence, and the mechanism operates through over-trading rather than through any systematic gender difference in stock-selection skill.

Dependent variable	Mean monthly turnover (%)	Own-benchmark abnormal net return	Portfolio volatility
Intercept	6.269***	−0.321***	11.466***
		(−11.47)	(58.85)
Single	0.483***	0.002	0.320***
	(4.24)	(0.14)	(3.40)
Woman	−1.461***	0.058***	−0.689***
	(−12.76)	(4.27)	(−7.27)
Single × woman	−0.733***	0.027	−0.439**
	(−3.38)	(1.08)	(−2.45)



Figure 4.3: Boys will be boys. Regression evidence from Barber and Odean (2001): men, and single men in particular, trade at higher monthly turnover, earn lower net returns, and hold more volatile portfolios than women—the pattern predicted if men are the more overconfident sex (Lundeberg, Fox, and Punćcohaŕ, 1994). *Source:* Barber and Odean (2001) and Lundeberg, Fox, and Punćcohaŕ (1994).

4.2.5 Momentum and Overconfidence

Daniel, Hirshleifer, and Subrahmanyam (1998) develop a formal model in which overconfidence interacts with *self-attribution bias* to produce momentum. Investors are overconfident about private signals they themselves generate (e.g., their own research) but less so about public signals. Positive returns reinforce their overconfidence through self-attribution—a streak of good calls is attributed to skill, a streak of bad calls to bad luck—and they continue to trade on their private signals with growing positions. Eventually, the public information that contradicts their private view arrives, and prices partially correct. The dynamics produce short-horizon continuation (momentum) followed by partial reversal as information is revealed.

The model predicts that momentum should be stronger in environments where overconfidence is culturally or institutionally more pronounced. Chui, Titman, and Wei (2010) test this prediction using Hofstede's individualism index as a proxy for cultural variation in overconfidence. Individualism, in Hofstede's framework, captures the extent to which a society rewards individual initiative and self-promotion relative to group cohesion—a pattern that maps naturally onto overplacement. The data confirm the prediction: over February 1984–June 2003, the winner-minus-loser momentum profit for country-average portfolios rises with individualism, from the bottom 30% to the top 30% of the individualism ranking, with a high-minus-low spread of 0.65% per month ($t = 4.3$).

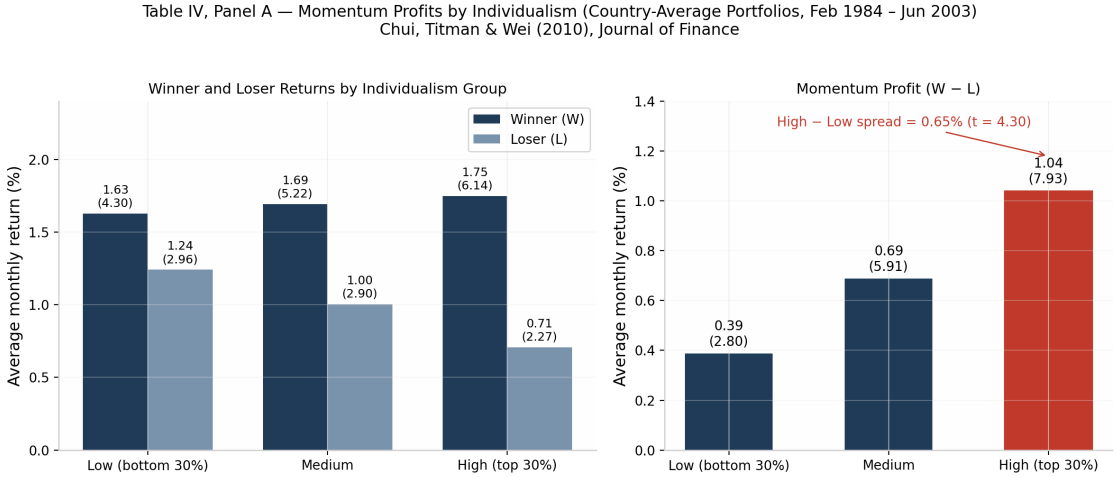


Figure 4.4: Momentum and individualism. Winner and loser returns and winner-minus-loser momentum profits for country-average portfolios grouped by Hofstede's individualism index (bottom 30%, medium, top 30%), February 1984–June 2003. Momentum profits rise with individualism; the high-minus-low spread in the momentum profit is 0.65% per month ($t = 4.30$). *Source:* EF8083 course materials, based on Chui, Titman, and Wei (2010).

4.3 Over-Extrapolation

Definition 4.3 (Over-extrapolation)**.** *The estimate of the future value of a quantity is a positive function of the recent past values of that quantity. The mechanism is the representativeness heuristic: a recent history of, say, increasing prices is judged representative of a generally increasing trend, so the next observation is expected to be larger still.*

Over-extrapolation is the most widely applied idea in behavioral finance. It is the source of return extrapolation (predicting future returns from past returns), of fundamentals extrapolation

(predicting future earnings from past growth), and, in aggregate, of the survey-implied "expected returns" that move with valuation ratios in ways rational discount-rate models cannot easily reproduce. The applications span almost every asset pricing puzzle of the past twenty years.

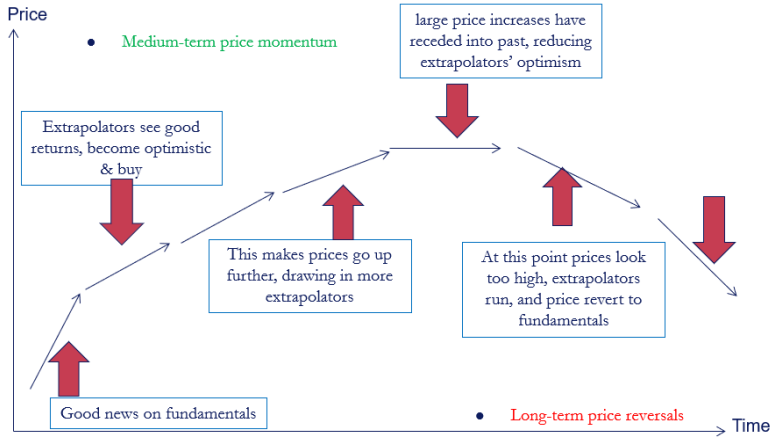


Figure 4.5: Over-extrapolation in pictures. Good news lifts recent observations, which are extrapolated to lift beliefs about future values; the higher beliefs in turn push prices further up; the price rise confirms the recent observations. The loop runs until some shock breaks it. *Source:* EF8083 course materials.

4.3.1 Survey Evidence on Extrapolation

Greenwood and Shleifer (2014) assemble survey expectations of one-year-ahead stock market returns from six sources, including the Gallup poll, the University of Michigan's survey of consumer sentiment, the Survey of Professional Forecasters, and several investor surveys. They compare the survey-implied expected return to realized subsequent returns over the same horizon and find a strongly negative correlation: the higher investors' expected returns, the lower realized returns are subsequently. The sign is the central fact of the paper. Survey expectations rise after market run-ups—they extrapolate recent returns—and high expectations therefore coincide with the conditions that precede low subsequent returns, the opposite of what rational time-varying-discount-rate models imply for measured expected returns. The authors then connect survey expectations to investor flows into equity mutual funds and show that the flows chase the same states of the world: inflows rise when expected returns are high, exactly the pattern that produces overvaluation when sentiment reverses.

4.3.2 Measuring the Degree of Extrapolation: DOX

Cassella and Gulen (2018) propose a formal measure of the weight investors place on recent returns. Their measure, called *DOX* for degree of extrapolation, is the complement of a parameter λ in a forecaster's representation of next-period expected return as a weighted average of past returns,

$$Exp_t = a + b \sum_{i=0}^{\infty} w_i \, R_{t-(i+1)\Delta t,\, t-i\Delta t}, \qquad w_i = \frac{\lambda^i}{\sum_{k=0}^{\infty} \lambda^k}, \quad 0 \le \lambda < 1, \tag{4.1}$$

so that DOX $= 1 - \lambda$. A lower λ places more weight on the most recent observations; earlier observations contribute less and less to the extrapolator's expectation. Estimating λ recursively

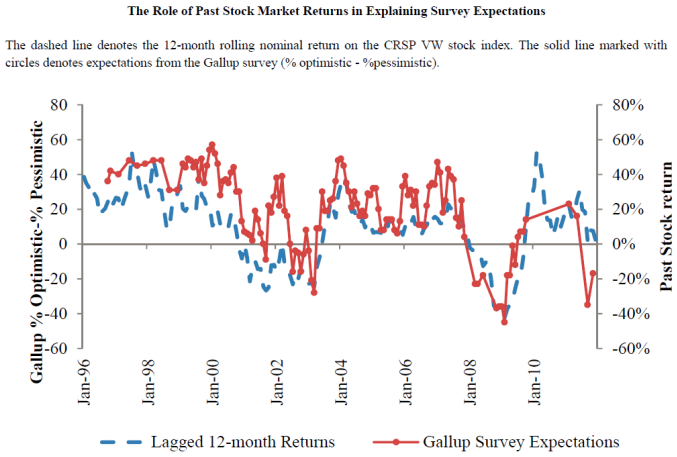


Figure 4.6: Survey expectations of stock returns. Survey measures of one-year-ahead return expectations are negatively correlated with realized returns over the same horizon. The pattern is consistent with extrapolation: when recent returns are high, expected returns are also high, and subsequent returns are low—the expectations are inverted by the next period, not vindicated. *Source:* Greenwood and Shleifer (2014).

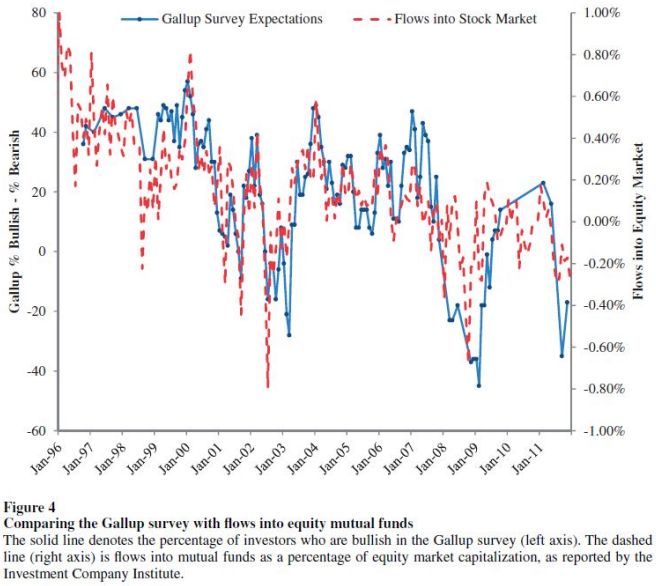


Figure 4
Comparing the Gallup survey with flows into equity mutual funds
The solid line denotes the percentage of investors who are bullish in the Gallup survey (left axis). The dashed line (right axis) is flows into mutual funds as a percentage of equity market capitalization, as reported by the Investment Company Institute.

Figure 4.7: Investors act on extrapolative beliefs. Net flows into equity mutual funds correlate with survey expectations of one-year-ahead returns. Investors buy after high expected returns (which follow high recent returns) and thereby deepen the overvaluation that the expectations forecast. *Source:* Greenwood and Shleifer (2014).

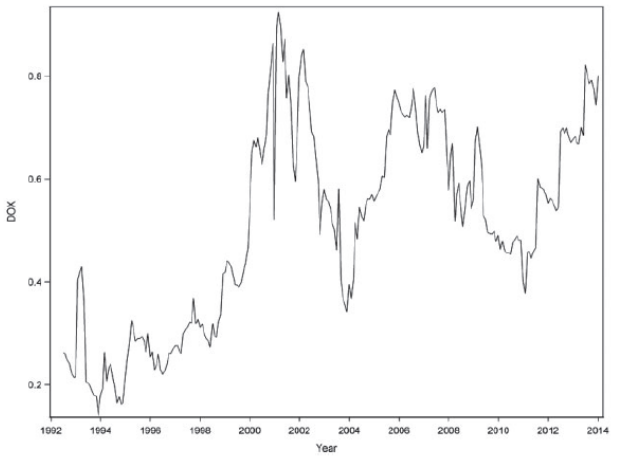


Figure 2
Time-varying DOX
The figure plots the recursively estimated DOX time series for the principal component (PC) of the II and AA surveys, in the sample period 1992:06–2013:12. (Details on estimation are in Section 2.)

Figure 4.8: Degree of extrapolation. The recursively estimated DOX time series for the principal component of the survey return expectations, 1992:1–2013:12. DOX varies over time: the weight investors place on past returns is a state variable, not a fixed parameter. *Source:* Cassella and Gulen (2018).

from survey data yields a time series for DOX rather than a constant: investors extrapolate more in some periods than in others (Figure 4.8).

DOX enters asset prices through the predictability of returns by price-scaled fundamentals (dividend yield, book-to-market, earnings-to-price). The predictability is contingent on the DOX. When DOX is high, investors more quickly "forget" the positive price changes that drove up valuations, and their beliefs mean-revert faster; as a result, high valuations are *more* strongly associated with low future returns. When DOX is low, extrapolative beliefs persist longer and valuation ratios forecast less. The DOX measure is therefore not just a description of beliefs but a state variable that conditions the predictability of returns—and Cassella and Gulen (2018) document that the conditional predictability is present in the data.

4.3.3 Fundamentals Extrapolation

A parallel stream of evidence concerns the extrapolation of fundamentals rather than prices. Bordalo, Gennaioli, La Porta, O'Brien, et al. (2024) show that long-term growth (LTG) expectations of corporate earnings overreact to good and bad news: a positive earnings surprise shifts the distribution of LTG forecasts upward by more than is justified by the signal alone. The authors track an aggregate LTG-expectation measure over time and show that its deviations from a fundamental benchmark are highly correlated with the price–dividend ratio and predict low subsequent market returns over a five-year horizon. The mechanism is the same as in return extrapolation: investors apply a diagnostic forecasting rule that overweights representative recent observations, and the implied LTG co-moves with prices and reverts.

Bordalo, Gennaioli, La Porta, and Shleifer (2025) take the logic to the cross-section. Every test of an asset pricing model faces a joint hypothesis problem: a return spread can reflect either distorted beliefs *or* a premium for some exotic risk. Their resolution is to shut the risk channel down by construction. They build a firm-level measure of *expectations based returns* (EBRs) from analyst forecast errors and revisions—the return each stock would earn if all cross-sectional variation in

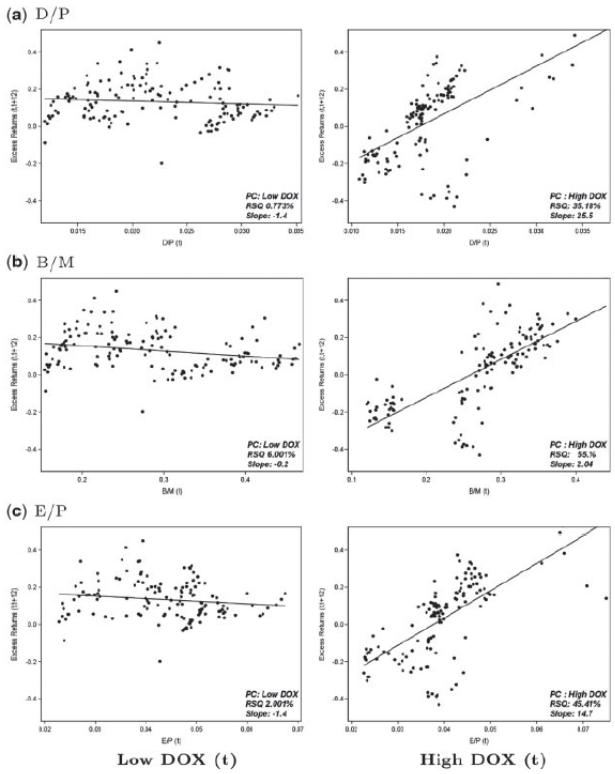


Figure 4.9: DOX and predictability. Predictive regressions of subsequent returns on price-scaled variables, conditional on the estimated DOX measure. The slope is steeper (more predictability) when DOX is high: investors forget the price run-up faster, beliefs mean-revert faster, and high valuations more strongly forecast low future returns. *Source:* Cassella and Gulen (2018).



Figure 4.10: Fundamentals extrapolation. The figure overlays a measure of long-term earnings-growth (LTG) expectations (red) with the price–dividend ratio (green) and five-year-ahead realized stock returns (black). LTG expectations track the price–dividend ratio and predict low subsequent returns, consistent with extrapolative beliefs. *Source:* Bordalo, Gennaioli, La Porta, O'Brien, et al. (2024).

expected returns came from beliefs about future cash flows and none from differences in required returns. If pure expectation-driven returns of this kind account for the observed factor spreads, the spreads do not require exotic risk premia. They do: variation in EBRs accounts for a large share of the return spreads on value (HML), size (SMB), investment (CMA), profitability (RMW), and momentum (WML) portfolios (Figure 4.11). The paper's title states the conclusion—*finance without exotic risk*—and it marks a shift in the extrapolation literature from documenting belief distortions to showing that measured beliefs, alone, quantitatively carry the cross-section.

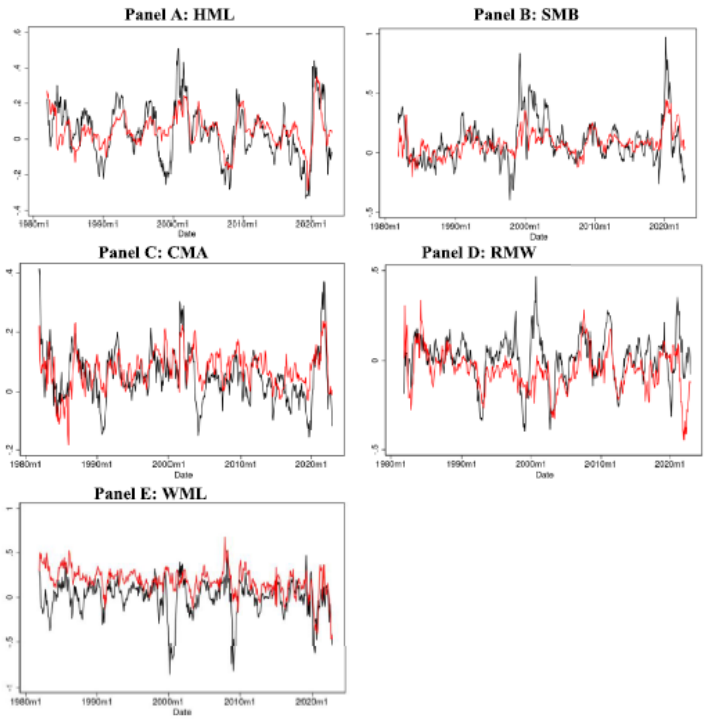


Fig. 1. Actual and expectations-based return spreads.
Note: Panel A plots one-year log returns (black line) and expectations-based returns (EBRs, red line) for the long-short value minus growth (HML) portfolio, where EBRs are computed following Eq. (13) in the text. Panel B plots one-year log returns (black line) and EBRs (red line) for the long-short small minus big (SMB) portfolio. Panel C plots one-year log returns (black line) and EBRs (red line) for the long-short conservative minus aggressive investment (CMA) portfolio. Panel D plots one-year log returns (black line) and EBRs (red line) for the long-short robust minus weak profitability (RMW) portfolio. Panel E plots one-year log returns (black line) and EBRs (red line) for the long-short winners minus losers (WML) portfolio. The sample period is December 1981 to December 2023. See Table 1 for a description of the portfolios.

Figure 4.11: Expectations-based returns. One-year log returns (black) and expectations-based returns (EBRs, red) for the long-short value (HML), size (SMB), investment (CMA), profitability (RMW), and momentum (WML) portfolios, December 1981–December 2023. EBRs are computed from analyst forecast errors and revisions and shut down cross-sectional differences in required returns; variation in EBRs accounts for a large share of the realized factor spreads. *Source:* Bordalo, Gennaioli, La Porta, and Shleifer (2025), Figure 1.

4.3.4 My Research: Naïve Earnings Growth Extrapolation

Cui, Li, and Zhang (2021) study fundamentals extrapolation in a setting where the extrapolated quantity is unusually visible: Chinese earnings announcements. China reports quarterly earnings on a year-to-date basis, which makes cumulative YTD growth highly salient to investors even though the YTD measure contains stale earnings information—earlier quarters' results roll mechanically into the cumulative number. Stocks with the strongest past YTD earnings growth run up in the five days before announcements and reverse afterward: investors extrapolate the salient YTD growth into the

upcoming announcement and are disappointed when the "growth" turns out to be old news. The pattern concentrates in small, retail-heavy, low-analyst-coverage stocks, and retail buying demand rises in the run-up to announcements. The paper is the cross-sectional counterpart of Bordalo, Gennaioli, La Porta, O'Brien, et al. (2024) on the aggregate market: fundamentals extrapolation that distorts the prices of individual stocks around earnings news rather than the valuation of the market as a whole.

4.4 Other Biases: Conservatism, Anchoring, and Feelings

The biases catalogued so far all push investors to *overreact* to new information—to be more confident in their forecasts, more certain about the trend, and more aggressive in their trading. The literature also identifies biases that push in the opposite direction, underreacting to news.

4.4.1 Conservatism

Definition 4.4 (Conservatism)**.** Conservatism *is the tendency to update beliefs too slowly in response to new information (Edwards, 1968). In the canonical Bayesian account, agents place posterior probability somewhere between the prior and the signal-consistent likelihood; the conservatism bias is the finding that the updating weight on the signal is smaller than the Bayesian formula implies.*

The conservatism bias is the natural complement to over-extrapolation. Conservatism by itself would predict slow price adjustment to news and the post-earnings-announcement drift of Section 1.5.7; over-extrapolation by itself would predict sharp initial reaction followed by reversal. Barberis, Shleifer, and Vishny (1998) build a model in which conservatism at short horizons and representativeness at long horizons combine to produce exactly the joint pattern of momentum and long-term reversal that the data show.

4.4.2 Belief Perseverance and Confirmation Bias

Two related biases affect how investors *search* for and *interpret* evidence. *Belief perseverance* is the tendency, once an opinion is formed, to be slow to revise it on receipt of new evidence; subjects often continue to endorse a belief even after the evidence that produced it has been fully discredited. *Confirmation bias* is the active counterpart: investors seek out evidence that supports their prior and interpret ambiguous evidence as supportive. The two effects compound: investors both fail to look for falsifying evidence and discount it when they encounter it. Stanovich, West, and Toplak (2013) show that higher cognitive ability can amplify confirmation bias, because more capable reasoners are better at constructing arguments for their existing position.

4.4.3 Anchoring

Definition 4.5 (Anchoring)**.** Anchoring *is the tendency, in forming numerical estimates of uncertain quantities, to adjust insufficiently from some initial value (the "anchor"), even when the anchor is generated randomly and is clearly irrelevant (Kahneman and Tversky, 1974).*

In the canonical demonstration, Kahneman and Tversky (1974) spin a wheel of fortune in front of subjects to produce a random number, ask subjects whether the percentage of African nations in the United Nations is greater or less than that number, and then ask them to estimate the actual percentage. The random anchor moves the estimates even though the subjects watch it being generated: those who compared their estimate to 10 subsequently estimated 25 percent, while

those who compared it to 60 estimated 45 percent. The financial applications include IPO pricing (where the initial price range functions as an anchor for analysts), analyst price targets, and merger valuations (where the announced offer price anchors subsequent negotiation).

The application with the cleanest evidence is anchoring on the 52-week high. George and Hwang (2004) show that nearness to the 52-week high predicts future returns: a stock trading close to its high earns anomalously high subsequent returns, because traders are slow to react to good news when the price is already near a salient reference point. The same anchor contributes to the momentum effect—prices drift upward as the anchor itself rises.

4.4.4 The Effect of Feelings

A more recent strand of research investigates how feelings affect beliefs and preferences. *Affective feelings*—mood and emotion—shift judgments in ways unrelated to the evidence: a happier mood produces more optimistic assessments, a sadder mood more pessimistic. Edmans, Garcia, and Norli (2007) document the case in finance: when the national soccer team loses a World Cup elimination match, the national stock market falls significantly the next trading day, with the effect concentrated in countries where soccer is more important. The economic mechanism is not information about fundamentals but mood: disappointing sports news worsens the national mood and depresses risk-taking.

Two environmental results reinforce the mood channel. Hirshleifer and Shumway (2003) find higher stock market returns on sunnier days: sunshine improves mood, improved mood raises optimism about future cash flows, and the optimism is priced. Kamstra, Kramer, and Levi (2003) link the seasonal pattern in returns to Seasonal Affective Disorder: fewer hours of daylight induce a mild depression, which raises risk aversion and lowers returns through the winter months.

Wang and Young (2020) extend the argument to a starker exogenous shock. Terrorist attacks raise aggregate risk aversion, as proxied by flows out of equity mutual funds and into government bond funds. The effect is larger in the days immediately following the attack and decays over time. Both findings point to a direct channel from affective feelings to beliefs about risk and, through beliefs, to portfolio choices. They also suggest a mechanism that can amplify market fluctuations: if the stock market rises, investors' mood improves, their judgments about the economy's prospects become more positive, and they push the market up still further.

4.4.5 My Research: Air Pollution and Carbon Divestment

Belief shifts triggered by exogenous stimuli reach professionals, and they can overshoot. Huynh, Li, and Xia (2025) study mutual fund managers exposed to local air pollution. Managers experiencing pollution underweight the stocks of high-emission firms, and the effect strengthens for environmentally conscious managers and for managers surprised by the pollution—that is, when the stimulus is salient, managers revise their beliefs about carbon risk. The divestment is costly: the carbon-intensive stocks managers sell subsequently outperform the stocks bought in their place. Like soccer losses and terrorist attacks, pollution is an exogenous stimulus that shifts beliefs; the difference is that here the professionals *overreact*, selling carbon-intensive stocks by more than fundamentals justify.

4.4.6 Cognitive Feelings

Cognitive feelings are about the ease with which information is processed rather than the mood that processing leaves behind. Zajonc (1968) and Bornstein (1989) document the basic result: people prefer stimuli that are familiar, easily processed, and fluently retrieved, even when the preference has no informational basis. Uppal and Wang (2003) formalize the asset pricing version: under

ambiguity aversion or model misspecification, investors view stocks whose payoffs they can better estimate as less ambiguous, and overweight them. The applications are *home bias* (domestic stocks are more familiar than foreign stocks) and *local bias* (stocks headquartered nearby are more familiar than distant ones), both of which Chapter 1 documents.

4.5 Heterogeneous Beliefs: Sources

The biases of Sections 4.1–4.4 all keep the *representative investor* assumption intact: a single agent, possibly with systematically distorted beliefs, faces the asset pricing problem. The next two sections drop that assumption and work with heterogeneous beliefs—different agents holding different views about asset values, with the average belief unbiased. The departure is consequential. Asymmetric information alone cannot sustain rational disagreement: under common priors, a Bayesian agent who is offered a trade recognizes that the offer itself reflects information he does not have, and revises his beliefs until the apparent gains from trade vanish. The Milgrom and Stokey (1982) no-trade theorem formalizes the result: with common priors and rational updating, agents who observe the same signals agree to agree, and no trade takes place. Heterogeneous beliefs in the rational sense must therefore come from somewhere beyond asymmetric information: from differences in prior beliefs, which Morris (1995) argues are reasonable for genuinely new phenomena, or from differences in how agents process the same information. The empirically important sources of such differences are the subject of this section.

The magnitude of the phenomenon is easy to underestimate. Figure 4.12 plots disagreement among respondents to the Survey of Professional Forecasters: the interquartile range across forecasters of one-quarter-ahead forecasts of annualized quarterly growth in real GDP, corporate profits, and CPI inflation. These are professionals, surveyed at the same moment, answering the same question with the same public data—and the disagreement is wide in every period. People regularly hold different beliefs about almost everything, from political views and sports events to economic outcomes and asset returns.

4.5.1 Experience Effects

One important source of heterogeneous beliefs is personal experience. When forming beliefs, a rational individual should take account of all available information; mounting evidence instead suggests that people overweight what they have lived through. Malmendier and Nagel (2011) show that whether a household participates in the stock market, and how much it invests, depends on the stock market returns the household experienced over its lifetime: cohorts that lived through poor markets stay underweight equities for decades, and the effect of experienced returns fades only slowly as new experience accumulates.

Malmendier and Nagel (2016) document a clean case in inflation expectations. Survey respondents overweight the inflation they experienced during their own lifetimes when forming expectations of future inflation: respondents born in the 1930s report higher inflation expectations in 2010 than respondents born in the 1970s, even though the cohort born later has experienced more inflation in absolute terms. Younger respondents update their beliefs more quickly in response to recent news; older respondents update more slowly and rely more heavily on their formative experiences. Kuchler and Zafar (2019) extend the experience logic with individual-level data: people's beliefs about aggregate housing prices and unemployment depend strongly on the economic conditions they have experienced locally, even though the aggregate outcomes are public information. The same mechanism—overweighting personally experienced events—produces persistent cross-sectional dispersion in beliefs about any quantity for which the historical record is long.

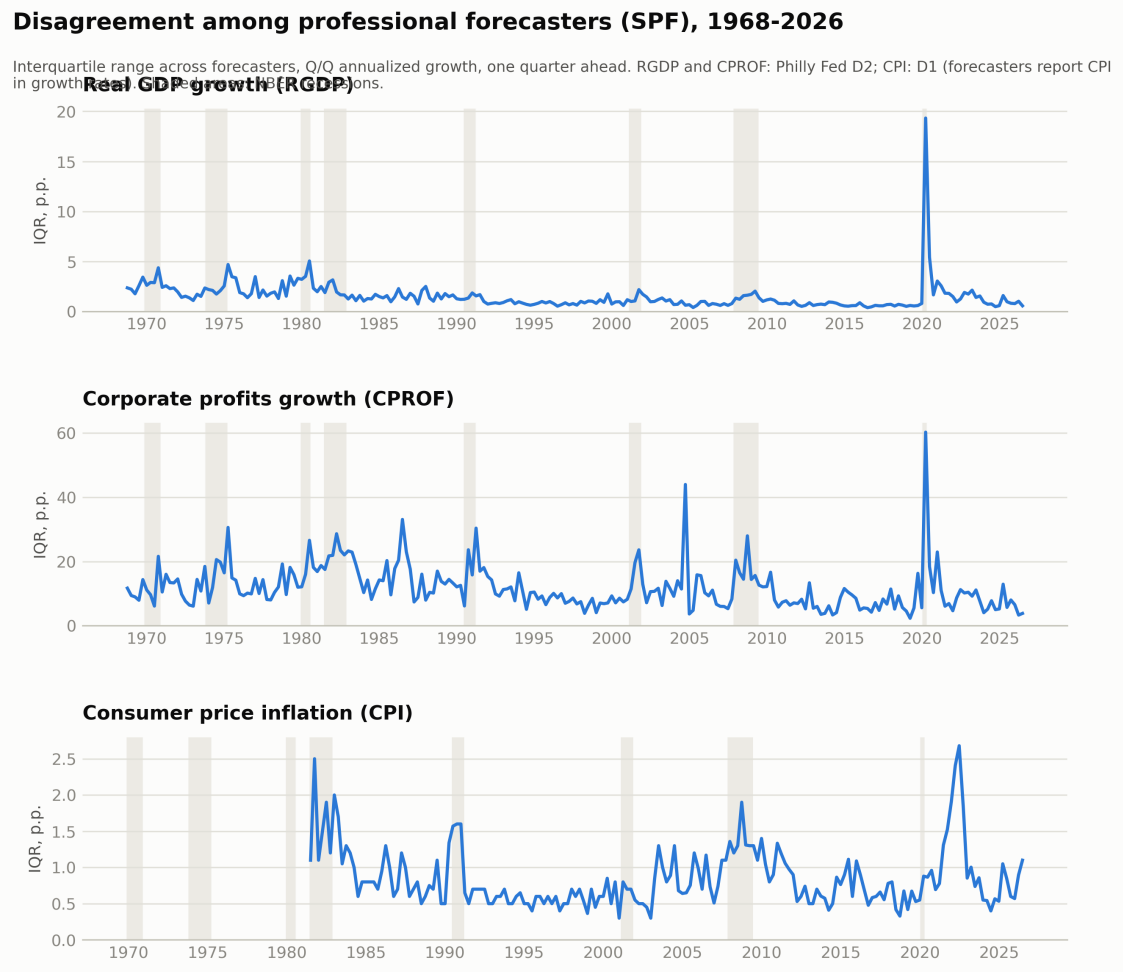


Figure 4.12: Even professionals disagree. Interquartile range across respondents to the Survey of Professional Forecasters of one-quarter-ahead forecasts of annualized quarterly growth in real GDP, corporate profits, and CPI inflation (Philly Fed variables D2 and D1), 1968–2026. Forecasters answering the same question at the same time, with the same public data, disagree widely. *Source:* Author's calculations based on the Federal Reserve Bank of Philadelphia Survey of Professional Forecasters.

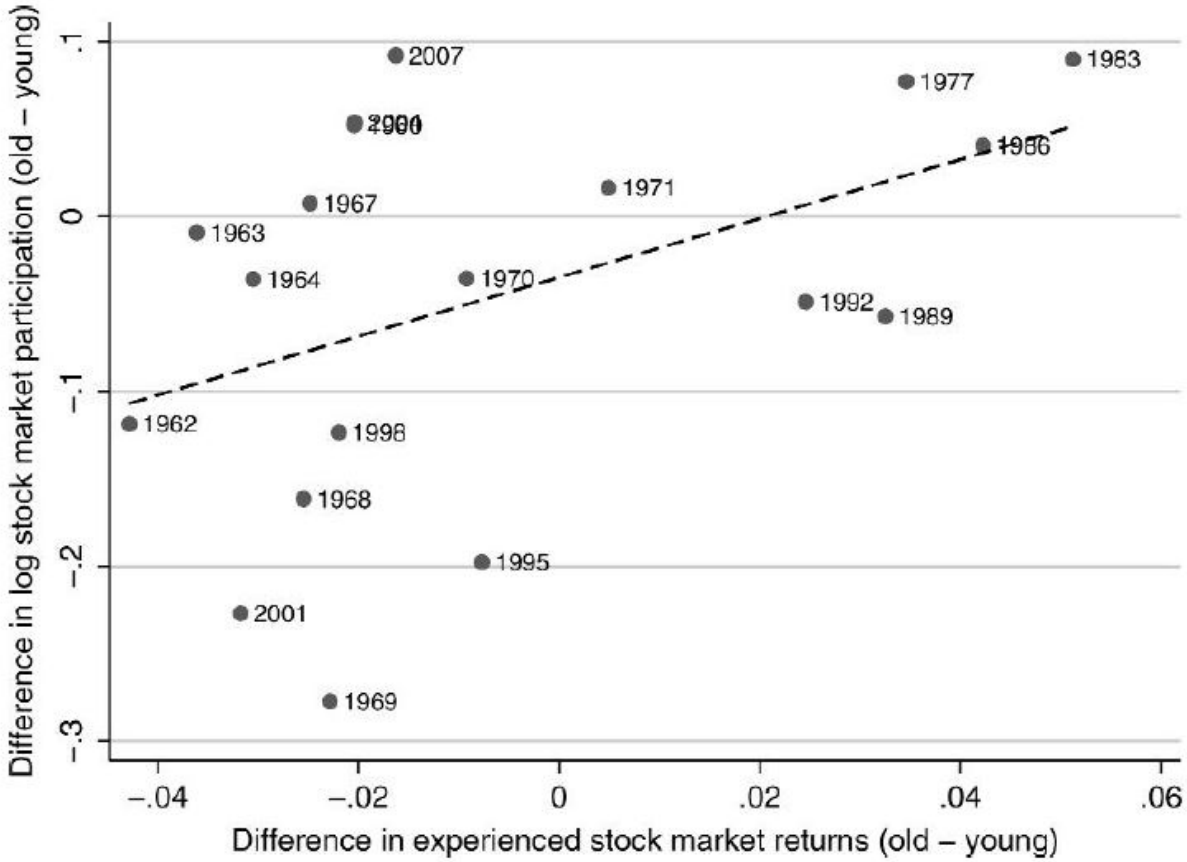


Figure 4.13: Experience effects. The figure illustrates how personal experience shapes beliefs: people place disproportionate weight on events they have lived through and form reference points from those events. The same statistical evidence can produce different posterior beliefs for agents with different life histories. *Source:* EF8083 course materials.

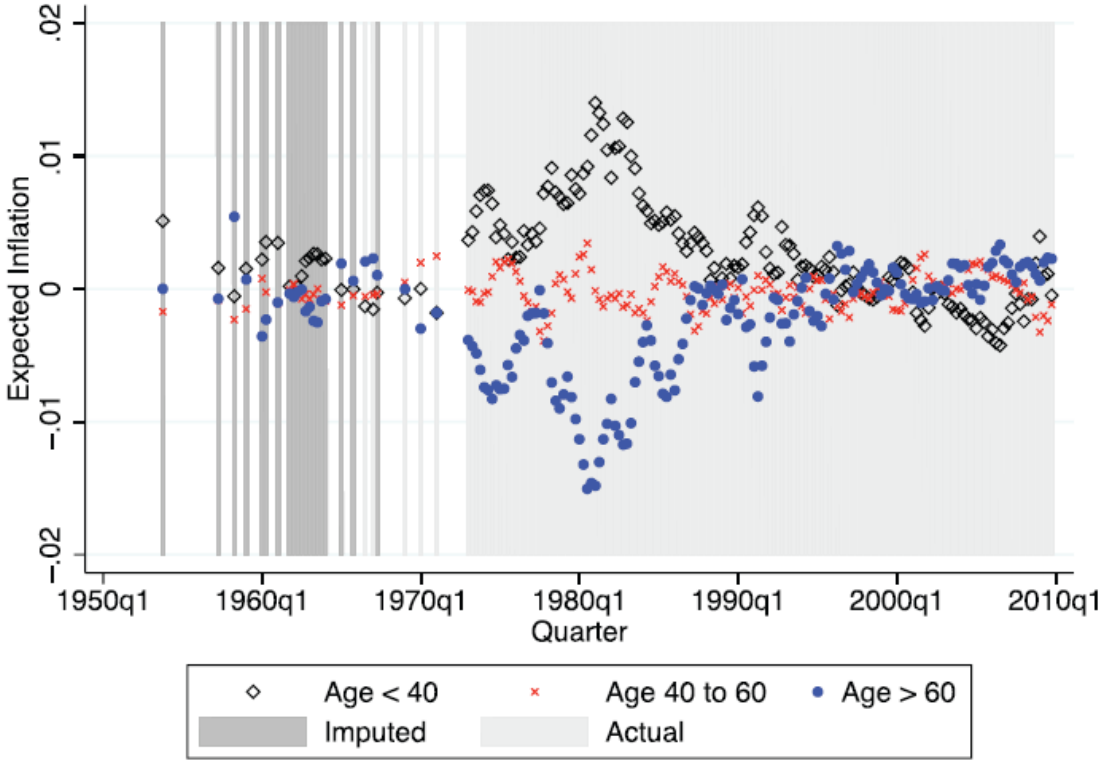


FIGURE I

Inflation Expectations by Age Group Relative to Cross-Sectional Mean

Figure 4.14: Inflation expectations. Survey respondents' expectations of long-run inflation overweight the inflation experienced during their own lifetimes. The cohort born in the 1930s reports higher expected inflation in 2010 than the cohort born in the 1970s, opposite to what the realized record implies. *Source:* Malmendier and Nagel (2016).

4.5.2 Political Polarization

Goldman, Gupta, and Israelsen (2024) document a more recent source of belief dispersion in financial news. The authors compare the same corporate financial news covered by the conservative *Wall Street Journal* and the liberal *New York Times*. The two newspapers use systematically more positive language for firms aligned with their political leanings, and the gap is wider for politically sensitive topics (climate, labor, executive compensation) than for neutral ones. Investors who consume one source rather than the other form different beliefs about the same companies, and the dispersion persists because the consumers do not switch sources. The finding is part of a broader pattern: media fragmentation along political lines is a quantitatively significant source of disagreement about financial facts.

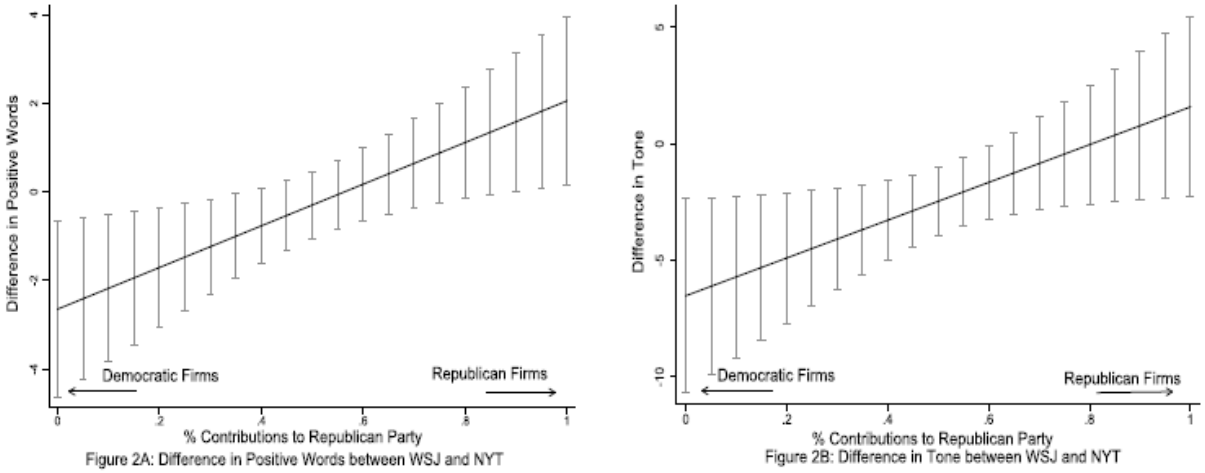


Figure 4.15: Political polarization in financial news. Tone-of-language scores for corporate financial coverage by the *Wall Street Journal* (conservative) and *New York Times* (liberal), separated by firm-level political alignment. Both newspapers use systematically more positive language for firms aligned with their political leanings, with the gap larger for politically sensitive topics. *Source:* Goldman, Gupta, and Israelsen (2024).

4.5.3 Overconfidence in Private Signals

The third source is the one emphasized by Daniel, Hirshleifer, and Subrahmanyam (1998). Two agents with the same prior can observe the same public signal and apply different likelihoods because they differ in their confidence about their own private information. The agent with higher confidence in his private signal will place less weight on the public signal and more on his own view; the agent with lower confidence will do the reverse. Both are acting rationally, given their subjective confidence, but they hold heterogeneous beliefs about the asset, and the dispersion in beliefs is larger when the agents are overconfident in their private signals. The mechanism is the bridge from the overconfidence of Section 4.2 to the heterogeneous-beliefs models of the next section.

4.6 Heterogeneous Beliefs with Short-Sale Constraints

Heterogeneous beliefs alone do not have interesting pricing effects. If two agents disagree about an asset's value, the optimist wants to buy and the pessimist wants to sell. Without frictions, the

two positions can be unwound and the price lies at a weighted average of the two beliefs. The overpricing and underpricing cancel out, and the equilibrium expected return is the rational expected return under the average belief. What dispersion does generate is volume: investors trade because they disagree about asset values, and higher dispersion of beliefs means higher trading volume. Grinblatt and Keloharju (2009), using Finnish data that link brokerage records to aptitude tests, show that more overconfident individuals—those whose self-reported confidence exceeds what their test performance justifies—trade more. Heterogeneous beliefs become interesting for *prices* when pessimists cannot fully express their view—in the canonical case, because short-selling is constrained.

4.6.1 Static Argument: Miller (1977)

Miller (1977) observed that if pessimists face binding short-sale constraints, only the optimists' valuation is reflected in the price. The equilibrium price lies above the average belief and above the asset's fundamental value, with the magnitude of the overpricing determined by the dispersion of beliefs. The implication is a sharp empirical prediction: assets that investors disagree about more should be more overpriced, and should earn lower subsequent returns.

The argument is static, in the sense that it does not require beliefs to change over time. It is enough that beliefs differ across agents in the cross-section at a given date, and that the pessimists cannot fully short. The mechanism is also asymmetric across the two legs of the trade: correcting overpricing requires shorting, which is constrained, while correcting underpricing requires only cash. The asymmetry is a deep source of pricing distortion in the heterogeneous-beliefs model.

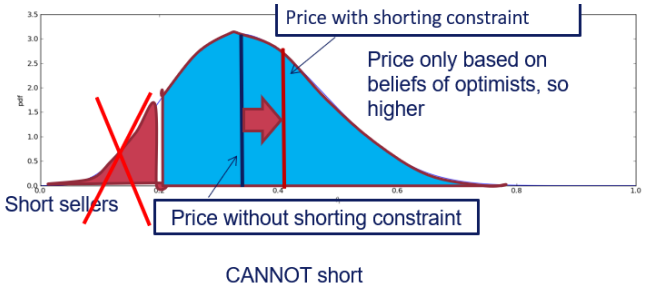


Figure 4.16: Miller (1977) in pictures. The figure illustrates the static argument: with short-sale constraints, only the optimist's valuation is reflected in the price, and the equilibrium price exceeds the average belief. The magnitude of the overpricing scales with the dispersion of beliefs. *Source:* EF8083 course materials.

4.6.2 Dynamic Argument: Harrison and Kreps (1978)

The dynamic version of the argument adds a *resale option*. Harrison and Kreps (1978) show that an agent with optimistic beliefs about an asset is willing to pay more than his buy-and-hold valuation, because there is a positive probability that future information will make his view the prevailing one, at which point he can resell to a future optimist at a still higher price. The resale option is valuable precisely because beliefs are heterogeneous: in a homogeneous-belief world there is no one to resell to. With short-sale constraints preventing pessimists from driving the price down, the resale option adds a *bubble component* to the price that can persist even when all agents know the asset is overvalued relative to fundamental value.

The following example, drawn from the Harrison and Kreps (1978) framework, illustrates the magnitude of the resale-option effect (Figure 4.17).

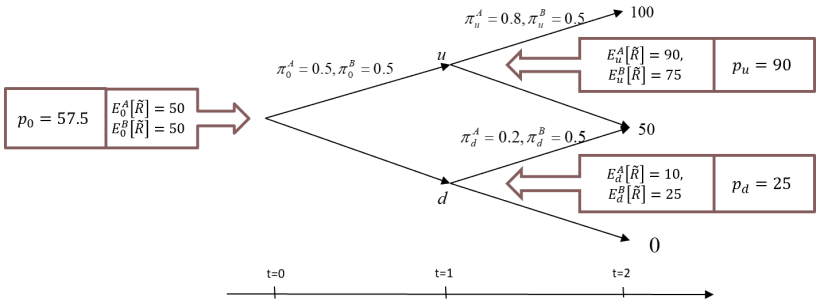


Figure 4.17: The resale-option example. Valuations and prices in the two-agent, two-state example. Both agents' buy-and-hold valuations at $t = 0$ are 50, yet both are willing to pay 57.5, because each expects to resell to the more optimistic agent at $t = 1$: at 90 in state u, above B's valuation of 75, or at 25 in state d, above A's valuation of 10. *Source:* EF8083 course materials.

Worked example. Two risk-neutral agents A and B trade a risky asset at $t = 0, 1, 2$. The interest rate is zero and short-selling is prohibited. At $t = 1$ a public signal arrives—good news (state u) or bad news (state d), each with probability $1/2$—and the two agents interpret it asymmetrically. After good news, agent A values the asset at 90 and agent B at 75; after bad news, agent A values it at 10 and agent B at 25. Each agent's buy-and-hold valuation at $t = 0$ is therefore $(90 + 10)/2 = 50$ for A and $(75 + 25)/2 = 50$ for B: without the option to resell, the two agents agree the asset is worth 50. With short-selling prohibited, the $t = 1$ price is set by the more optimistic agent: 90 in state u and 25 in state d. At $t = 0$, both agents are therefore willing to pay

$$\tfrac{1}{2} \cdot 90 + \tfrac{1}{2} \cdot 25 \;=\; 57.5,$$

above either agent's buy-and-hold valuation. The excess 7.5 is the resale-option premium. Each agent expects to resell the asset to the other at a price above his own valuation: B expects to resell to A at 90 in state u, which exceeds his valuation of 75, and A expects to resell to B at 25 in state d, which exceeds his valuation of 10. The resale option enters both agents' initial willingness to pay: a speculative bubble component.

4.6.3 Bubble Characteristics

The bubble component of Harrison and Kreps (1978) has properties that match observed bubble episodes. Scheinkman and Xiong (2003) formalize the result. First, the resale-option premium is determined by fluctuations in agents' beliefs: the more likely beliefs are to flip between optimists and pessimists, the more the agents trade and the more valuable the resale option—which is why bubbles tend to coincide with trading frenzies. Second, the premium depends on asset float: with more shares available for trading, the bubble is smaller. The mechanism runs through downward-sloping demand curves. An owner can sell his share only if a buyer's belief passes his own by a sufficient margin; a larger float makes it less likely that enough sufficiently optimistic buyers exist, and the resale option is worth less. The predictions are exactly the dimensions on which Xiong and Yu (2011) take the model to Chinese warrant data (Section 4.7).

4.7 Heterogeneous Beliefs and Asset Prices

4.7.1 High Valuations and Heavy Trading

Models of heterogeneous beliefs with short-sale constraints are popular partly because they explain not just overpricing but also *the coincidence of high valuations and heavy trading*, which is one of the empirical regularities of bubble episodes (Chapter 1). With short-sale constraints, the optimists trade against each other as their beliefs diverge; pessimists who would otherwise trade to express their views are sidelined. The trading volume and the price level both rise, and they do so together. Hong and Stein (2007) document this co-movement in U.S. stock market data.

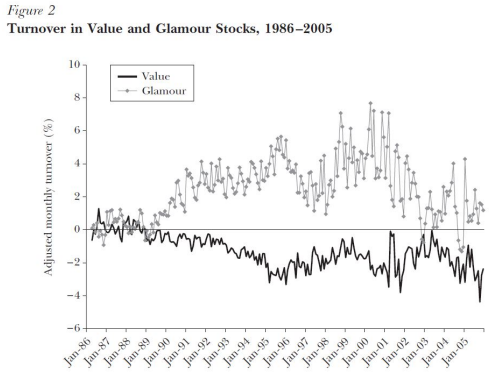


Figure 4.18: High valuations and heavy trading. The figure plots turnover against cumulative returns in the cross-section of U.S. stocks. Stocks with the highest turnover also exhibit the highest cumulative returns, consistent with heterogeneous-beliefs models in which optimists trade against each other while pessimists are constrained. *Source:* Hong and Stein (2007), reproduced from the original article.

4.7.2 The Chinese Warrant Bubble

The most rigorous empirical test of the heterogeneous-beliefs-with-short-sale-constraints theory is Xiong and Yu (2011) on Chinese warrants. Between 2005 and 2008, more than a dozen put warrants traded on the Chinese exchanges at intrinsic values that were essentially zero (the underlying stocks were deep out of the money relative to exercise prices). Yet the warrants traded at high prices and with enormous turnover. The fundamental value of a deep-out-of-the-money put warrant is very small; the observed prices are an order of magnitude larger.

Xiong and Yu (2011) document three regularities consistent with the resale-option theory of bubbles. The size of the bubble is increasing in turnover (consistent with high dispersion of beliefs driving both prices and volume); increasing in the volatility of the underlying asset (consistent with the resale-option value rising in the volatility of beliefs); and decreasing in the asset float (consistent with tighter short-sale constraints when fewer shares are available). The warrant data are an unusually clean test because the fundamental value is essentially known and constant, leaving little room for the rational discount-rate counter-story.

4.7.3 Cross-Sectional Stock Returns

The heterogeneous-beliefs model with short-sale constraints also predicts a cross-sectional return pattern: assets that investors disagree about more should be more overpriced, and should earn lower subsequent returns. Diether, Malloy, and Scherbina (2002) test the prediction using analyst forecast dispersion as the disagreement measure. They form portfolios sorted on the cross-sectional

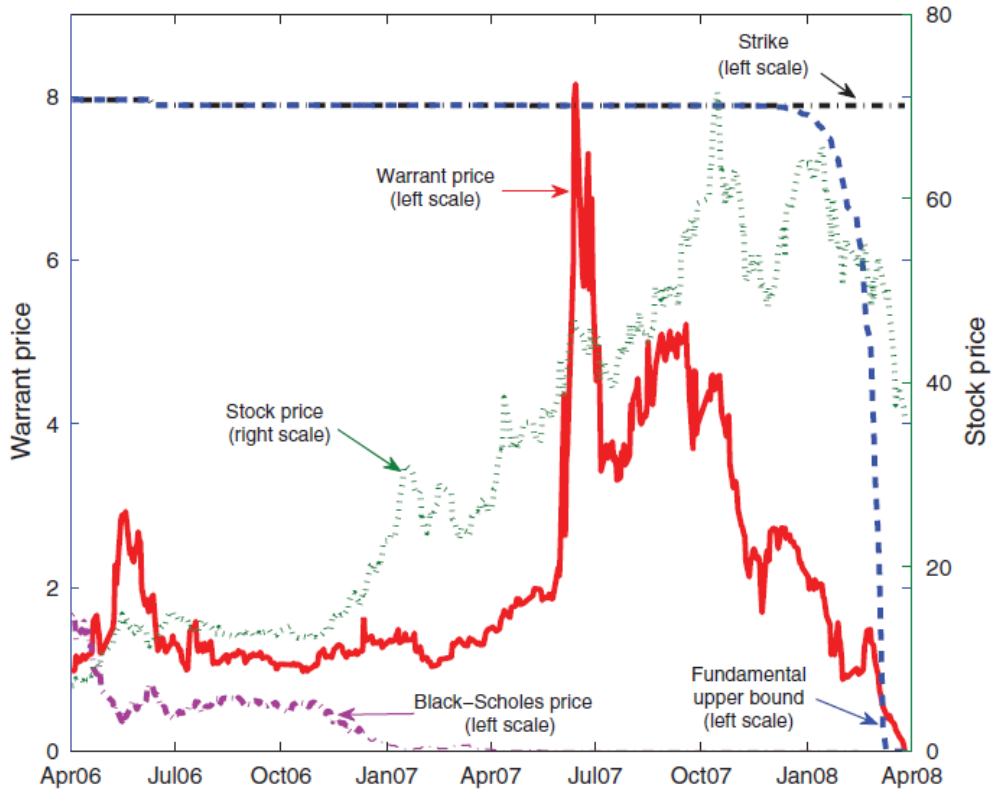


FIGURE 1. PRICES OF WULIANG PUT WARRANT

Notes: This figure shows the daily closing prices of WuLiang stock and its put warrant, along with WuLiang warrant strike price, upper bound of its fundamental value assuming WuLiang stock price drops 10 percent every day before expiration (maximum allowed per day in China's stock market), and its Black-Scholes price using WuLiang stock's previous one-year rolling daily return volatility.

Figure 4.19: Chinese warrant bubble. The figure shows price and turnover for a representative deep-out-of-the-money put warrant traded in China, 2005–2008. Intrinsic value is essentially zero throughout the sample; the warrant trades at high prices and with very high turnover. *Source:* Xiong and Yu (2011), reproduced from the original article.

TABLE 5—DETERMINANTS OF SIZE OF THE WARRANTS BUBBLE

	(1)	(2)	(3)	(4)	(5)	(6)
TURNOVER	0.203			0.158		0.105
(*t*-stat)	(8.91)			(5.46)		(3.80)
VOL		21.76		14.12		13.99
(*t*-stat)		(5.38)		(2.97)		(3.05)
FLOAT			−0.302	−0.324		
(*t*-stat)			(12.62)	(12.31)		
FIRMISSUE					−1.302	−1.260
(*t*-stat)					(25.03)	(23.53)
BROKERAGEISSUE					0.496	0.419
(*t*-stat)					(15.84)	(12.60)

Notes: This table includes the regression results of daily warrant closing prices on *TURNOVER* (daily warrant turnover) in column 1, on *VOL* (5min warrant return volatility) in column 2, on FLOAT (the total number of warrants outstanding, in billions) in column 3, on *TURNOVER*, *VOL*, and *FLOAT* together in column 4, on FIRMISSUE (the total amount of warrants issued by the underlying firm from Table 1, in billions) and *BROKERAGEISSUE* (net total warrant issued by brokerage firms, in billions) in column 5, and on *TURNOVER*, *VOL*, *FIRMISSUE*, and *BROKERAGEISSUE* together in column 6. The regressions use the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike) and include warrant maturity fixed effect. The *t*-statistics, in the parentheses, are adjusted for heteroskedasticity and correlation within a trading day.

Figure 4.20: Warrant bubble evidence. The bubble component of warrant prices increases with turnover and volatility and decreases with asset float, consistent with the resale-option theory of speculative bubbles. *Source:* Xiong and Yu (2011).

standard deviation of analyst earnings forecasts and find that high-dispersion stocks earn significantly lower returns than low-dispersion stocks over the subsequent twelve months, even after controlling for size, book-to-market, and momentum. The return spread is consistent with the overpricing mechanism: disagreement is high precisely for stocks where some analysts are optimistic and others are pessimistic, and the short-sale constraint prevents the pessimists from fully expressing their view. Figure 4.21 shows the pattern in the Open Source Asset Pricing data: over 1976–2024, the low-dispersion quintile out-earns the high-dispersion quintile by a statistically significant spread.

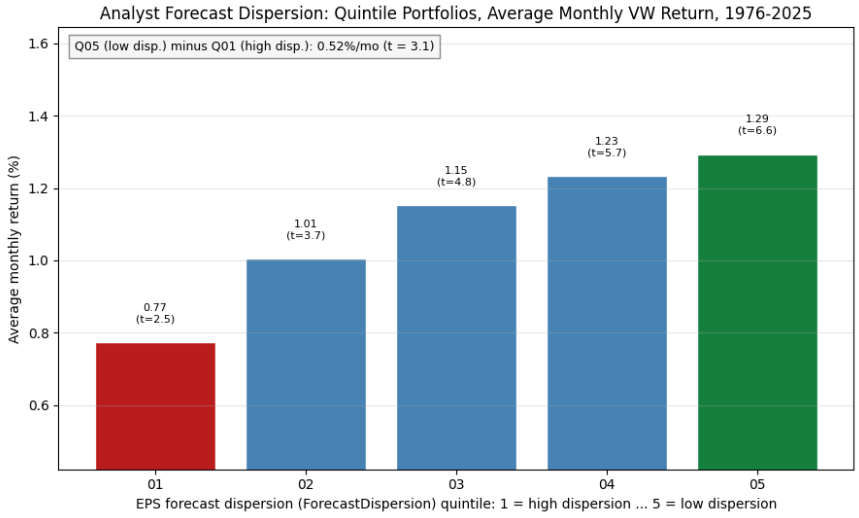


Figure 4.21: Cross-sectional disagreement. Average monthly value-weighted returns (%) of quintile portfolios sorted on analyst earnings-forecast dispersion, 1976–2024; quintile 1 holds the highest-dispersion stocks. Low-dispersion stocks outperform high-dispersion stocks (spread $t = 3.2$), consistent with the overpricing prediction of the heterogeneous-beliefs model with short-sale constraints. *Source:* Author's calculations based on Open Source Asset Pricing quintile portfolio data (Chen and Zimmermann, 2022), after Diether, Malloy, and Scherbina (2002).

4.7.4 Machine Forecast Disagreement

Disagreement itself is unobservable; the analyst-dispersion proxy inherits the limits of the I/B/E/S data—coverage is thin outside the large caps, so small and micro-cap stocks are effectively unmeasured, and the dispersion of human forecasts is itself contaminated by the biases of Section 4.2. Bali, Kelly, et al. (2026) propose measuring disagreement by *simulating* the investors. Each of $K = 100$ "investors" is a random forest that forecasts next-month returns from a common information set of 152 stock characteristics, but each draws its own random 50% subset of the features (differences in information) and its own bootstrap and dropout draws (differences in priors). Between them, the 100 models span the three canonical sources of heterogeneous beliefs—different priors, different information, and different processing of the same information—while each individual model remains sophisticated. The dispersion of the 100 forecasts is the stock's *machine forecast disagreement* (MFD), computed on a rolling ten-year window refit annually and available for the full cross-section over the entire sample.

MFD prices exactly as the Miller mechanism predicts. High-MFD stocks earn significantly lower subsequent returns: the decile long-short portfolio that buys low-MFD and shorts high-MFD stocks earns about 13% per year value-weighted, and the result is robust to 19 variations of the simulation design. The mechanism evidence identifies the discount as mispricing rather than risk compensation. The high-minus-low MFD return spread is far more negative among stocks that are expensive to short (-2.36% per month among high-borrow-fee stocks versus -0.12% among low-fee stocks; difference -2.24, $t = -6.06$), is amplified among stocks with high retail ownership (-1.10% versus -0.31%, $t = 4.53$) and among stocks with high Stambaugh, Yu, and Yuan (2012) mispricing scores, and the correction concentrates on earnings announcement days, consistent with limits to arbitrage delaying the repricing. An earnings-based variant—dispersion of machine forecasts of earnings rather than returns—also predicts returns and correlates more strongly with analyst dispersion.

The methodological point outlasts the specific paper. Belief dispersion has always been measured by asking people; Bali, Kelly, et al. (2026) show it can be *generated*—a population of heterogeneous yet individually reasonable forecasting models produces a belief distribution whose cross-sectional pricing signature matches what survey-based disagreement measures deliver on their best-covered subsamples. As machine-learned and language-model-based forecasts become standard inputs to research, simulated disagreement is likely to become a standard instrument for the unobservable it proxies.

4.7.5 Aggregate Stock Market

The same mechanism predicts a relation at the aggregate level: when aggregate disagreement is high, the market as a whole should be more overpriced and earn lower subsequent returns. Yu (2011) measures aggregate disagreement as the value-weighted average of stock-level forecast dispersion and shows that high-dispersion periods are followed by low returns on the aggregate market index.

4.7.6 Risk–Return Tradeoff

Another important application of the same logic is the risk–return tradeoff. Standard theory predicts that higher risk is rewarded with higher expected return: Sharpe (1964) and the CAPM that follows from it imply an upward-sloping security market line. Empirically, the relation between risk and return is often negative instead. Heterogeneous-beliefs models explain the anomaly: when investors disagree about a stock, the optimists set the price and the pessimists are sidelined by the short-sale constraint, so higher disagreement is associated with a *higher* price and a *lower* expected return. The negative risk–return relation is one of the empirical regularities of Chapter 1. Figure 4.23

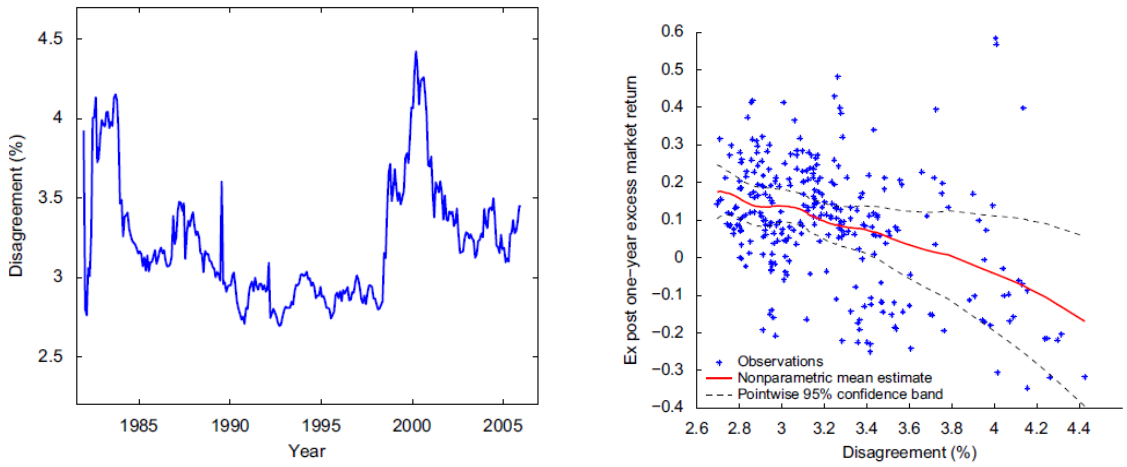


Figure 4.22: Aggregate disagreement. Left: the time series of aggregate analyst forecast dispersion, measured as the value-weighted average of stock-level dispersion. Right: a nonparametric estimate of expected market returns against aggregate disagreement, with pointwise 95% confidence bands; expected returns decline as disagreement rises. High-disagreement periods are followed by low market returns, consistent with the heterogeneous-beliefs overpricing mechanism operating at the aggregate level. *Source:* Yu (2011), reproduced from the original article.

shows the pattern in beta-sorted portfolios: average CAPM alphas decline with beta across the ten BetaFP decile portfolios, with cross-sectional fit $\alpha = 1.178 - 0.178\,\beta$ and $R^2 = 0.95$. High-beta stocks do not earn commensurately high returns; measured against the CAPM, they are overpriced.

Hong and Sraer (2016) turn the observation into a model of the *high-beta–low-alpha puzzle*. Write each asset's payoff as its beta times the aggregate cash flow plus idiosyncratic noise,

$$d_i = \beta_i(Z) + e_i.$$

Suppose agents disagree about the aggregate cash flow Z: optimists expect $Z + \delta$, pessimists $Z - \delta$. The disagreement passes through beta. High-beta assets are more sensitive to disagreement about aggregate cash flows than low-beta assets, so optimists expect $\beta_i(Z + \delta)$ and pessimists $\beta_i(Z - \delta)$: high-beta assets experience more divergence of opinion. With short-sale constraints, the amplified divergence overprices high-beta assets relative to low-beta assets. In a slogan, *beta amplifies aggregate disagreement*.

The model delivers sharp predictions about the shape of the security market line. When aggregate disagreement is low, the CAPM holds and the SML slopes upward. When aggregate disagreement is high, the SML takes an inverted-U shape—expected returns initially increase with beta and then decrease as overpricing of high-beta assets dominates. Stock-level disagreement should increase with beta, especially when aggregate disagreement is high. Hong and Sraer (2016) measure aggregate disagreement as the lagged-beta-weighted average of stock-level analyst forecast dispersion (Figure 4.24) and test the predictions in beta-decile portfolios split by high and low disagreement months (Figure 4.25).

4.7.7 Macro Factors and Disagreement

Li (2016) extends the logic from individual stocks to the pricing of macroeconomic factors. If investors disagree about the prospects of consumption growth or inflation, the resulting overpricing should be more pronounced for factor portfolios whose payoffs load more heavily on the most-disputed macro

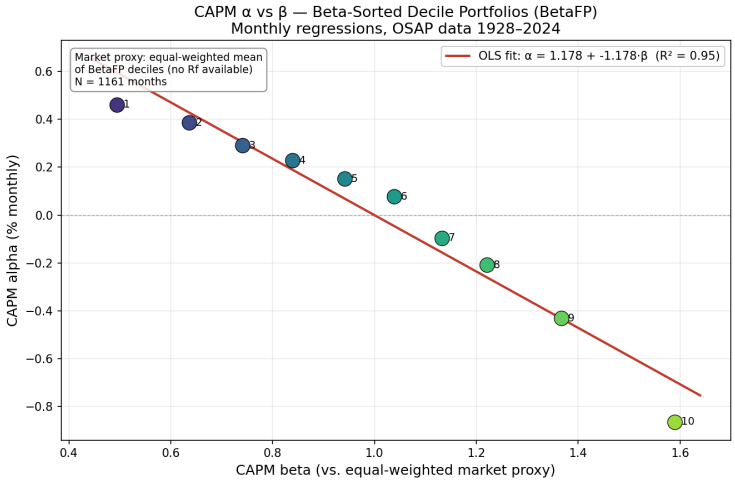
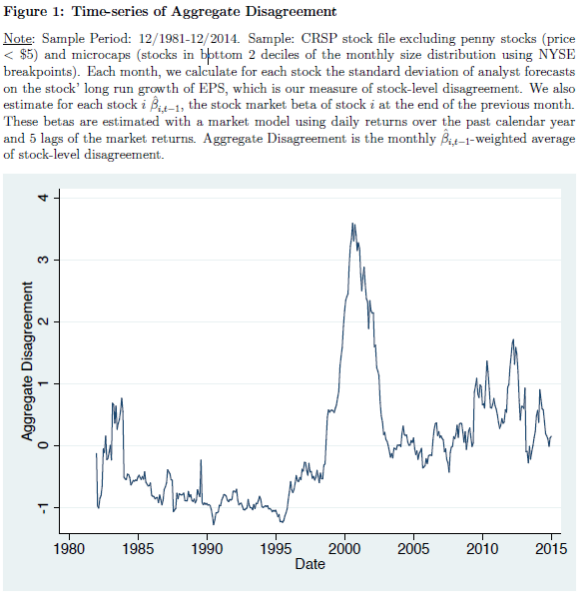


Figure 4.23: Alpha versus beta. Average monthly CAPM alphas against CAPM betas for the ten beta-sorted (BetaFP) decile portfolios, Open Source Asset Pricing data, 1928–2024 ($N = 1{,}161$ months; market proxy: equal-weighted mean of the deciles, no risk-free rate available). The fitted line $\alpha = 1.178 - 0.178\,\beta$ ($R^2 = 0.95$) slopes down: high-beta portfolios earn lower alphas, the high-beta–low-alpha pattern that standard theory cannot produce. *Source:* Author's calculations based on Open Source Asset Pricing decile portfolio data (Chen and Zimmermann, 2022).


Figure 1: Time-series of Aggregate Disagreement

Note: Sample Period: 12/1981-12/2014. Sample: CRSP stock file excluding penny stocks (price < \$5) and microcaps (stocks in bottom 2 deciles of the monthly size distribution using NYSE breakpoints). Each month, we calculate for each stock the standard deviation of analyst forecasts on the stock' long run growth of EPS, which is our measure of stock-level disagreement. We also estimate for each stock i $\hat{\beta}_{i,t-1}$, the stock market beta of stock i at the end of the previous month. These betas are estimated with a market model using daily returns over the past calendar year and 5 lags of the market returns. Aggregate Disagreement is the monthly $\hat{\beta}_{i,t-1}$-weighted average of stock-level disagreement.

Figure 4.24: Aggregate disagreement. The time series of aggregate disagreement from Hong and Sraer (2016): each month, the lagged-beta-weighted average across stocks of the standard deviation of analyst forecasts of the stock's long-run EPS growth, December 1981–February 2014, CRSP stocks excluding penny stocks and microcaps. *Source:* Hong and Sraer (2016), reproduced from the original article.

Figure: Average 3-months excess returns for equal-weighted beta decile portfolios in low and high aggregate disagreement months.

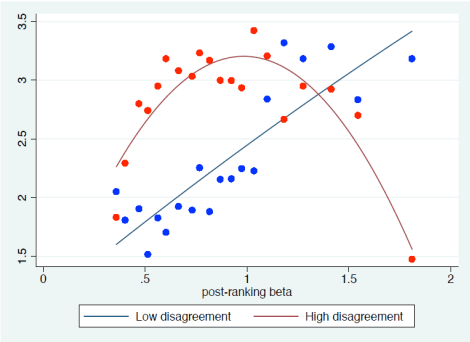


Figure: Equal-weighted average of dispersion of analyst earnings forecasts by beta deciles during low and high aggregate disagreement months.

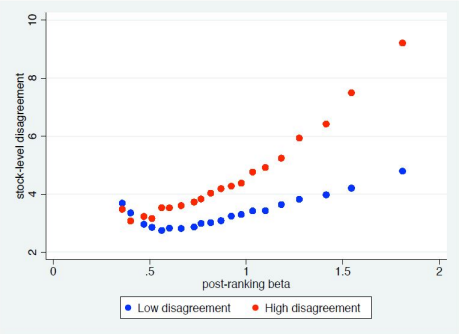


Figure 4.25: Speculative betas in the data. Left: average three-month excess returns of equal-weighted beta-decile portfolios during low and high aggregate-disagreement months. Right: the equal-weighted average dispersion of analyst earnings forecasts by beta decile during low and high aggregate-disagreement months. The panels are the empirical counterpart of the model's predictions that stock-level disagreement rises with beta and that the shape of the security market line depends on aggregate disagreement. *Source:* Hong and Sraer (2016).

states. The data are consistent with the prediction: factor returns are lower when disagreement about the underlying macro state is higher, and the effect is stronger for factor portfolios more exposed to that state.

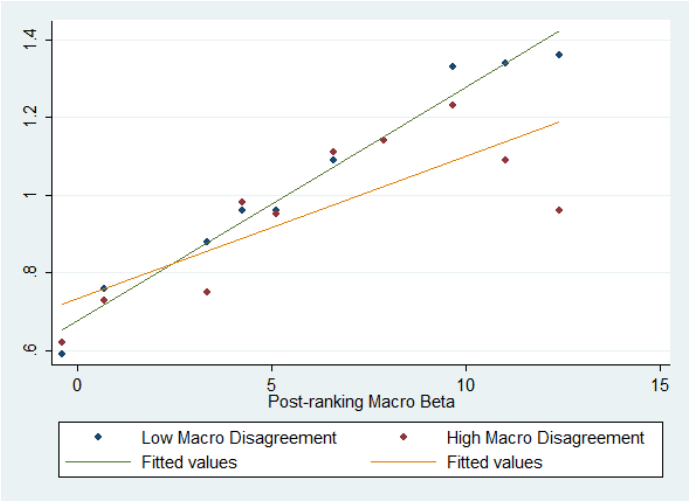


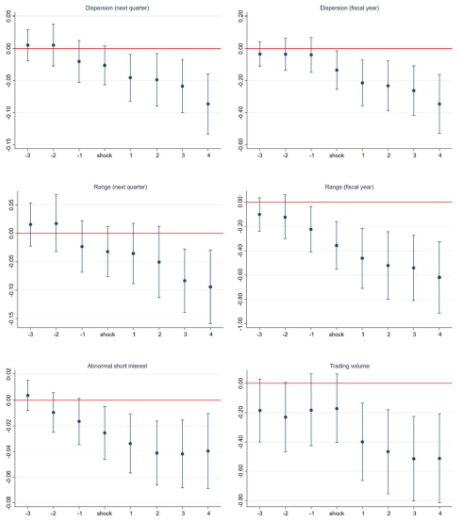
Figure 4.26: Macro factors and disagreement. The figure shows that the returns on factor portfolios more exposed to highly disputed macro states are lower in periods of high disagreement about those states. *Source:* Li (2016), reproduced from the original article.

4.8 Causal Evidence: The EDGAR Identification

The cross-sectional and time-series correlations of Section 4.7 are consistent with the heterogeneous-beliefs model, but the same correlations admit other readings. A stock with high analyst dispersion is also a stock with high uncertainty, high trading volume, and high media attention; the overpricing

interpretation must compete with risk-based and information-based alternatives. Distinguishing the channels requires exogenous variation in disagreement.

Chang et al. (2022) provide such variation by exploiting the staggered rollout of the SEC's EDGAR electronic disclosure system. Before EDGAR, investors could access firms' mandatory filings only at high cost, either by subscribing to commercial data providers or by physically visiting the SEC's reference rooms. Beginning in April 1993, the SEC required U.S. firms to file their mandatory disclosures electronically through the EDGAR system, and it randomly assigned firms to one of ten implementation waves, staggering inclusion over the three-year period between 1993 and 1996. The rollout therefore moves information-acquisition costs exogenously across firms and time. The authors argue that lower information costs reduce the dispersion of beliefs: as more investors access the same public information, disagreement about what the firm is worth shrinks. EDGAR inclusion helps resolve disagreement around information events, leading to price corrections and lower crash risk.



	Skewness (*NCSKEW*) (1)	Down-to-up volatility (*DUVOL*) (2)	Extreme negative returns, 0.01% (*CRASH001*) (3)
Quarter of EDGAR inclusion	0.007	0.000	-0.003
	0.022	*0.014*	*0.008*
Next four quarters	-0.049**	-0.031**	-0.027***
	0.020	*0.012*	*0.007*
Controls?	yes	yes	yes
Calendar quarter FE?	yes	yes	yes
Fiscal quarter FE?	yes	yes	yes
Firm FE?	yes	yes	yes
R-squared	17.4%	19.1%	15.8%
Pre-trends (*p*-value)	0.562	0.141	0.957
No. of firms	3,366	3,366	3,366
No. of firm-quarters	28,652	28,652	28,652

Figure 4.27: Causal evidence on disagreement. Evidence from the EDGAR quasi-experiment of Chang et al. (2022): as staggered EDGAR inclusion lowers information-acquisition costs, disagreement resolves around information events, prices correct, and crash risk falls (right: crash-risk regressions using NCSKEW). *Source:* Chang et al. (2022), reproduced from the original article.

The central finding is that the rollout of EDGAR, by lowering information costs and disagreement, also lowered stock prices—the sign predicted by the overpricing interpretation. The magnitude is consistent with a quantitatively meaningful share of the variation in prices being driven by disagreement, and the design rules out the alternative interpretations that rely on time-invariant firm characteristics. The result is one of the cleanest causal pieces of evidence in the heterogeneous-beliefs literature, and it sharpens the case for treating disagreement as a priced source of risk in its own right rather than as a proxy for some other dimension of firm fundamentals.

4.9 Do People Ever Learn?

The biases of Sections 4.1–4.4 are persistent; they are part of the cognitive and emotional machinery that evolution built into human decision-making. Many of the most persistent behavioral biases are mental shortcuts with survival value: loss aversion increases survival probability, and overconfidence

increases effort and trust within one's community. The kinds of financial decisions in which the biases matter most are recent on evolutionary timescales. Do people simply learn their way out of them? Only to a limited extent.

Do highly trained professionals exhibit the biases? Yes, and sometimes professionals do worse. CEOs and founders are more overconfident than the average person—selection into those roles rewards confidence, and success at the top of the distribution reinforces it. Expertise does not immunize judgment; it supplies the wrong mental models with more resources.

The most direct evidence compares investor behavior under different stakes. Sui and Wang (2025) use a unique setting in which investors trade stocks in real accounts with their own money and simultaneously in a simulated setting where the personal stakes are lower. Within-investor estimation shows that the same investors exhibit stronger biases—disposition effect, lottery preference, extrapolation, underdiversification, local bias—and perform worse in the higher-stakes real accounts than in the simulated accounts. Investors do better when they care less. The explanation is uncomfortable for the debiasing view: individuals carry incorrect mental models, and greater attention and effort in the high-stakes setting lead to the more careful *application* of those incorrect models. Trying harder makes the biases worse, not better.

Table 6
Differences between the real and simulated accounts. In this table, we report differences between real and simulated accounts. See Section 4.1 for the bias definitions. For each type of bias we report the mean of the real accounts, the mean of the simulated accounts, and the mean of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column "N" we report the number of investors.

		Real	Simulated	Difference		
	N	Mean	Mean	Mean	*t*-stat	Wilcoxon *p*
Disposition effect	4371	4.163	0.489	3.674	(32.43)	0.000
Lottery preference	4369	6.491	4.264	2.228	(13.88)	0.000
Extrapolation	4373	15.599	12.66	2.939	(6.49)	0.006
Underdiversification	4394	−1.006	−1.473	0.386	(32.67)	0.000
Local Bias	1739	0.630	0.645	−0.015	(−0.64)	0.810
Turnover	4386	5.960	0.839	5.121	(47.44)	0.000
Performance	4375	−0.656	−0.148	−0.509	(−4.95)	0.001

Figure 4.28: Investors do better when they care less. For each bias—disposition effect, lottery preference, extrapolation, underdiversification, local bias—the table reports the mean across investors' real accounts, the mean in their simulated accounts, and the within-investor difference with t-statistics and Wilcoxon p-values (N: number of investors). Biases are systematically stronger in the real, higher-stakes accounts. *Source:* Sui and Wang (2025), reproduced from the original article.

The finding has two implications for the behavioral asset pricing program. First, the biases are not diluted by stakes; they are magnified. The emotional salience of real money leads investors to apply their (incorrect) mental models more carefully, so raising the stakes deepens rather than corrects the distortion. Second, it raises a methodological point that recurs in the chapter: behavioral biases are not fixed properties of the investor. They are properties of the investor *in context*, and the context matters for how strongly they affect trading and prices.

4.10 What Remains Contested

Two strands of evidence keep the behavioral reading honest.

First, the cognitive biases catalogued in Sections 4.1–4.4 are individually well-documented in laboratory data, but their *quantitative* contribution to asset prices is harder to pin down. The disposition effect of Chapter 3 and the trading-cost evidence of Section 4.2 are reasonably direct: investors' biases translate into overtrading and into measurable return patterns. The over-extrapolation evidence is stronger at the survey level than at the trading level: the Greenwood–Shleifer correlations are tight, but the contribution of extrapolative beliefs to the cross-section of stock returns is harder to estimate cleanly. The bias-by-bias literature must be aggregated across biases that may or may not interact, and the empirical case is uneven.

Second, the heterogeneous-beliefs models with short-sale constraints are competitive with rational explanations in several applications. The cross-sectional disagreement result of Diether, Malloy, and Scherbina (2002) admits a risk-based interpretation: high-dispersion stocks are also stocks with high fundamental uncertainty, and the low returns may be compensation for that uncertainty rather than the result of overpricing. The warrant-bubble evidence of Xiong and Yu (2011) is sharper, but it remains a single market; the macro-factor extension of Li (2016) is suggestive rather than conclusive. The causal EDGAR evidence of Chang et al. (2022) is the strongest single piece of evidence, but it speaks to one channel (information costs and disagreement) rather than to the broader class of bias-driven models. The machine-forecast-disagreement evidence of Bali, Kelly, et al. (2026) extends the overpricing result to a simulated measure available for the entire cross-section, but its mechanism tests remain conditional correlations rather than experiments.

The methodological lesson is the same as in the earlier chapters: a behavioral model earns credibility when it predicts something not used to fit it, and when that prediction survives out-of-sample tests. The strongest out-of-sample predictions of the biased-beliefs program are the disposition effect and the trading-cost evidence of overconfidence, the Greenwood–Shleifer correlations of survey expectations and returns, the warrant-bubble evidence on resale-option values, and the EDGAR causal evidence on disagreement. The rest of the literature is competitive with rational alternatives.

4.11 The Road Ahead

This chapter has developed the second of the two departures from the rational benchmark—biased beliefs—together with the cross-sectional dispersion in beliefs that the heterogeneous-beliefs model exploits. Combined with the prospect-theory departures of Chapter 3, the book now has the two psychological ingredients needed for most of behavioral finance: a non-standard theory of preferences under risk, and a non-standard theory of belief formation. The combination is what most applications in the chapters that follow deploy.

The remaining chapters assemble these ingredients into specific empirical and theoretical applications. Chapter 5 develops *bounded rationality*: limited attention, category-based thinking, and psychology-free models of noise trading and sentiment. A further chapter develops *memory-based* approaches—the way investors form beliefs from the cases they remember rather than from a Bayesian aggregation of evidence. These models retain rational beliefs in a thin sense but relax the assumption that all available information is processed.

Together with Chapter 1 on the empirical facts, Chapter 2 on limits to arbitrage, Chapter 3 on preferences, and the present chapter on biased beliefs, the book develops the building blocks of behavioral asset pricing. The remaining chapters apply them to specific settings—the role of attention, the formation of investor sentiment, and the pricing of memory-driven beliefs—in each case tracing the path from a documented cognitive bias to a measurable asset-pricing implication.

Chapter 5

Bounded Rationality and Psychology-free Models

The behavioral theories developed in Chapters 3 and 4 modify either investors' preferences or their beliefs. They keep intact the assumption that investors process whatever information they do attend to, completely and correctly. This chapter relaxes that assumption in two different ways.

The first half (Sections 5.1–5.4) develops *bounded rationality*: investors have limited capacity for gathering and processing information, and they end up with a stylized version of the data—one in which salient news is incorporated promptly and subtle news is incorporated slowly or not at all. The systematic consequence is that prices underreact to news, with the degree of underreaction depending on whether the news is prominent or quiet. The behavioral signature is not a wrong belief about cash flows but a slow belief about what has just been released.

The second half (Sections 5.5–5.11) develops *psychology-free models*: behavioral predictions that require no psychological assumption at all, beyond the central observation that some investors trade on noisy signals while others have limited capital and short horizons. The standard architecture is De Long et al. (1990): less than fully rational *noise traders* interact with more rational *arbitrageurs* whose corrections of mispricing are themselves imperfect and informative. The framework makes contact with a wide range of facts—the trades of sophisticated agents forecast returns in the direction opposite to the trades of unsophisticated agents, and aggregate *investor sentiment* moves with the cross-sectional and time-series properties of returns—without requiring any assumption about why noise traders are wrong. Two extensions close the empirical arc. Language models trained on the same news human forecasters read can be prompted for their beliefs, which produces a sentiment measure that runs for a century at daily frequency and identifies bubbles that survey-based indices could not reach (Bybee, 2025). And the credit market, where sentiment is a measure of perceived default risk rather than of expected growth, carries the larger macroeconomic signal (Krishnamurthy and Muir, 2025). Section 5.12 catalogues what remains contested, and Section 5.13 closes the chapter.

5.1 Limited Attention: The General Pattern

Definition 5.1 (Limited attention)**.** *An investor is said to have limited attention when the cognitive resources available to acquire and process information are scarce relative to the volume of information available. As a consequence, some news items are processed promptly while others are processed slowly, partially, or not at all.*

The asset pricing prediction of limited attention is sharp: prices should react more weakly to

news that is harder to attend to, and more strongly to news that is salient. The underreaction is not a property of investors' beliefs about fundamentals but of their beliefs about what has been released. Limited attention is therefore the natural complement to the Bayesian benchmark: with rational beliefs but bounded processing, prices converge to rational posteriors, only slowly.

The strongest empirical evidence for limited attention in finance comes from the post-earnings-announcement drift (PEAD) literature. Stocks that announce positive earnings surprises drift up after the announcement; stocks that announce negative earnings surprises drift down. The drift begins immediately after the announcement but persists for many months, a pattern that is hard to reconcile with a market that fully and promptly incorporates the information content of the surprise. Two influential studies sharpen the inference by showing that the drift is larger exactly when attention is plausibly scarcer.

Hirshleifer, Lim, and Teoh (2009) show that PEAD is stronger for firms that announce earnings at the same time as many other firms. When several firms report on the same day, investors' attention is divided across many announcements, and the immediate market reaction to any single announcement is muted; the drift that follows is correspondingly larger. DellaVigna and Pollet (2009) show that PEAD is stronger for firms that announce on Fridays. Friday announcements attract less attention because Friday afternoons compete with the weekend, and the immediate price reaction is weaker; the post-announcement drift is correspondingly larger.

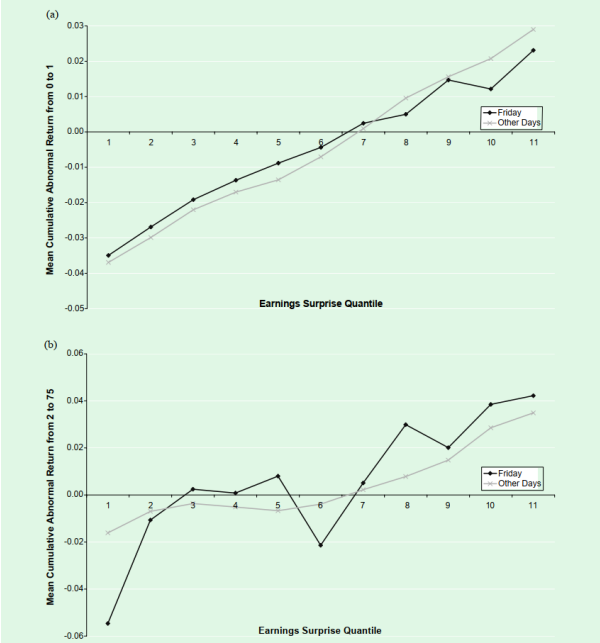


Figure 1a and b. Immediate response to earnings surprises and delayed response to earnings surprises. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year. Announcements made on Saturday or Sunday are excluded from the sample.

Figure 5.1: Friday announcements drift more. The figure illustrates the Della Vigna–Pollet evidence: earnings announcements made on Fridays exhibit a weaker immediate market reaction but a stronger post-earnings announcement drift than announcements on other weekdays, because Friday announcements attract less investor attention. *Source:* EF8083 course materials, after DellaVigna and Pollet (2009).

The two findings help discipline the interpretation. An alternative explanation for PEAD is that earnings surprises contain information that the market rationally discounts as noisy, and that the drift is the slow resolution of that uncertainty rather than the slow incorporation of clear news. The Hirshleifer–Lim–Teoh and Della Vigna–Pollet designs vary attention while holding the information content of the surprise roughly constant, which weakens the noisy-information reading: if attention were irrelevant, varying it should not change the size of the drift. The evidence is consistent with limited attention as the binding constraint.

5.1.1 The "Frog in the Pan" Hypothesis

A further pattern points to a structural feature of attention: investors notice discrete, salient events better than they notice continuous streams of similar events. Da, Gurun, and Warachka (2014) propose the *frog in the pan* hypothesis: investors are more likely to be inattentive to information that arrives in a stream of small increments than to information that arrives in a single large shock. The reasoning is that a series of small changes is easier to miss than a single large change of the same total magnitude—the same effect that makes it possible to drop a frog into water that is slowly heated, while dropping it into already-boiling water produces an immediate reaction. Momentum, on this account, is the empirical signature of inattention to continuous news: investors fail to update on each small step, the price moves slowly relative to the news, and the next step arrives before the previous one is fully incorporated. The hypothesis predicts that momentum should be stronger when information arrives continuously rather than in discrete announcements, and the data support the prediction: momentum profits are larger in periods of small incremental earnings surprises than in periods of large discrete surprises.

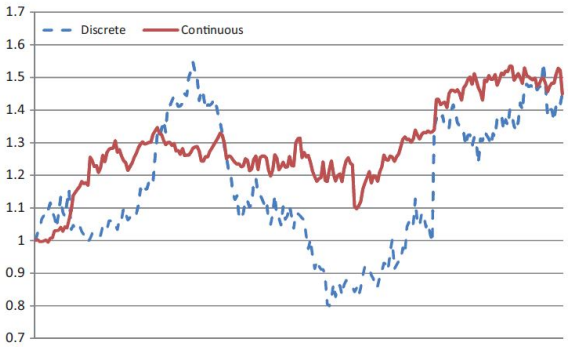


Figure 5.2: The frog in the pan. Information arriving as a series of frequent small changes attracts less attention than the same information arriving as an infrequent large change. Investors are inattentive to the continuous stream, so the price drifts with the news instead of jumping, and momentum emerges. *Source:* EF8083 course materials, after Da, Gurun, and Warachka (2014).

5.1.2 Underreaction to Subtle News

Underreaction is most likely when the news is less salient or more subtle. Three cases illustrate the principle.

Demographic information. DellaVigna and Pollet (2007) document that investors are slow to recognize the impact of demographic shifts on the future profitability of firms with age-sensitive products. A baby boom today predicts higher demand for toys in six years; in an efficient market

the prices of toy-making firms should rise immediately, but they rise only gradually. The information is publicly available and straightforward to compute; the failure to act on it is the predictable signature of attention scarcity rather than of any inference failure.

Technological information. Hirshleifer, Hsu, and Li (2013) use citation patterns in patent data to measure firms' innovative efficiency, and find that investors underreact to the implied information about future productivity. Cohen, Diether, and Malloy (2013) use the release dates of the innovation scores that *BusinessWeek* published in its (now-discontinued) annual ranking of the world's most innovative companies to identify when the information became public, and show that the market is slow to incorporate it—an unusually clean case of bounded attention tied to a known publication event.

The absence of news. A more surprising version of the pattern is that the *absence* of news is itself information. Giglio and Shue (2014) observe that the passage of time after an announced merger is informative about the probability that the deal will close: as time passes without disruption, the conditional probability of completion rises. Investors should therefore bid up target-firm prices gradually after an announcement; in practice, the price moves sharply on the announcement and drifts up only weakly thereafter.

5.1.3 My Research: The Information Content of Sudden Insider Silence

Hong and Li (2019) study the same logic—underreaction to the absence of an expected event—in the setting of corporate insider trading. Some insiders trade on a regular calendar: they buy or sell in the same months year after year, for liquidity and diversification reasons. The paper identifies these *routine* insiders from their calendar trading patterns and treats a sudden stop in their trading as a deviation from routine—an event that carries information precisely because nothing happened. The premise is that routine insiders strategically stay silent when they hold private information: an insider who would ordinarily sell, but knows bad news is coming, has a reason to stop selling.

The empirical patterns bear out the interpretation. Silence after routine selling predicts positive future returns and fundamentals, and silence after routine buying predicts negative ones; a long–short strategy that trades on sudden silence earns 6–10% annually. The returns concentrate at earnings announcements—when the private information the silence concealed is revealed—and are stronger where information environments are poor and arbitrage is costly, exactly the conditions under which underreaction should survive. The result extends the subtle-news evidence of this section: what investors fail to process is not a buried disclosure but the nonoccurrence of an expected trade, which is hard to recognize as informative without deliberate attention to the insider's own history.

5.1.4 Momentum Spillover: Economic Linkages

A growing literature documents that investors underreact to news of economically linked peers. The pattern is often called the *momentum spillover effect*: positive news for a customer spills over to the supplier and vice versa, but the spillover is incomplete and gradual.

The structure of the argument generalizes beyond any single linkage. For an underreaction to appear, three conditions must hold, and each identifies a separate research design. First, the two firms must be linked economically, so that news about one is news about the other—this is what makes the spillover a mispricing rather than two unrelated price adjustments. Second, the linkage must be costly to compute, since a linkage that investors can assemble at no effort is one they will have assembled. Third, the linkage must be hard for arbitrageurs to exploit, since mispricing

that can be arbitraged away is not mispricing. The first condition is established by the economic argument for the link; the second by the difficulty of assembling the data (customer identities come from 10-K filings, peer identities from annual-report text, establishment locations from regulatory filings at the plant level); the third by the characteristics of the firms that are hard to arbitrage and costly to short.

Customer–supplier links. Cohen and Frazzini (2008) identify firms' major customers using the regulatory requirement that firms report any customer accounting for more than 10% of sales. They then show that when a customer announces good news, the supplier's price moves only weakly on the announcement and continues to drift upward for months afterwards. The information is publicly available and easy to compute—the linkage is disclosed in 10-K filings—but the market underreacts. The customer-supplier case is the cleanest of the linkages because the economic link is contractual and the data are disclosed by the customer itself, which also means some sophisticated investors do recognize the link; the underreaction is a matter of degree, with the supplier's return catching up to the customer's over roughly a six-month window rather than immediately.

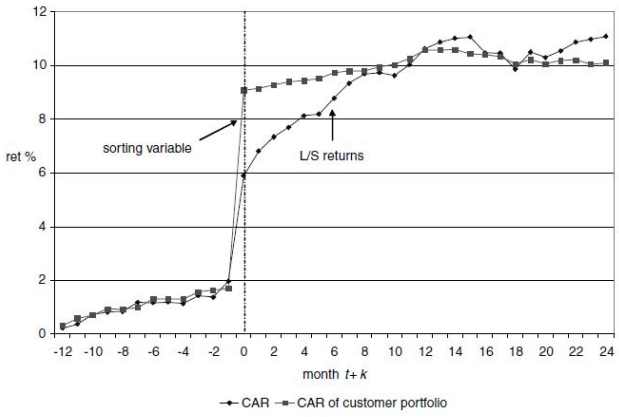


Figure 2. Customer momentum, event-time CAR. This figure shows the average cumulative return in month $t+k$ on a long–short portfolio formed on the firm's customer return in month t. At the beginning of every calendar month, stocks are ranked in ascending order based on the return of a portfolio of its major customers at the end of the previous month. Stocks are assigned to one of five quintile portfolios. The figure shows average cumulative returns (in %) over time of a zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer returns stocks.

Figure 5.3: Spillover across customer–supplier links. The figure illustrates the central finding of Cohen and Frazzini (2008): when a major customer announces positive earnings news, the supplier's price moves only weakly on impact and continues to drift upward for months. *Source:* EF8083 course materials, after Cohen and Frazzini (2008).

Other economic linkages. The same pattern holds for other types of economic linkages. Moskowitz and Grinblatt (1999) document industry momentum, in which firms in the same industry comove beyond what their cash flows can justify. Cohen and Lou (2012) show that conglomerate firms are slower than standalone firms to reflect industry news, consistent with the interpretation that complex structures draw less attention. Huang (2015) document that multinational firms underreact to foreign industry news; Hoberg and Phillips (2016) use product market peers identified through textual analysis of annual reports; Parsons, Sabbatucci, and Titman (2020) identify geographic peers as firms headquartered in the same state; Jin and Li (2024) use firms with overlapping geographic establishments. Lee, Sun, et al. (2019) identify technological peers through shared technology

classifications, and Ali and Hirshleifer (2020) show that firms followed by the same analysts comove more than fundamentals predict. The list is long and the pattern is uniform: when two firms are linked through any economically meaningful channel, the price of one moves only gradually in response to news about the other.

5.1.5 My Research: Geographic and Scope Links

Two of my own papers push the momentum-spillover literature one step further by asking whether investors underreact even when the linkage itself is the subtle information. Jin and Li (2024) link firms through overlapping establishment locations—the places where sales are actually generated—rather than through headquarters. Because firms report establishment-level activity only in dispersed regulatory filings, computing a firm's geographic footprint requires assembling information across sources, which is exactly the kind of processing to which investors are inattentive. Returns and fundamentals of geography-linked peers predict focal-firm returns, and the predictability is stronger for low-attention firms, for costly-to-arbitrage stocks, and during high-sentiment periods—the same conditioning variables that amplify the underreaction patterns documented throughout this section.

A companion working paper with Li Jin asks the same question about *scope similarity*: firms whose product-market scope overlaps, in the sense of Hoberg and Phillips (2016), but that sit in different industries and therefore never appear in the same industry list. Investors are slow to incorporate news about scope peers, and a long–short portfolio that buys firms whose scope peers recently outperformed earns 8.9% per year in five-factor alpha. The common lesson of both papers is that the named linkages of the spillover literature—customers, industries, states—are only the easiest cases; the underlying phenomenon is that any economically meaningful connection whose computation is costly will be processed slowly.

5.1.6 Cross-Asset Predictability

Limited attention also predicts that prices will underreact to information that arrives in markets different from the one being studied. The clearest case is predictability between the debt and equity of the same firm.

Han, Subrahmanyam, and Zhou (2017) document that the term structure of firms' credit spreads forecasts future firm fundamentals and stock returns: a flatter credit-spread curve predicts higher future earnings and higher subsequent stock returns. Addoum and Murfin (2020) document that the prices of syndicated loans carry information for equity prices; investors in the equity market appear to fail to unravel the information embedded in loan prices, and predictability flows from the loan market to the equity market.

The source of underreaction across asset classes is plausibly *market segmentation*. Both debt and equity are claims to the same underlying cash flows, but the investor populations are different: many investors trade in equity markets but not in the leveraged loan market, and vice versa. The information that arrives in one market is not promptly incorporated into the other because the two markets are dominated by different sets of participants. Han, Subrahmanyam, and Zhou (2017) provide supporting evidence: the predictability from credit spreads to stock returns is weaker for firms whose equity is held by mutual funds that also trade debt securities, where the overlap in investor populations reduces the segmentation.

Other asset classes. Bae, Da, and Zurita (2024) show that U.S. equity investors of multinational firms underreact to currency shocks in the countries where the firms operate. Ho and Lauwers (2023) document that aggregate positions of traders in the commodity futures market can predict the

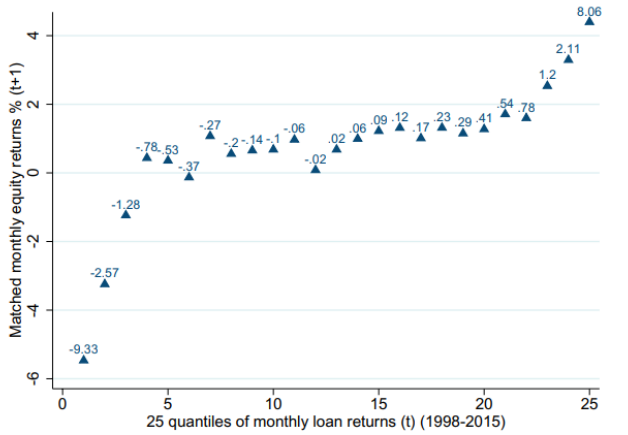


Figure 1
Equity returns versus lagged loan returns
The above plot combines both time-series and cross-sectional relationships between loan returns and linked equity returns in the next month. Loan returns are sorted into twenty-five quantiles over the entire sample period and plotted against the next month's equity returns for the same borrowers. Average loan returns for each quantile are reported next to the triangles.

Figure 5.4: Loan prices forecast equity returns. The figure illustrates the central finding of Addoum and Murfin (2020): prices of syndicated loans contain useful information about the borrowing firms, but equity investors fail to incorporate the information, generating predictability from loan markets to equity markets. *Source:* Addoum and Murfin (2020), reproduced from the original article.

cross-section of stock returns of commodity producers. The unifying theme is that any information that arrives in a market other than the one being studied is incorporated slowly: the investor populations are different, the data feeds are different, and the cognitive bandwidth to process them all is finite.

5.2 Buying versus Selling: Asymmetric Attention

A second set of facts concerns the differential role of attention in buying versus selling decisions. The argument is intuitive: an investor who is not already holding a stock must decide whether to buy, and buying decisions are driven by salience and attention; an investor who already holds a stock can sell at any time without having to search for the stock, and selling decisions are driven by liquidity needs and private considerations rather than by salience. Barber and Odean (2008) test this prediction directly. The data come from a large discount brokerage account: individuals are heavy net buyers of attention-grabbing stocks (those with extreme returns, high volume, or recent news), but their selling decisions show essentially no relation to these same attention measures.

The implication is that attention matters asymmetrically for prices. The strongest price effects should be on the buy side—the side where investors must actively choose to enter—and the implication extends to asset pricing more broadly: any mechanism that operates through increased demand will be amplified by attention, while any mechanism that operates through increased selling pressure will be relatively muted.

5.3 Slow Information Diffusion: The Hong–Stein Mechanism

A structural model of limited attention is Hong and Stein (1999). The model combines two types of investors who each process only part of the available information:

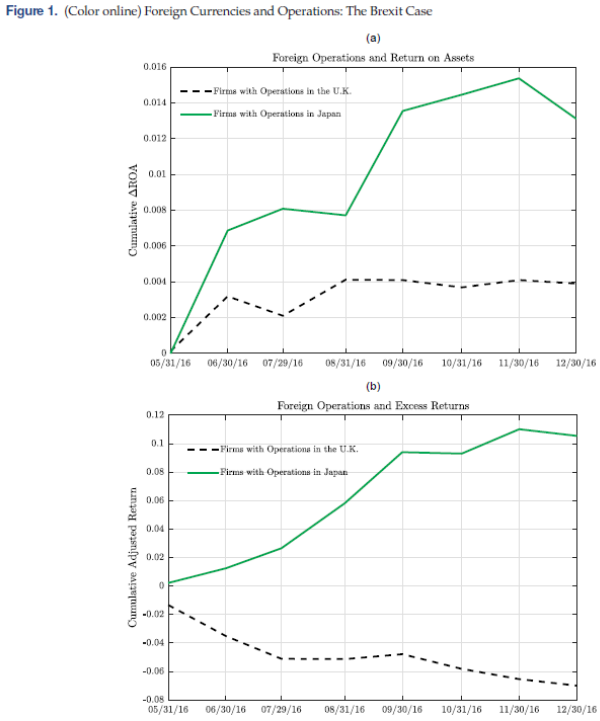


Figure 5.5: Commodity futures forecast commodity-producer returns. The figure illustrates the central finding of Ho and Lauwers (2023): aggregate positions of commodity futures traders predict the cross-section of stock returns for commodity-producing firms. *Source:* Ho and Lauwers (2023), reproduced from the original article.

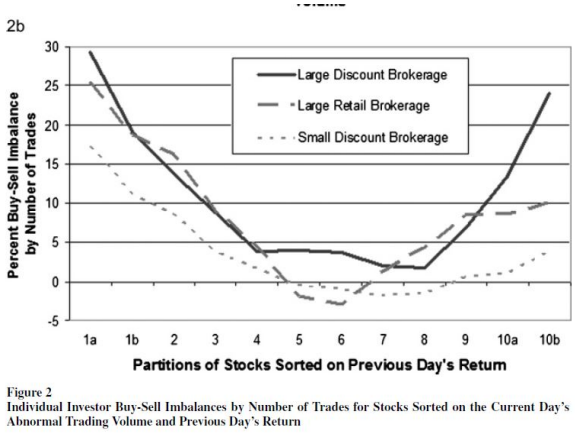


Figure 5.6: Attention and individual buying vs. selling. Stocks with extreme returns, high volume, or news announcement draw strong net buying interest from individual investors but no comparable selling interest. The asymmetry reflects the role of attention in search for new positions. *Source:* EF8083 course materials, after Barber and Odean (2008).

- *Newswatchers*, who process only private firm-specific signals.

- *Momentum traders*, who process only past price changes and whose demand is an increasing function of past returns.

Newswatchers update gradually because information diffuses slowly through the newswatcher population: at any date only a fraction of newswatchers have observed the signal, so the price response is partial, and the next newswatcher to observe it pushes the price further in the same direction. In an economy with only newswatchers, the dynamics would produce pure momentum: prices drift in the direction of the news until all newswatchers have updated. The momentum traders add a second effect: they buy as prices rise, and because they continue to buy even after the private information is fully incorporated, the price eventually overshoots. The combination produces both short-horizon momentum (the newswatchers' gradual incorporation) and long-horizon reversal (the momentum traders' overshoot).

The two investor types generate these effects for different reasons, and the distinction is what makes the model's predictions testable. The newswatchers process only private firm-specific signals and cannot read prices; their demand is a function of what they know, which is what makes the diffusion of that knowledge into prices gradual. The momentum traders process only past price changes and cannot distinguish fresh news from the price momentum itself; their demand is a purely univariate function of the past return, which is what allows them to keep buying into a price that has already moved past its fundamental value. The externality that generates reversal is therefore not a mistake in either group's information. It is a consequence of two groups holding different information and of one group's trading rule being blind to the distinction between news and trading.

Because the momentum traders trade on a signal that is a linear function of the past return, their aggressiveness is determined in equilibrium rather than imposed from outside. Each generation of momentum traders holds its position for j periods and trades on the univariate signal $F_t = A + \phi\,\Delta P_{t-1}$, and the aggressiveness ϕ solves a fixed-point condition: it equals the risk-tolerance-scaled profitability of trend-chasing given the price dynamics that trend-chasing itself produces,

$$\phi \;=\; \frac{\gamma\,\mathrm{cov}(P_{t+j} - P_t,\; \Delta P_{t-1})}{\mathrm{var}(\Delta P)\;\mathrm{var}_M(P_{t+j} - P_t)},$$

where γ is aggregate risk tolerance and var_M is the variance the momentum traders perceive. The numerator is positive whenever the newswatchers' slow diffusion leaves predictable drift to capture, so $\phi > 0$ in equilibrium: rational momentum traders must trend-chase, and their buying then feeds the price further in the same direction.

The resulting impulse response is hump-shaped (Figure 5.7). With newswatchers alone, the price moves smoothly toward its long-run value and stays there. With momentum traders, the price rises above the long-run value at an intermediate horizon and then converges back through damped oscillations, and the positive short-run autocorrelation of returns implies eventual negative autocorrelation under risk neutrality.

The comparative statics follow from the model's structure. A longer holding horizon of the momentum traders delays the first negative autocorrelation and ties the timing of the peak to that horizon. Greater aggregate risk tolerance raises the aggressiveness of momentum trading, which raises volatility and the size of the overshoot, and makes the pricing error U-shaped in the aggressiveness parameter. Slower diffusion of news creates larger momentum opportunities, stronger continuation, and more pronounced reversals. And a contrarian or smart-money arbitrageur attenuates the dynamics without eliminating them unless arbitrage capacity is unlimited.

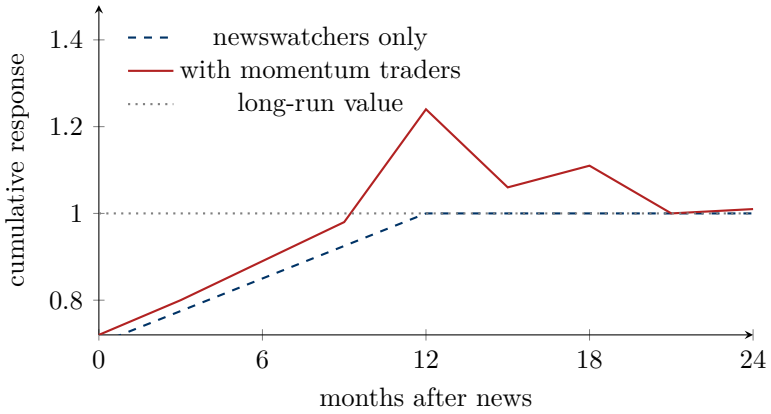


Figure 5.7: Underreaction, overshoot, and reversal. Cumulative price response to a news shock in Hong and Stein (1999). With newswatchers alone (dashed), the price adjusts gradually to the new fundamental value (dotted). Adding momentum traders (solid) accelerates the initial adjustment but overshoots the long-run value at an intermediate horizon before converging back through damped oscillations. *Source:* schematic, after Hong and Stein (1999).

The cross-sectional implications are the sharpest, because they identify where the mechanism must be strongest if it is the one at work. Momentum should be strongest among small, low-analyst-coverage stocks, where the newswatcher population is thinnest and information diffuses most slowly, and the same stocks should be the most reversal-prone at longer horizons. Public news may drift without producing the same overreaction, since traders who can condition on announcement dates can distinguish news from momentum trades. And higher trading costs, or longer horizons, keep return autocorrelations positive for longer.

Hong, Lim, and Stein (2000) test the central prediction. If momentum arises from slow diffusion of private information, the effect should be stronger where newswatcher coverage is thinner. The data confirm the prediction: momentum is stronger among small stocks and stocks with low analyst coverage, exactly the stocks where the newswatcher population is plausibly smaller and information diffuses more slowly.

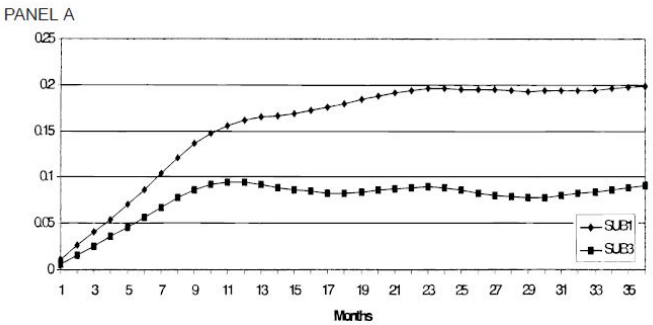


Figure 5.8: Momentum is stronger where news is thinner. The figure illustrates the central evidence of Hong, Lim, and Stein (2000): momentum profits are larger for small stocks and stocks with low analyst coverage, consistent with the slow-diffusion hypothesis of Hong and Stein (1999). *Source:* EF8083 course materials, after Hong, Lim, and Stein (2000).

5.4 The Causal Effect of Attention on Asset Prices

The evidence reviewed so far establishes that attention is correlated with underreaction, but correlation is not causation. Investor attention is endogenous to stock fundamentals—the same firms that attract attention may also be the ones with the strongest news—and identifying a causal effect requires an exogenous source of variation in attention.

Chen, An, et al. (2023) provide such variation by exploiting a peculiarity of the Chinese stock market. On Chinese trading screens, the order in which stocks are displayed is determined by the stock's listing code, which is essentially randomly assigned at the time of listing. Investors are more likely to pay attention to neighboring stocks if they have positive investment experience with the focal stock—a behavioral spillover identified in earlier work—and the authors use the random listing code as an instrument for the set of stocks that any given investor observes.

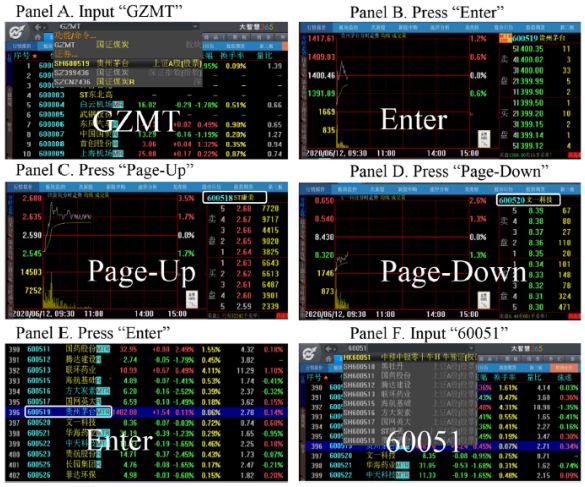


Figure 5.9: Attention spillover in China. The figure illustrates the identification strategy of Chen, An, et al. (2023): listing codes are randomly assigned in China, which randomizes the stocks displayed next to any given focal stock and creates plausibly exogenous variation in the attention paid to neighbours. *Source:* Chen, An, et al. (2023), reproduced from the original article.

Because attention leads to excessive buying demand, stocks displayed next to those with higher returns in the past two weeks earn higher returns in the following week, with the effect reverting over the long run. The pattern is exactly what limited attention predicts: more attention raises the price in the short run, and the price subsequently corrects. The random listing-code variation is the cleanest identification of the causal effect of attention on prices in the literature.

5.5 Category Thinking and Excess Comovement

A basic feature of human cognition is that we put things into categories and form beliefs at the level of categories rather than at the level of individual objects (Rosch and Lloyd, 1978). With thousands of individual securities and mutual funds, investors plausibly simplify their decision-making by putting financial assets into categories. We routinely use labels such as "value stocks," "growth stocks," "small-cap stocks," and "Chinese internet stocks," and the labels structure beliefs, expectations, and flows. The asset pricing question is whether category-based thinking affects prices.

Barberis and Shleifer (2003) develop a model in which some investors put assets into categories and form beliefs about the assets' future returns at the category level. A news shock that affects a

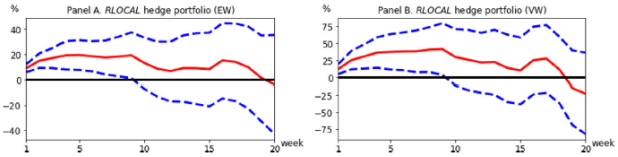


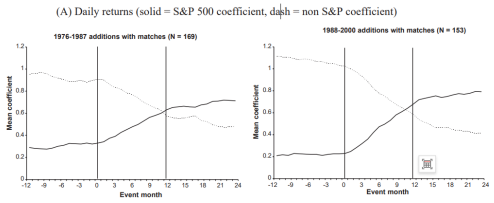
Figure 5. Cumulative return of the long-short portfolio based on *RLOCAL*. This figure shows the annualized cumulative CH4-alpha (in percentage points) to the equal- and value-weighted long-short portfolios (P5-P1) based on *RLOCAL* from week $t+1$ to week $t+20$, as well as the 95% confidence interval. For each stock, *LOCAL* is calculated as the value-weighted average return over the past two weeks of the 10 closest stocks on screen display, and *RLOCAL* is the residual of the cross-sectional regression of *LOCAL* on the focal stock's return over the past two weeks. Portfolios are formed at the end of week t based on *RLOCAL*. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 5.10: More attention, higher price, subsequent reversal. The figure shows that stocks displayed next to high-return neighbours earn higher returns in the following week (consistent with attention-driven buying) and revert over longer horizons (consistent with bounded rationality rather than new information). *Source:* Chen, An, et al. (2023), reproduced from the original article.

category—say, news that "value stocks are good buys"—pushes demand up for all stocks in the category, even though the news is not directly informative about any individual member. The novel prediction is *excess return comovement*: returns of assets in the same category comove more than can be explained by fundamentals, driven by correlated demand shocks that share a category label but no information.

5.5.1 Evidence for Excess Comovement

A sizeable literature supports the prediction. Barberis, Shleifer, and Wurgler (2005) document comovement among stocks in the S&P 500 index: a stock that is added to the index experiences a large increase in its loading on S&P returns—a change in covariance that is hard to explain on fundamental grounds, since the stock's cash flows are unchanged by inclusion. Greenwood (2008) and Boyer (2011) confirm the pattern with different samples and identification strategies. Pirinsky and Wang (2006) document comovement among firms headquartered in the same state; Green and Hwang (2009) among stocks with similar price levels; Israelsen (2016) among firms covered by similar sets of analysts; Anton and Polk (2014) among stocks held by a common set of mutual funds; and Hameed and Xie (2019) among dividend-paying stocks. The list is long enough that category thinking has acquired a status comparable to limited attention: it is one of the few behavioral mechanisms for which the evidence is broad, replicated, and quantitatively meaningful.



$$R_{j,t} = \alpha_j + \beta_{j,\mathrm{SP500}} R_{\mathrm{SP500},t} + \beta_{j,\mathrm{nonSP500}} R_{\mathrm{nonSP500},t} + v_{j,t}$$

Figure 5.11: Index inclusion raises covariance with the index. The figure illustrates the central finding of Barberis, Shleifer, and Wurgler (2005): when a stock is added to the S&P 500, its loading on S&P returns increases sharply—a change in covariance that cannot be explained by fundamentals. *Source:* Barberis, Shleifer, and Wurgler (2005), reproduced from the original article.

5.6 Psychology-free Approaches: The De Long–Shleifer–Summers–Waldmann Framework

The approaches developed so far have all been rooted in a specific psychology: a cognitive bias, an emotional response, or a category. Some behavioral finance models produce testable predictions without appealing to any particular psychology, and these models form a distinct—and in many ways the most theoretically parsimonious—branch of the literature.

The standard architecture is De Long et al. (1990). There are two types of investors. Some are *noise traders*—investors whose demand for an asset is driven by factors other than its fundamental value (sentiment, fashion, liquidity shocks, idiosyncratic beliefs). Others are more rational and are conventionally called *arbitrageurs*. Arbitrageurs observe the noise traders' demand and trade against it, but their ability to do so is constrained by the limits of Chapter 2: noise trader risk, fundamental risk, horizon risk, and the costs of trading.

The framework yields a set of testable predictions without further psychological assumptions:

1. **Level of mispricing.** Mispricing is larger for assets whose valuation is more uncertain—the more the fundamental value is in dispute, the more room noise-trader demand has to push prices away from it.

2. **Arbitrage difficulty.** Mispricing is larger for assets whose characteristics make arbitrage more difficult—stocks with high idiosyncratic volatility, low institutional ownership, or expensive short selling.

3. **Informative arbitrage.** Arbitrageurs' actions in correcting mispricing are themselves informative about future returns. When arbitrageurs buy an underpriced stock, they expect the price to rise as the mispricing closes; their trades forecast returns.

Each prediction has generated a large empirical literature. Sections 5.7–5.9 develop the sophisticated-trader, noise-trader, and sentiment branches in turn.

5.7 Actions of Sophisticated Agents

If some assets are mispriced, the trades of sophisticated investors who act against the mispricing should predict returns. The framework has been tested using four classes of sophisticated agents: firm managers, short sellers, hedge funds, and analysts.

Firm managers. The key assumption is that firm managers can detect and respond to market mispricing. Two types of managerial action generate testable predictions. *Issuance and repurchase* decisions change the supply of shares in response to investor demand: when a stock is overpriced, managers issue more shares, and when it is underpriced, they repurchase. *Corporate insider trading*—managers' personal transactions in their own firms' stocks—similarly predicts future returns, with insider purchases forecasting positive returns and insider sales forecasting negative returns.

Short sellers. Short sellers are classic arbitrageurs in the De Long et al. (1990) framework. Asquith, Pathak, and Ritter (2005) document that stocks with heavy short interest have low subsequent returns, consistent with the overpricing being identified (and partly corrected) by sophisticated traders.

Hedge funds. Chen, Da, and Huang (2019) construct a measure of *net arbitrage trading* (NAT) by hedge funds and show that NAT strongly predicts stock returns in the cross-section. The pattern is consistent with hedge funds betting against overvalued stocks and earning profits as the overpricing subsequently closes.

Analysts. Lee and So (2017) document that abnormal analyst coverage—sudden additions of analyst coverage—positively predicts subsequent returns, consistent with the interpretation that analyst attention identifies underpriced stocks that subsequently revert as the information is incorporated.

Analysts also face an incentive conflict that constrains what their published recommendations can reveal. A recommendation carries the firm's name, and the brokerage earns revenue from the covered issuer, so the analyst's public statements are filtered by the issuer's interest in them. The coverage decisions themselves escape that filter, and they carry information the ratings do not. McNichols and O'Brien (1997) document the self-selection: analysts concentrate coverage on firms with favorable prospects and drop firms with poor ones, so the set of firms an analyst covers and the set it abandons each carry a signal. An analyst who stops covering a firm has revealed information without publishing a rating, which is why the analyst's actions rather than the analyst's words are the more informative object for a sophisticated investor to track.

5.7.1 Aggregate Predictions

The same logic applies at the aggregate level. When sophisticated agents' actions are aggregated to the market level, they should predict market returns in the same direction as at the stock level. Baker and Wurgler (2000) show that the share of equity issues in total equity and debt issues negatively predicts market returns—when firms issue equity to take advantage of perceived overpricing, the aggregate issuance ratio is high and the subsequent market return is low. Seyhun (1998) document that aggregate insider trading positively forecasts market returns. Rapach, Ringgenberg, and Zhou (2016) document that aggregate short interest negatively forecasts market returns.

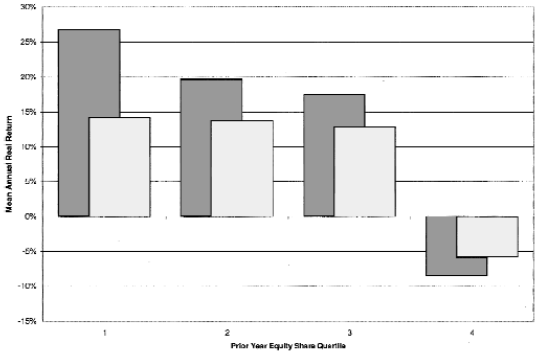


Figure 2. Mean equity returns by prior-year equity share in new issues, 1928–1997. Mean annual real returns on the CRSP value-weighted (light) and equal-weighted (solid) indexes by quartile of the prior-year share of equity issues in total equity and debt issues. Real returns are created using the Consumer Price Index from Ibbotson Associates (1998).

Figure 5.12: Equity issuance forecasts market returns. The figure illustrates the central finding of Baker and Wurgler (2000): the share of equity issues in total equity and debt issues negatively predicts future market returns, consistent with managers timing the market by issuing equity when the market is overvalued. *Source:* EF8083 course materials, after Baker and Wurgler (2000).

5.8 Actions of Noise Traders

If some assets are mispriced, the trades of unsophisticated agents should *negatively* predict returns: the noise traders buy at the peak and sell at the trough, and the resulting patterns reverse. Three influential studies establish the prediction.

Frazzini and Lamont (2008) document that stocks held by mutual funds with large inflows over the previous three years have low subsequent returns. The mechanism is that fund inflows push up the prices of the stocks held by the recipient funds; the inflow-driven price increase subsequently reverses as flows subside.

Barber, Odean, and Zhu (2009) and Hvidkjaer (2008) document the same pattern at the individual investor level. Stocks heavily bought by individual investors in the previous year perform poorly in the following year. The key data innovation is the use of small trades in the TAQ dataset to identify individual investors and the Lee–Ready algorithm to classify trades as buys or sells. The implied flow–return relation is large and statistically robust.

Da, Engelberg, and Gao (2011) document the same pattern using Google search volume as a measure of attention. Higher attention, as captured by aggregate search volume for a stock's ticker, correlates with abnormal retail buying and elevated initial prices, with the price subsequently reverting.

The pattern across the three studies is consistent and unified: any measure of unsophisticated buying pressure predicts subsequent low returns, and any measure of unsophisticated selling pressure predicts subsequent high returns. The mirror-image pattern in Section 5.7—sophisticated trades predict returns in the opposite direction—closes the loop. Mispricing is identified by sophisticated traders and unwound as the noise traders' demand fades.

5.8.1 My Research: Information Acquisition via EDGAR Search

Li and Sun (2022) study the same cross-section of demand, but from the information-acquisition side rather than the trading side. SEC EDGAR server logs record the number of distinct IP addresses that search each firm's filings, which makes observable—for the entire market, at firm-month frequency—a direct measure of costly information acquisition. When abnormal EDGAR search is high, more investors are paying the cost of reading a firm's filings; the paper shows that abnormal EDGAR search predicts future returns and fundamentals, and that a long–short portfolio sorted on it earns 80 basis points per month with no long-run reversal. Predictability is stronger for larger and lengthier filings, the filings that cost more to process, consistent with endogenous information acquisition: investors search more when the payoff to processing is higher.

The comparison with the Google-search evidence of Da, Engelberg, and Gao (2011) in this section is instructive. Google search volume measures what grabs retail investors' attention—a stock that is already salient—and the attention-driven buying it flags reverses. EDGAR search measures deliberate, costly processing by investors who choose to read filings, and the demand it foreshadows does not reverse. Attention and acquisition are two sides of bounded rationality operating in opposite directions on prices: the first pushes them away from fundamentals, the second pulls them back.

5.9 Investor Sentiment and the Stock Market

A more aggregate version of the noise-trader argument is *investor sentiment*. Sentiment is a market-wide, time-varying component of demand that is not driven by fundamentals. The empirical

challenge is that sentiment is unobservable, and the literature has converged on a small set of proxies.

5.9.1 The Baker–Wurgler Sentiment Index

Baker and Wurgler (2006) construct a sentiment index as the principal component of six proxies drawn from market data:

- **Trading volume** (TTURN): the natural logarithm of the NYSE turnover ratio, detrended by its five-year moving average.
- **Closed-end fund discount** (CEFD): the average difference between the net asset values of closed-end stock fund shares and their market prices.
- **IPO activity** (NIPO): the number of IPOs.
- **First-day returns** (RIPO): the average first-day return on IPOs.
- **Equity issuance** (EIT): the ratio of equity issues to total equity and debt issues by all corporations.
- **Dividend premium** (DP): the difference between the average market-to-book ratios of dividend payers and nonpayers.

Each of these measures has been proposed by prior studies as capturing investor sentiment; the principal-component construction aggregates them into a single monthly index.

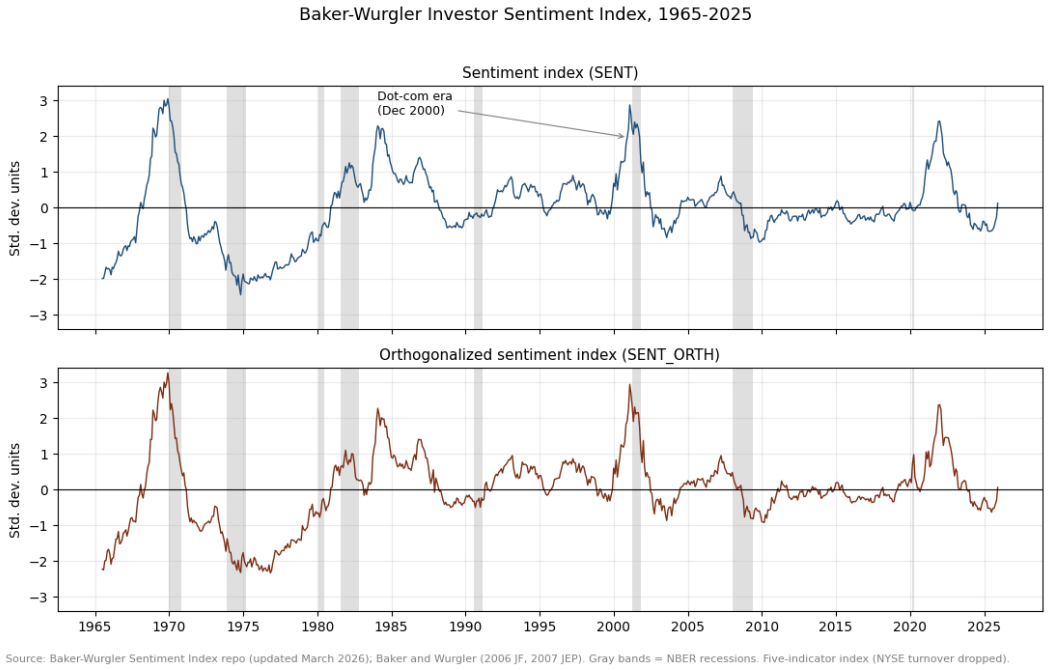


Figure 5.13: The Baker–Wurgler sentiment index. Two-panel monthly time series, 1965–2025, with NBER recession shading: the raw sentiment index (top panel) and the orthogonalized index that purges macroeconomic conditions (bottom panel). *Source:* EF8083 course materials, using the updated Baker–Wurgler index series after Baker and Wurgler (2006) and Huang et al. (2015).

Huang et al. (2015) refine the construction by replacing the principal-component step with a partial least squares (PLS) regression. PLS extracts the linear combination of the six proxies that best predicts one-year-ahead aggregate market returns, which is the economic target the index is supposed to capture. The PLS index has stronger in-sample and out-of-sample predictive power than the original sentiment indices.

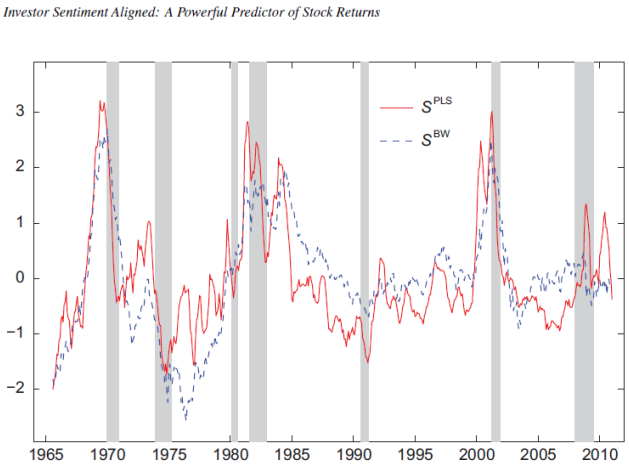


Figure 5.14: The PLS sentiment index. The refined index of Huang et al. (2015) uses partial least squares to extract the linear combination of the six proxies that most efficiently forecasts one-year-ahead market returns. The refined index has greater predictive power than the original. *Source:* Huang et al. (2015), reproduced from the original article.

5.9.2 Sentiment and Aggregate Market Returns

Baker and Wurgler (2007) show that sentiment predicts market returns with a negative sign: high sentiment today forecasts low market returns over the next one to three years. The pattern is consistent with the mispricing interpretation: high sentiment is associated with elevated prices that subsequently revert, and the reversion shows up as low future returns.

The time-series evidence is the weaker half of the sentiment result, and the weakness is informative. The predictive coefficient on the index is negative and economically large, but its statistical precision in the full sample is poor, because the sentiment index is built from proxies that each carry noise of their own and because the aggregate signal it contains varies only slowly. The cross-sectional evidence is the stronger half: the same index sorts stocks by subsequent return with a monotone gradient. The interpretation that reconciles the two is that the index measures a slow-moving component of mispricing—valuation uncertainty that resolves over long horizons—so its information shows up in relative prices across stocks far more cleanly than in the level of the index over time.

5.9.3 The Role of Uncertainty

Birru and Young (2022) sharpen the inference by showing that the return predictability of investor sentiment is much greater when uncertainty is high. The mechanism is that sentiment has the largest effect on asset prices precisely when valuations are most subjective, which is precisely when disagreement about the fundamental is most pronounced. The finding is consistent with the De Long–Shleifer–Summers–Waldmann prediction that mispricing is larger when valuation is more uncertain.

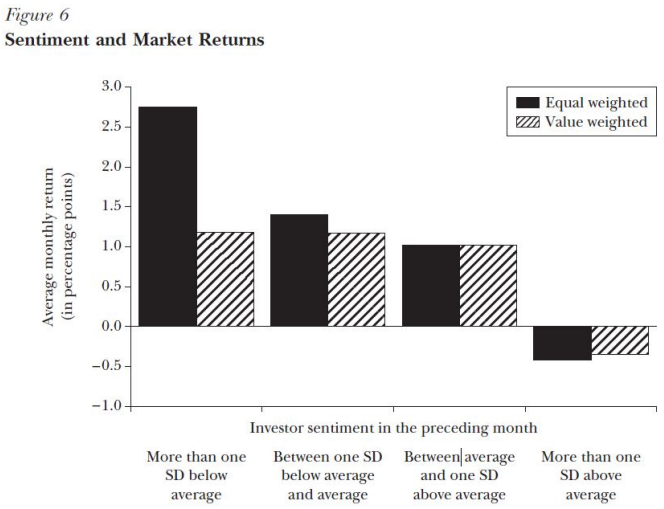


Figure 5.15: Sentiment predicts low future market returns. The figure illustrates the central finding of Baker and Wurgler (2007): periods of high sentiment forecast low market returns over the subsequent one to three years. *Source:* Baker and Wurgler (2007), reproduced from the original article.

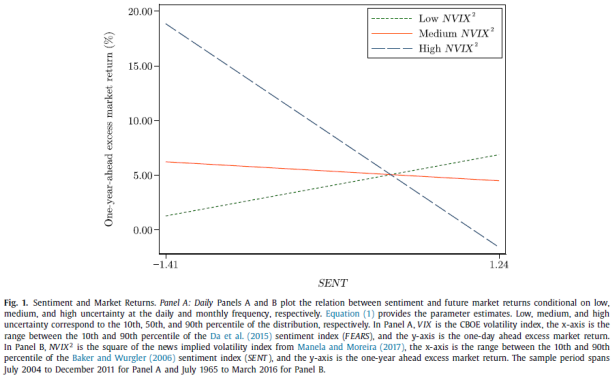


Fig. 1. Sentiment and Market Returns. *Panel A: Daily* Panels A and B plot the relation between sentiment and future market returns conditional on low, medium, and high uncertainty at the daily and monthly frequency, respectively. Equation (1) provides the parameter estimates. Low, medium, and high uncertainty correspond to the 10th, 50th, and 90th percentile of the distribution, respectively. In Panel A, *VIX* is the CBOE volatility index, the x-axis is the range between the 10th and 90th percentile of the Da et al. (2015) sentiment index (*FEARS*), and the y-axis is the one-day ahead excess market return. In Panel B, $NVIX^2$ is the square of the news implied volatility index from Manela and Moreira (2017), the x-axis is the range between the 10th and 90th percentile of the Baker and Wurgler (2006) sentiment index (*SENT*), and the y-axis is the one-year ahead excess market return. The sample period spans July 2004 to December 2011 for Panel A and July 1965 to March 2016 for Panel B.

Figure 5.16: Sentiment matters more in uncertain times. The figure illustrates the central finding of Birru and Young (2022): the predictive power of the sentiment index for market returns is sharply larger in periods of high uncertainty, consistent with the interpretation that mispricing is largest when fundamental value is most disputed. *Source:* Birru and Young (2022), reproduced from the original article.

5.9.4 Cross-Sectional Implications

The De Long–Shleifer–Summers–Waldmann framework also predicts a cross-sectional implication. When market-wide sentiment is high, stocks that are hard to arbitrage and difficult to value should be overpriced most, and should earn the lowest subsequent returns. Baker and Wurgler (2006) test the prediction by sorting stocks into hard-to-arbitrage categories (small, young, volatile, unprofitable, non-dividend-paying) and find that the predictive coefficient of sentiment is largest for these categories and near zero for the easy-to-arbitrage categories (large, mature, stable, profitable, dividend-paying).

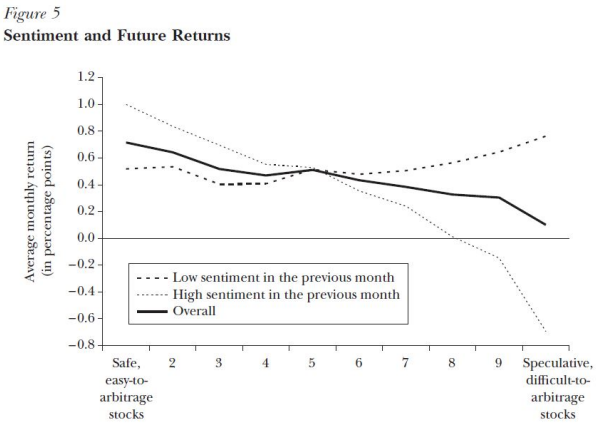


Figure 5.17: Sentiment effects concentrate in hard-to-arbitrage stocks. The figure illustrates the cross-sectional prediction: the predictive coefficient of sentiment for subsequent returns is large and significant for small, young, volatile, unprofitable, and non-dividend-paying stocks, and close to zero for the easy-to-arbitrage counterparts. *Source:* EF8083 course materials, after Baker and Wurgler (2006).

5.9.5 Sentiment and Anomalies

Stambaugh, Yu, and Yuan (2012) sharpen the cross-sectional argument with a theoretical and empirical point: anomalies should be amplified following high sentiment. The theoretical reason is asymmetry: the effect of sentiment should be stronger for overpricing than for underpricing, because short-sale constraints make the short leg harder to correct. Empirically, the authors show that long–short anomaly returns are significantly larger following periods of high sentiment, with the effect concentrated on the short leg.

5.9.6 The Day-of-the-Week Effect

Birru (2018) provides a more granular look at the same mechanism. Long–short anomaly returns are systematically related to the day of the week: they are largest on Monday and smallest on Friday. The pattern is most pronounced when the short leg is speculative—that is, when the overpricing is most sensitive to investor mood. The result is consistent with a psychology literature documenting that mood rises over the weekend and falls at the start of the work week; the day-of-week effect on anomalies is the asset-pricing counterpart of mood-induced variation in risk-taking.

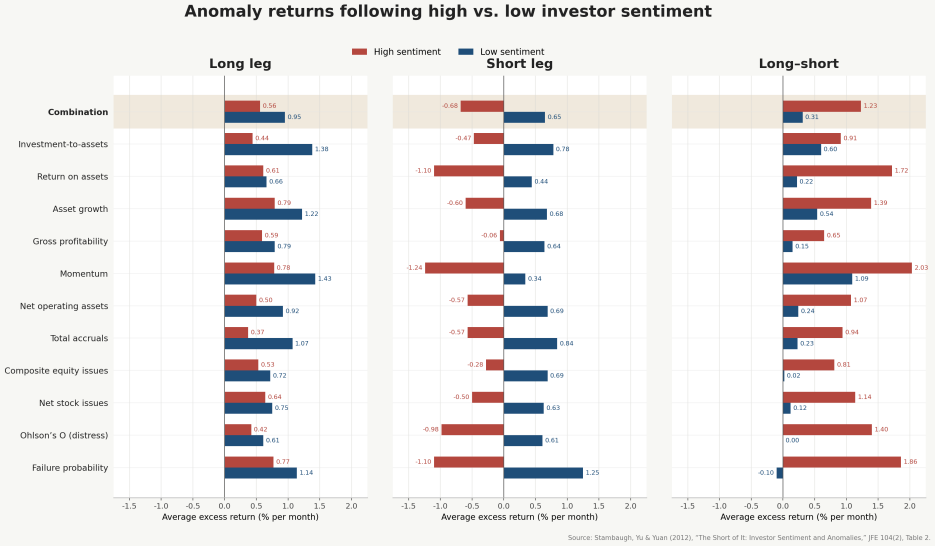


Figure 5.18: Sentiment amplifies anomalies on the short side. The figure illustrates the central finding of Stambaugh, Yu, and Yuan (2012): long–short anomaly returns are larger following high-sentiment periods than following low-sentiment periods, and the amplification concentrates in the short legs—the overpriced stocks that short-sale constraints prevent arbitrageurs from correcting. *Source:* EF8083 course materials, after Stambaugh, Yu, and Yuan (2012).

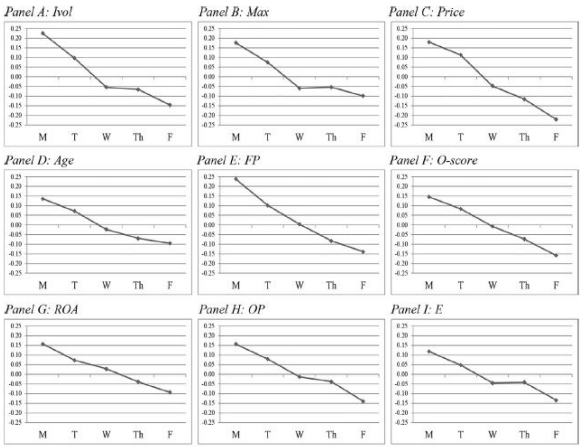


Figure 5.19: Monday up, Friday down. The figure illustrates the day-of-the-week effect on long–short anomaly returns documented by Birru (2018): returns are largest on Mondays and smallest on Fridays, with the effect strongest when the short leg is most speculative. *Source:* Birru (2018), reproduced from the original article.

5.9.7 The Risk–Return Trade-off

Shen, Yu, and Zhao (2017) document that sentiment also affects the cross-sectional risk–return trade-off. When sentiment is high, sentiment-driven investors tend to require a smaller compensation for the risk they bear—they are more willing to bear risk because they are optimistic about it. In addition, riskier stocks are more subject to sentiment shifts and become more overpriced. Together with short-sale constraints, this mechanism attenuates or even reverses the risk–return trade-off during high-sentiment periods.

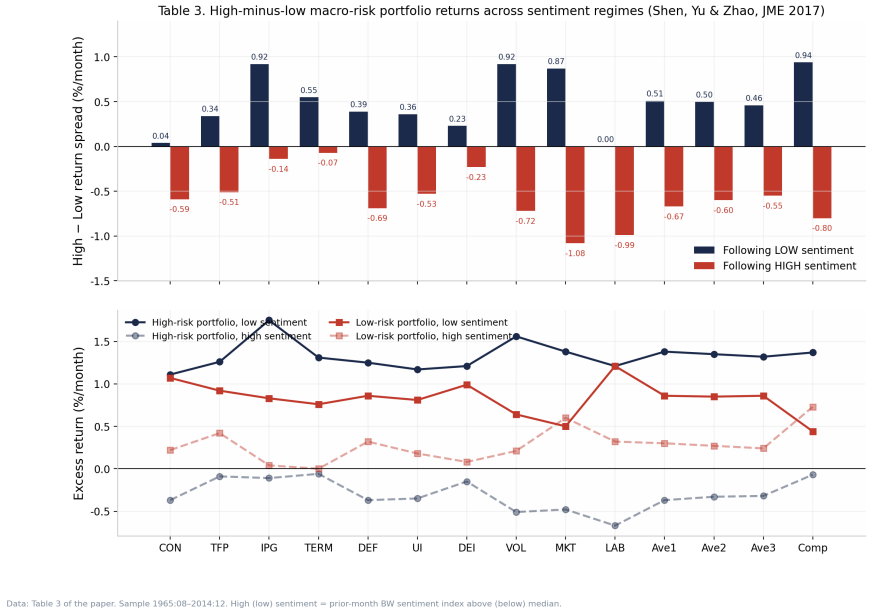


Figure 5.20: The risk–return trade-off weakens when sentiment is high. The figure illustrates the mechanism and evidence of Shen, Yu, and Zhao (2017): during high-sentiment periods, sentiment-driven investors require less compensation for bearing risk and riskier stocks are more subject to sentiment-driven overpricing, so the cross-sectional relation between risk and expected returns flattens or even reverses. *Source:* EF8083 course materials, after Shen, Yu, and Zhao (2017).

A related strand of evidence concerns characteristic-based factor betas. Chen, Liu, et al. (2025) examine characteristic-beta-sorted portfolios (e.g., momentum beta) and find that high-beta portfolios yield higher returns than low-beta portfolios following high-sentiment periods. The result suggests that characteristic-based factor beta is more likely a proxy for the level of mispricing than for exposure to systematic risk.

5.9.8 Other Measures of Aggregate Sentiment

The Baker–Wurgler index and its refinements are the canonical sentiment measure, but a number of alternative constructions have appeared in the literature. They share a common feature: each attempts to capture investor mood or belief using a source of data other than equilibrium asset prices.

FEARS. Da, Engelberg, and Gao (2015) aggregate the volume of Google search queries related to household concerns and construct the Financial and Economic Attitudes Revealed by Search (FEARS) index. The advantages over the Baker–Wurgler proxies are threefold. The index is more

transparent (the underlying search terms are visible). It is higher frequency (daily, rather than monthly). It reveals attitudes rather than inquiring about them: the index is based on what investors choose to search for, not on what they report in a survey.

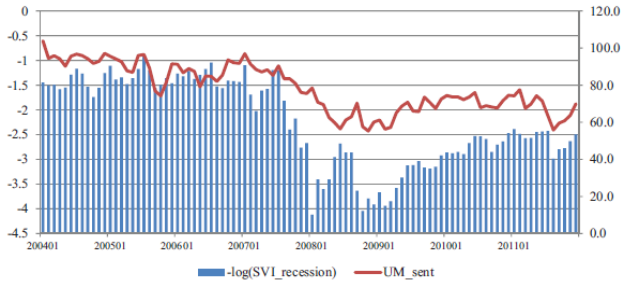


Figure 5.21: FEARS. The figure illustrates the central patterns documented by Da, Engelberg, and Gao (2015): FEARS is negatively related to contemporaneous returns and positively related to future returns; it also predicts temporary increases in volatility and mutual fund flows out of equity funds and into bond funds. *Source:* EF8083 course materials, after Da, Engelberg, and Gao (2015).

Manager sentiment. Jiang et al. (2019) construct a manager sentiment index from the aggregated textual tone in 10-K filings, 10-Q filings, and conference call transcripts. Manager sentiment is a strong negative predictor of future aggregate stock market returns—a finding that is interesting precisely because managers are sophisticated agents whose communications are observable to investors. The result suggests that even sophisticated agents' revealed sentiment (as opposed to their actions) carries incremental information for prices.

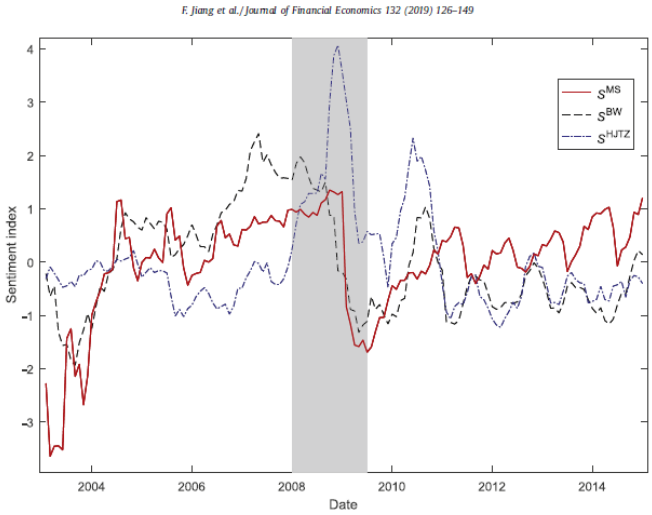


Figure 5.22: Manager sentiment forecasts aggregate returns. The figure illustrates the central finding of Jiang et al. (2019): a sentiment index built from the textual tone of 10-K filings, 10-Q filings, and conference call transcripts is a strong negative predictor of future aggregate stock market returns. *Source:* Jiang et al. (2019), reproduced from the original article.

Aggregate short interest. Short sellers are sophisticated investors who can correct mispricing induced by sentiment traders; the aggregate short-interest position is therefore a contrarian signal.

Rapach, Ringgenberg, and Zhou (2016) and others document that short interest aggregated to the market level is a strong contrarian predictor of future market returns.

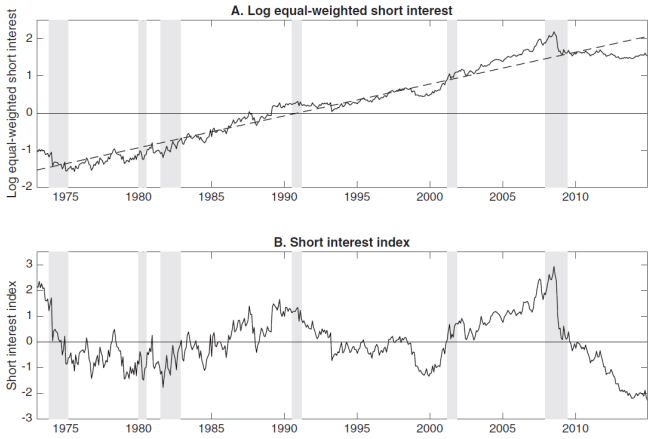


Figure 5.23: Aggregate short interest is contrarian. The figure illustrates the central finding of Rapach, Ringgenberg, and Zhou (2016): the aggregate short-interest position is a strong contrarian predictor of future market returns, consistent with short sellers as sophisticated investors who correct sentiment-induced mispricing. *Source:* Rapach, Ringgenberg, and Zhou (2016), reproduced from the original article.

Leveraged ETF flows. Davies (2022) constructs a Speculative Sentiment Index (SSI) based on the difference between share change in leveraged-long ETFs and share change in leveraged-short ETFs. The index captures the speculative component of market demand: leveraged ETFs are heavily traded by investors who bet on the direction of future market movement, and the difference in flows between the long and short sides is a clean measure of speculative demand. A one-standard-deviation increase in the monthly index is associated with a 1.14–1.67% decline in broad-market stock returns over the next month.

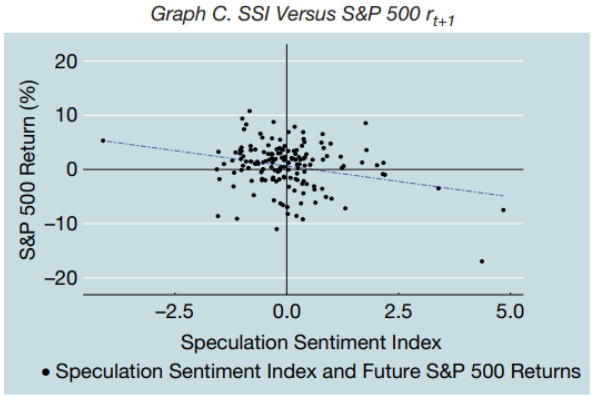


Figure 5.24: Leveraged ETF flows measure speculative demand. The figure illustrates the central finding of Davies (2022): the Speculative Sentiment Index, built from the difference between share changes in leveraged-long and leveraged-short ETFs, predicts lower broad-market stock returns over the following month. *Source:* Davies (2022), reproduced from the original article.

Music sentiment. Edmans, Fernandez-Perez, et al. (2022) propose a real-time, continuous measure of national sentiment based on the positivity of songs that individuals choose to listen to. Music sentiment has three attractive properties as a sentiment measure: it is real-time, language-free, and comparable across countries, and exogenous to financial market events. Music sentiment is positively correlated with same-week equity market returns and negatively correlated with next-week returns; it also predicts increases in net mutual fund flows and a rise in stock market volatility. Music sentiment is negatively associated with government bond returns, consistent with a flight to safety when sentiment deteriorates.

5.9.9 LLM-Generated Beliefs

The survey measures that anchor the sentiment literature share a limitation that matters for long-horizon applications: they cover a short period and a small set of respondents, so a sentiment index built from them cannot say much about episodes before the survey began. Bybee (2025) propose a way around the constraint. Language models trained on large corpora absorb the same world models and the same biases that human forecasters carry, so a model queried with a news article can be asked for its beliefs about the direction, confidence, and magnitude of the change in a set of financial and macroeconomic variables. The article-level beliefs are then aggregated into a belief series.

The construction has three steps. From each month of *Wall Street Journal* articles over 1984–2021, 300 articles are sampled at random and fed to GPT-3.5 with a prompt requesting a direction, a confidence, a magnitude, and an explanation for each of a set of macro and financial series. The resulting beliefs are collapsed into a balance statistic for each series, weighting by the model's stated confidence. Because a GPT-3.5 query returns one response, the model cannot cover the full corpus on its own, so a BERT encoder with ridge transfer learning infills expectations for the articles not queried; a *New York Times* corpus then extends the sample back to 1900.

The validation is what makes the approach more than a demonstration. Generated expectations correlate with the AAII and Duke CFO Survey return-expectation measures at 0.47 on average over a three-month aggregation window, against 0.43 for the correlation among the surveys themselves, and they track Survey of Professional Forecasters revisions for all but two of the thirteen macroeconomic series examined. The generated expectations also reproduce the deviations from full-information rational expectations that the surveys display: they are extrapolative, they correlate with equity fund flows, they are disconnected from objective expected-return proxies such as the dividend-price ratio and the consumption-wealth ratio, and they are negatively correlated with subsequent realized returns. The errors have a traceable origin. The model expects high future returns because it expects high future earnings, a neglect of general equilibrium of the kind that human survey participants show as well (Andre, Schirmer, and Wohlfart, 2026). Using articles that postdate the model's training window leaves the correlations with survey measures intact, which is the evidence that the model generalizes rather than recalls.

Extending the sample backward yields 120 years of economic sentiment. The extraction proceeds from a factor model of expectations in which each series' expectation decomposes into a rational component and a sentiment component. The sentiment component is recovered by orthogonalizing each expectation series against a rational forecast and then taking the first principal component of the orthogonalized series. The resulting measure behaves as a sentiment index should: it responds positively to past returns and negatively predicts future returns, with a correlation of -0.41 at the 24-month horizon in the recent sample. Its turning points line up with history. The spikes are negative at the 1929 crash, at the collapse of Lehman Brothers, and at Covid; they are positive at the dot-com bubble and at the 1936 landslide election of Franklin Roosevelt. The series correlates

0.54 with Shiller's CAPE, and the two measures catch many of the same episodes.

5.9.10 Bubbles through the Lens of Generated Beliefs

The extended sample makes it possible to ask whether sentiment helps identify bubbles, which is the application that survey-based sentiment indices could not support. Bybee (2025) construct a mispricing proxy by running rolling regressions of industry returns on daily sentiment and extracting the time-varying sentiment beta of each industry. The test uses the run-up dataset of Greenwood, Shleifer, and You (2019), which identifies 40 industry-level price run-ups between 1926 and 2014, of which 21 end in a crash in the form of a 40% drawdown over the following two years.

Two findings carry the application. First, the sentiment betas are significantly higher during the run-ups that crash than during those that do not, so an industry's exposure to economic sentiment is a marker of the mispricing that precedes its collapse. Second, the feedback between returns and sentiment is stronger in the crashing run-ups: the response of sentiment to a return shock is significantly larger there, which identifies return extrapolation as a key mechanism behind the bubbles. The economic payoff is a threshold strategy that holds the industry portfolios the sentiment beta predicts will not crash; it predicts roughly 80% of the crashes correctly and earns significant returns against the strategy that holds all run-ups.

The methodological point generalizes beyond this paper. Once the mechanism that produces human sentiment is understood well enough to specify, a language model can be asked to reproduce it. That extends a sentiment measure to any market with a news archive, over any period the archive covers, and it does so at daily frequency rather than monthly. The tool is new enough that its validation rests on agreement with existing survey measures rather than on independent evidence, and the extrapolation of a trained model's biases is a replication of human biases rather than a test of them.

5.10 Credit Market Sentiment

Equity market sentiment is one thing; credit market sentiment is another. The two are conceptually distinct because debt and equity are claims to different parts of the cash-flow distribution and trade in markets with different investor populations.

Greenwood and Hanson (2013) construct a sentiment measure for the credit market by comparing the credit quality of high-issuance firms with that of low-issuance firms. The intuition is that a significant decline in issuers' credit quality is a reliable signal of credit market overheating. The implementation uses the issuers' credit ratings: high-issuance firms whose credit quality has fallen relative to low-issuance firms are the firms most likely to have been financed by overheated credit markets.

The economic magnitudes are substantial. A one-standard-deviation increase in the issuer credit-quality differential implies cumulative excess bond returns that fall by 7.30 percentage points over the following two years.

5.10.1 Credit Sentiment and Economic Growth

Credit market overheating is associated with a higher probability of a future financial crisis and a subsequent economic downturn. Schularick and Taylor (2012) document that credit growth is a robust predictor of financial crises in a long historical sample. Lopez-Salido, Stein, and Zakrajšek (2017) document the same pattern in U.S. postwar data. Baron and Xiong (2017) show that bank credit expansion predicts increased bank equity crash risk: despite the elevated crash risk,

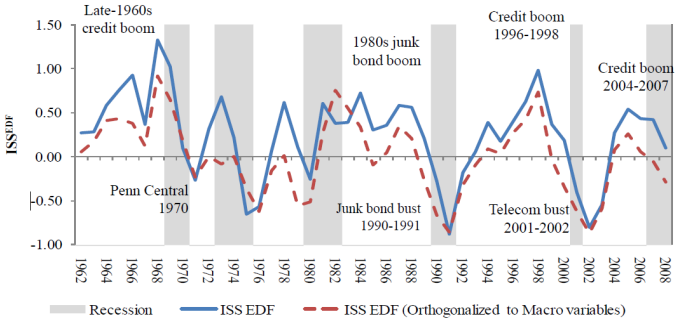


Figure 5.25: Credit market sentiment. The figure illustrates the central finding of Greenwood and Hanson (2013): periods in which firms with falling credit quality are heavily issuing debt are periods of credit market overheating, with subsequent low excess bond returns. *Source:* Greenwood and Hanson (2013), reproduced from the original article.

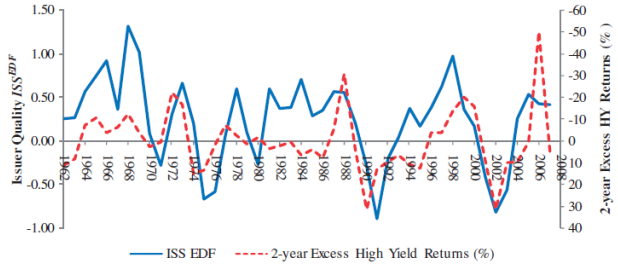


Figure 3
Issuer quality and subsequent high-yield excess returns
Issuer quality (left axis) plotted alongside cumulative excess high-yield bond returns for the following two years (right axis). Returns are plotted in reverse scale, so the negative correlation appears positive visually. Issuer quality is measured with ISS^{EDF}, the difference between the average *EDF* decile of high- and low-debt issuers from 1962–2008.

Figure 5.26: Credit sentiment forecasts excess bond returns. The figure illustrates the central finding of Greenwood and Hanson (2013): periods in which firms with falling credit quality are heavily issuing debt are periods of credit market overheating, with subsequent low excess bond returns. *Source:* Greenwood and Hanson (2013), reproduced from the original article.

credit expansion predicts lower mean bank equity returns over the subsequent one to three years. Bank credit expansion is distinct from equity market sentiment as captured by the dividend yield, suggesting that credit and equity markets are only partially integrated. The framing is one of neglected risk: the risk of a banking crisis is underweighted by bank equity investors during a credit expansion, and the mispricing of that neglected risk shows up in the crash risk that follows.

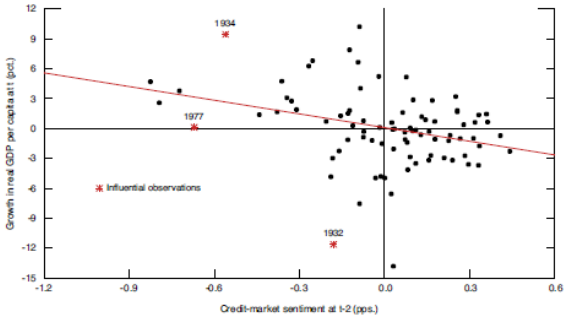


FIGURE II

Credit-Market Sentiment and Economic Growth, 1929–2015

The scatter plot depicts a visual representation of the relationship between credit-market sentiment in year $t-2$ and the growth in real GDP per capita from year $t-1$ to year t, implied by the specification in column (1) of Table II. See the text and Figure III for the definition of influential observations

Figure 5.27: Credit expansion forecasts economic downturns. The figure illustrates the central finding of Lopez-Salido, Stein, and Zakrajšek (2017): rapid credit expansion forecasts subsequent economic downturns with high probability, and credit-based sentiment measures have greater macroeconomic significance than equity-based measures. *Source:* Lopez-Salido, Stein, and Zakrajšek (2017), reproduced from the original article.

5.10.2 Predictable Financial Crises

Greenwood, Hanson, Shleifer, et al. (2022) formalize the crisis-prediction argument using data on financial crises around the world. The combination of rapid credit and asset price growth over the prior three years—the *R-zone*—is associated with a 40% probability of entering a financial crisis within the next three years. The pattern is robust across countries and over more than a century of data.

Panel A: Business Sector

	Crisis within One Year				Crisis within Two Years				Crisis within Three Years				Crisis within Four Years			
	(1.1)	(1.2)	(1.3)	(1.4)	(2.1)	(2.2)	(2.3)	(2.4)	(3.1)	(3.2)	(3.3)	(3.4)	(4.1)	(4.2)	(4.3)	(4.4)
High Debt Growth$^{Bus.}$ (β^h)	6.9**		5.3**		11.6***		9.5**		16.8***		11.5**		15.6**		10.3*	
	[2.3]		[2.1]		[3.0]		[2.5]		[3.3]		[2.7]		[2.7]		[2.2]	
High Price Growth$^{Bus.}$ (δ^h)		0.4	−0.4			4.8	3.8			10.5	7.4			10.7	7.6	
		[0.1]	[−0.2]			[0.9]	[0.8]			[1.4]	[1.1]			[1.5]	[1.2]	
R-Zone$^{Bus.}$ (γ^h)			5.3	9.0			7.8	17.9*			19.4**	33.7***			19.4**	33.0**
			[0.8]	[1.1]			[1.3]	[2.1]			[2.8]	[3.3]			[2.6]	[3.1]
Sum of coefficients ($\beta^h + \delta^h + \gamma^h$)	6.9	0.4	10.2	9.0	11.6	4.8	21.1	17.9	16.8	10.5	38.2	33.7	15.6	10.7	37.3	33.0
t-Statistic ($\beta^h + \delta^h + \gamma^h$)			1.2				2.1				3.2				3.1	
R^2 *(within)*	1.6	0.0	1.9	1.1	2.5	0.7	3.6	2.3	3.8	2.4	7.8	6.1	2.8	2.1	6.2	4.8
N	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258

Figure 5.28: The R-zone predicts crises. The figure illustrates the central finding of Greenwood, Hanson, Shleifer, et al. (2022): countries in the R-zone (rapid credit growth and asset price growth over the prior three years) face a 40% probability of entering a financial crisis within the next three years. *Source:* Greenwood, Hanson, Shleifer, et al. (2022), reproduced from the original article.

5.10.3 Credit versus Equity Sentiment

Empirical measures of credit market sentiment are only modestly correlated with measures of equity market sentiment. The low correlation suggests that equity and credit markets should not be thought of as fully integrated: each market has its own investor population, its own information flows, and its own sentiment dynamics. The macroeconomic implication is important: measures of expected returns on the stock market (such as Shiller's CAPE) do not predict future GDP growth, while measures of credit market overheating do. Credit market sentiment therefore has potentially greater macroeconomic significance than equity market sentiment.

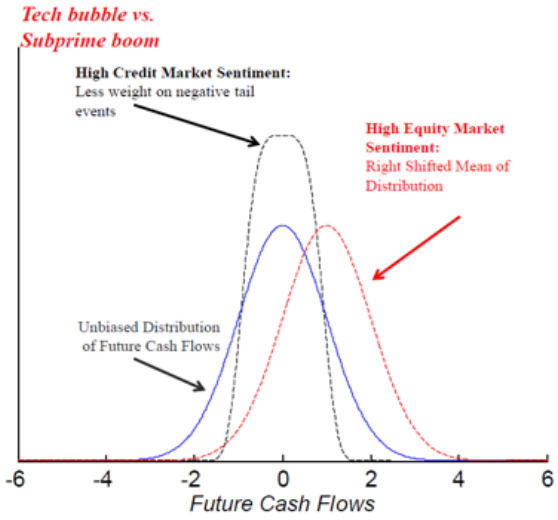


Figure 5.29: Credit and equity sentiment are distinct measures. The figure illustrates the low correlation between empirical measures of credit market sentiment and measures of equity market sentiment, consistent with the interpretation that the two markets have their own investor populations and their own sentiment dynamics. *Source:* EF8083 course materials, after Lopez-Salido, Stein, and Zakrajšek (2017).

The reason for the difference in macroeconomic power lies in the payoff structure of the two claims. Equity investors care about the upside—how good the state of the world will be—and a change in that assessment moves prices in either direction. Creditors do not: the promised coupon and principal are fixed, so what matters to a creditor is the probability of default. Credit sentiment is therefore a measure of perceived tail risk rather than of expected growth, and the variable that damages the real economy is the collapse of a tail rather than the failure of an expansion to arrive. This is also why leverage amplifies credit sentiment into real activity. Firms that have levered up have promised their creditors a fixed payment out of a cyclical cash flow stream; a small negative shock raises the probability that the payment is missed, and the resulting defaults cut credit supply, which forces other levered borrowers to cut back as well.

5.10.4 Credit Cycles across a Financial Crisis

Krishnamurthy and Muir (2025) characterize the credit cycle in the data rather than in narrative, using credit spreads and credit growth around financial crises in a long international sample. Their evidence reverses the natural reading of the price and quantity dynamics. Spreads fall ahead of a crisis and appear too low relative to the credit being extended, and credit grows as the crisis approaches. A rise in spreads at the crisis itself marks a dramatic shift in expectations that arrives as a surprise rather than as a gradual repricing. The severity of the crisis that follows is forecast

by the size of the credit losses, proxied by the change in spreads, together with the fragility of the financial sector, proxied by pre-crisis credit growth, and the interaction of the two is what distinguishes crises that are severe from those that are not. Recessions in the aftermath of financial crises are both severe and protracted.

The reading these results support is that credit supply expansion is the precursor to a crisis, not the response to one. Investors extend credit on terms that look attractive because prices are high and defaults have been few; the credit that is extended is increasingly held by borrowers whose own positions have deteriorated; the boom sustains itself until a shock arrives that the accumulated leverage cannot absorb.

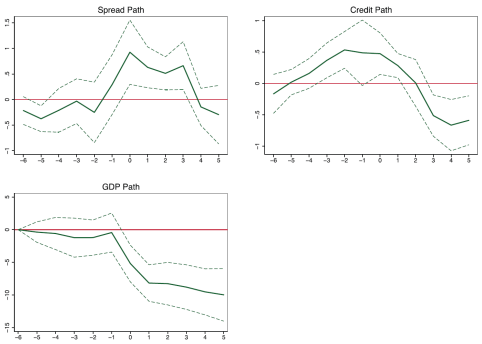


Figure 1. Behavior of spreads, credit, and GDP. This figure plots the behavior of credit spreads, GDP, and the quantity of credit around crises, with date 0 as the JST mod. date associated with the 40 JST crises. GDP is expressed as the percentage fall from year $t-6$, after demeaning by the growth rate in GDP for country-i. Credit is the three-year growth in credit-to-GDP, demeaned by the country-i growth rate and normalized by the standard deviation of credit growth for the country. Spreads are normalized by dividing by the unconditional mean consistent with our normalization in the main text and plotted as deviations from the country-i mean. See the text for further details. The dashed lines indicate 95% confidence bands. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 5.30: Spreads fall and credit grows ahead of a crisis. The figure illustrates the central finding of Krishnamurthy and Muir (2025): credit spreads drop to low levels ahead of a financial crisis while credit growth accelerates, so credit supply expansion is a precursor to the crisis rather than a response to it. *Source:* Krishnamurthy and Muir (2025), reproduced from the original article.

5.10.5 Determinants of Credit Market Sentiment

A number of variables track the credit sentiment cycle. Short-term rates are low or have fallen in the recent past, consistent with "reaching for yield." Term and credit spreads are low or have fallen. Past default rates have been low or falling. Investors flow into high-yield bond funds. Intermediary balance sheets are strong. The pattern is consistent with both frictional and behavioral interpretations: low rates push intermediaries into riskier assets, but the same low rates also feed the optimistic beliefs that justify the risk-taking.

5.10.6 An Early Indicator

Ben-Rephael, Choi, and Goldstein (2021) document an early indicator of credit booms based on intra-family flow shifts towards high-yield bond mutual funds. The advantage of intra-family flows is that they capture investors' deliberate asset allocation decisions rather than saving or investment decisions: when an investor moves money from one fund in a family to a higher-yielding fund in the same family, the move is more likely to reflect an active allocation choice. The intra-family flow indicator leads other measures of credit overheating (growth in financial intermediary balance sheets, increases in the share of high-yield issuers, downturns in credit spreads), suggesting that the credit boom originates in an increase in investors' demand for high-yield bonds, possibly due to biased perceptions of default risk.

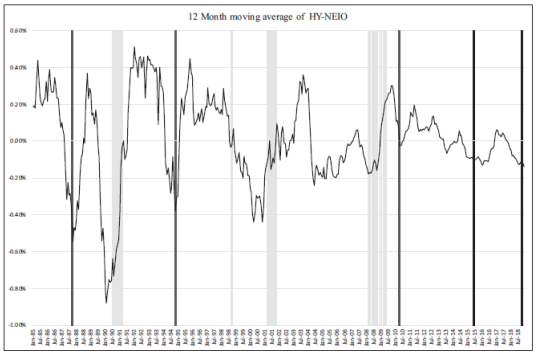


Fig. 1. 12-month moving averages of HY-NEIO. In this figure we plot the 12-month moving averages of HY-NEIO from January 1985 (i.e., February 84–January 85) through December 2018. HY-NEIO is net exchanges (exchanges-in minus exchanges-out) from high-yield corporate bond funds normalized by the end-of-previous-month assets. The data are obtained from the ICI and run from February 1984 through December 2018. The three light gray columns represent NBER recession periods. The dark gray columns represent the 1987 market crash, the Mexican Peso crisis in 1994, the European sovereign debt crisis in early 2010, the high-yield spread hike during June 2015–February 2016, and the market crash in the last quarter of 2018.

Figure 5.31: Intra-family flows predict credit booms. The figure illustrates the central finding of Ben-Rephael, Choi, and Goldstein (2021): intra-family flow shifts towards high-yield bond mutual funds lead other indicators of credit-market overheating, suggesting that the credit cycle originates in investors' demand for high-yield bonds. *Source:* Ben-Rephael, Choi, and Goldstein (2021), reproduced from the original article.

5.11 A Behavioral Account of the Credit Cycle

The credit cycle has a long historical narrative. Kindleberger (1978) describes the Minsky–Kindleberger cycle of credit expansion and reversal, in which investor optimism grows during the boom and turns to panic during the bust. The full behavioral model is developed in Bordalo, Gennaioli, and Shleifer (2018) and Greenwood, Hanson, and Jin (2024).

The key feature of the model is *reflexivity*—the feedback loop between investors' biased perceptions and market outcomes. The mechanism is the following. Investors hold extrapolative beliefs based on borrowers' recent default history: during a boom, investors extend credit on attractive terms because borrowers have paid back in the past. Even if borrowers' fundamental position deteriorates, this may not be directly visible for a while because borrowers roll over their debt at low rates. The terms at which new credit is extended remain favorable, and the market perceives the issuers as healthy. When the reckoning eventually comes—when refinancing conditions tighten, when a default actually arrives, or when a fundamental shock hits—markets overreact in the opposite direction, and the recovery is delayed by the gradual unfreezing of credit.

The behavioral account of the credit cycle has the structure of a bounded-rationality model in which beliefs adapt slowly and credit conditions are persistent. It predicts the leading indicators documented above: rapid credit and asset price growth, followed by an elevated probability of crisis, followed by slow recovery. It also predicts the cross-asset pattern of returns during the boom and bust phases, with bond and equity returns both reversing sharply at the turning point.

5.12 What Remains Contested

Three strands of evidence keep the bounded-rationality and psychology-free programs honest.

First, the causal evidence is thinner than the correlational evidence. The Hirshleifer–Lim–Teoh and Della Vigna–Pollet designs on PEAD, the Chen–An–Wang–Yu identification on attention spillover, and the Cohen–Frazzini customer–supplier spillover—each provides a quasi-experimental wedge on a particular mechanism. The wedges are convincing for the specific mechanisms they target, but they do not aggregate into a comprehensive identification of all bounded-rationality phenomena. The bulk of the evidence remains correlational, and the rational-alternative interpretations have to

be ruled out by argument rather than by design.

Second, the bounded-rationality reading competes with rational information-based alternatives in several applications. The PEAD evidence, on the noisy-information reading, is the slow resolution of uncertainty rather than the slow incorporation of clear news; the Hong–Stein model itself, while parsimonious, is structurally similar to a model in which information diffuses slowly through a fully rational population. The bounded-rationality reading is favored by the attention-manipulation evidence (Hirshleifer–Lim–Teoh, Della Vigna–Pollet, Chen–An–Wang–Yu) but is not uniquely supported by it.

Third, the psychology-free framework is silent on the source of noise-trader demand. The De Long–Shleifer–Summers–Waldmann architecture makes contact with a wide range of facts without requiring any assumption about why noise traders are wrong. The strength of the framework is also its limitation: it does not adjudicate among the competing sources of noise-trader demand, and the predictions of the framework depend on how the noise-trader component is modeled. When the framework is used to interpret sentiment indices (Baker–Wurgler, FEARS, manager sentiment), the inference is from the predictive coefficient on subsequent returns; when it is used to interpret sophisticated-agent trades (insiders, short sellers, hedge funds), the inference is from the correlation of trades with subsequent returns. The two inferences are correlated and reinforce one another, but neither pins down the underlying source of noise-trader demand.

The methodological lesson is the same as in earlier chapters: a behavioral or psychology-free model earns credibility when it predicts something not used to fit it, and when that prediction survives out-of-sample tests. The strongest out-of-sample predictions of this chapter are the cross-sectional amplification of anomalies following high sentiment (Stambaugh, Yu, and Yuan, 2012), the predictability of credit returns from credit-market sentiment measures (Greenwood and Hanson, 2013), the EDGAR-based evidence on information acquisition and disagreement (Li and Sun, 2022; Chang et al., 2022), and the causal evidence on attention from the Chinese listing-code instrument (Chen, An, et al., 2023). The rest of the literature is competitive with rational alternatives.

A fourth consideration is specific to the LLM-generated belief evidence, and it cuts differently from the three above. The validation of Bybee (2025) against survey measures is an out-of-sample test in the sense that the generated expectations were never fitted to the surveys. But the model used to generate them was trained on the same news that human forecasters read, so the agreement with human biases is closer to a replication than to an independent test of the mechanism. What the evidence establishes is that human sentiment can be reproduced and extended in time and cross-section; what it does not establish is that the reproduced biases are the biases a specific human forecaster held, and the replication itself carries the extrapolation of the training corpus into the measurement.

5.13 The Road Ahead

This chapter has developed the third psychological ingredient of behavioral asset pricing: bounded rationality. Limited attention produces underreaction to subtle news; category thinking produces excess comovement; and the psychology-free framework produces a wide range of sentiment-driven return patterns with a minimum of psychological assumptions. Combined with the prospect-theory departures of Chapter 3 and the biased-beliefs departures of Chapter 4, the book now has the three psychological ingredients needed for most of behavioral finance.

The final chapter assembles these ingredients into a specific class of models that have come to play an outsized role in modern behavioral asset pricing: *memory-based* approaches. The premise is that investors form views about future cash flows not from a Bayesian aggregation of all available

evidence but from the cases they most easily retrieve from memory. The same bounded-rationality machinery that produced limited attention in Chapter 5 produces memory-based beliefs when the bias is in what investors remember rather than in how they process what they attend to. Memory-based models are the natural endpoint of a behavioral program that began with departures from expected utility (Chapter 3) and continued through biased beliefs (Chapter 4) and bounded attention (Chapter 5). The final chapter develops the memory-based models and the asset pricing evidence they generate.

Together with the four preceding chapters, the book develops the building blocks of behavioral asset pricing. The remaining chapter applies them to one of the most recent and rapidly expanding branches of the field: the role of memory in forming beliefs and the implications for prices.

Bibliography

Addoum, J. M. and J. R. Murfin (2020). "Equity Price Discovery with Informed Private Debt". In: *Review of Financial Studies* 33.8, pp. 3766–3803.

Ali, U. and D. Hirshleifer (2020). "Shared Analyst Coverage: Unifying Momentum Spillover Effects". In: *Journal of Financial Economics* 136.3, pp. 776–797.

Allais, M. (1953). "Le Comportement de l'Homme Rationnel devant le Risque: Critique des Postulats et Axiomes de l'École Américaine". In: *Econometrica* 21.4, pp. 503–546.

Alpert, M. and H. Raiffa (1982). "A Progress Report on the Training of Probability Assessors". In: *Judgment under Uncertainty: Heuristics and Biases*. Cambridge University Press, pp. 294–305.

Andre, P., P. Schirmer, and J. Wohlfart (2026). "Mental Models of the Stock Market". In: *Quarterly Journal of Economics* 141.4. Originally circulated as a working paper, pp. 2925–2978.

Ang, A., R. J. Hodrick, Y. Xing, and X. Zhang (2006). "The Cross-Section of Volatility and Expected Returns". In: *Journal of Finance* 61.1, pp. 259–299.

Anton, J. J. and C. Polk (2014). "Connected Stocks". In: *Journal of Finance* 69.3, pp. 1099–1127.

Asness, C., A. Frazzini, and L. H. Pedersen (2019). "Quality Minus Junk". In: *Review of Accounting Studies* 24.1, pp. 34–112.

Asness, C., T. Moskowitz, and L. H. Pedersen (2013). "Value and Momentum Everywhere". In: *Journal of Finance* 68.3, pp. 929–985.

Asquith, P., P. A. Pathak, and J. R. Ritter (2005). "Short Interest, Institutional Ownership, and Stock Returns". In: *Journal of Financial Economics* 78.2, pp. 243–276.

Bae, J. W., Z. Da, and V. Zurita (2024). "Digesting FOREXS: Information Transmission Across Asset Classes and Return Predictability". In: *Management Science* 70.3, pp. 1943–1969.

Baker, M., B. Bradley, and J. Wurgler (2011). "Benchmarks as Limits to Arbitrage: Understanding the Low-Volatility Anomaly". In: *Financial Analysts Journal* 67.1, pp. 40–54.

Baker, M. and J. Wurgler (2000). "The Equity Share in New Issues and Aggregate Stock Returns". In: *Journal of Finance* 55.5, pp. 2219–2257.

Baker, M. and J. Wurgler (2006). "Investor Sentiment and the Cross-Section of Stock Returns". In: *Journal of Finance* 61.4, pp. 1645–1680.

Baker, M. and J. Wurgler (2007). "Investor Sentiment in the Stock Market". In: *Journal of Economic Perspectives* 21.2, pp. 129–151.

Bali, T. G., S. J. Brown, S. Murray, and Y. Tang (2017). "A Lottery Demand-Based Explanation of the Beta Anomaly". In: *Journal of Financial and Quantitative Analysis* 52.6, pp. 2369–2397.

Bali, T. G., N. Cakici, and R. F. Whitelaw (2011). "Maxing Out: Stocks as Lotteries and the Cross-Section of Expected Returns". In: *Journal of Financial Economics* 101.3, pp. 592–616.

Bali, T. G., B. T. Kelly, M. Mörke, and J. Rahman (2026). "Machine Forecast Disagreement". In: *Review of Financial Studies*. Forthcoming.

Ball, R. and P. Brown (1968). "An Empirical Evaluation of Accounting Income Numbers". In: *Journal of Accounting Research* 6.2, pp. 159–178.

Bansal, R. and A. Yaron (2004). "Risks for the Long Run: A Potential Resolution of Asset Pricing Puzzles". In: *Journal of Finance* 59.4, pp. 1481–1509.

Barber, B. M. and T. Odean (2000). "Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors". In: *Journal of Finance* 55.2, pp. 773–806.

Barber, B. M. and T. Odean (2001). "Boys Will Be Boys: Gender, Overconfidence, and Common Stock Investment". In: *Quarterly Journal of Economics* 116.1, pp. 261–292.

Barber, B. M. and T. Odean (2008). "All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors". In: *Review of Financial Studies* 21.2, pp. 785–818.

Barber, B. M., T. Odean, and N. Zhu (2009). "Do Retail Trades Move Markets?" In: *Review of Financial Studies* 22.1, pp. 151–186.

Barberis, N. and M. Huang (2008). "Stocks as Lotteries: The Implications of Probability Weighting for Security Prices". In: *American Economic Review* 98.5, pp. 2066–2100.

Barberis, N., M. Huang, and T. Santos (2001). "Prospect Theory and Stock Market Prices". In: *Quarterly Journal of Economics* 116.1, pp. 1–53.

Barberis, N., M. Huang, and R. H. Thaler (2006). "Individual Preferences, Monetary Gambles, and Stock Market Participation: A Case for Narrow Framing". In: *American Economic Review* 96.4, pp. 1069–1090.

Barberis, N., L. J. Jin, and B. Wang (2021). "Prospect Theory and Stock Market Anomalies". In: *Journal of Finance* 76.5, pp. 2639–2687.

Barberis, N., A. Mukherjee, and B. Wang (2016). "Prospect Theory and Stock Returns: An Empirical Test". In: *Review of Financial Studies* 29.11, pp. 3068–3107.

Barberis, N. and A. Shleifer (2003). "Style Investing". In: *Journal of Financial Economics* 68.2, pp. 161–199.

Barberis, N., A. Shleifer, and R. Vishny (1998). "A Model of Investor Sentiment". In: *Journal of Financial Economics* 49.3, pp. 307–343.

Barberis, N., A. Shleifer, and J. Wurgler (2005). "Comovement". In: *Journal of Financial Economics* 75.3, pp. 523–565.

Barberis, N. and R. Thaler (2003). "A Survey of Behavioral Finance". In: *Handbook of the Economics of Finance*. Ed. by G. M. Constantinides, M. Harris, and R. M. Stulz. Vol. 1. Elsevier, pp. 1053–1128.

Baron, M. and W. Xiong (2017). "Credit Expansion and Neglected Crash Risk". In: *Quarterly Journal of Economics* 132.2, pp. 713–764.

Barro, R. J. (2006). "Rare Disasters and Asset Markets in the Twentieth Century". In: *Quarterly Journal of Economics* 121.3, pp. 823–866.

Ben-David, I., J. R. Graham, and C. R. Harvey (2013). "Managerial Miscalibration". In: *Quarterly Journal of Economics* 128.4, pp. 1547–1584.

Ben-Rephael, A., J. Choi, and I. Goldstein (2021). "Mutual Fund Flows and Fluctuations in Credit and Business Cycles". In: *Journal of Financial Economics* 139.1, pp. 84–108.

Benartzi, S. (2001). "Excessive Extrapolation and the Allocation of 401(k) Accounts to Company Stock". In: *Journal of Finance* 56.5, pp. 1747–1764.

Benartzi, S. and R. H. Thaler (1995). "Myopic Loss Aversion and the Equity Premium Puzzle". In: *Quarterly Journal of Economics* 110.1, pp. 73–92.

Berk, J. B. and R. C. Green (2004). "Mutual Fund Flows and Performance in Rational Markets". In: *American Economic Review* 94.5, pp. 1269–1282.

Birru, J. (2018). "Day of the Week and the Cross-Section of Returns". In: *Journal of Financial Economics* 130.1, pp. 182–214.

Birru, J. and T. R. Young (2022). "Sentiment and Uncertainty". In: *Journal of Financial Economics* 145.2, pp. 461–488.

Bordalo, P., N. Gennaioli, R. La Porta, M. O'Brien, and A. Shleifer (2024). "Long-Term Expectations and Aggregate Fluctuations". In: *NBER Macroeconomics Annual* 38, pp. 311–347.

Bordalo, P., N. Gennaioli, R. La Porta, and A. Shleifer (2025). "Finance Without Exotic Risk". In: *Journal of Financial Economics* 173, p. 104145.

Bordalo, P., N. Gennaioli, and A. Shleifer (2018). "Diagnostic Expectations and Credit Cycles". In: *Journal of Finance* 73.1, pp. 199–227.

Bornstein, R. F. (1989). "Exposure and Affect: Overview and Meta-Analysis of Research". In: *Psychological Bulletin* 106.2, pp. 265–289.

Bowles, B., A. V. Reed, M. C. Ringgenberg, and J. R. Thornock (2024). "Anomaly Time". In: *Journal of Finance* 79.5, pp. 3543–3579.

Boyer, B. H. (2011). "Style-Related Comovement: Fundamentals or Labels?" In: *Journal of Finance* 66.1, pp. 307–332.

Boyer, B. H., T. Mitton, and K. Vorkink (2010). "Expected Idiosyncratic Skewness". In: *Review of Financial Studies* 23.1, pp. 169–202.

Boyer, B. H. and K. Vorkink (2014). "Stock Options as Lotteries". In: *Journal of Finance* 69.4, pp. 1485–1527.

Brunnermeier, M. K. and S. Nagel (2004). "Hedge Funds and the Technology Bubble". In: *Journal of Finance* 59.5, pp. 2013–2040.

Bybee, J. L. (2025). "The Ghost in the Machine: Generating Beliefs with Large Language Models". Working paper, University of Chicago Booth School of Business; earlier version published as arXiv:2305.02823. URL: https://lelandbybee.com/files/LLM.pdf.

Campbell, J. Y. and J. H. Cochrane (1999). "By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior". In: *Journal of Political Economy* 107.2, pp. 205–251.

Campbell, J. Y., J. Hilscher, and J. Szilagyi (2008). "In Search of Distress Risk". In: *Journal of Finance* 63.6, pp. 2899–2939.

Campbell, J. Y. and R. J. Shiller (1988). "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors". In: *Review of Financial Studies* 1.3, pp. 195–228.

Cassella, S. and H. Gulen (2018). "Extrapolation Bias and the Predictability of Stock Returns by Price-Scaled Variables". In: *Review of Financial Studies* 31.11, pp. 4345–4397.

Chang, Y.-C., P.-J. Hsiao, A. Ljungqvist, and K. Tseng (2022). "Testing Disagreement Models". In: *Journal of Finance* 77.4, pp. 2239–2285.

Chen, A. Y. and T. Zimmermann (2022). "Open Source Cross-Sectional Asset Pricing". In: *Critical Finance Review* 11.2. Open Source Asset Pricing project, https://www.openassetpricing.com, pp. 207–264.

Chen, H., B.-H. Hwang, and Z. Peng (2025). "Why Do Investors Like Short-Leg Securities? Evidence from a Textual Analysis of Buy Recommendations". In: *Review of Financial Studies* 38.12, pp. 3729–3767.

Chen, J., Z. Da, and D. Huang (2019). "Arbitrage Trading: The Long and the Short of It". In: *Review of Financial Studies* 32.4, pp. 1608–1646.

Chen, X., L. An, Z. Wang, and J. Yu (2023). "Attention Spillover in Asset Pricing". In: *Journal of Finance* 78.6, pp. 3515–3559.

Chen, Z., B. Liu, H. Wang, Z. Wang, and J. Yu (2025). "Investor Sentiment and the Pricing of Characteristics-Based Factors". In: *Review of Financial Studies* 38.12, pp. 3580–3625.

Chu, Y., D. Hirshleifer, and L. Ma (2020). "The Causal Effect of Limits to Arbitrage on Asset Pricing Anomalies". In: *Journal of Finance* 75.5, pp. 2631–2672.

Chui, A. C. W., S. Titman, and K. C. J. Wei (2010). "Individualism and Momentum around the World". In: *Journal of Finance* 65.1, pp. 361–392.

Cohen, L., K. Diether, and C. Malloy (2013). "Misvaluing Innovation". In: *Review of Financial Studies* 26.1, pp. 18–51.

Cohen, L. and A. Frazzini (2008). "Economic Links and Predictable Returns". In: *Journal of Finance* 63.4, pp. 1977–2011.

Cohen, L. and D. Lou (2012). "Complicated Firms". In: *Journal of Financial Economics* 104.2, pp. 383–400.

Conrad, J., R. F. Dittmar, and E. Ghysels (2013). "Ex Ante Skewness and Expected Stock Returns". In: *Journal of Finance* 68.1, pp. 85–124.

Cooper, M. J., H. Gulen, and M. J. Schill (2008). "Asset Growth and the Cross-Section of Stock Returns". In: *Journal of Finance* 63.4, pp. 1609–1651.

Cui, C., F. W. Li, J. Pang, and D. Xie (2026). "A Behavioral Signaling Explanation for Stock Splits". In: *Financial Management*. Advance online publication, https://doi.org/10.1111/fima.70027.

Cui, C., F. W. Li, and X. Zhang (2021). "Naïve Earnings Growth Extrapolation". Revise and resubmit, Review of Accounting Studies; SSRN Working Paper No. 3897761. URL: https://ssrn.com/abstract=3897761.

Curcuru, S., J. Heaton, D. Lucas, and D. Moore (2004). *Heterogeneity and Portfolio Choice: Theory and Evidence*. Working Paper 10889. NBER.

Da, Z., J. Engelberg, and P. Gao (2011). "In Search of Attention". In: *Journal of Finance* 66.5, pp. 1461–1499.

Da, Z., J. Engelberg, and P. Gao (2015). "The Sum of All FEARS: Investor Sentiment and Asset Prices". In: *Review of Financial Studies* 28.1, pp. 1–32.

Da, Z., U. G. Gurun, and M. Warachka (2014). "Frog in the Pan: Continuous Information and Momentum". In: *Review of Financial Studies* 27.7, pp. 2171–2218.

Daniel, K., D. Hirshleifer, and A. Subrahmanyam (1998). "Investor Psychology and Security Market Under- and Overreactions". In: *Journal of Finance* 53.6, pp. 1839–1885.

Daniel, K. and S. Titman (2006). "Market Reactions to Tangible and Intangible Information". In: *Journal of Finance* 61.4, pp. 1605–1663.

Davies, S. W. (2022). "Speculation Sentiment". In: *Journal of Financial and Quantitative Analysis* 57.7, pp. 2485–2515.

De Bondt, W. F. M. and R. Thaler (1985). "Does the Stock Market Overreact?" In: *Journal of Finance* 40.3, pp. 793–805.

De Giorgi, E. and S. Legg (2012). "A Model of Myopic Loss Aversion with Cross-Sectional Applications". In: *Working Paper*. University of Zurich and University of Technology Sydney.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990). "Noise Trader Risk in Financial Markets". In: *Journal of Political Economy* 98.4, pp. 703–738.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1991). "The Survival of Noise Traders in Financial Markets". In: *Journal of Financial Economics* 31.1, pp. 95–132.

DellaVigna, S. and J. M. Pollet (2007). "Demographics and Industry Returns". In: *American Economic Review* 97.5, pp. 1667–1702.

DellaVigna, S. and J. M. Pollet (2009). "Investor Inattention and Friday Earnings Announcements". In: *Journal of Finance* 64.2, pp. 709–749.

Dhar, R. and N. Zhu (2006). "Up Close and Personal: Investor Sophistication and the Disposition Effect". In: *Management Science* 52.5, pp. 726–740.

Dichev, I. D. (1998). "Is the Risk of Bankruptcy a Systematic Risk?" In: *Journal of Finance* 53.3, pp. 1131–1147.

Dichev, I. D. (2007). "What Are Stock Investors' Actual Historical Returns? Evidence from Dollar-Weighted Returns". In: *American Economic Review* 97.1, pp. 386–401.

Diether, K. B., C. J. Malloy, and A. Scherbina (2002). "Differences of Opinion and the Cross-Section of Stock Returns". In: *Journal of Finance* 57.5, pp. 2113–2141.

Dow, J. and S. R. d. C. Werlang (1992). "Uncertainty Aversion, Risk Aversion, and the Optimal Choice of Portfolio". In: *Econometrica* 60.1, pp. 197–204.

Edelen, R. M., O. S. Ince, and G. B. Kadlec (2016). "Institutional Investors and Stock Return Anomalies". In: *Journal of Financial Economics* 119.3, pp. 472–488.

Edmans, A., A. Fernandez-Perez, A. Garel, and I. Indriawan (2022). "Music Sentiment and Stock Returns Around the World". In: *Journal of Financial Economics* 145.2, pp. 234–254.

Edmans, A., D. Garcia, and Ö. Norli (2007). "Sports Sentiment and Stock Returns". In: *Journal of Finance* 62.4, pp. 1967–1998.

Edwards, W. (1968). "Conservatism in Human Information Processing". In: *Formal Representation of Human Judgment*. Wiley, pp. 17–52.

Ellsberg, D. (1961). "Risk, Ambiguity, and the Savage Axioms". In: *Quarterly Journal of Economics* 75.4, pp. 643–669.

Epstein, L. G. and M. Schneider (2010). "Ambiguity and Asset Markets". In: *Annual Review of Financial Economics* 2, pp. 315–346.

Eraker, B. and M. J. Ready (2015). "Do Investors Overpay for Stocks with Lottery-Like Payoffs? An Examination of the Returns on OTC Stocks". In: *Journal of Financial Economics* 115.2, pp. 261–282.

Fama, E. F. (1998). "Market Efficiency, Long-Term Returns, and Behavioral Finance". In: *Journal of Financial Economics* 49.3, pp. 283–306.

Fama, E. F. and K. R. French (1988). "Dividend Yields and Expected Stock Returns". In: *Journal of Political Economy* 96.1, pp. 246–273.

Fama, E. F. and K. R. French (1993). "Common Risk Factors in the Returns on Stocks and Bonds". In: *Journal of Finance* 48.1, pp. 33–56.

Fama, E. F. and K. R. French (1996). "Multifactor Explanations of Asset Pricing Anomalies". In: *Journal of Finance* 51.1, pp. 55–84.

Fama, E. F. and K. R. French (2004). "The Capital Asset Pricing Model: Theory and Evidence". In: *Journal of Economic Perspectives* 18.3, pp. 25–46.

Fama, E. F. and K. R. French (2015). "A Five-Factor Asset Pricing Model". In: *Journal of Financial Economics* 116.1, pp. 1–22.

Frazzini, A. (2006). "The Disposition Effect and Underreaction to News". In: *Journal of Finance* 61.4, pp. 2017–2046.

Frazzini, A. and O. A. Lamont (2008). "Dumb Money: Mutual Fund Flows and the Cross-Section of Stock Returns". In: *Journal of Financial Economics* 88.2, pp. 299–322.

Frazzini, A. and L. H. Pedersen (2014). "Betting Against Beta". In: *Journal of Financial Economics* 111.1, pp. 1–25.

French, K. R. and J. M. Poterba (1991). "Investor Diversification and International Equity Markets". In: *American Economic Review* 81.2, pp. 222–226.

Friedman, M. (1953). "The Case for Flexible Exchange Rates". In: *Essays in Positive Economics*. Chicago: University of Chicago Press, pp. 157–203.

Froot, K. A. and E. M. Dabora (1999). "How Are Stock Prices Affected by the Location of Trade?" In: *Journal of Financial Economics* 53.2, pp. 189–216.

Genesove, D. and C. Mayer (2001). "Loss Aversion and Seller Behavior: Evidence from the Housing Market". In: *Quarterly Journal of Economics* 116.4, pp. 1233–1260.

George, T. J. and C.-Y. Hwang (2004). "The 52-Week High and Momentum Investing". In: *Journal of Finance* 59.5, pp. 2145–2176.

Giglio, S. and K. Shue (2014). "No News Is News: Do Markets Underreact to Nothing?" In: *Review of Financial Studies* 27.12, pp. 3468–3501.

Gilboa, I. and D. Schmeidler (1989). "Maxmin Expected Utility with Non-Unique Prior". In: *Journal of Mathematical Economics* 18.2, pp. 141–153.

Gilovich, T., R. Vallone, and A. Tversky (1985). "The Hot Hand in Basketball: On the Misperception of Random Sequences". In: *Cognitive Psychology* 17.3, pp. 295–314.

Goldman, E., N. Gupta, and R. D. Israelsen (2024). "Political Polarization in Financial News". In: *Journal of Financial Economics* 155, p. 103816.

Green, T. C. and B.-H. Hwang (2009). "Price-Based Return Comovement". In: *Journal of Financial Economics* 93.1, pp. 37–50.

Green, T. C. and B.-H. Hwang (2012). "Initial Public Offerings as Lotteries: Skewness Perception and First-Day Returns". In: *Management Science* 58.2, pp. 432–444.

Greenwood, R. (2008). "Excess Comovement of Stock Returns: Evidence from Cross-Sectional Variation in Nikkei 225 Weights". In: *Review of Financial Studies* 21.3, pp. 1153–1186.

Greenwood, R. and S. G. Hanson (2013). "Issuer Quality and Corporate Bond Returns". In: *Review of Financial Studies* 26.6, pp. 1483–1525.

Greenwood, R., S. G. Hanson, and L. J. Jin (2024). "Reflexivity in Credit Markets". In: *Journal of Finance* 79.4, pp. 2473–2503.

Greenwood, R., S. G. Hanson, A. Shleifer, and J. A. Sorensen (2022). "Predictable Financial Crises". In: *Journal of Finance* 77.2, pp. 975–1026.

Greenwood, R. and M. Sammon (2025). "The Disappearing Index Effect". In: *Journal of Finance* 80.2, pp. 657–698.

Greenwood, R. and A. Shleifer (2014). "Expectations of Returns and Expected Returns". In: *Review of Financial Studies* 27.3, pp. 714–746.

Greenwood, R., A. Shleifer, and Y. You (2019). "Bubbles for Fama". In: *Journal of Financial Economics* 131.1, pp. 20–43.

Grinblatt, M. and B. Han (2005). "Prospect Theory, Mental Accounting, and Momentum". In: *Journal of Financial Economics* 78.2, pp. 311–339.

Grinblatt, M. and M. Keloharju (2009). "Sensation Seeking, Overconfidence, and Trading Activity". In: *Journal of Finance* 64.2, pp. 549–578.

Gromb, D. and D. Vayanos (2010). "Limits of Arbitrage". In: *Annual Review of Financial Economics* 2, pp. 251–275.

Guo, L., F. W. Li, and K. C. J. Wei (2020). "Security Analysts and Capital Market Anomalies". In: *Journal of Financial Economics* 137.1, pp. 204–230.

Hameed, A. and J. Xie (2019). "Preference for Dividends and Return Comovement". In: *Journal of Financial Economics* 132.1, pp. 103–125.

Han, B., A. Subrahmanyam, and Y. Zhou (2017). "The Term Structure of Credit Spreads, Firm Fundamentals, and Expected Stock Returns". In: *Journal of Financial Economics* 124.1, pp. 147–171.

Han, B., P. Sui, and W. Yang (2026). "Prospect Theory in the Field: Revealed Preferences from Mutual Fund Flows". In: *Journal of Financial Economics* 176, p. 104221.

Harris, L. and E. Gurel (1986). "Price and Volume Effects Associated with Changes in the S&P 500 List: New Evidence for the Existence of Price Pressures". In: *Journal of Finance* 41.4, pp. 815–829.

Harrison, J. M. and D. M. Kreps (1978). "Speculative Investor Behavior in a Stock Market with Heterogeneous Expectations". In: *Quarterly Journal of Economics* 92.2, pp. 323–336.

Harvey, C. R., Y. Liu, and H. Zhu (2016). "... and the Cross-Section of Expected Returns". In: *Review of Financial Studies* 29.1, pp. 5–68.

Hirshleifer, D., K. Hou, S. H. Teoh, and F. Zhang (2004). "Do Investors Overvalue Firms with Bloated Balance Sheets?" In: *Journal of Accounting and Economics* 38.1–3, pp. 297–331.

Hirshleifer, D., P.-H. Hsu, and D. Li (2013). "Innovative Efficiency and Stock Returns". In: *Journal of Financial Economics* 107.3, pp. 632–654.

Hirshleifer, D., S. S. Lim, and S. H. Teoh (2009). "Driven to Distraction: Extraneous Events and Underreaction to Earnings News". In: *Journal of Finance* 64.5, pp. 2289–2325.

Hirshleifer, D. and T. Shumway (2003). "Good Day Sunshine: Stock Returns and the Weather". In: *Journal of Finance* 58.3, pp. 1009–1032.

Ho, S. W. and A. R. Lauwers (2023). "Is There Smart Money? How Information in the Commodity Futures Market Is Priced into the Cross-Section of Stock Returns with Delay". In: *Journal of Financial and Quantitative Analysis* 58.8, pp. 3201–3230.

Hoberg, G. and G. Phillips (2016). "Text-Based Network Industries and Endogenous Product Differentiation". In: *Journal of Political Economy* 124.5, pp. 1423–1465.

Hong, C. Y. and F. W. Li (2019). "The Information Content of Sudden Insider Silence". In: *Journal of Financial and Quantitative Analysis* 54.4, pp. 1499–1538.

Hong, H., T. Lim, and J. C. Stein (2000). "Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies". In: *Journal of Finance* 55.1, pp. 265–295.

Hong, H. and D. A. Sraer (2016). "Speculative Betas". In: *Journal of Finance* 71.5, pp. 2095–2144.

Hong, H. and J. C. Stein (1999). "A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets". In: *Journal of Finance* 54.6, pp. 2143–2184.

Hong, H. and J. C. Stein (2007). "Disagreement and the Stock Market". In: *Journal of Economic Perspectives* 21.2, pp. 109–128.

Hou, K., C. Xue, and L. Zhang (2015). "Digesting Anomalies: An Investment Approach". In: *Review of Financial Studies* 28.3, pp. 650–705.

Hou, K., C. Xue, and L. Zhang (2020). "Replicating Anomalies". In: *Review of Financial Studies* 33.5, pp. 2019–2133.

Huang, D., F. Jiang, J. Tu, and G. Zhou (2015). "Investor Sentiment Aligned: A Powerful Predictor of Stock Returns". In: *Review of Financial Studies* 28.3, pp. 791–837.

Huang, X. (2015). "Thinking Outside the Borders: Investors' Underreaction to Foreign Operations Information". In: *Review of Financial Studies* 28.11, pp. 3109–3152.

Huynh, T. D., F. W. Li, and Y. Xia (2025). "Something in the Air: Does Air Pollution Affect Fund Managers' Carbon Divestment?" In: *Review of Accounting Studies* 30, pp. 2607–2634.

Hvidkjaer, S. (2008). "Small Trades and the Cross-Section of Stock Returns". In: *Review of Financial Studies* 21.3, pp. 1123–1151.

Ikenberry, D., J. Lakonishok, and T. Vermaelen (1995). "Market Underreaction to Open Market Share Repurchases". In: *Journal of Financial Economics* 39.2–3, pp. 181–208.

Israelsen, R. D. (2016). "Does Common Analyst Coverage Explain Excess Comovement?" In: *Journal of Financial and Quantitative Analysis* 51.4, pp. 1193–1229.

Ivković, Z. and S. Weisbenner (2005). "Local Does as Local Is: Information Content of the Geography of Individual Investors' Common Stock Investments". In: *Journal of Finance* 60.1, pp. 267–306.

Jegadeesh, N. (1990). "Evidence of Predictable Behavior of Security Returns". In: *Journal of Finance* 45.3, pp. 881–898.

Jegadeesh, N. and S. Titman (1993). "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency". In: *Journal of Finance* 48.1, pp. 65–91.

Jensen, T. I., B. T. Kelly, and L. H. Pedersen (2023). "Is There a Replication Crisis in Finance?" In: *Journal of Finance* 78.5, pp. 2465–2518.

Jiang, F., J. Lee, X. Martin, and G. Zhou (2019). "Manager Sentiment and Stock Returns". In: *Journal of Financial Economics* 132.1, pp. 126–149.

Jin, Z. and F. W. Li (2024). "Geographic Links and Predictable Returns". In: *Journal of Business Finance & Accounting* 51.7–8, pp. 2239–2274.

Kahneman, D. and A. Tversky (1974). "Judgment under Uncertainty: Heuristics and Biases". In: *Science* 185.4157, pp. 1124–1131.

Kahneman, D. and A. Tversky (1979). "Prospect Theory: An Analysis of Decision Under Risk". In: *Econometrica* 47.2, pp. 263–292.

Kamstra, M. J., L. A. Kramer, and M. D. Levi (2003). "Winter Blues: A SAD Stock Market Cycle". In: *American Economic Review* 93.1, pp. 324–343.

Kindleberger, C. P. (1978). *Manias, Panics, and Crashes: A History of Financial Crises*. Basic Books.

Klibanoff, P., M. Marinacci, and S. Mukerji (2005). "A Smooth Model of Decision Making Under Ambiguity". In: *Econometrica* 73.6, pp. 1849–1892.

Knight, F. H. (1921). *Risk, Uncertainty, and Profit*. Boston: Houghton Mifflin.

Krishnamurthy, A. and T. Muir (2025). "How Credit Cycles across a Financial Crisis". In: *Journal of Finance* 80.3, pp. 1339–1378.

Kuchler, T. and B. Zafar (2019). "Personal Experiences and Expectations about Aggregate Outcomes". In: *American Economic Review* 109.5, pp. 1700–1737.

Kumar, A. (2009). "Who Gambles in the Stock Market?" In: *Journal of Finance* 64.4, pp. 1889–1933.

Lakonishok, J., A. Shleifer, and R. W. Vishny (1994). "Contrarian Investment, Extrapolation, and Risk". In: *Journal of Finance* 49.5, pp. 1541–1578.

Lamont, O. A. and R. H. Thaler (2003). "Can the Market Add and Subtract? Mispricing in Tech Carve-Outs". In: *Journal of Political Economy* 111.2, pp. 227–268.

Lee, C. M. C. and E. C. So (2017). "Uncovering Expected Returns: Information in Analyst Coverage Proxies". In: *Journal of Financial Economics* 124.2, pp. 331–348.

Lee, C. M. C., S. T. Sun, R. Wang, and R. Zhang (2019). "Technological Links and Predictable Returns". In: *Journal of Financial Economics* 132.3, pp. 76–96.

Lehmann, B. N. (1990). "Fads, Martingales, and Market Efficiency". In: *Quarterly Journal of Economics* 105.1, pp. 1–28.

Lewellen, J. (2011). "Institutional Investors and the Limits of Arbitrage". In: *Journal of Financial Economics* 102.1, pp. 62–80.

Li, F. W. (2016). "Macro Disagreement and the Cross-Section of Stock Returns". In: *Review of Asset Pricing Studies* 6.1, pp. 1–45.

Li, F. W. and C. Sun (2022). "Information Acquisition and Expected Returns: Evidence from EDGAR Search Traffic". In: *Journal of Economic Dynamics and Control* 141, p. 104384.

Lopez-Salido, D., J. C. Stein, and E. Zakrajšek (2017). "Credit-Market Sentiment and the Business Cycle". In: *Quarterly Journal of Economics* 132.3, pp. 1373–1426.

Loughran, T. and J. R. Ritter (1995). "The New Issues Puzzle". In: *Journal of Finance* 50.1, pp. 23–51.

Lundeberg, M. A., P. W. Fox, and J. Punćcohaŕ (1994). "Highly Confident but Wrong: Gender Differences and Similarities in Confidence Judgments". In: *Journal of Educational Psychology* 86.1, pp. 114–121.

Maenhout, P. J. (2004). "Robust Portfolio Rules and Asset Pricing". In: *Review of Financial Studies* 17.4, pp. 951–983.

Malmendier, U. and S. Nagel (2011). "Depression Babies: Do Macroeconomic Experiences Affect Risk-Taking?" In: *Quarterly Journal of Economics* 126.1, pp. 373–416.

Malmendier, U. and S. Nagel (2016). "Learning from Inflation Experiences". In: *Quarterly Journal of Economics* 131.1, pp. 53–87.

Mankiw, N. G. and S. P. Zeldes (1991). "The Consumption of Stockholders and Nonstockholders". In: *Journal of Financial Economics* 29.1, pp. 97–112.

McLean, R. D. and J. Pontiff (2016). "Does Academic Research Destroy Stock Return Predictability?" In: *Journal of Finance* 71.1, pp. 5–32.

McLean, R. D., J. Pontiff, and C. Reilly (2025). "Taking Sides on Return Predictability". In: *Journal of Financial Economics* 173, p. 104158.

McNichols, M. and P. C. O'Brien (1997). "Self-Selection and Analyst Coverage". In: *Journal of Accounting Research* 35.2, pp. 167–199.

Mehra, R. and E. C. Prescott (1985). "The Equity Premium: A Puzzle". In: *Journal of Monetary Economics* 15.2, pp. 145–161.

Milgrom, P. and N. Stokey (1982). "Information, Trade and Common Knowledge". In: *Journal of Economic Theory* 26.1, pp. 17–27.

Miller, E. M. (1977). "Risk, Uncertainty, and Divergence of Opinion". In: *Journal of Finance* 32.4, pp. 1151–1168.

Miller, J. B. and A. Sanjurjo (2018). "Surprised by the Hot Hand Fallacy? A Truth of Probability Theory with Implications for the Hot Hand Fallacy". SSRN Working Paper No. 2622995. URL: https://ssrn.com/abstract=2622995.

Mitchell, M., T. Pulvino, and E. Stafford (2002). "Limited Arbitrage in Equity Markets". In: *Journal of Finance* 57.2, pp. 551–584.

Mitton, T. and K. Vorkink (2007). "Equilibrium Underdiversification and the Preference for Skewness". In: *Review of Financial Studies* 20.4, pp. 1255–1288.

Morris, S. (1995). "The Common Prior Assumption in Economic Theory". In: *Economics and Philosophy* 11.2, pp. 227–253.

Moskowitz, T. J. and M. Grinblatt (1999). "Do Industries Explain Momentum?" In: *Journal of Finance* 54.4, pp. 1249–1290.

Nagel, S. (2005). "Short Sales, Institutional Investors and the Cross-Section of Stock Returns". In: *Journal of Financial Economics* 78.2, pp. 277–309.

Nagel, S. (2012). "Evaporating Liquidity". In: *Review of Financial Studies* 25.7, pp. 2005–2039.

Neumann, J. von and O. Morgenstern (1944). *Theory of Games and Economic Behavior*. Princeton: Princeton University Press.

Odean, T. (1998). "Are Investors Reluctant to Realize Their Losses?" In: *Journal of Finance* 53.5, pp. 1775–1798.

Odean, T. (1999). "Do Investors Trade Too Much?" In: *American Economic Review* 89.5, pp. 1279–1298.

Parsons, C. A., R. Sabbatucci, and S. Titman (2020). "Geographic Lead-Lag Effects". In: *Review of Finance* 24.5, pp. 965–1000.

Pirinsky, C. and Q. Wang (2006). "Does Corporate Headquarters Location Matter for Stock Returns?" In: *Journal of Finance* 61.4, pp. 1991–2015.

Pontiff, J. and A. Woodgate (2008). "Share Issuance and Cross-Sectional Returns". In: *Journal of Finance* 63.2, pp. 921–945.

Rapach, D. E., M. C. Ringgenberg, and G. Zhou (2016). "Short Interest and Aggregate Stock Returns". In: *Journal of Financial Economics* 121.1, pp. 46–65.

Rendleman, R. J., C. P. Jones, and H. A. Latané (1982). "Empirical Anomalies Based on Unexpected Earnings and the Importance of Risk Adjustments". In: *Journal of Financial Economics* 10.3, pp. 269–287.

Rietz, T. A. (1988). "The Equity Risk Premium: A Solution". In: *Journal of Monetary Economics* 22.1, pp. 117–131.

Ritter, J. R. (2003). "Behavioral Finance". In: *Pacific-Basin Finance Journal* 11.4, pp. 429–437.

Rosch, E. and B. B. Lloyd (1978). *Cognition and Categorization*. Lawrence Erlbaum.

Scheinkman, J. A. and W. Xiong (2003). "Overconfidence and Speculative Bubbles". In: *Journal of Political Economy* 111.6, pp. 1183–1220.

Schmeidler, D. (1989). "Subjective Probability and Expected Utility without Additivity". In: *Econometrica* 57.3, pp. 571–587.

Schularick, M. and A. M. Taylor (2012). "Credit Booms Gone Bust: Monetary Policy, Leverage Cycles, and Financial Crises, 1870–2008". In: *American Economic Review* 102.2, pp. 1029–1061.

Seasholes, M. S. and N. Zhu (2010). "Individual Investors and Local Bias". In: *Journal of Finance* 65.5, pp. 1987–2010.

Seyhun, H. N. (1998). *Investment Intelligence from Insider Trading*. Cambridge, MA: MIT Press.

Sharpe, W. F. (1964). "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk". In: *Journal of Finance* 19.3, pp. 425–442.

Shen, J., J. Yu, and S. Zhao (2017). "Investor Sentiment and Economic Forces". In: *Journal of Monetary Economics* 86, pp. 1–21.

Shiller, R. J. (1981). "Do Stock Prices Move Too Much to Be Justified by Subsequent Changes in Dividends?" In: *American Economic Review* 71.3, pp. 421–436.

Shleifer, A. (1986). "Do Demand Curves for Stocks Slope Down?" In: *Journal of Finance* 41.3, pp. 579–590.

Shleifer, A. and R. W. Vishny (1997). "The Limits of Arbitrage". In: *Journal of Finance* 52.1, pp. 35–55.

Sloan, R. G. (1996). "Do Stock Prices Fully Reflect Information in Accruals and Cash Flows About Future Earnings?" In: *The Accounting Review* 71.3, pp. 289–315.

Stambaugh, R. F., J. Yu, and Y. Yuan (2012). "The Short of It: Investor Sentiment and Anomalies". In: *Journal of Financial Economics* 104.2, pp. 288–302.

Stambaugh, R. F., J. Yu, and Y. Yuan (2015). "Arbitrage Asymmetry and the Idiosyncratic Volatility Puzzle". In: *Journal of Finance* 70.5, pp. 1903–1948.

Stanovich, K. E., R. F. West, and M. E. Toplak (2013). "Individual Differences in Cognitive Reasoning: A Darwinian Perspective". In: *The Oxford Handbook of Thinking and Reasoning*. Oxford University Press, pp. 665–685.

Sui, P. and B. Wang (2025). "Stakes and Investor Behaviors". In: *Journal of Financial Economics* 172, p. 104146.

Svenson, O. (1981). "Are We All Less Risky and More Skillful than Our Fellow Drivers?" In: *Acta Psychologica* 47.2, pp. 143–148.

Titman, S., K. C. J. Wei, and F. Xie (2004). "Capital Investments and Stock Returns". In: *Journal of Financial and Quantitative Analysis* 39.4, pp. 677–700.

Tversky, A. and D. Kahneman (1992). "Advances in Prospect Theory: Cumulative Representation of Uncertainty". In: *Journal of Risk and Uncertainty* 5.4, pp. 297–323.

Uppal, R. and T. Wang (2003). "Model Misspecification and Underdiversification". In: *Journal of Finance* 58.6, pp. 2465–2486.

Wang, A. Y. and M. Young (2020). "Terrorist Attacks and Investor Risk Preference: Evidence from Mutual Fund Flows". In: *Journal of Financial Economics* 137.2, pp. 491–514.

Wang, H., J. Yan, and J. Yu (2017). "Reference-Dependent Preferences and the Risk–Return Trade-Off". In: *Journal of Financial Economics* 123.2, pp. 395–414.

Wurgler, J. and E. Zhuravskaya (2002). "Does Arbitrage Flatten Demand Curves for Stocks?" In: *Journal of Finance* 57.2, pp. 583–608.

Xiong, W. and J. Yu (2011). "The Chinese Warrants Bubble". In: *American Economic Review* 101.6, pp. 2723–2753.

Yu, J. (2011). "Disagreement and Return Predictability of Stock Portfolios". In: *Journal of Financial Economics* 99.1, pp. 162–183.

Zajonc, R. B. (1968). "Attitudinal Effects of Mere Exposure". In: *Journal of Personality and Social Psychology* 9.2p2, pp. 1–27.